

APPENDIX 2

Part 3 of 3

providing power to operate the third control circuit and third switch from across the first switch.

7. – 10. (Canceled)

11. (Previously Presented) The method of claim 6 further comprising:
providing a second control circuit to operate the second switch;
providing power to operate the second control circuit and second switch from across the fourth switch;
providing a fourth control circuit to operate the fourth switch; and
providing power to operate the fourth control circuit and fourth switch from across the second switch.

12. – 15. (Cancelled).

16. (Previously Presented) The method of claim 1 further comprising:
forming a second series circuit with a first capacitor connected to a second capacitor at a second junction, the second series circuit having two end terminals;
connecting the second terminal of the secondary winding to the second junction; and
connecting the second series circuit across the output with each of the two end terminals connected to a respective terminal of the output.

17. (Previously Presented) The method of claim 16 further comprising:
sensing across each of the switches, a respective voltage polarity; and
turning each of the switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.

18. (Previously Presented) A method of providing fault tolerance in a power converter comprising:

supplying an alternating current from a secondary winding to an input of a full-bridge rectifier circuit having switches,

operating the switches as rectifiers to provide full-wave rectification of the alternating current at an output of the full-bridge rectifier circuit,

connecting the output of the full-bridge rectifier circuit to an output of the power converter,

monitoring selected circuit conditions;

determining from the selected circuit conditions whether a predetermined fault is present, disabling the switches in response to a determination that the predetermined fault is present, and

providing a first one of the switches connected in series with a second one of the switches across the output of the full-bridge rectifier circuit with the first switch connected to a first terminal of the output of the full-bridge rectifier circuit and the second switch connected to a second terminal of the output of the full-bridge rectifier circuit; and

sensing an output of the second switch and preventing the first switch from turning ON if the sensing indicates that the second switch is not open; and

sensing an output of the second switch and preventing the second switch from turning ON if the sensing indicates that the first switch is not open.

19. (Previously Presented) The method of claim 18 further comprising:
sensing across each of the first and second switches, a respective voltage polarity; and
turning each of the first and second switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.

20. (Previously Presented) The method of claim 18 further comprising:
providing a third one of the switches and a fourth one of the switches each adapted to block current in an OFF state in at least one direction;

forming a second series circuit with the third switch connected to the fourth switch at a second junction, the second series circuit having two end terminals;

connecting the second terminal of the secondary winding to the second junction; and

connecting the second series circuit across the output of the full-bridge rectifier circuit with each of the two end terminals connected to a respective terminal of the output of the full-bridge rectifier circuit and with each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output of the full-bridge rectifier circuit, wherein a short across one of the third or fourth switches does not short the output of the full-bridge rectifier circuit.

21. (Previously Presented) The method of claim 20 further comprising:

preventing the third switch from turning ON if the fourth switch fails to open; and

preventing the fourth switch from turning ON if the third switch fails to open.

22. (Previously Presented) The method of claim 20 further comprising:

sensing across each of the third and fourth switches, a respective voltage polarity; and

turning each of the third and fourth switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.

23. (Previously Presented) The method of claim 20 further comprising:

providing a first control circuit to operate the first switch;

providing power to operate the first control circuit and first switch from across the third switch;

providing a third control circuit to operate the third switch; and

providing power to operate the third control circuit and third switch from across the first switch.

24. – 27. (Cancelled)

28. (Previously Presented) The method of claim 20 further comprising:
providing a second control circuit to operate the second switch;
providing power to operate the second control circuit and second switch from across the fourth switch;
providing a fourth control circuit to operate the fourth switch; and
providing power to operate the fourth control circuit and fourth switch from across the second switch.

29. – 43. (Cancelled).

44. (Previously Presented) A fault-tolerant converter comprising:
a transformer having a primary winding and a secondary winding, the secondary winding having a first terminal and a second terminal;
a first series circuit having a first switch and a second switch each adapted to block current flow in an OFF state in at least one direction, the first switch being connected to the second switch at a first junction, the first series circuit having two end terminals, the first junction being connected to the first terminal of the secondary winding, the first series circuit being connected across an output of the converter with each of the two end terminals connected to a respective terminal of the output and with each of the first and second switches being arranged in the OFF state to block current flow through the first series circuit in a direction that would short the output, wherein a short across one of the first or second switches does not short the output; and
a switch control circuit configured to sense an output of the second switch and prevent the first switch from turning ON if the sensing indicates that the second switch is not open, and to sense an output of the first switch and prevent the second switch from turning ON if the sensing indicates that the first switch is not open.

45. (Previously Presented) The apparatus of claim 44 wherein the switch control circuit is configured to sense across each of the first and second switches, a respective voltage polarity,

and to turn each of the first and second switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.

46. (Previously Presented) The apparatus of claim 44, comprising a second series circuit that comprises a third switch and a fourth switch each adapted to block current in an OFF state in at least one direction, the third switch being connected to the fourth switch at a second junction, the second series circuit having two end terminals, the second terminal of the secondary winding being connected to the second junction, the second series circuit being connected across the output with each of the two end terminals connected to a respective terminal of the output and with each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output, wherein a short across one of the third or fourth switches does not short the output.

47. (Previously Presented) The apparatus of claim 46 wherein the switch control circuit comprises a first control circuit and a third control circuit, the first control circuit operates the first switch, the third control circuit operates the third switch, the first control circuit and the first switch are powered using a voltage across the third switch, and the third control circuit and third switch are powered using a voltage across the first switch.

48. – 51. (Cancelled).

52. (Previously Presented) The apparatus of claim 47 wherein the switch control circuit comprises a second control circuit and a fourth control circuit, the second control circuit operates the second switch, the fourth control circuit operates the fourth switch, the second control circuit and the second switch are powered using a voltage across the fourth switch, and the fourth control circuit and fourth switch are powered using a voltage across the second switch.

53. – 56. (Cancelled).

57. (Previously Presented) The apparatus of claim 44, comprising a second series circuit having a first capacitor connected to a second capacitor at a second junction that is connected to the second terminal of the secondary winding, the second series circuit having two end terminals connected across the output of the converter with each of the two end terminals connected to a respective terminal of the output of the converter.

58. (Previously Presented) The apparatus of claim 57 wherein the switch control circuit is configured to sense across each of the first and second switches, a respective voltage polarity, and to turn each of the first and second switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.

59. – 88. (Cancelled).

89. (Previously Presented) A method of delivering power to an output, comprising:
providing a transformer having a secondary winding, the secondary winding having a first terminal and a second terminal;
providing a first switch, a second switch, a third switch and a fourth switch, each adapted to block current flow in an OFF state in at least one direction;
forming a first series circuit with the first switch connected to the second switch at a first junction, the first series circuit having two end terminals;
forming a second series circuit with the third switch connected to the fourth switch at a second junction, the second series circuit having two end terminals;
connecting the first terminal of the secondary winding to the first junction and the second terminal of the secondary winding to the second junction;
connecting the first series circuit across the output with each of the two end terminals connected to a respective terminal of the output, sensing outputs of the first and second switches, arranging the first switch in the OFF state if the sensing indicates that the second switch is not

open, and arranging the second switch in the OFF state if the sensing indicates that the first switch is not open to block current flow through the first series circuit in a direction that would short the output, wherein a short across one of the first or second switch does not short the output;

connecting the second series circuit across the output with each of the two end terminals connected to a respective terminal of the output, sensing outputs of the third and fourth switches, arranging the third switch in the OFF state if the sensing indicates that the fourth switch is not open, and arranging the fourth switch in the OFF state if the sensing indicates that the third switch is not open to block current flow through the second series circuit in a direction that would short the output, wherein a short across one of the third or fourth switch does not short the output;

converting power via the transformer to deliver a unipolar voltage of five Volts or less to the output.

90. (Previously Presented) The method of claim 89 wherein the unipolar output is approximately 1 Volt.

91. (New) The method of claim 6 further comprising:
providing a common-source dual MOSFET semiconductor device including a first MOSFET and a third MOSFET integrated on a semiconductor die;
providing a common source terminal for connecting to the first and third MOSFETs in the common-source dual MOSFET semiconductor device; and
using the first MOSFET and the third MOSFET of the common-source dual MOSFET semiconductor device for the first and third switches.

92. (New) The method of claim 91 further comprising integrating the first and third control circuits in the common-source dual MOSFET semiconductor device.

93. (New) The method of claim 92 further comprising:

providing a plurality of individual first and third MOSFET elements on the semiconductor die;

arranging the individual first and third MOSFET elements in an alternating pattern on the semiconductor die;

connecting the first MOSFET elements together to form the first MOSFET;

connecting the third MOSFET elements together to form the third MOSFET; and

using a common source metallization to interconnect the source terminals of the first and third MOSFET elements.

94. (New) The method of claim 93 wherein the common-source dual MOSFET semiconductor device has exactly three terminals.

95. (New) The method of claim 11 further comprising:

providing a first common-source dual MOSFET semiconductor device including a first MOSFET and a third MOSFET integrated on a first semiconductor die;

providing a common source terminal for connecting to the first and third MOSFETs in the first common-source dual MOSFET semiconductor device;

using the first MOSFET and the third MOSFET of the first common-source dual MOSFET semiconductor device for the first and third switches;

providing a second common-source dual MOSFET semiconductor device including a second MOSFET and a fourth MOSFET integrated on a second semiconductor die;

providing a common source terminal for connecting to the second and fourth MOSFETs in the second common-source dual MOSFET semiconductor device;

using the second MOSFET and the fourth MOSFET of the second common-source dual MOSFET semiconductor device for the second and fourth switches.

96. (New) The method of claim 95 further comprising:

integrating the first and third control circuits in the first common-source dual MOSFET semiconductor device; and

integrating the second and fourth control circuits in the second common-source dual MOSFET semiconductor device.

97. (New) The method of claim 96 further comprising:
providing a plurality of individual first and third MOSFET elements on the first semiconductor die;
arranging the individual first and third MOSFET elements in an alternating pattern on the first semiconductor die;
connecting the first MOSFET elements together to form the first MOSFET;
connecting the third MOSFET elements together to form the third MOSFET;
using a common source metallization to interconnect the source terminals of the first and third MOSFET elements;
providing a plurality of individual second and fourth MOSFET elements on the second semiconductor die;
arranging the individual second and fourth MOSFET elements in an alternating pattern on the second semiconductor die;
connecting the second MOSFET elements together to form the second MOSFET;
connecting the fourth MOSFET elements together to form the fourth MOSFET; and
using a common source metallization to interconnect the source terminals of the second and fourth MOSFET elements.

98. (New) The method of claim 97 wherein the first and second common-source dual MOSFET semiconductor devices each have exactly three terminals.

99. (New) The method of claim 20 further comprising:
providing a common-source dual MOSFET semiconductor device including a first MOSFET and a third MOSFET integrated on a semiconductor die;
providing a common source terminal for connecting to the first and third MOSFETs in the common-source dual MOSFET semiconductor device; and

using the first MOSFET and the third MOSFET of the common-source dual MOSFET semiconductor device for the first and third switches.

100. (New) The method of claim 99 further comprising integrating the first and third control circuits in the common-source dual MOSFET semiconductor device.

101. (New) The method of claim 99 further comprising:
providing a plurality of individual first and third MOSFET elements on the semiconductor die;
arranging the individual first and third MOSFET elements in an alternating pattern on the semiconductor die;
connecting the first MOSFET elements together to form the first MOSFET;
connecting the third MOSFET elements together to form the third MOSFET; and
using a common source metallization to interconnect the source terminals of the first and third MOSFET elements.

102. (New) The method of claim 99 wherein the common-source dual MOSFET semiconductor device has exactly three terminals.

103. (New) The method of claim 20 further comprising:
providing a first common-source dual MOSFET semiconductor device including a first MOSFET and a third MOSFET integrated on a first semiconductor die;
providing a common source terminal for connecting to the first and third MOSFETs in the first common-source dual MOSFET semiconductor device;
using the first MOSFET and the third MOSFET of the first common-source dual MOSFET semiconductor device for the first and third switches;
providing a second common-source dual MOSFET semiconductor device including a second MOSFET and a fourth MOSFET integrated on a second semiconductor die;
providing a common source terminal for connecting to the second and fourth MOSFETs in the second common-source dual MOSFET semiconductor device; and

using the second MOSFET and the fourth MOSFET of the second common-source dual MOSFET semiconductor device for the second and fourth switches.

104. (New) The method of claim 103 further comprising:
integrating the first and third control circuits in the first common-source dual MOSFET semiconductor device; and
integrating the second and fourth control circuits in the second common-source dual MOSFET semiconductor device.

105. (New) The method of claim 103 further comprising:
providing a plurality of individual first and third MOSFET elements on the first semiconductor die;
arranging the individual first and third MOSFET elements in an alternating pattern on the first semiconductor die;
connecting the first MOSFET elements together to form the first MOSFET;
connecting the third MOSFET elements together to form the third MOSFET;
using a common source metallization to interconnect the source terminals of the first and third MOSFET elements;
providing a plurality of individual second and fourth MOSFET elements on the second semiconductor die;
arranging the individual second and fourth MOSFET elements in an alternating pattern on the second semiconductor die;
connecting the second MOSFET elements together to form the second MOSFET;
connecting the fourth MOSFET elements together to form the fourth MOSFET; and
using a common source metallization to interconnect the source terminals of the second and fourth MOSFET elements.

106. (New) The method of claim 103 wherein the first and second common-source dual MOSFET semiconductor devices each have exactly three terminals.

107. (New) The apparatus of claim 46, comprising a common-source dual MOSFET semiconductor device having a first MOSFET and a third MOSFET that are integrated on a semiconductor die, the first MOSFET is used for the first switch, the third MOSFET is used for the third switch, and the common-source dual MOSFET semiconductor device has a source terminal connected to the source of the first MOSFET and the source of the third MOSFET.

108. (New) The apparatus of claim 107 wherein the switch control circuit comprises a first control circuit and a third control circuit, the first control circuit operates the first switch, the third control circuit operates the third switch, the first control circuit and the first switch are powered using a voltage across the third switch, the third control circuit and the third switch are powered using a voltage across the first switch, and the first and third control circuits are integrated in the semiconductor die of the common-source dual MOSFET semiconductor device.

109. (New) The apparatus of claim 108 wherein the first MOSFET comprises a plurality of individual first MOSFET elements on the semiconductor die, the third MOSFET comprises a plurality of individual third MOSFET elements on the semiconductor die, the individual first and third MOSFET elements are arranged in an alternating pattern on the semiconductor die, and a common source metallization interconnects the source terminals of the first and third MOSFET elements.

110. (New) The apparatus of claim 109 wherein the common-source dual MOSFET semiconductor device has exactly three terminals that includes the source terminal, a terminal connected to the drain of the first MOSFET, and a terminal connected to the drain of the third MOSFET.

111. (New) The apparatus of claim 52, comprising:
a first common-source dual MOSFET semiconductor device that comprises a first MOSFET and a third MOSFET integrated on a first semiconductor die, the first common-source dual MOSFET semiconductor device comprising a common source terminal for connecting to

the first and third MOSFETs, wherein the first MOSFET and the third MOSFET are used for the first and third switches; and

a second common-source dual MOSFET semiconductor device that comprises a second MOSFET and a fourth MOSFET integrated on a second semiconductor die, the second common-source dual MOSFET semiconductor device comprising a common source terminal for connecting to the second and fourth MOSFETs, wherein the second MOSFET and the fourth MOSFET are used for the second and fourth switches.

112. (New) The apparatus of claim 111 wherein the first and third control circuits are integrated on the first semiconductor die, and the second and fourth control circuits are integrated on the second semiconductor die.

113. (New) The apparatus of claim 112 wherein the first MOSFET comprises a plurality of individual first MOSFET elements on the first semiconductor die, the third MOSFET comprises a plurality of individual third MOSFET elements on the first semiconductor die, the individual first and third MOSFET elements are arranged in an alternating pattern on the first semiconductor die, a common source metallization interconnects the source terminals of the first and third MOSFET elements, the second MOSFET comprises a plurality of individual second MOSFET elements on the second semiconductor die, the individual second and fourth MOSFET elements are arranged in an alternating pattern on the second semiconductor die, and a common source metallization interconnects the source terminals of the second and fourth MOSFET elements.

114. (New) The apparatus of claim 113 wherein the first common-source dual MOSFET semiconductor device has exactly three terminals that includes the common source terminal, a terminal connected to the drain of the first MOSFET, and a terminal connected to the drain of the third MOSFET, and

the second common-source dual MOSFET semiconductor device has exactly three terminals that include the common source terminal, a terminal connected to the drain of the second MOSFET, and a terminal connected to the drain of the fourth MOSFET.

First Named Inventor : Patrizio Vinciarelli
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Attorney Docket: 00614-0166001

REMARKS

Claim 1 has been amended. New claims 91-144 have been added.

Claims 91-94 correspond to original claims 7-10, which depend on allowable claim 1.

Claims 95-98 correspond to original claims 12-15, which depend on allowable claim 1.

Claims 99-102 correspond to original claims 24-27, which depend on allowable claim 18.

Claims 103-106 correspond to original claims 29-32, which depend on allowable claim 18.

Claims 107-110 correspond to original claims 48-51, which depend on allowable claim 44.

Claims 111-114 correspond to original claims 53-56, which depend on allowable claim 44.

Applicant asks that all claims be allowed in view of the amendment to the claims. Please apply any necessary charges or credits to Deposit Account 06-1050, referencing the above attorney docket number.

Respectfully submitted,

Date: March 3, 2015 _____

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NOTICE OF ALLOWANCE AND FEE(S) DUE

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PAPER NUMBER

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DATE MAILED: 02/20/2015

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13/044,063

03/09/2011

Patrizio Vinciarelli

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1081

TITLE OF INVENTION: FAULT TOLERANT POWER CONVERTER

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	05/20/2015

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

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(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/044,063	03/09/2011	Patrizio Vinciarelli	00614-0166001	1081

TITLE OF INVENTION: FAULT TOLERANT POWER CONVERTER

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	05/20/2015

EXAMINER	ART UNIT	CLASS-SUBCLASS
LEE, JYE-JUNE	2838	363-017000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363). ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached. ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.	2. For printing on the patent front page, list (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____ (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____ 3 _____
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3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

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4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

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☐ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

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- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
☐ Applicant asserting small entity status. See 37 CFR 1.27
☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____ Date _____
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/044,063	03/09/2011	Patrizio Vinciarelli	00614-0166001	1081
26214 7590 02/20/2015 FISH & RICHARDSON P.C. P.O. BOX 1022 MINNEAPOLIS, MN 55440-1022			EXAMINER LEE, JYE-JUNE	
			ART UNIT 2838	PAPER NUMBER
DATE MAILED: 02/20/2015				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 13/044,063	Applicant(s) VINCIARELLI, PATRIZIO	
	Examiner JYE-JUNE LEE	Art Unit 2838	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to the amendment filed on 10/23/2014.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1-6,11,16-23,28,44-47,52,57,58,89 and 90. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|--|--|
| 1. ☐ Notice of References Cited (PTO-892) | 5. ☒ Examiner's Amendment/Comment |
| 2. ☐ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date _____ | 6. ☒ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | 7. ☐ Other _____. |
| 4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date _____. | |

/ADOLF BERHANE/
Primary Examiner, Art Unit 2838

DETAILED ACTION

1. The present application is being examined under the pre-AIA first to invent provisions.
2. This action is in response to the amendment filed on 10/23/2014.

EXAMINER'S AMENDMENT

3. An examiner's amendment to the record appears below. Should the changes and/or additions be unacceptable to applicant, an amendment may be filed as provided by 37 CFR 1.312. To ensure consideration of such an amendment, it MUST be submitted no later than the payment of the issue fee.

Authorization for this examiner's amendment was given by Applicants representative Rex Huang on 02/10/2015.

The application has been amended as follows:

Regarding the **Abstract**, the abstract is amended to read as follows.

Low-voltage outputs are provided by full-bridge rectification using controlled switches with fault detection monitoring of circuit conditions and disabling switches upon detection of a fault to decouple the converter from the system. Common-source dual MOSFET devices include elements arranged in alternating patterns on the die. Common-source dual synchronous rectifiers include control circuitry powered from the voltage across the complementary switch. A DC-to-DC transformer converts power using a fixed voltage transformation ratio. A clamp phase may be used to reduce power

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losses, control the output resistance, effectively regulate the voltage transformation ratio, provide narrow band output regulation, and control the rate of change of output voltage.

A new point of load converter includes input driver circuitry removed from and output circuitry located at the point of load, with a transformer located near the output circuit and an AC bus between the driver circuit and the primary winding of the transformer.

Regarding the Claims, **claims 7-10, 12-15, 24-27, 29-43, 48-51, 53-56, 59-88** are cancelled.

Claims 1, 18, 42, 44, and 89 are amended so the claims would read as follows.

1. A method of converting power via a transformer for supplying a low voltage output comprising:

providing a secondary winding having a first terminal and a second terminal;

providing a first switch and a second switch each adapted to block current flow in an OFF state in at least one direction;

forming a first series circuit with the first switch connected to the second switch at a first junction, the first series circuit having two end terminals;

connecting the first terminal of the secondary winding to the first junction;

connecting the first series circuit across an output with each of the two end terminals connected to a respective terminal of the output and with each of the first and second switches being arranged in the OFF state to block current flow through the first

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series circuit in a direction that would short the output, wherein a short across one of the first or second switches does not short the output;

sensing an output of the second switch and preventing the first switch from turning ON if the sensing indicates that the second switch is not open; and

sensing an output of the second switch and preventing the second switch from turning ON if the sensing indicates that the first switch is not open.

18. A method of providing fault tolerance in a power converter comprising:

supplying an alternating current from a secondary winding to an input of a full-bridge rectifier circuit having switches,

operating the switches as rectifiers to provide full-wave rectification of the alternating current at an output of the full-bridge rectifier circuit,

connecting the output of the full-bridge rectifier circuit to an output of the power converter,

monitoring selected circuit conditions;

determining from the selected circuit conditions whether a predetermined fault is present,

disabling the switches in response to a determination that the predetermined fault is present, and

providing a first one of the switches connected in series with a second one of the switches across the output of the full-bridge rectifier circuit with the first switch connected to a first terminal of the output of the full-bridge rectifier circuit and the

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second switch connected to a second terminal of the output of the full-bridge rectifier circuit; and

sensing an output of the second switch and preventing the first switch from turning ON if the sensing indicates that the second switch is not open; and

sensing an output of the second switch and preventing the second switch from turning ON if the sensing indicates that the first switch is not open.

42. Cancelled.

44. A fault-tolerant converter comprising:

a transformer having a primary winding and a secondary winding, the secondary winding having a first terminal and a second terminal;

a first series circuit having a first switch and a second switch each adapted to block current flow in an OFF state in at least one direction, the first switch being connected to the second switch at a first junction, the first series circuit having two end terminals, the first junction being connected to the first terminal of the secondary winding, the first series circuit being connected across an output of the converter with each of the two end terminals connected to a respective terminal of the output and with each of the first and second switches being arranged in the OFF state to block current flow through the first series circuit in a direction that would short the output, wherein a short across one of the first or second switches does not short the output; and

a switch control circuit configured to sense an output of the second switch and prevent the first switch from turning ON if the sensing indicates that the second switch is not open, and to sense an output of the first switch and prevent the second switch from turning ON if the sensing indicates that the first switch is not open.

89. A method of delivering power to an output, comprising:

providing a transformer having a secondary winding, the secondary winding having a first terminal and a second terminal;

providing a first switch, a second switch, a third switch and a fourth switch, each adapted to block current flow in an OFF state in at least one direction;

forming a first series circuit with the first switch connected to the second switch at a first junction, the first series circuit having two end terminals;

forming a second series circuit with the third switch connected to the fourth switch at a second junction, the second series circuit having two end terminals;

connecting the first terminal of the secondary winding to the first junction and the second terminal of the secondary winding to the second junction;

connecting the first series circuit across the output with each of the two end terminals connected to a respective terminal of the output, sensing outputs of the first and second switches, arranging the first switch in the OFF state if the sensing indicates that the second switch is not open, and arranging the second switch in the OFF state if the sensing indicates that the first switch is not open to block current flow through the

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first series circuit in a direction that would short the output, wherein a short across one of the first or second switch does not short the output;

connecting the second series circuit across the output with each of the two end terminals connected to a respective terminal of the output, sensing outputs of the third and fourth switches, arranging the third switch in the OFF state if the sensing indicates that the fourth switch is not open, and arranging the fourth switch in the OFF state if the sensing indicates that the third switch is not open to block current flow through the second series circuit in a direction that would short the output, wherein a short across one of the third or fourth switch does not short the output;

converting power via the transformer to deliver a unipolar voltage of five Volts or less to the output.

Reasons For Allowance

4. **Claims 1-6, 11, 16-23, 28, 44-47, 52, 57, 58, 89, and 90** are allowed.

The following is an examiner's statement of reasons for allowance:

Regarding **claim 1**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which "...sensing an output of the second switch and preventing the first switch from turning ON if the sensing indicates that the second

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switch is not open; and sensing an output of the second switch and preventing the second switch from turning ON if the sensing indicates that the first switch is not open.”.

Regarding **claim 18**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which “...sensing an output of the second switch and preventing the first switch from turning ON if the sensing indicates that the second switch is not open; and sensing an output of the second switch and preventing the second switch from turning ON if the sensing indicates that the first switch is not open.”.

Regarding **claim 44**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which “...a switch control circuit configured to sense an output of the second switch and prevent the first switch from turning ON if the sensing indicates that the second switch is not open, and to sense an output of the first switch and prevent the second switch from turning ON if the sensing indicates that the first switch is not open.”.

Regarding **claim 89**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which “...sensing outputs of the first and second switches, arranging the first switch in the OFF state if the sensing indicates that the second switch is not open, and arranging the second switch in the OFF state if the sensing indicates that the first switch is not open...sensing outputs of the third and fourth switches, arranging the third switch in the OFF state if the sensing indicates that the fourth switch is not open, and arranging the fourth switch in the OFF state if the sensing indicates that the third switch is not open...”.

The aforementioned limitations in combination with all remaining limitations of the

Art Unit: 2838

respective claims are believed to render the aforementioned indicated claim and any dependent claims thereof patentable over the art of record.

5. Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to JYE-JUNE LEE whose telephone number is (571)270-7726. The examiner can normally be reached on Monday to Thursday 9:00 AM to 7:30 PM EST.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Monica Lewis can be reached on 571-272-1838. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Art Unit: 2838

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/JYE-JUNE LEE/

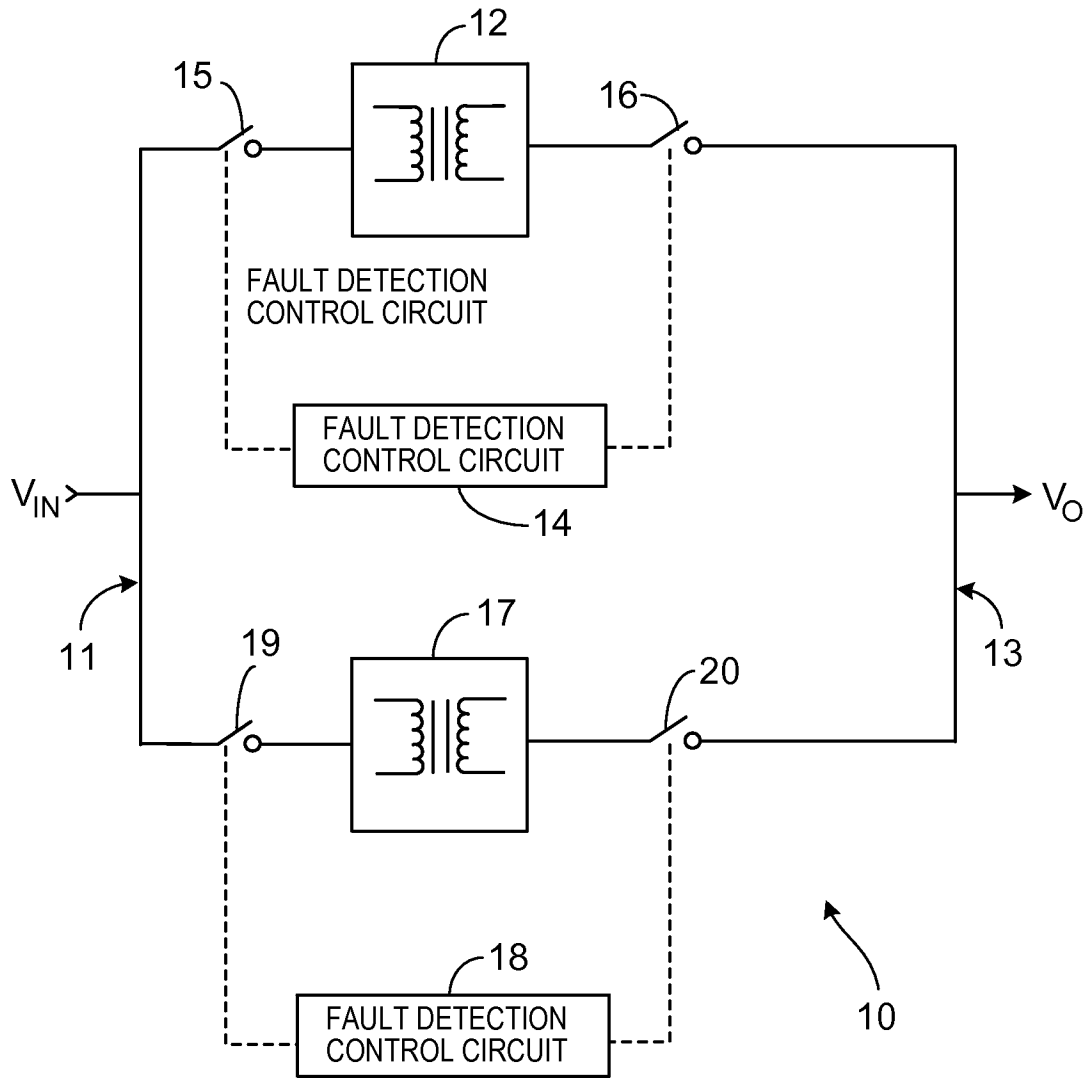
Examiner, Art Unit 2838

/ADOLF BERHANE/

Primary Examiner, Art Unit 2838

OK TO ENTER: /J.L./

02/08/2015



(PRIOR ART)

FIG. 1



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

26214 7590 06/05/2015
FISH & RICHARDSON P.C.
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

LEE, JYE-JUNE

ART UNIT

PAPER NUMBER

2838

DATE MAILED: 06/05/2015

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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13/044,063

03/09/2011

Patrizio Vinciarelli

00614-0166001

1081

TITLE OF INVENTION: FAULT TOLERANT POWER CONVERTER

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	09/08/2015

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

26214 7590 06/05/2015
FISH & RICHARDSON P.C.
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/044,063	03/09/2011	Patrizio Vinciarelli	00614-0166001	1081

TITLE OF INVENTION: FAULT TOLERANT POWER CONVERTER

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	09/08/2015

EXAMINER	ART UNIT	CLASS-SUBCLASS
LEE, JYE-JUNE	2838	363-017000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363). ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached. ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.	2. For printing on the patent front page, list (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____ (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____ 3 _____
---	---

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
☐ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
☐ Payment by credit card. Form PTO-2038 is attached.
☐ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
☐ Applicant asserting small entity status. See 37 CFR 1.27
☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____ Date _____
 Typed or printed name _____ Registration No. _____



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UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/044,063	03/09/2011	Patrizio Vinciarelli	00614-0166001	1081

26214 7590 06/05/2015
FISH & RICHARDSON P.C.
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

LEE, JYE-JUNE

ART UNIT	PAPER NUMBER
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2838

DATE MAILED: 06/05/2015

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

DETAILED ACTION

1. The present application is being examined under the pre-AIA first to invent provisions.
2. This action is in response to the RCE filed on 05/18/2015.

Continued Examination Under 37 CFR 1.114

3. A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after allowance or after an Office action under *Ex Parte Quayle*, 25 USPQ 74, 453 O.G. 213 (Comm'r Pat. 1935). Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, prosecution in this application has been reopened pursuant to 37 CFR 1.114. Applicant's submission filed on 05/18/2015 has been entered.

Reasons For Allowance

4. **Claims 1-6, 11, 16-23, 28, 44-47, 52, 57, 58, 89, 90-114** are allowed.

The following is an examiner's statement of reasons for allowance:

Regarding **claim 1**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which "...sensing an output of the second switch and preventing the first switch from turning ON if the sensing indicates that the second switch is not open; and sensing an output of the second switch and preventing the second switch from turning ON if the sensing indicates that the first switch is not open.".

Regarding **claim 18**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which "...sensing an output of the second switch and preventing the first switch from turning ON if the sensing indicates that the second switch is not open; and sensing an output of the second switch and preventing the second switch from turning ON if the sensing indicates that the first switch is not open.".

Regarding **claim 44**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which "...a switch control circuit configured to sense an output of the second switch and prevent the first switch from turning ON if the sensing indicates that the second switch is not open, and to sense an output of the first switch and prevent the second switch from turning ON if the sensing indicates that the first switch is not open.".

Regarding **claim 89**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which "...sensing outputs of the first and second switches, arranging the first switch in the OFF state if the sensing indicates that the second switch is not open, and arranging the second switch in the OFF state if the sensing indicates that the first switch is not open...sensing outputs of the third and fourth switches, arranging the third switch in the OFF state if the sensing indicates that the fourth switch is not open, and arranging the fourth switch in the OFF state if the sensing indicates that the third switch is not open...".

The aforementioned limitations in combination with all remaining limitations of the respective claims are believed to render the aforementioned indicated claim and any dependent claims thereof patentable over the art of record.

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5. Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to JYE-JUNE LEE whose telephone number is (571)270-7726. The examiner can normally be reached on Monday to Thursday 9:00 AM to 7:30 PM EST.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Monica Lewis can be reached on 571-272-1838. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Art Unit: 2838

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/JYE-JUNE LEE/

Examiner, Art Unit 2838

/ADOLF BERHANE/

Primary Examiner, Art Unit 2838

Notice of Allowability	Application No. 13/044,063	Applicant(s) VINCIARELLI, PATRIZIO	
	Examiner JYE-JUNE LEE	Art Unit 2838	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to the RCE filed on 05/18/2015.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1-6, 11, 16-23, 28, 44-47, 52, 57, 58 and 89-114. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying Indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|--|--|
| 1. ☐ Notice of References Cited (PTO-892) | 5. ☐ Examiner's Amendment/Comment |
| 2. ☒ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date <u>05/18/2015</u> | 6. ☒ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | 7. ☐ Other _____. |
| 4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date _____. | |

/ADOLF BERHANE/
Primary Examiner, Art Unit 2838

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	2838
Serial No. :	13/044,063	Examiner :	Jye-June Lee
Filed :	March 9, 2011	Conf. No. :	1081
Title :	FAULT TOLERANT POWER CONVERTER		

MAIL STOP RCE

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Please consider the references listed on the enclosed PTO-SB-08 or Disclosure Form.

Cited U.S. patents will be provided on request.

This filing is being made with the filing of a Request for Continued Examination. No fee is required.

Apply any necessary charges or credits to deposit account 06-1050, referencing the above attorney docket number.

Respectfully submitted,

Date: May 18, 2015_____

/Rex I. Huang/_____
Rex I. Huang
Reg. No. 57,661

Customer Number 26214
Fish & Richardson P.C.
Telephone: (617) 542-5070
Facsimile: (877) 769-7945

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0166001	Application No. 13/044,063
	Applicant Patrizio Vinciarelli	
	Filing Date March 9, 2011	Group Art Unit 2838

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
/JL/	1	6,031,743	02/29/2000	Carpenter et al.			
/JL/	2	6,035,261	03/07/2000	Carpenter et al.			
/JL/	3	6,275,958	08/14/2001	Carpenter et al.			
/JL/	4	6,466,458	10/15/2002	Zhang et al.			

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No
	5							

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	6	

Examiner Signature /JYE-JUNE LEE/	Date Considered 05/28/2015
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EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.



UNITED STATES PATENT AND TRADEMARK OFFICE

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United States Patent and Trademark Office
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P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/044,063	03/09/2011	Patrizio Vinciarelli	00614-0166001	1081

26214 7590 05/23/2014
FISH & RICHARDSON P.C.
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

LEE, JYE-JUNE

ART UNIT	PAPER NUMBER
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2838

NOTIFICATION DATE	DELIVERY MODE
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05/23/2014

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

PATDOCTC@fr.com

Office Action Summary	Application No. 13/044,063	Applicant(s) VINCIARELLI, PATRIZIO	
	Examiner JYE-JUNE LEE	Art Unit 2838	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
 - If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
 - Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133).
- Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 05/02/2014.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

- 5) ☒ Claim(s) 1-90 is/are pending in the application.
5a) Of the above claim(s) 7-10, 12-15, 24-27, 29-43, 48-51, 53-56 and 59-88 is/are withdrawn from consideration.
- 6) ☐ Claim(s) _____ is/are allowed.
- 7) ☒ Claim(s) 1, 3, 4, 16, 18-23, 28, 44, 46, 57, 58, 89 and 90 is/are rejected.
- 8) ☒ Claim(s) 2, 5, 6, 11, 17, 45, 47, and 52 is/are objected to.
- 9) ☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☒ The drawing(s) filed on 05/20/2011 is/are: a) ☐ accepted or b) ☒ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|--|---|
| 1) ☒ Notice of References Cited (PTO-892) | 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. _____ |
| 2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date <u>03/09/2011</u> . | 4) ☐ Other: _____ |

DETAILED ACTION

1. The present application is being examined under the pre-AIA first to invent provisions.
2. This action is in response to the application filed on 03/09/2011, the preliminary amendment filed on 07/06/2012, and the response to restriction requirement filed on 05/02/2014.

Election/Restrictions

3. Applicant's election without traverse of claims 1-6, 11, 16-23, 28, 44-47, 52, 57, 78, 89, and 90 in the reply filed on 05/02/2014 is acknowledged.

Drawings

4. Figures 1, 2A, 2B and 9 should be designated by a legend such as --Prior Art-- because only that which is old is illustrated. See MPEP § 608.02(g). Corrected drawings in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application. The replacement sheet(s) should be labeled "Replacement Sheet" in the page header (as per 37 CFR 1.84(c)) so as not to obstruct any portion of the drawing figures. If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

5. The drawings are objected to because of the following informalities.

Regarding Fig. 1, 5, and 6, the unlabeled rectangular box(es) shown in the drawings should be provided with descriptive text labels.

Corrected drawing sheets in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application. Any amended replacement drawing sheet should include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. The figure or figure number of an amended drawing should not be labeled as "amended." If a drawing figure is to be canceled, the appropriate figure must be removed from the replacement sheet, and where necessary, the remaining figures must be renumbered and appropriate changes made to the brief description of the several views of the drawings for consistency. Additional replacement sheets may be necessary to show the renumbering of the remaining figures. Each drawing sheet submitted after the filing date of an application must be labeled in the top margin as either "Replacement Sheet" or "New Sheet" pursuant to 37 CFR 1.121(d). If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

Specification

6. The lengthy specification has not been checked to the extent necessary to determine the presence of all possible minor errors. Applicant's cooperation is

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requested in correcting any errors of which applicant may become aware in the specification.

7. The abstract of the disclosure is objected to due to its undue length.

Applicant is reminded of the proper language and format for an abstract of the disclosure

The abstract should be in narrative form and generally limited to a single paragraph on a separate sheet within the range of 50 to 150 words. The form and legal phraseology often used in patent claims, such as "means" and "said," should be avoided. The abstract should describe the disclosure sufficiently to assist readers in deciding whether there is a need for consulting the full patent text for details.

The language should be clear and concise and should not repeat information given in the title. It should avoid using phrases which can be implied, such as, "The disclosure concerns," "The disclosure defined by this invention," "The disclosure describes," etc.

Correction is required. See MPEP § 608.01(b).

Claim Objections

8. **Claims 1, 2 and 5** are objected to because of the following informalities:

regarding **claim 1**, in line 9, "the output" should read as "an output";

regarding **claim 2**, in line 5, "the second polarity is opposite the ..." should read as "the second polarity is opposite of the ...";

regarding **claim 5**, in line 5, "the second polarity is opposite the ..." should read as "the second polarity is opposite of the ...";

Appropriate correction is required.

Claim Rejections - 35 USC § 112

9. The following is a quotation of 35 U.S.C. 112(b):

(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:

The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

10. **Claims 18, 20-23, 28, and 58** are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor, or for pre-AIA the applicant regards as the invention.

Regarding **claim 18**, in lines 13 and 14, “the output” renders the claims to be indefinite as it is unclear whether the term “the output” refers to “the output of the full-bridge rectifier circuit” or “the output of power converter”. For examining purposes, the office interprets “the output” as “the output of the full-bridge rectifier circuit”.

Regarding **claim 20**, in lines 7, 8, 10, and 11, “the output” renders the claims to be in definite as it is unclear whether the term “the output” refers to “the output of the full-bridge rectifier circuit” or “the output of power converter”. For examining purposes, the office interprets “the output” as “the output of the full-bridge rectifier circuit”.

Regarding **claim 58**, in lines 2 and 3, the bold underlined part in “**each of the** first, second, **third and fourth switches**” lacks antecedent basis.

Claims 19, 21-23, and 28 depend directly or indirectly from a rejected claim above and are therefore also rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-

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AIA), second paragraph for the reasons set above.

Claim Rejections - 35 USC § 103

11. The following is a quotation of 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

(a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negated by the manner in which the invention was made.

12. **Claims 1, 3, 4, 16, 44, 46, and 57** are rejected under 35 U.S.C. 103(a) as being unpatentable over Chen et al. (US Patent 7,768,807 B2, hereafter referred to as "Chen") in view of ISL6210 (Datasheet ISL6210, "Dual Synchronous Rectified MOSFET Drivers," Intersil Americas Inc., December 2008).

Regarding **claim 1**, as best understood, Chen discloses (see Fig. 2 and Fig. 3) a method of converting power via a transformer (transformer of Fig. 2) for supplying a low voltage output (V_{hi}) comprising: providing a secondary winding (right hand side winding of transformer) having a first terminal (top-side output of right hand side winding of transformer) and a second terminal (bottom-side output of right hand side winding of transformer); providing a first switch (S1) and a second switch (S3) each adapted to block current flow in an OFF state in at least one direction; forming a first series circuit with the first switch connected to the second switch at a first junction (common node of S1 and S3), the first series circuit having two end terminals (top node of S1 and bottom node of S3); connecting the first terminal of the secondary winding to the first junction; connecting the first series circuit across *an* output (top and bottom nodes of C_o) with

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each of the two end terminals connected to a respective terminal of the output (see connection) and with each of the first and second switches being arranged in the OFF state to block current flow through the first series circuit in a direction that would short the output (when S1 and S3 are OFF, it prevents current flowing from the top node of S1 to the bottom node of S3, see Fig. 3), wherein a short across one of the first or second switches does not short the output (S1 and S3 are never simultaneously ON, thus the output is not shorted (see Fig. 3)).

Chen does not disclose preventing the first switch from turning ON if the second switch fails to open; and preventing the second switch from turning ON if the first switch fails to open.

However, ISL6210 discloses (see p. 6) preventing a first switch from turning ON if a second switch fails to open; and preventing the second switch from turning ON if the first switch fails to open (see Adaptive Shoot-Through Protection).

Therefore, it would have been obvious to one having ordinary skill in the art at the time of the invention was made to modify the method of Chen to include the method of preventing the first switch from turning ON if the second switch fails to open and preventing the second switch from turning ON if the first switch fails to open as disclosed in ISL6210, because it can help prevent current shoot through between the two switches.

Regarding **claim 3**, as best understood, Chen discloses (see Fig. 2 and Fig. 3) further comprising: providing a third switch (S2) and a fourth switch (S4) each adapted to block current in an OFF state in at least one direction; forming a second series circuit

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with the third switch connected to the fourth switch at a second junction (common node of S2 and S4), the second series circuit having two end terminals (top node of S2 and bottom node of S4); connecting the second terminal of the secondary winding to the second junction; and connecting the second series circuit across the output with each of the two end terminals connected to a respective terminal of the output (see connection) and with each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output (when S2 and S4 are OFF, it prevents current flowing from the top node of S2 to the bottom node of S4, see Fig. 3), wherein a short across one of the third or fourth switches does not short the output (S2 and S4 are never simultaneously ON, thus the output is not shorted (see Fig. 3)).

Regarding **claim 4**, as best understood, Chen does not disclose preventing the third switch from turning ON if the fourth switch fails to open; preventing the fourth switch from turning ON if the third switch fails to open.

However, ISL6210 discloses (see p. 6) preventing a third switch from turning ON if a fourth switch fails to open; and preventing the fourth switch from turning ON if the third switch fails to open (see Adaptive Shoot-Through Protection).

Therefore, it would have been obvious to one having ordinary skill in the art at the time of the invention was made to modify the method of Chen to include the method of preventing the third switch from turning ON if the fourth switch fails to open; preventing the fourth switch from turning ON if the third switch fails to open as disclosed in ISL6210, because it can help prevent current shoot through between the two

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switches.

Regarding **claim 16**, as best understood, Chen discloses (see Fig. 2 and Fig. 3) forming a second series circuit with a first capacitor (C1) connected to a second capacitor (C2) at a second junction (common node of C1 and C2), the second series circuit having two end terminals (top node of C1 and bottom node of C2); connecting the second terminal of the secondary winding to the second junction; and connecting the second series circuit across the output with each of the two end terminals connected to a respective terminal of the output (see connection).

Regarding **claim 44**, Chen discloses (see Fig. 2 and Fig. 3) a fault-tolerant converter comprising: a transformer (transformer of Fig. 2) having a primary winding (left hand side winding of transformer) and a secondary winding (right hand side winding of transformer), the secondary winding having a first terminal (top-side output of right hand side winding of transformer) and a second terminal (bottom-side output of right hand side winding of transformer); a first series circuit having a first switch (S1) and a second switch (S3) each adapted to block current flow in an OFF state in at least one direction; the first switch being connected to the second switch at a first junction (common node of S1 and S3), the first series circuit having two end terminals (top node of S1 and bottom node of S3); the first junction being connected to the first terminal of the secondary winding; the first series circuit being connected across an output (top and bottom nodes of Co) of the converter with each of the two end terminals connected to a respective terminal of the output (see connection) and with each of the first and second switches being arranged in the OFF state to block current flow through the first series

circuit in a direction that would short the output (when S1 and S3 are OFF, it prevents current flowing from the top node of S1 to the bottom node of S3, see Fig. 3), wherein a short across one of the first or second switches does not short the output (S1 and S3 are never simultaneously ON, thus the output is not shorted (see Fig. 3)).

Chen does not disclose a switch control circuit configured to prevent the first switch from turning ON if the second switch fails to open; and to prevent the second switch from turning ON if the first switch fails to open.

However, ISL6210 discloses (see p. 6) a switch control circuit (ISL6210) configured to prevent a first switch from turning ON if a second switch fails to open; and preventing the second switch from turning ON if the first switch fails to open (see Adaptive Shoot-Through Protection).

Therefore, it would have been obvious to one having ordinary skill in the art at the time of the invention was made to modify the converter of Chen to include the switch control circuit configured to prevent the first switch from turning ON if the second switch fails to open; and to prevent the second switch from turning ON if the first switch fails to open as disclosed in ISL6210, because it can help prevent current shoot through between the two switches.

Regarding **claim 46**, Chen discloses (see Fig. 2 and Fig. 3) comprising a second series circuit that comprises a third switch (S2) and a fourth switch (S4) each adapted to block current in an OFF state in at least one direction, the third switch being connected to the fourth switch at a second junction (common node of S2 and S4), the second series circuit having two end terminals (top node of S2 and bottom node of S4), the

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second terminal of the secondary winding being connected to the second junction, the second series circuit being connected across the output with each of the two end terminals connected to a respective terminal of the output (see connection) and with each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output (when S2 and S4 are OFF, it prevents current flowing from the top node of S2 to the bottom node of S4, see Fig. 3), wherein a short across one of the third or fourth switches does not short the output (S2 and S4 are never simultaneously ON, thus the output is not shorted (see Fig. 3)).

Regarding **claim 57**, Chen discloses (see Fig. 2 and Fig. 3) comprising a second series circuit having a first capacitor (C1) connected to a second capacitor (C2) at a second junction (common node of C1 and C2) that is connected to the second terminal of the secondary winding, the second series circuit having two end terminals (top node of C1 and bottom node of C2) connected across the output of the converter with each of the two end terminals connected to a respective terminal of the output of the converter (see connections).

13. **Claims 18, and 20** are rejected under 35 U.S.C. 103(a) as being unpatentable over Chen in view of AN2738 (Application Note AN2738, "L6390 half-bridge gate driver," STMicroelectronics, August 2009).

Regarding **claim 18**, as best understood, Chen discloses (see Fig. 2 and Fig. 3) a method of providing fault tolerance in a power converter (converter of Fig. 2)

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comprising: supplying an alternating current (AC in transformer) from a secondary winding (right hand side winding of transformer) to an input of a full-bridge rectifier circuit (circuit comprising S1, S2, S3, S4, C1, C2) having switches (S1, S2, S3, S4), operating the switches as rectifiers to provide full-wave rectification of the alternating current (see Fig. 3) at an output of the full-bridge rectifier circuit (top node of S2 and bottom node of S4), connecting the output of the full-bridge rectifier circuit to an output of the power converter (top and bottom nodes of Co), and providing a first one of the switches connected in series with a second one of the switches across the output with the first switch connected to a first terminal of the output and the second switch connected to a second terminal of the output.

Chen does not disclose monitoring selected circuit conditions; determining from the selected circuit conditions whether a predetermined fault is present, disabling the switches in response to a determination that the predetermined fault is present.

However, AN2738 discloses (see p. 7) monitoring selected circuit conditions (VCC); determining from the selected circuit conditions whether a predetermined fault is present (when VCC goes below a threshold), disabling the switches in response to a determination that the predetermined fault is present (turns off IC outputs).

Therefore, it would have been obvious to one having ordinary skill in the art at the time of the invention was made to modify the method of Chen to include the method of monitoring selected circuit conditions; determining from the selected circuit conditions whether a predetermined fault is present, disabling the switches in response to a determination that the predetermined fault is present as disclosed in AN2738, because

it can help prevent faults in the circuit which can help protect the circuit elements from excessive electrical stress.

Regarding **claim 20**, as best understood, Chen discloses (see Fig. 2 and Fig. 3) further comprising: providing a third one of the switches and a fourth one of the switches each adapted to block current in an OFF state in at least one direction; forming a second series circuit with the third switch connected to the fourth switch at a second junction (common node of S2 and S4), the second series circuit having two end terminals (top node of S2 and bottom node of S4); connecting the second terminal of the secondary winding to the second junction; and connecting the second series circuit across the output with each of the two end terminals connected to a respective terminal of the output (see connection) and with each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output (when S2 and S4 are OFF, it prevents current flowing from the top node of S2 to the bottom node of S4, see Fig. 3), wherein a short across one of the third or fourth switches does not short the output (S2 and S4 are never simultaneously ON, thus the output is not shorted (see Fig. 3)).

14. **Claim 21** is rejected under 35 U.S.C. 103(a) as being unpatentable over Chen in view of AN2738, and further in view of ISL6210.

Regarding **claim 21**, as best understood, Chen does not disclose preventing the third switch from turning ON if the fourth switch fails to open; preventing the fourth switch from turning ON if the third switch fails to open.

However, ISL6210 discloses (see p. 6) preventing a third switch from turning ON if a fourth switch fails to open; and preventing the fourth switch from turning ON if the third switch fails to open (see Adaptive Shoot-Through Protection).

Therefore, it would have been obvious to one having ordinary skill in the art at the time of the invention was made to modify the method of Chen to include the method of preventing the third switch from turning ON if the fourth switch fails to open; preventing the fourth switch from turning ON if the third switch fails to open as disclosed in ISL6210, because it can help prevent current shoot through between the two switches.

15. **Claims 89 and 90** are rejected under 35 U.S.C. 103(a) as being unpatentable over Chen.

Regarding **claim 89**, Chen discloses (see Fig. 2 and Fig. 3) a method of delivering power to an output (V_{hl}), comprising: providing a secondary winding (right hand side winding of transformer) having a first terminal (top-side output of right hand side winding of transformer) and a second terminal (bottom-side output of right hand side winding of transformer); providing a first switch (S1), a second switch (S3), a third switch (S2) and a fourth switch (S4) each adapted to block current flow in an OFF state in at least one direction; forming a first series circuit with the first switch connected to the second switch at a first junction (common node of S1 and S3), the first series circuit having two end terminals (top node of S1 and bottom node of S3); forming a second series circuit with the third switch connected to the fourth switch at a second junction

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(common node of S2 and S4), the second series circuit having two end terminals (top node of S2 and bottom node of S4); connecting the first series circuit across an output (top and bottom nodes of Co) with each of the two end terminals connected to a respective terminal of the output (see connection) and with each of the first and second switches being arranged in the OFF state to block current flow through the first series circuit in a direction that would short the output (when S1 and S3 are OFF, it prevents current flowing from the top node of S1 to the bottom node of S3, see Fig. 3), wherein a short across one of the first or second switches does not short the output (S1 and S3 are never simultaneously ON, thus the output is not shorted (see Fig. 3)); connecting the second series circuit across the output with each of the two end terminals connected to a respective terminal of the output (see connection) and with each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output (when S2 and S4 are OFF, it prevents current flowing from the top node of S2 to the bottom node of S4, see Fig. 3), wherein a short across one of the third or fourth switches does not short the output (S2 and S4 are never simultaneously ON, thus the output is not shorted (see Fig. 3)); converting power via the transformer to deliver a unipolar voltage (V_{hl}) to the output.

Chen does not disclose wherein the unipolar voltage is five Volts or less.

However, it would have been obvious to one of ordinary skill in the art at the time the invention was made to have the unipolar voltage be five Volts or less, since it has been held that where the general conditions of a claim are disclosed in the prior art, discovering the optimum or working ranges involves only routine skill in the art. In re

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Aller, 105 USPQ 233.

Regarding **claim 90**, Chen does not disclose wherein the unipolar output is approximately 1 Volt.

However, it would have been obvious to one of ordinary skill in the art at the time the invention was made to have the unipolar voltage be approximately 1 Volt, since it has been held that discovering an optimum value of a result effective variable involves only routine skill in the art. In re Boesch, 617 F.2d 272, 205 USPQ 215 (CCPA 1980).

Allowable Subject Matter

16. **Claims 2, 5, 6, 11, 17, 19, 22, 23, 28, 45, 47, 52, and 58** are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

The following is a statement of reasons for the indication of allowable subject matter:

Regarding **claims 2, 17, 19, 45, and 58**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which “sensing across each of the first and second switches, a respective voltage polarity; and turning each of the first and second switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.”.

Regarding **claims 5 and 22**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which “sensing across each of the third and fourth switches, a respective voltage polarity; and turning each of the third and fourth switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.”.

Regarding **claims 23 and 47**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which “providing a first control circuit to operate the first switch; providing power to operate the first control circuit and first switch from across the third switch; providing a third control circuit to operate the third switch; and providing power to operate the third control circuit and third switch from across the first switch.”.

Regarding **claims 28 and 52**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which “providing a second control circuit to operate the second switch; providing power to operate the second control circuit and second switch from across the fourth switch; providing a fourth control circuit to operate the fourth switch; and providing power to operate the fourth control circuit and fourth switch from across the second switch.”.

17. Regarding **claims 6, 11, and 52**, claims 6 and 11 are objected to due to their dependency on claim 5, and claim 52 is objected to due to its dependency on claim 47.

18. **Claim 58** would be allowable if rewritten to overcome the rejection(s) under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), 2nd paragraph, set forth in this Office action

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and to include all of the limitations of the base claim and any intervening claims.

The following is a statement of reasons for the indication of allowable subject matter:

Regarding **claim 58**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which “wherein the switch control circuit is configured to sense across each of the first, second, third, and fourth switches, a respective voltage polarity, and to turn each of the first, second, third, and fourth switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.”.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to JYE-JUNE LEE whose telephone number is (571)270-7726. The examiner can normally be reached on Monday to Thursday 9:00 AM to 7:30 PM EST.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Monica Lewis can be reached on 571-272-1838. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/MONICA LEWIS/
Supervisory Patent Examiner, Art Unit 2838

/JYE-JUNE LEE/

Examiner, Art Unit 2838

Notice of References Cited	Application/Control No. 13/044,063	Applicant(s)/Patent Under Reexamination VINCIARELLI, PATRIZIO	
	Examiner JYE-JUNE LEE	Art Unit 2838	Page 1 of 1

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	D	US-			
	E	US-			
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NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	AN2738 (Application Note AN2738, "L6390 half-bridge gate driver," STMicroelectronics, August 2009)
	V	ISL6210 (Datasheet ISL6210, "Dual Synchronous Rectified MOSFET Drivers," Intersil Americas Inc., December 2008)
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Receipt date: 03/09/2011

Doc code: IDS

Doc description: Information Disclosure Statement (IDS) Filed

13044063 - GAI: 2838

Pat. Sec. 101-19

Approved for use through 07/31/2012. OMB 0651-0031

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	First Named Inventor	Vinciarelli	
	Art Unit		
	Examiner Name		
	Attorney Docket Number	00614-0166001	

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Receipt date: 03/09/2011 INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		13044063 - GAU: 2838	
	Filing Date			
	First Named Inventor	Vinciarelli		
	Art Unit			
	Examiner Name			
	Attorney Docket Number	00614-0166001		

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	2838
Serial No. :	13/044,063	Examiner :	Jye-June Lee
Filed :	March 9, 2011	Conf. No. :	1081
Title :	FAULT TOLERANT POWER CONVERTER		

Mail Stop Amendment

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

REPLY TO ACTION OF MAY 23, 2014

Please consider the following reply.

Amendment to the Abstract:

Please replace the abstract at page 66 with the following amended abstract:

ABSTRACT

A power converter provides a low-voltage output using a full-bridge fault-tolerant rectification circuit. The output circuit uses controlled switches as rectifiers. A fault detection circuit monitors circuit conditions. Upon detection of a fault, the switches are disabled decoupling the power converter from the system.

A common-source dual MOSFET device includes a plurality of elements arranged in alternating patterns on a semiconductor die. A common-source dual synchronous rectifier includes control circuitry powered from the drain to source voltage of the complementary switch.

A DC-to-DC transformer converts power from an input source to a load using a fixed voltage transformation ratio. A clamp phase may be used to reduce power losses in the converter at light loads, control the effective output resistance of the converter, effectively regulate the voltage transformation ratio, provide narrow band output regulation, and control the rate of change of output voltage for example during start up. ~~One or more of the transformer windings may be clamped. The converter may use the sine amplitude converter topology. The converter may use common source dual MOSFET devices and fault detection.~~

~~The density of point of load power conversion may be increased and the associated power dissipation reduced by removing the input driver circuitry from the point of load where it is not necessary. An output circuit may be located at the point of load providing fault tolerant rectification of the AC power from the secondary winding of a power transformer which may be located nearby the output circuit. The resonant voltage and current waveforms on the primary side of the transformer are readily communicated via an AC bus between the driver circuit and the primary winding of the power transformer. The driver circuit may drive a plurality of transformer output circuit pairs. The transformer and output circuit may be combined in a single module at the point of load. Alternatively, the output circuit may be integrated into point of load circuitry such as a processor core. The transformer may be deployed near the output circuit.~~

List of claims (replacing prior versions).

1. (Currently Amended) A method of converting power via a transformer for supplying a low voltage output comprising:

- providing a secondary winding having a first terminal and a second terminal;
- providing a first switch and a second switch each adapted to block current flow in an OFF state in at least one direction;
- forming a first series circuit with the first switch connected to the second switch at a first junction, the first series circuit having two end terminals;
- connecting the first terminal of the secondary winding to the first junction;
- connecting the first series circuit across ~~[[the]]~~ an output with each of the two end terminals connected to a respective terminal of the output and with each of the first and second switches being arranged in the OFF state to block current flow through the first series circuit in a direction that would short the output, wherein a short across one of the first or second switches does not short the output;
- preventing the first switch from turning ON if the second switch fails to open; and
- preventing the second switch from turning ON if the first switch fails to open.

2. (Currently Amended) The method of claim 1 further comprising:

- sensing across each of the first and second switches, a respective voltage polarity; and
- turning each of the first and second switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite of the first polarity.

3. (Original) The method of claim 1 further comprising:

- providing a third switch and a fourth switch each adapted to block current in an OFF state in at least one direction;
- forming a second series circuit with the third switch connected to the fourth switch at a second junction, the second series circuit having two end terminals;
- connecting the second terminal of the secondary winding to the second junction; and

connecting the second series circuit across the output with each of the two end terminals connected to a respective terminal of the output and with each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output, wherein a short across one of the third or fourth switches does not short the output.

4. (Original) The method of claim 3 further comprising:
preventing the third switch from turning ON if the fourth switch fails to open;
preventing the fourth switch from turning ON if the third switch fails to open.

5. (Currently Amended) The method of claim 4 further comprising:
sensing across each of the third and fourth switches, a respective voltage polarity; and
turning each of the third and fourth switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite of the first polarity.

6. (Original) The method of claim 5 further comprising:
providing a first control circuit to operate the first switch;
providing power to operate the first control circuit and first switch from across the third switch;
providing a third control circuit to operate the third switch; and
providing power to operate the third control circuit and third switch from across the first switch.

7. (Withdrawn) The method of claim 6 further comprising:
providing a common-source dual MOSFET semiconductor device including a first MOSFET and a third MOSFET integrated on a semiconductor die;
providing a common source terminal for connecting to the first and third MOSFETs in the common-source dual MOSFET semiconductor device; and

using the first MOSFET and the third MOSFET of the common-source dual MOSFET semiconductor device for the first and third switches.

8. (Withdrawn) The method of claim 7 further comprising integrating the first and third control circuits in the common-source dual MOSFET semiconductor device.

9. (Withdrawn) The method of claim 8 further comprising:
providing a plurality of individual first and third MOSFET elements on the semiconductor die;
arranging the individual first and third MOSFET elements in an alternating pattern on the semiconductor die;
connecting the first MOSFET elements together to form the first MOSFET;
connecting the third MOSFET elements together to form the third MOSFET; and
using a common source metallization to interconnect the source terminals of the first and third MOSFET elements.

10. (Withdrawn) The method of claim 9 wherein the common-source dual MOSFET semiconductor device has exactly three terminals.

11. (Original) The method of claim 6 further comprising:
providing a second control circuit to operate the second switch;
providing power to operate the second control circuit and second switch from across the fourth switch;
providing a fourth control circuit to operate the fourth switch; and
providing power to operate the fourth control circuit and fourth switch from across the second switch.

12. (Withdrawn) The method of claim 11 further comprising:
providing a first common-source dual MOSFET semiconductor device including a first MOSFET and a third MOSFET integrated on a first semiconductor die;

providing a common source terminal for connecting to the first and third MOSFETs in the first common-source dual MOSFET semiconductor device;

using the first MOSFET and the third MOSFET of the first common-source dual MOSFET semiconductor device for the first and third switches;

providing a second common-source dual MOSFET semiconductor device including a second MOSFET and a fourth MOSFET integrated on a second semiconductor die;

providing a common source terminal for connecting to the second and fourth MOSFETs in the second common-source dual MOSFET semiconductor device;

using the second MOSFET and the fourth MOSFET of the second common-source dual MOSFET semiconductor device for the second and fourth switches.

13. (Withdrawn) The method of claim 12 further comprising:

integrating the first and third control circuits in the first common-source dual MOSFET semiconductor device; and

integrating the second and fourth control circuits in the second common-source dual MOSFET semiconductor device.

14. (Withdrawn) The method of claim 13 further comprising:

providing a plurality of individual first and third MOSFET elements on the first semiconductor die;

arranging the individual first and third MOSFET elements in an alternating pattern on the first semiconductor die;

connecting the first MOSFET elements together to form the first MOSFET;

connecting the third MOSFET elements together to form the third MOSFET;

using a common source metallization to interconnect the source terminals of the first and third MOSFET elements;

providing a plurality of individual second and fourth MOSFET elements on the second semiconductor die;

arranging the individual second and fourth MOSFET elements in an alternating pattern on the second semiconductor die;

connecting the second MOSFET elements together to form the second MOSFET;
connecting the fourth MOSFET elements together to form the fourth MOSFET; and
using a common source metallization to interconnect the source terminals of the second and fourth MOSFET elements.

15. (Withdrawn) The method of claim 14 wherein the first and second common-source dual MOSFET semiconductor devices each have exactly three terminals.

16. (Original) The method of claim 1 further comprising:
forming a second series circuit with a first capacitor connected to a second capacitor at a second junction, the second series circuit having two end terminals;
connecting the second terminal of the secondary winding to the second junction; and
connecting the second series circuit across the output with each of the two end terminals connected to a respective terminal of the output.

17. (Original) The method of claim 16 further comprising:
sensing across each of the switches, a respective voltage polarity; and
turning each of the switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.

18. (Currently Amended) A method of providing fault tolerance in a power converter comprising:

supplying an alternating current from a secondary winding to an input of a full-bridge rectifier circuit having switches,

operating the switches as rectifiers to provide full-wave rectification of the alternating current at an output of the full-bridge rectifier circuit,

connecting the output of the full-bridge rectifier circuit to an output of the power converter,

monitoring selected circuit conditions;

determining from the selected circuit conditions whether a predetermined fault is present, disabling the switches in response to a determination that the predetermined fault is present, and

providing a first one of the switches connected in series with a second one of the switches across the output of the full-bridge rectifier circuit with the first switch connected to a first terminal of the output of the full-bridge rectifier circuit and the second switch connected to a second terminal of the output of the full-bridge rectifier circuit.

19. (Original) The method of claim 18 further comprising:
sensing across each of the first and second switches, a respective voltage polarity; and
turning each of the first and second switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.

20. (Currently Amended) The method of claim 18 further comprising:
providing a third one of the switches and a fourth one of the switches each adapted to block current in an OFF state in at least one direction;
forming a second series circuit with the third switch connected to the fourth switch at a second junction, the second series circuit having two end terminals;
connecting the second terminal of the secondary winding to the second junction; and
connecting the second series circuit across the output of the full-bridge rectifier circuit with each of the two end terminals connected to a respective terminal of the output of the full-bridge rectifier circuit and with each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output of the full-bridge rectifier circuit, wherein a short across one of the third or fourth switches does not short the output of the full-bridge rectifier circuit.

21. (Original) The method of claim 20 further comprising:
preventing the third switch from turning ON if the fourth switch fails to open; and

preventing the fourth switch from turning ON if the third switch fails to open.

22. (Original) The method of claim 20 further comprising:
sensing across each of the third and fourth switches, a respective voltage polarity; and
turning each of the third and fourth switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.

23. (Original) The method of claim 20 further comprising:
providing a first control circuit to operate the first switch;
providing power to operate the first control circuit and first switch from across the third switch;
providing a third control circuit to operate the third switch; and
providing power to operate the third control circuit and third switch from across the first switch.

24. (Withdrawn) The method of claim 20 further comprising:
providing a common-source dual MOSFET semiconductor device including a first MOSFET and a third MOSFET integrated on a semiconductor die;
providing a common source terminal for connecting to the first and third MOSFETs in the common-source dual MOSFET semiconductor device; and
using the first MOSFET and the third MOSFET of the common-source dual MOSFET semiconductor device for the first and third switches.

25. (Withdrawn) The method of claim 24 further comprising integrating the first and third control circuits in the common-source dual MOSFET semiconductor device.

26. (Withdrawn) The method of claim 24 further comprising:

providing a plurality of individual first and third MOSFET elements on the semiconductor die;

arranging the individual first and third MOSFET elements in an alternating pattern on the semiconductor die;

connecting the first MOSFET elements together to form the first MOSFET;

connecting the third MOSFET elements together to form the third MOSFET; and

using a common source metallization to interconnect the source terminals of the first and third MOSFET elements.

27. (Withdrawn) The method of claim 24 wherein the common-source dual MOSFET semiconductor device has exactly three terminals.

28. (Original) The method of claim 20 further comprising:

providing a second control circuit to operate the second switch;

providing power to operate the second control circuit and second switch from across the fourth switch;

providing a fourth control circuit to operate the fourth switch; and

providing power to operate the fourth control circuit and fourth switch from across the second switch.

29. (Withdrawn) The method of claim 20 further comprising:

providing a first common-source dual MOSFET semiconductor device including a first MOSFET and a third MOSFET integrated on a first semiconductor die;

providing a common source terminal for connecting to the first and third MOSFETs in the first common-source dual MOSFET semiconductor device;

using the first MOSFET and the third MOSFET of the first common-source dual MOSFET semiconductor device for the first and third switches;

providing a second common-source dual MOSFET semiconductor device including a second MOSFET and a fourth MOSFET integrated on a second semiconductor die;

providing a common source terminal for connecting to the second and fourth MOSFETs in the second common-source dual MOSFET semiconductor device; and

using the second MOSFET and the fourth MOSFET of the second common-source dual MOSFET semiconductor device for the second and fourth switches.

30. (Withdrawn) The method of claim 29 further comprising:

integrating the first and third control circuits in the first common-source dual MOSFET semiconductor device; and

integrating the second and fourth control circuits in the second common-source dual MOSFET semiconductor device.

31. (Withdrawn) The method of claim 29 further comprising:

providing a plurality of individual first and third MOSFET elements on the first semiconductor die;

arranging the individual first and third MOSFET elements in an alternating pattern on the first semiconductor die;

connecting the first MOSFET elements together to form the first MOSFET;

connecting the third MOSFET elements together to form the third MOSFET;

using a common source metallization to interconnect the source terminals of the first and third MOSFET elements;

providing a plurality of individual second and fourth MOSFET elements on the second semiconductor die;

arranging the individual second and fourth MOSFET elements in an alternating pattern on the second semiconductor die;

connecting the second MOSFET elements together to form the second MOSFET;

connecting the fourth MOSFET elements together to form the fourth MOSFET; and

using a common source metallization to interconnect the source terminals of the second and fourth MOSFET elements.

32. (Withdrawn) The method of claim 29 wherein the first and second common-source dual MOSFET semiconductor devices each have exactly three terminals.

33. (Withdrawn) A semiconductor device comprising:
a first electronic switch having a first control terminal, a first switched terminal, and a first common terminal,
a second electronic switch having a second control terminal, a second switched terminal, and a second common terminal,
a first control circuit having an input connected to the first switched terminal and the first common terminal and an output connected to the first control terminal, the first control circuit being adapted to turn the first switch ON and OFF in response to a first voltage across the first switched terminal and the first common terminal;
a second control circuit having an input connected to the second switched terminal and the second common terminal and an output connected to the second control terminal, the second control circuit being adapted to turn the second switch ON and OFF in response to a second voltage across the second switched terminal and the second common terminal; and
a package having at least three terminals including one terminal connected to the first switched terminal, another terminal connected to the second switched terminal, and a common terminal connected to the first common terminal and the second common terminal;
wherein the first control circuit is connected to receive power from the second switched terminal; and the second control circuit is connected to receive power from the first switched terminal.

34. (Withdrawn) The apparatus of claim 33 wherein
the first and second electronic switches are formed in a common monolithic semiconductor die;
the first and second electronic switches respectively comprise a plurality of first and second switch elements;
the first and second switch elements each have a respective G terminal, S terminal, and D terminal; and

the S terminals of the first and second switch elements are connected together on the semiconductor die.

35. (Withdrawn) The apparatus of claim 33 wherein the first and second switch elements are arranged in an alternating pattern on the semiconductor die.

36. (Withdrawn) The apparatus of claim 34 wherein the first and second control circuits are also formed in the common monolithic semiconductor die.

37. (Withdrawn) The apparatus of claim 33 wherein each of the first control circuit has an offset and a gain that are configured to ensure that the first switch is turned ON when the voltage across the first switched terminal and the first common terminal is above a first predetermined threshold, and fully ON when the voltage across the first switched terminal and the first common terminal is above a second predetermined threshold.

38. (Withdrawn) A semiconductor device comprising:
a semiconductor die comprising a multiplicity of switch elements each having a respective G terminal, S terminal, and D terminal;
a first electronic switch and a second electronic switch respectively comprising a first plurality and a second plurality of the switch elements,
the switch elements of the first plurality being arranged in an alternating pattern with the switch elements of the second plurality on the semiconductor die;
the S terminals of the switch elements of the first and second pluralities being connected together on the semiconductor die;
the first electronic switch having a first control terminal connected to the G terminals of the switch elements of the first plurality and a first switched terminal connected to the D terminals of the switch elements of the first plurality;
a second electronic switch having a second control terminal connected to the G terminals of the switch elements of the second plurality and a second switched terminal connected to the D terminals of the switch elements of the second plurality;

a first control circuit having an input connected to the first switched terminal and the common terminal and an output connected to the first control terminal, the first control circuit being adapted to turn the first switch ON and OFF in response to a first voltage across the first switched terminal and the common terminal; and

a second control circuit having an input connected to the second switched terminal and the common terminal and an output connected to the second control terminal, the second control circuit being adapted to turn the second switch ON and OFF in response to a second voltage across the second switched terminal and the common terminal.

39. (Withdrawn) The apparatus of claim 38 wherein the first and second control circuits are also formed in the common monolithic semiconductor die.

40. (Withdrawn) The apparatus of claim 38 further comprising:

a package including one terminal connected to the first switched terminal, another terminal connected to the second switched terminal, and a third terminal connected to the common terminal.

41. (Withdrawn) The apparatus of claim 40 wherein the package further comprises a fourth terminal connected to supply power to the first and second control circuits.

42. (Original) The apparatus of claim 40 wherein

the first control circuit is connected to receive power from the second switched terminal; and the second control circuit is connected to receive power from the first switched terminal.

43. (Withdrawn) The apparatus of claim 42 wherein the package comprises exactly three terminals.

44. (Original) A fault-tolerant converter comprising:

a transformer having a primary winding and a secondary winding, the secondary winding having a first terminal and a second terminal;

a first series circuit having a first switch and a second switch each adapted to block current flow in an OFF state in at least one direction, the first switch being connected to the second switch at a first junction, the first series circuit having two end terminals, the first junction being connected to the first terminal of the secondary winding, the first series circuit being connected across an output of the converter with each of the two end terminals connected to a respective terminal of the output and with each of the first and second switches being arranged in the OFF state to block current flow through the first series circuit in a direction that would short the output, wherein a short across one of the first or second switches does not short the output; and

a switch control circuit configured to prevent the first switch from turning ON if the second switch fails to open, and to prevent the second switch from turning ON if the first switch fails to open.

45. (Original) The apparatus of claim 44 wherein the switch control circuit is configured to sense across each of the first and second switches, a respective voltage polarity, and to turn each of the first and second switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.

46. (Original) The apparatus of claim 44, comprising a second series circuit that comprises a third switch and a fourth switch each adapted to block current in an OFF state in at least one direction, the third switch being connected to the fourth switch at a second junction, the second series circuit having two end terminals, the second terminal of the secondary winding being connected to the second junction, the second series circuit being connected across the output with each of the two end terminals connected to a respective terminal of the output and with each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output, wherein a short across one of the third or fourth switches does not short the output.

47. (Original) The apparatus of claim 46 wherein the switch control circuit comprises a first control circuit and a third control circuit, the first control circuit operates the first switch, the third control circuit operates the third switch, the first control circuit and the first switch are powered using a voltage across the third switch, and the third control circuit and third switch are powered using a voltage across the first switch.

48. (Withdrawn) The apparatus of claim 46, comprising a common-source dual MOSFET semiconductor device having a first MOSFET and a third MOSFET that are integrated on a semiconductor die, the first MOSFET is used for the first switch, the third MOSFET is used for the third switch, and the common-source dual MOSFET semiconductor device has a source terminal connected to the source of the first MOSFET and the source of the third MOSFET.

49. (Withdrawn) The apparatus of claim 48 wherein the switch control circuit comprises a first control circuit and a third control circuit, the first control circuit operates the first switch, the third control circuit operates the third switch, the first control circuit and the first switch are powered using a voltage across the third switch, the third control circuit and the third switch are powered using a voltage across the first switch, and the first and third control circuits are integrated in the semiconductor die of the common-source dual MOSFET semiconductor device.

50. (Withdrawn) The apparatus of claim 49 wherein the first MOSFET comprises a plurality of individual first MOSFET elements on the semiconductor die, the third MOSFET comprises a plurality of individual third MOSFET elements on the semiconductor die, the individual first and third MOSFET elements are arranged in an alternating pattern on the semiconductor die, and a common source metallization interconnects the source terminals of the first and third MOSFET elements.

51. (Withdrawn) The apparatus of claim 50 wherein the common-source dual MOSFET semiconductor device has exactly three terminals that includes the source terminal, a terminal connected to the drain of the first MOSFET, and a terminal connected to the drain of the third MOSFET.

52. (Original) The apparatus of claim 47 wherein the switch control circuit comprises a second control circuit and a fourth control circuit, the second control circuit operates the second switch, the fourth control circuit operates the fourth switch, the second control circuit and the second switch are powered using a voltage across the fourth switch, and the fourth control circuit and fourth switch are powered using a voltage across the second switch.

53. (Withdrawn) The apparatus of claim 52, comprising:

a first common-source dual MOSFET semiconductor device that comprises a first MOSFET and a third MOSFET integrated on a first semiconductor die, the first common-source dual MOSFET semiconductor device comprising a common source terminal for connecting to the first and third MOSFETs, wherein the first MOSFET and the third MOSFET are used for the first and third switches; and

a second common-source dual MOSFET semiconductor device that comprises a second MOSFET and a fourth MOSFET integrated on a second semiconductor die, the second common-source dual MOSFET semiconductor device comprising a common source terminal for connecting to the second and fourth MOSFETs, wherein the second MOSFET and the fourth MOSFET are used for the second and fourth switches.

54. (Withdrawn) The apparatus of claim 53 wherein the first and third control circuits are integrated on the first semiconductor die, and the second and fourth control circuits are integrated on the second semiconductor die.

55. (Withdrawn) The apparatus of claim 54 wherein the first MOSFET comprises a plurality of individual first MOSFET elements on the first semiconductor die, the third MOSFET comprises a plurality of individual third MOSFET elements on the first semiconductor die, the individual first and third MOSFET elements are arranged in an alternating pattern on the first semiconductor die, a common source metallization interconnects the source terminals of the first and third MOSFET elements, the second MOSFET comprises a plurality of individual second MOSFET elements on the second semiconductor die, the individual second and fourth MOSFET

elements are arranged in an alternating pattern on the second semiconductor die, and a common source metallization interconnects the source terminals of the second and fourth MOSFET elements.

56. (Withdrawn) The apparatus of claim 55 wherein the first common-source dual MOSFET semiconductor device has exactly three terminals that includes the common source terminal, a terminal connected to the drain of the first MOSFET, and a terminal connected to the drain of the third MOSFET, and

the second common-source dual MOSFET semiconductor device has exactly three terminals that include the common source terminal, a terminal connected to the drain of the second MOSFET, and a terminal connected to the drain of the fourth MOSFET.

57. (Original) The apparatus of claim 44, comprising a second series circuit having a first capacitor connected to a second capacitor at a second junction that is connected to the second terminal of the secondary winding, the second series circuit having two end terminals connected across the output of the converter with each of the two end terminals connected to a respective terminal of the output of the converter.

58. (Currently Amended) The apparatus of claim 57 wherein the switch control circuit is configured to sense across each of the first and second ~~first, second, third, and fourth~~ switches, a respective voltage polarity, and to turn each of the first and second ~~first, second, third, and fourth~~ switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.

59. (Withdrawn) A multi-cell power converter comprising:
a plurality of input cells having inputs connected in parallel with an input, and connected to drive parallel respective primary windings of a transformer with power received from the input, the input cells forming a power-sharing array of cells in which if one of the input cells fails, the converter continues to process power through remaining functional input cell or cells;

a plurality of output circuits to receive power from respective secondary windings of the transformer, the output circuits having outputs that are connected in parallel to provide an output voltage to a load, the output circuits forming a power-sharing array in supplying power to the output in which if one of the output circuits fails, the converter continues to process power through remaining functional output circuit or circuits; and

a switch control circuit;

wherein each of the output circuits comprises a first series circuit having a first switch and a second switch each adapted to block current flow in an OFF state in at least one direction, the first switch is connected to the second switch at a first junction, the first series circuit has two end terminals, the first junction is connected to a first terminal of the respective secondary winding, the first series circuit is connected across an output of the converter with each of the two end terminals connected to a respective terminal of the output and with each of the first and second switches being arranged in the OFF state to block current flow through the first series circuit in a direction that would short the output, and a short across one of the first or second switches does not short the output; and

wherein the switch control circuit is configured to prevent the first switch in each output circuit from turning ON if the second switch in the output circuit fails to open, and to prevent the second switch in the output circuit from turning ON if the first switch in the output circuit fails to open.

60. (Withdrawn) The apparatus of claim 59 wherein the switch control circuit is configured to sense across each of the first and second switches in each of the output circuits, a respective voltage polarity, and to turn each of the first and second switches ON when the sensed voltage polarity across the respective switch comprises a first polarity and OFF when the sensed voltage polarity across the respective switch comprises a second polarity, wherein the second polarity is opposite the first polarity.

61. (Withdrawn) A method of converting power received from an input source at an input voltage via a transformer for delivery via an output at an output voltage to a load, where the load may vary over a normal operating range, comprising:

using an input circuit having a plurality of primary switches to drive a primary winding of the transformer with power received from the input source;

using output circuitry having a plurality of secondary switches for rectifying power received from a secondary winding of the transformer to conduct current from the secondary winding to the output;

operating the primary switches in a series of converter operating cycles, each converter operating cycle characterized by

(a) two power transfer phases of essentially equal duration during which (i) one or more of the primary switches are ON, (ii) a primary winding of the transformer is driven with power received from the input, (iii) a magnetizing current flows through the primary winding and increases in magnitude, and (iv) power is transferred between the input and the output via the transformer; and

(b) a clamp phase during which a winding of the transformer is clamped, the clamp phase being characterized by (i) essentially zero voltage across the clamped winding and (ii) an average value of current flowing in the clamped winding, the average value of current having a polarity, wherein the average is taken over the duration of the clamp phase, and the polarity is the same as a polarity of an average value of current flowing in the clamped winding during the power transfer phase immediately preceding the clamp phase, and (iii) essentially zero current is conducted between the secondary winding and the output.

62. (Withdrawn) The method of claim 61 further comprising:

forming a resonant circuit including the transformer and having a characteristic resonant frequency and period; wherein a resonant current flows in the primary winding during the power transfer phases; and

turning one or more primary switches OFF essentially at times when the resonant current returns to zero.

63. (Withdrawn) The method of claim 61 wherein each converter operating cycle further comprises a plurality of energy-recycling intervals during which at least one primary switch transitions to OFF; and

further comprising using the magnetizing current to charge and discharge capacitances during the energy-recycling intervals.

64. (Withdrawn) The method of claim 61 further comprising using an inductance to limit the rate of change of current in the transformer; and wherein

the primary and secondary switches are operated in the series of converter operating cycles and each converter operating cycle further comprises:

a freewheel phase during which a circuit including a winding of the transformer is clamped, the freewheel phase being characterized by (i) essentially zero voltage across the clamped circuit and (ii) an average value of current flowing in the clamped circuit, the average value of current having a polarity, wherein the average is taken over the duration of the initial clamp phase, and the polarity is the same as a polarity of an average value of current flowing in the clamped winding during the power transfer interval immediately preceding the freewheel phase, (iii) a current flowing in the inductance declines in magnitude to essentially zero current, and (iv) the secondary switches conduct current between the secondary winding and the output.

65. (Withdrawn) The method of claim 64 wherein each clamp phase occurs after a respective freewheel phase; and the freewheel phase ends, and the secondary switches are turned OFF, essentially when the current between the secondary winding and the output returns to zero.

66. (Withdrawn) The method of claim 65 wherein each converter operating cycle further comprises a plurality of energy-recycling intervals during which at least one primary switch transitions to OFF; and

further comprising using the magnetizing current to charge and discharge capacitances during the energy-recycling intervals.

67. (Withdrawn) The method of claim 61, 62, or 65 further comprising:
varying the duration of the clamp phase to control an effective output resistance of the power conversion.

68. (Withdrawn) The method of claim 61, 62, or 65 further comprising:
varying the duration of the clamp phase to regulate conversion factor defined as the ratio of input voltage to output voltage.

69. (Withdrawn) The method of claim 61, 62, or 65 further comprising:
varying the duration of the clamp phase to reduce power losses at light loads.

70. (Withdrawn) The method of claim 61, 62 or 65 further comprising:
varying the duration of the clamp phase to control the rate of change of the output voltage.

71. (Withdrawn) The method of claim 61, 62, or 65 further comprising:
varying the duration of the clamp phase to regulate the output voltage.

72. (Withdrawn) A method of converting power received from an input source at an input voltage via transformer for supplying a load at a DC output voltage, comprising:
providing a driver circuit having an input for receiving power from the source and a driver output,
providing an output circuit for deployment near the load, the output circuit having an AC input and an output for supplying the DC output voltage to the load,
providing a power transformer having a first winding for electrical connection to the driver output and a second winding for electrical connection to the AC input of the output circuit;
forming a resonant circuit including a winding of the transformer, the resonant circuit being characterized by a characteristic resonant period;
providing a power bus between the driver circuit and the power transformer;
operating the driver circuit to drive the transformer in a series of converter operating cycles, each converter operating cycle characterized by two power transfer phases of essentially equal duration each having a duration less than the characteristic resonant period, during which one or more switches in the driver circuit are ON and power is transferred from the input to the output via the transformer; and

deploying the driver circuit as a self contained unit at a first location separated from the output circuit.

73. (Withdrawn) The method of claim 72, comprising deploying the output circuit as a self-contained unit that is separate from the self-contained unit of the driver circuit.

74. (Withdrawn) The method of claim 72 wherein providing an output circuit comprises providing an output circuit that comprises a first series circuit having a first switch and a second switch, the first switch being connected to the second switch at a first junction, the first series circuit having two end terminals that are connected to the output of the output circuit, and the first junction being connected to a first terminal of the second winding of the power transformer.

75. (Withdrawn) The method of claim 74 wherein providing an output circuit comprises providing an output circuit that comprises the first series circuit and a second series circuit having a third switch and a fourth switch, the third switch being connected to the fourth switch at a second junction, the second series circuit having two end terminals that are connected to the output of the output circuit, and the second junction being connected to a second terminal of the second winding of the power transformer.

76. (Withdrawn) The method of claim 74, comprising deploying the first and second switches of the output circuit on a die having the load, wherein the driver circuit is located outside of the die.

77. (Withdrawn) The method of claim 76 wherein the load comprises a processor core.

78. (Withdrawn) The method of claim 76, comprising providing control circuitry to operate the first and second switches of the output circuit, and deploying the control circuitry on the die having the load.

79. (Withdrawn) The method of claim 72 wherein the driver circuit comprises a plurality of primary switches to drive the first winding of the power transformer, and a resonant current flows in the first winding during the power transfer phases, and

the method comprises turning one or more of the primary switches OFF essentially at times when the resonant current returns to zero.

80. (Withdrawn) The method of claim 79 wherein a magnetizing current flows through the primary winding during the power transfer phases, and each converter operating cycle comprises a plurality of energy-recycling intervals during which at least one primary switch transitions to OFF, and

the method comprises using the magnetizing current to charge and discharge capacitances during the energy-recycling intervals.

81. (Withdrawn) An apparatus comprising:

a first package comprising a driver circuit having an input for receiving power from an input source and a driver output;

an output circuit having an AC input and an output for supplying a DC output voltage to a load, the output circuit being disposed outside of the first package;

a power transformer having a first winding for electrical connection to the driver output of the driver circuit and a second winding for electrical connection to the AC input of the output circuit;

a resonant circuit formed in part by one of the windings of the transformer, the resonant circuit having a characteristic resonant period; and

a power bus for electrical connection to the driver circuit and the power transformer;

wherein the driver circuit is configured to drive the transformer in a series of converter operating cycles, each converter operating cycle comprising two power transfer phases of essentially equal duration each having a duration less than the characteristic resonant period, during which one or more switches in the driver circuit are ON and power is transferred from the input of the driver circuit to the output of the output circuit via the transformer.

82. (Withdrawn) The apparatus of claim 81 wherein the output circuit and the power transformer are provided in a second package that is separate from the first package.

83. (Withdrawn) The apparatus of claim 81 wherein the output circuit comprises a first series circuit having a first switch and a second switch, the first switch being connected to the second switch at a first junction, the first series circuit having two end terminals that are connected to the output of the output circuit, and the first junction being connected to a first terminal of the second winding of the power transformer.

84. (Withdrawn) The apparatus of claim 83 wherein the output circuit comprises a second series circuit having a third switch and a fourth switch, the third switch being connected to the fourth switch at a second junction, the second series circuit having two end terminals that are connected to the output of the output circuit, and the second junction being connected to a second terminal of the second winding of the power transformer.

85. (Withdrawn) The apparatus of claim 83 wherein the first and second switches of the output circuit are provided on a die having the load.

86. (Withdrawn) The apparatus of claim 85 wherein the load comprises a processor core.

87. (Withdrawn) The apparatus of claim 85, comprising control circuitry to operate the first and second switches of the output circuit, wherein the control circuitry is provided on the die having the load.

88. (Withdrawn) The apparatus of claim 81, wherein the driver circuit comprises a plurality of primary switches to drive the first winding of the power transformer, and a resonant current flows in the first winding during the power transfer phases, and wherein the apparatus comprises a control circuit to operate the primary switches, and the control circuit is configured to turn OFF one or more of the primary switches essentially at times when the resonant current returns to zero.

89. (Previously Presented) A method of delivering power to an output, comprising:

providing a transformer having a secondary winding, the secondary winding having a first terminal and a second terminal;

providing a first switch, a second switch, a third switch and a fourth switch, each adapted to block current flow in an OFF state in at least one direction;

forming a first series circuit with the first switch connected to the second switch at a first junction, the first series circuit having two end terminals;

forming a second series circuit with the third switch connected to the fourth switch at a second junction, the second series circuit having two end terminals;

connecting the first terminal of the secondary winding to the first junction and the second terminal of the secondary winding to the second junction;

connecting the first series circuit across the output with each of the two end terminals connected to a respective terminal of the output and with each of the first and second switches being arranged in the OFF state to block current flow through the first series circuit in a direction that would short the output, wherein a short across one of the first or second switches does not short the output;

connecting the second series circuit across the output with each of the two end terminals connected to a respective terminal of the output and with each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output, wherein a short across one of the third or fourth switches does not short the output;

converting power via the transformer to deliver a unipolar voltage of five Volts or less to the output.

90. (Previously Presented) The method of claim 89 wherein the unipolar output is approximately 1 Volt.

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Serial No. : 13/044,063
Filed : March 9, 2011
Page : 27 of 33

Attorney's Docket No.: 00614-0166001

Amendments to the Drawings:

The attached replacement sheets of drawings include changes to FIGS. 1, 2A, 2B, 5, 6, and 9 and replaces the original sheets including FIGS. 1, 2A, 2B, 5, 6, and 9.

In Figures 1, 2A, 2B, and 9, the legend "Prior Art" has been added as suggested by the Examiner.

In Figures 1, 5, and 6, descriptive text labels have been added as suggested by the Examiner.

Attachments following last page of this Amendment:

Replacement Sheet (5 pages)

REMARKS

The comments of the Applicant below are each preceded by related comments of the Examiner (in small, bold type).

Drawings

**Figures 1, 2A, 2B and 9 should be designated by a legend such as -
-Prior Art -- because only that which is old is illustrated.**

**...
The drawings are objected to because of the following
informalities. Regarding Fig. 1, 5, and 6, the unlabeled rectangular box(es)
shown in the drawings should be provided with descriptive text labels.**

Figures 1, 2A, 2B, 5, 6, and 9 have been amended.

Specification

**The abstract of the disclosure is objected to due to its undue
length. Applicant is reminded of the proper language and format for an
abstract of the disclosure**

The Abstract has been amended.

Claim Objections

**Claims 1, 2 and 5 are objected to because of the following
informalities:
 regarding claim 1, in line 9, "the output" should read as "an
output";
 regarding claim 2, in line 5, "the second polarity is opposite the ...
" should read as "the second polarity is opposite of the ... ";
 regarding claim 5, in line 5, "the second polarity is opposite the ...
" should read as "the second polarity is opposite of the ... ";**

Claims 1, 2, and 5 have been amended.

Claim Rejections – 35 USC § 112

**Claims 18, 20-23, 28, and 58 are rejected under 35 U.S.C. 112(b)
or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for
failing to particularly point out and distinctly claim the subject matter
which the inventor or a joint inventor, or for pre-AIA the applicant
regards as the invention.**

Claims 18, 20, and 58 have been amended.

Claim Rejections – 35 USC §103

Claims 1, 3, 4, 16, 44, 46, and 57 are rejected under 35 U.S.C. 103(a) as being unpatentable over Chen et al. (US Patent 7,768,807 B2, hereafter referred to as "Chen") in view of ISL621 0 (Datasheet ISL621 0, "Dual Synchronous Rectified MOSFET Drivers," Intersil Americas Inc., December 2008).

Claim 1

The Examiner acknowledges that Chen does not disclose preventing the first switch from turning ON if the second switch fails to open; and preventing the second switch from turning ON if the first switch fails to open. The Applicant notes this feature is also not disclosed or suggested by ISL6210.

ISL6210 describes an adaptive shoot-through circuit that monitors the LGATE voltage, such that when LGATE has fallen below 1V, UGATE is allowed to turn ON (page 6). The LGATE voltage is the voltage of the lower gate drive output, which is connected to the gate of the low-side power N-Channel MOSFET (page 5). However, monitoring the LGATE voltage alone does not provide information on whether the MOSFET failed. For example, if the MOSFET malfunctions, the MOSFET may be shorted even when the LGATE voltage has fallen below 1V, and upon detecting that the LGATE voltage has fallen below 1V, the driver in ISL6210 will allow UGATE to turn on. The adaptive shoot-through circuit of ISL6210 prevents the lower and upper MOSFETs from conducting simultaneously if both lower and upper MOSFETs are functioning properly. The adaptive shoot-through circuit of ISL6210 does not prevent the lower MOSFET from turning ON if the upper MOSFET fails to open (e.g., due to failure of the device), and does not prevent the upper MOSFET from turning ON if the lower MOSFET fails to open (e.g., due to failure of the device). Therefore, ISL6210 does not disclose or suggest "preventing the first switch from turning ON if the second switch fails to open; and preventing the second switch from turning ON if the first switch fails to open," as recited in claim 1.

Because neither Chen nor ISL6210 discloses the feature of claim 1 recited above, claim 1 would not have been made been obvious by Chen and ISL6210. Claim 1 is patentable over Chen and ISL6201 for at least the above reasons.

Claim 44

The Examiner acknowledges that Chen does not disclose a switch control circuit configured to prevent the first switch from turning ON if the second switch fails to open, and to prevent the second switch from turning ON if the first switch fails to open. The Applicant notes this feature is also not disclosed or suggested by ISL6210.

ISL6210 describes an adaptive shoot-through circuit that monitors the LGATE voltage, such that when LGATE has fallen below 1V, UGATE is allowed to turn ON (page 6). The LGATE voltage is the voltage of the lower gate drive output, which is connected to the gate of the low-side power N-Channel MOSFET (page 5). However, monitoring the LGATE voltage alone does not provide information on whether the low-side MOSFET failed. For example, if the MOSFET malfunctions, the MOSFET may be shorted even when the LGATE voltage has fallen below 1V; upon detecting that the LGATE voltage has fallen below 1V, the driver in ISL6210 would allow UGATE to turn on. The adaptive shoot-through circuit of ISL6210 prevents the lower and upper MOSFETs from conducting simultaneously if both lower and upper MOSFETs are functioning properly. The adaptive shoot-through circuit of ISL6210 is not configured to prevent the lower MOSFET from turning ON if the upper MOSFET fails to open (e.g., due to failure of the device), and is not configured to prevent the upper MOSFET from turning ON if the lower MOSFET fails to open (e.g., due to failure of the device). Therefore, ISL6210 does not disclose or suggest “a switch control circuit configured to prevent the first switch from turning ON if the second switch fails to open, and to prevent the second switch from turning ON if the first switch fails to open,” as recited in claim 44.

Because neither Chen nor ISL6210 discloses the feature of claim 44 recited above, claim 44 would not have been made been obvious by Chen and ISL6210. Claim 44 is patentable over Chen and ISL6201 for at least the above reasons.

Claims 18, and 20 are rejected under 35 U.S.C. 103(a) as being unpatentable over Chen in view of AN2738 (Application Note AN2738, "L6390 half-bridge gate driver," STMicroelectronics, August 2009).

Claim 18

The Examiner acknowledges that Chen does not disclose monitoring selected circuit conditions; determining from the selected circuit conditions whether a predetermined fault is present, disabling the switches in response to a determination that that predetermined fault is present.

AN2738 discloses monitoring the supply voltage VCC using an under-voltage lockout circuitry that turns off the IC outputs when the supply voltage goes below a threshold and turns on the device when the supply voltage goes above the threshold (page 7). A person of ordinary skill in the art would have no reason to modify Chen's switching power conversion circuit to use the under-voltage lockout circuitry of AN2738. In Chen, the switches are not powered by a supply voltage, so there is no need to detect whether a supply voltage is below a threshold and then turn off the switches when the supply voltage is below the threshold.

Furthermore, in AN2738, L6390 is a half-bridge gate driver. The conditions for affecting a half-bridge gate driver are quite different from the conditions affecting the full-bridge gate driver described in Chen. It is unclear how the under-voltage lockout circuitry of the half-bridge gate driver of AN2738 can be used in Chen's full bridge switching power conversion circuit. It is also unclear from AN2738 what circuit conditions in Chen's full bridge switching power conversion circuit should be monitored to determine a predetermined fault, such that the switches should be disabled upon determining that the predetermined fault is present.

Because a person of ordinary skill in the art would not have modified Chen full bridge switching power conversion circuit based on the teachings of AN2738, claim 18 would not have been made obvious by the combination of Chen and AN2738.

Claim 18 is patentable over Chen and AN2738 for at least the above reasons.

Claims 89 and 90 are rejected under 35 U.S.C. 103(a) as being unpatentable over Chen.

Claim 89

Chen does not disclose and would not have made obvious “each of the first and second switches being arranged in the OFF state to block current flow through the first series circuit in a direction that would short the output, wherein a short across one of the first or second switches does not short the output,” as recited in claim 89.

In claim 89, each of the first and second switches is arranged in the OFF state, wherein a short across one of the first or second switches does not short the output. This condition is not disclosed in Chen. Chen discloses operation conditions in which the switches are all functioning properly, such that when a switch is arranged in the OFF state, there would not be a short across the switch. Chen does not describe a situation (e.g., where a switch fails) such that when the switch is arranged in the OFF state, there is still a short across the switch.

Similarly, Chen does not disclose or suggest “each of the third and fourth switches being arranged in the OFF state to block current flow through the second series circuit in a direction that would short the output, wherein a short across one of the third or fourth switches does not short the output,” as recited in claim 89.

Because Chen does not disclose or suggest the above features of claim 89, claim 89 is patentable over Chen.

Allowable Subject Matter

Claims 2, 5, 6, 11, 17, 19, 22, 23, 28, 45, 47, 52, and 58 are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

The Applicant acknowledges that claims 2, 5, 6, 11, 17, 19, 22, 23, 28, 45, 47, 52, and 58 are allowable, but notes that the other claims are also allowable for the reasons stated above.

All of the dependent claims are patentable for at least the reasons for which the claims on which they depend are patentable.

First Named Inventor : Patrizio Vinciarelli
Serial No. : 13/044,063
Filed : March 9, 2011
Page : 33 of 33

Attorney's Docket No.: 00614-0166001

Any circumstance in which the Applicant has addressed certain comments of the Examiner does not mean that the Applicant concedes other comments of the Examiner. Any circumstance in which the Applicant has made arguments for the patentability of some claims does not mean that there are not other good reasons for patentability of those claims and other claims. Any circumstance in which the Applicant has amended a claim does not mean that the Applicant concedes any of the Examiner's positions with respect to that claim or other claims.

The fees for excess claims and extension of time are being paid with this reply on the Electronic Filing System. Apply those fees and any other necessary charges or credits to Deposit Account 06-1050, referencing the above attorney docket number.

Respectfully submitted,

Date: October 23, 2014_____

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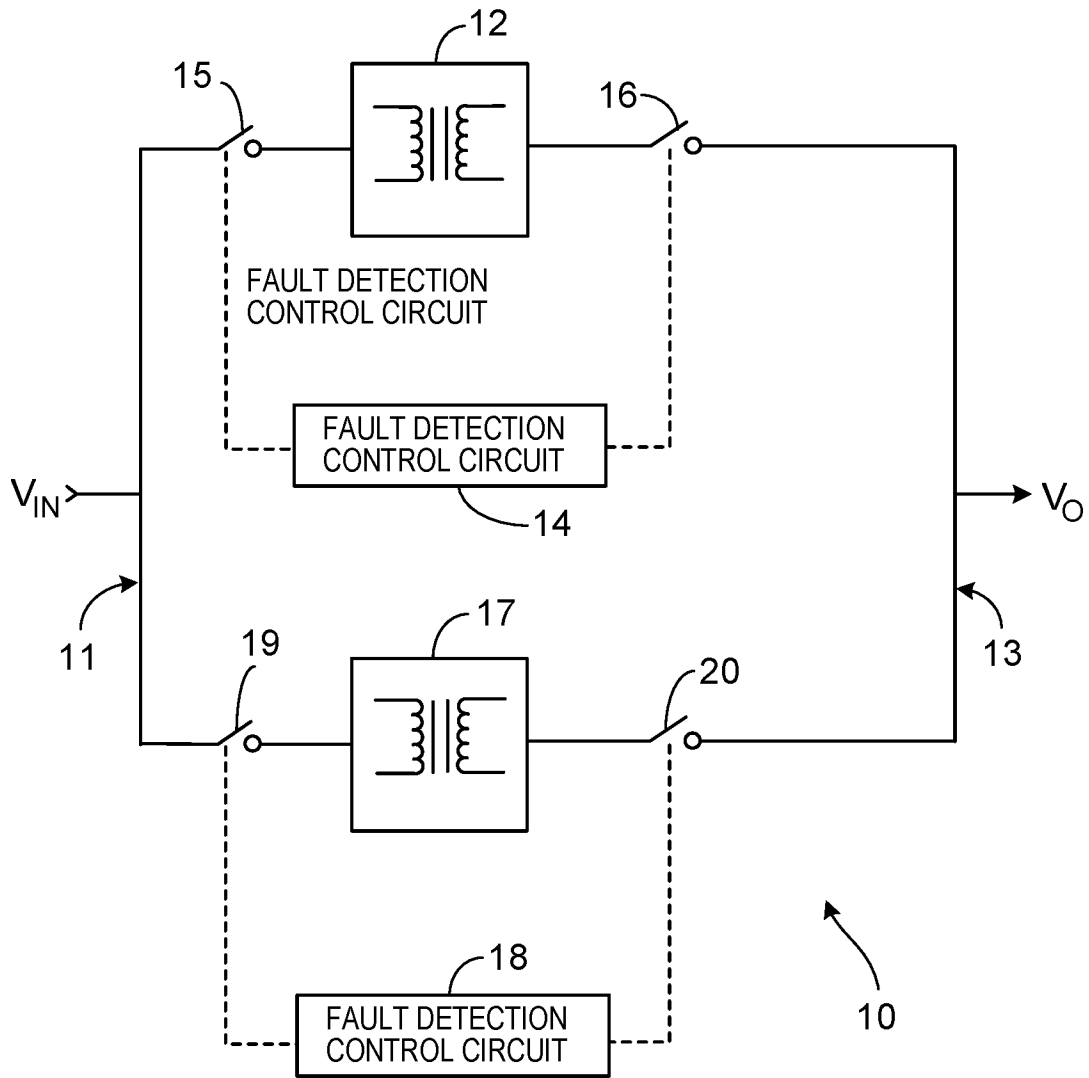
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ABSTRACT

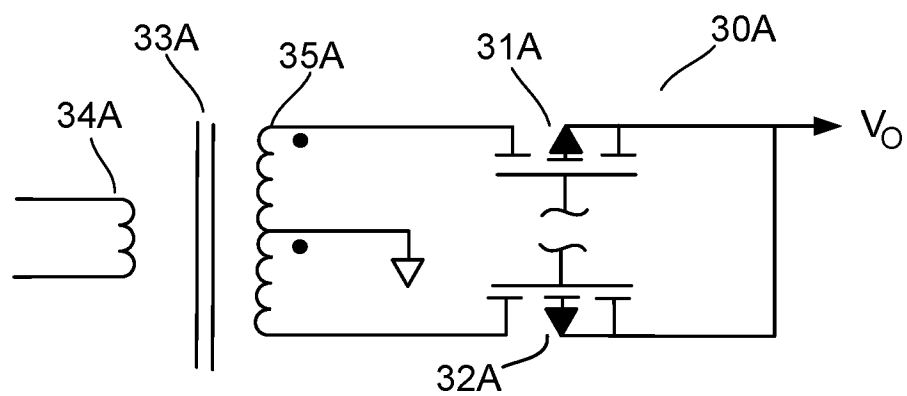
A power converter provides a low-voltage output using a full-bridge fault-tolerant rectification circuit. The output circuit uses controlled switches as rectifiers. A fault detection circuit monitors circuit conditions. Upon detection of a fault, the switches are disabled decoupling the power converter from the system.

A common-source dual MOSFET device includes a plurality of elements arranged in alternating patterns on a semiconductor die. A common-source dual synchronous rectifier includes control circuitry powered from the drain to source voltage of the complementary switch.

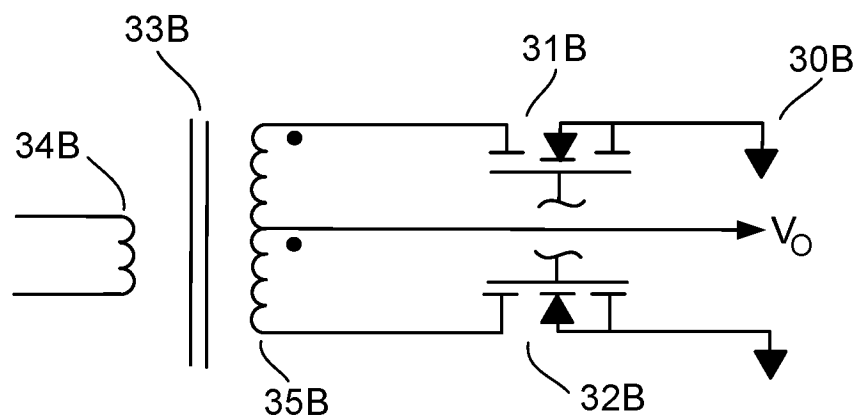
A DC-to-DC transformer converts power from an input source to a load using a fixed voltage transformation ratio. A clamp phase may be used to reduce power losses in the converter at light loads, control the effective output resistance of the converter, effectively regulate the voltage transformation ratio, provide narrow band output regulation, and control the rate of change of output voltage for example during start up.



(PRIOR ART)
FIG. 1



(PRIOR ART)
FIG. 2A



(PRIOR ART)
FIG. 2B

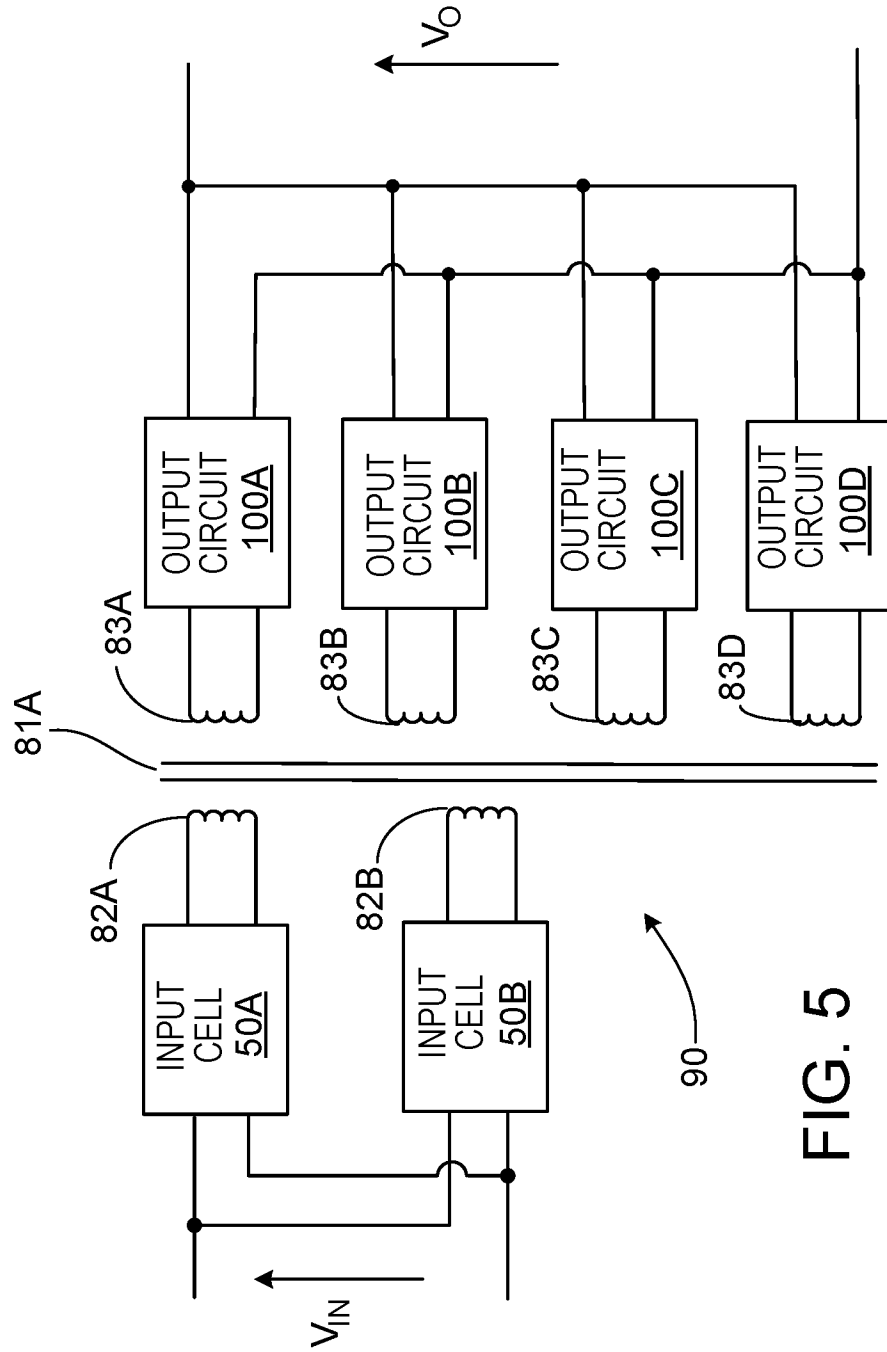


FIG. 5

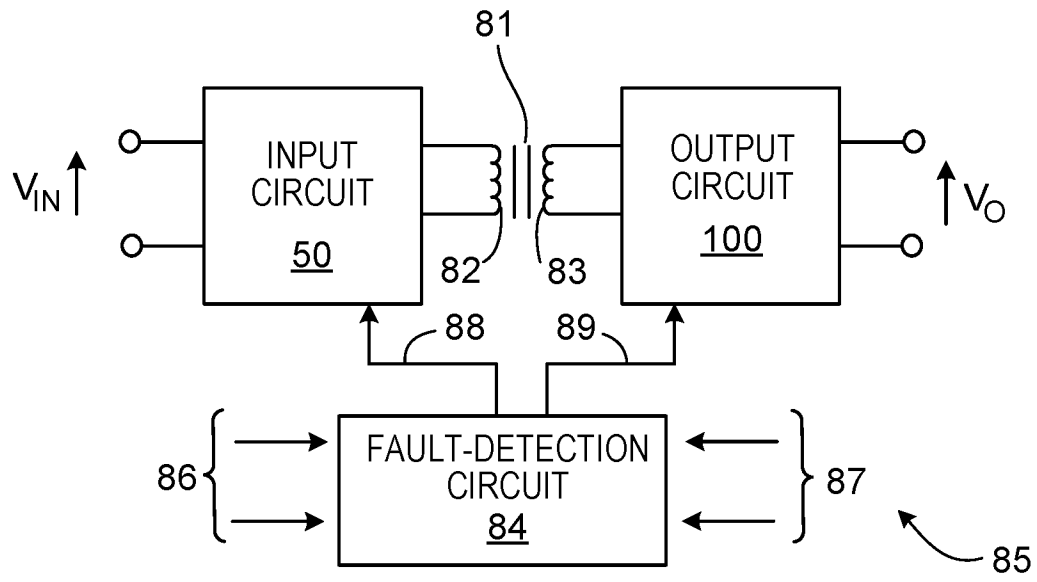
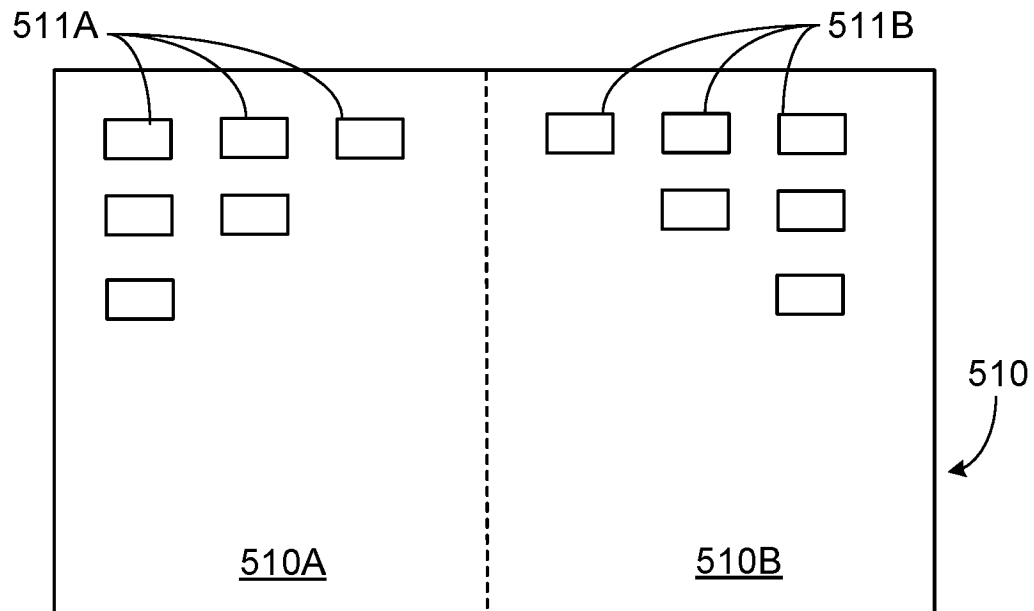


FIG. 6



(PRIOR ART)
FIG. 9

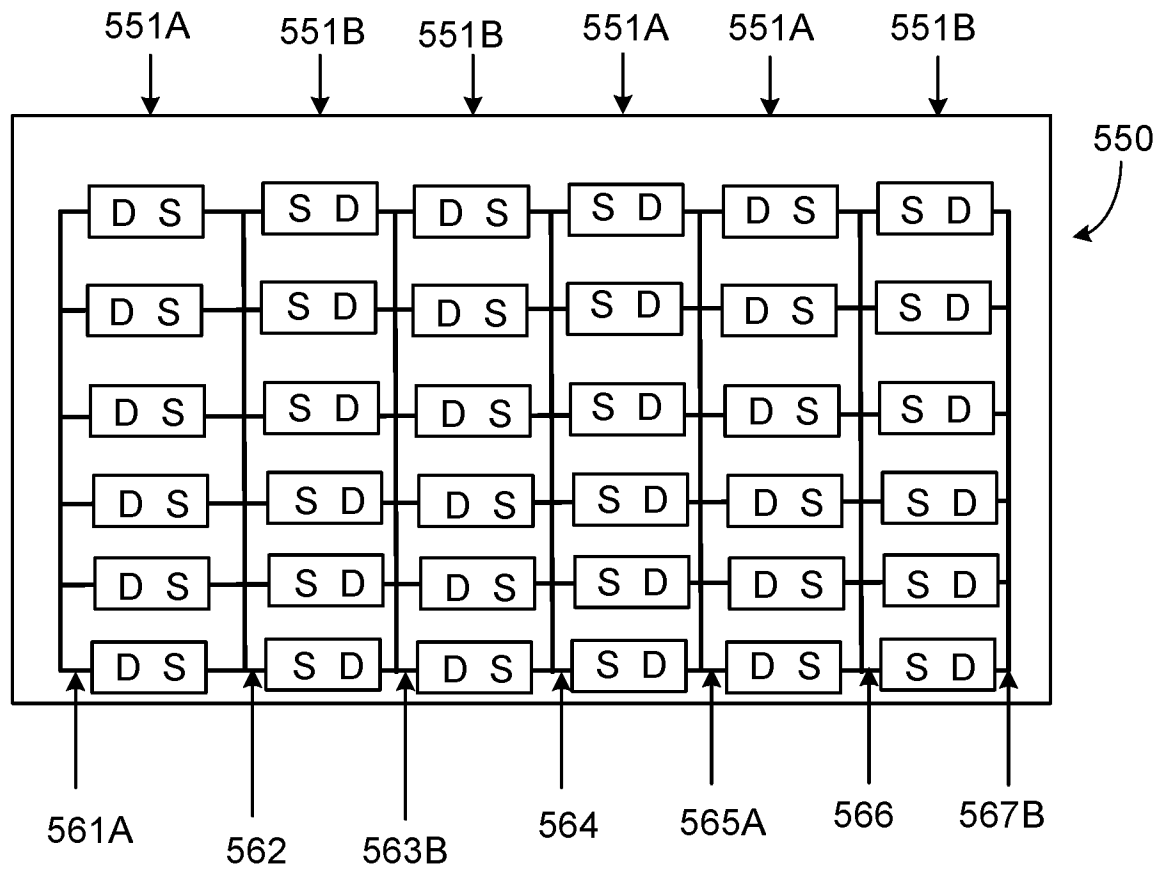


FIG. 10



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EXAMINER

LEE, JYE-JUNE

ART UNIT	PAPER NUMBER
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2838

NOTIFICATION DATE	DELIVERY MODE
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04/20/2015

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

PATDOCTC@fr.com

**Supplemental
Notice of Allowability**

Application No.
13/044,063

Examiner
JYE-JUNE LEE

Applicant(s)
VINCIARELLI, PATRIZIO

Art Unit
2838

**AIA (First Inventor to
File) Status**
No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to the amendment filed on 03/03/2015.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1-6,11,16-23,28,44-47,52,57,58 and 89-114. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying Indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892)
2. ☐ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date _____
3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material
4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date _____
5. ☐ Examiner's Amendment/Comment
6. ☒ Examiner's Statement of Reasons for Allowance
7. ☒ Other PTO-271.

/ADOLF BERHANE/
Primary Examiner, Art Unit 2838

DETAILED ACTION

1. The present application is being examined under the pre-AIA first to invent provisions.
2. This action is in response to the amendment filed on 03/03/2015.

Reasons For Allowance

3. **Claims 1-6, 11, 16-23, 28, 44-47, 52, 57, 58, 89, 90-114** are allowed.

The following is an examiner's statement of reasons for allowance:

Regarding **claim 1**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which "...sensing an output of the second switch and preventing the first switch from turning ON if the sensing indicates that the second switch is not open; and sensing an output of the second switch and preventing the second switch from turning ON if the sensing indicates that the first switch is not open.".

Regarding **claim 18**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which "...sensing an output of the second switch and preventing the first switch from turning ON if the sensing indicates that the second switch is not open; and sensing an output of the second switch and preventing the second switch from turning ON if the sensing indicates that the first switch is not open.".

Regarding **claim 44**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which "...a switch control circuit configured to sense an output of the second switch and prevent the first switch from turning ON if the sensing indicates that the second switch is not open, and to sense an output of the first

Art Unit: 2838

switch and prevent the second switch from turning ON if the sensing indicates that the first switch is not open.”.

Regarding **claim 89**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which “...sensing outputs of the first and second switches, arranging the first switch in the OFF state if the sensing indicates that the second switch is not open, and arranging the second switch in the OFF state if the sensing indicates that the first switch is not open...sensing outputs of the third and fourth switches, arranging the third switch in the OFF state if the sensing indicates that the fourth switch is not open, and arranging the fourth switch in the OFF state if the sensing indicates that the third switch is not open...”.

The aforementioned limitations in combination with all remaining limitations of the respective claims are believed to render the aforementioned indicated claim and any dependent claims thereof patentable over the art of record.

4. Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled “Comments on Statement of Reasons for Allowance.”

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to JYE-JUNE LEE whose telephone number is (571)270-

Art Unit: 2838

7726. The examiner can normally be reached on Monday to Thursday 9:00 AM to 7:30 PM EST.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Monica Lewis can be reached on 571-272-1838. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/JYE-JUNE LEE/

Examiner, Art Unit 2838

/ADOLF BERHANE/

Primary Examiner, Art Unit 2838

L6390 half-bridge gate driver

Introduction

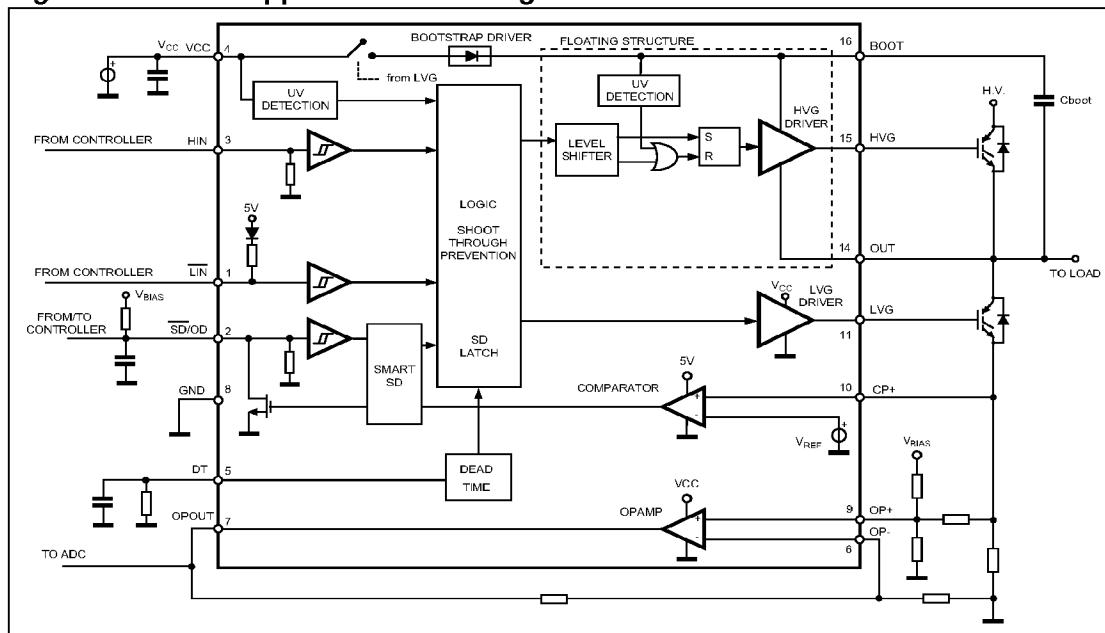
The L6390 is a versatile high voltage gate driver IC which is particularly suited for field oriented control (FOC) motor driving applications. It simplifies the design of control systems for a wide range of motor applications such as home appliances, industrial drives, DC motors and fans.

Designed using BCD off-line technology, this device is capable of operating with voltage rails up to 600 V. The gate driver provides all the functions and current capability necessary for high side and low side power MOSFET and IGBT driving.

The L639x series are high voltage half-bridge gate drivers. These devices can be used in all applications where high voltage shifted control is necessary. The devices have a driver current capability best suited for home appliance motor driving ratings, and they are also equipped with patented internal circuitry which replaces the external bootstrap diode. This feature is achieved by means of a high voltage DMOS synchronously driven with the low side gate driver.

The L6390 is a half-bridge driver with several functions such as externally adjustable dead-time, interlocking, smart shutdown (patented), fault comparator and a dedicated high performance op-amp for advanced current sensing. The outputs can be driven by two dedicated logic signals or, alternatively, only one logic signal by connecting the two inputs together. The device is available in the DIP16 or SO16 packages.

Figure 1. L6390 application block diagram



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1 Pin description

Table 1. Pin description

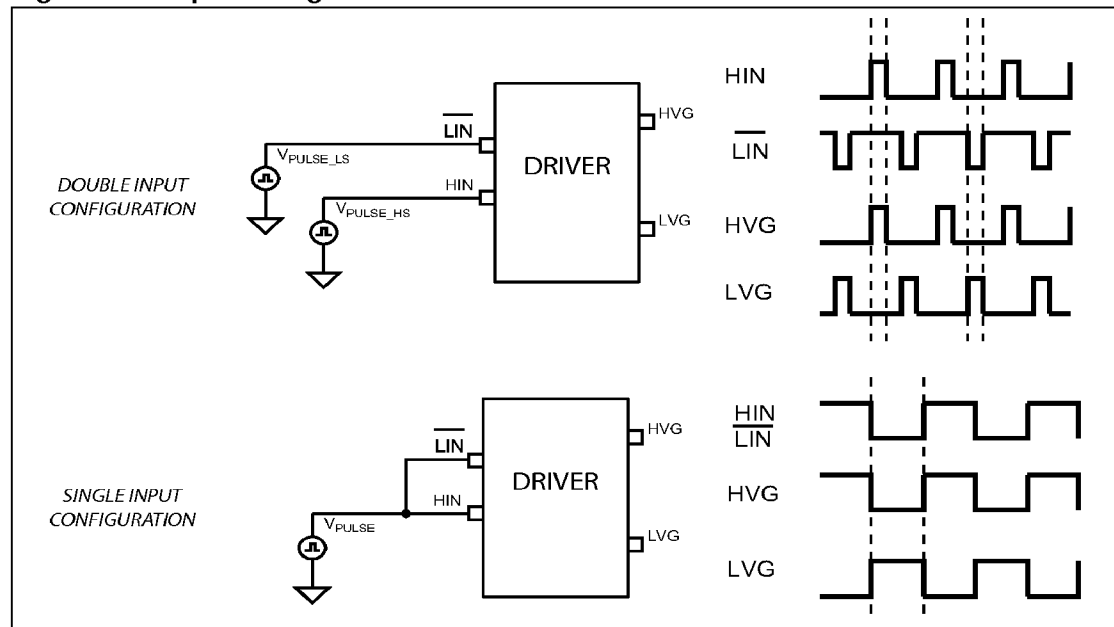
Pin n°	Pin name	Type	Function
1	$\overline{\text{LIN}}$	I	Low side driver logic input (active low)
2	$\overline{\text{SD}}/\text{OD}^{(1)}$	I/O	Shutdown logic input (active low)/open drain (comparator output)
3	HIN	I	High side driver logic input (active high)
4	VCC	P	Lower section supply voltage
5	DT	I	Dead time setting
6	OP-	I	Op-amp inverting input
7	OPOUT	O	Op-amp output
8	GND	P	Ground
9	OP+	I	Op-amp non inverting input
10	CP+	I	Comparator input
11	LVG ⁽¹⁾	O	Low side driver output
12,13	NC		Not connected
14	OUT	P	High side (floating) common voltage
15	HVG ⁽¹⁾	O	High side driver output
16	BOOT	P	Floating section (bootstrap) supply voltage

1. The circuit provides less than 1 V on the LVG and HVG pins (@ $I_{\text{sink}} = 10 \text{ mA}$), with $V_{\text{CC}} > 3 \text{ V}$. This allows the omission of the “bleeder” resistor connected between the gate and the source of the external MOSFET normally used to hold the pin low. The gate driver ensures low impedance in SD conditions also. See *Section 4*.

2 Logic inputs

The L6390 has two logic inputs, HIN and $\overline{\text{LIN}}$, to separately control the high side and low side outputs, HVG and LVG. HIN is in phase with HVG, while $\overline{\text{LIN}}$ is out of phase with LVG. The signal inversion on the low side input allows control of the half-bridge output with only one control signal (see *Figure 2*).

Figure 2. Input configuration



Note that by connecting the two logic input signals together, the resulting dead time is defined by the resistor connected between pin 5 and ground. The dead time can be set to a wide range of values from hundreds of nanoseconds to a few microseconds (see *Figure 5* or the L6390 datasheet). All the logic inputs are provided with hysteresis (~ 1 V) for low noise sensitivity and are TTL/CMOS 3.3 V compatible. Thanks to this low voltage interface logic compatibility, the L6390 can be used with any kind of high performance controller, such as microcontrollers, DSPs or FPGAs.

As shown in the block diagram in *Figure 1*, the logic inputs have internal pull-down (or pull-up) resistors. The purpose of these resistors is to set a proper logic level in case, for example, there is an interruption in the logic lines or the controller outputs are in tri-state conditions. If logic inputs are left floating, the gate driver outputs LVG and HVG are set to low level. The internal resistors are:

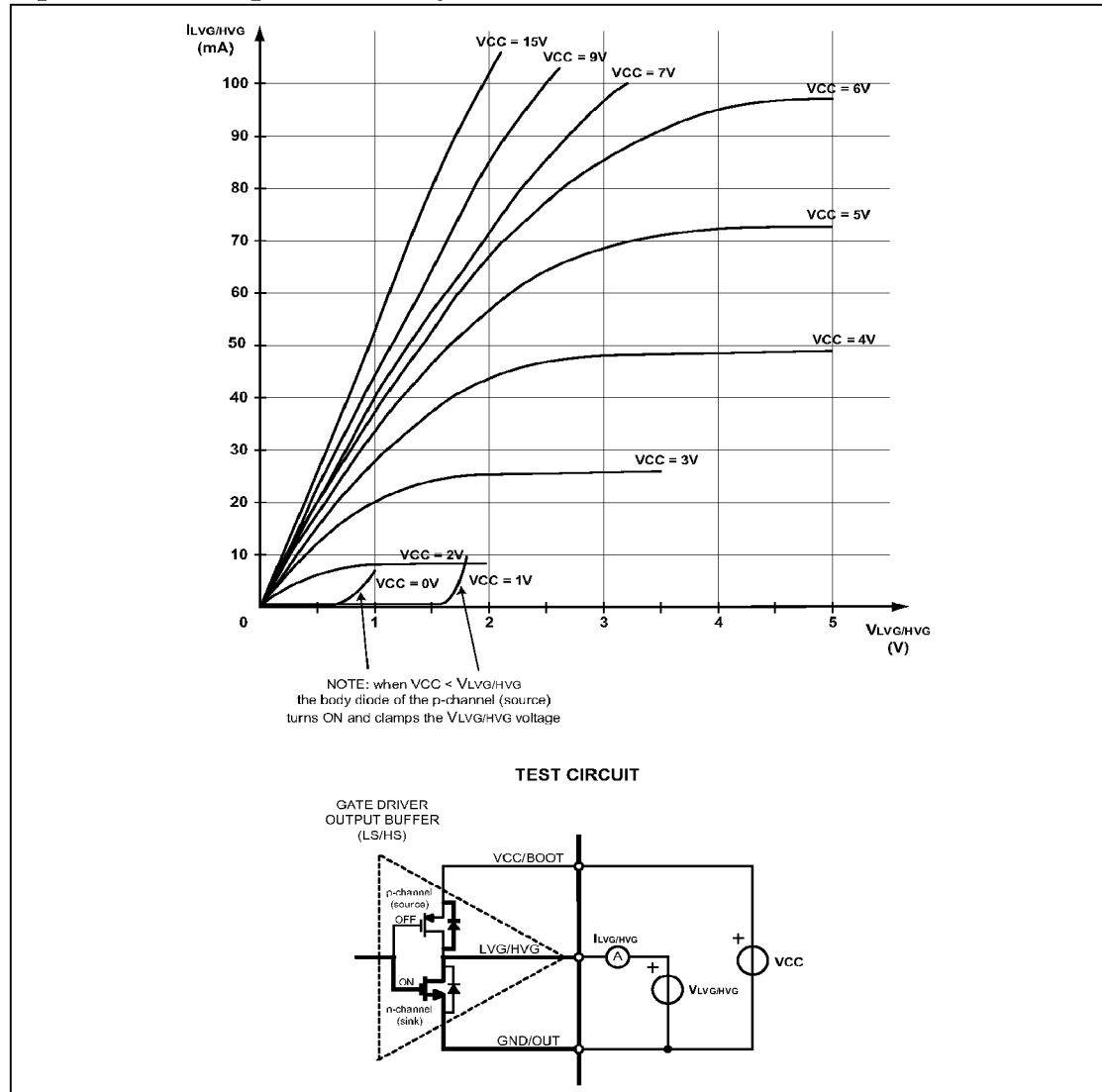
- HIN logic input: 85 k Ω (typ.) pull-down
- $\overline{\text{LIN}}$ logic input: 720 k Ω (typ.) pull-up connected to an internal 5 V regulator through a diode
- $\overline{\text{SD}}$ logic input: 375 k Ω (typ.) pull-down

If the logic inputs are connected together as in the single input configuration (*Figure 2*) and they are left floating, the internal pull-down and pull-up resistors form a resistive divider providing a voltage value (about 460 mV) which keeps the HVG off and LVG on, thus turning on the low side power switch.

3 UVLO function

The L6390 supply voltage V_{CC} is continuously monitored by an under-voltage lockout (UVLO) circuitry which turns off the IC outputs when the supply voltage goes below the V_{CC_thOFF} threshold (see L6390 datasheet for values) and turns on the device when the supply voltage goes above the V_{CC_thON} voltage. A hysteresis of about 1.5 V is provided for noise rejection purpose. The high voltage floating supply V_{BOOT} is provided with a similar under-voltage lockout circuitry also. When the L6390 is in UVLO condition, both gate driver outputs are set to low level, setting the half-bridge power stage output to high impedance. *Figure 3* below shows the I-V characteristics of the output buffers at different V_{CC} values.

Figure 3. L6390 gate driver outputs in UVLO condition



4 Dead time and interlocking function management

In order to avoid any possible cross-conduction between the power MOSFETs/IGBTs of the half-bridge, the L6390 provides both the dead time and the interlocking functions. The interlocking function is a logic operation which sets both the outputs to low level when the inputs are simultaneously active (HIN to high level and LIN to low level). The dead time function is a safety time introduced by the device between the falling edge transition of one driver output and the rising edge of the other output. If the rising edge set externally by the user occurs before the end of this dead time, it is ignored and results delayed until the end of the dead time. The dead time can be adjusted externally through the value of the DT resistor connected between pin 5 and pin 8 (see *Figure 5*). A 100 nF ceramic capacitor in parallel with this resistor is recommended for noise immunity. In *Figure 4* the details of dead time and interlocking function management are described.

Table 2. L6390 truth table

Inputs			Outputs	
SD	LIN	HIN	LVG	HVG
L	X ⁽¹⁾	X ⁽¹⁾	L	L
H	H	L	L	L
H	L	H	L	L
H	L	L	H	L
H	H	H	L	H

1. Don't care



Figure 4. Timing waveforms

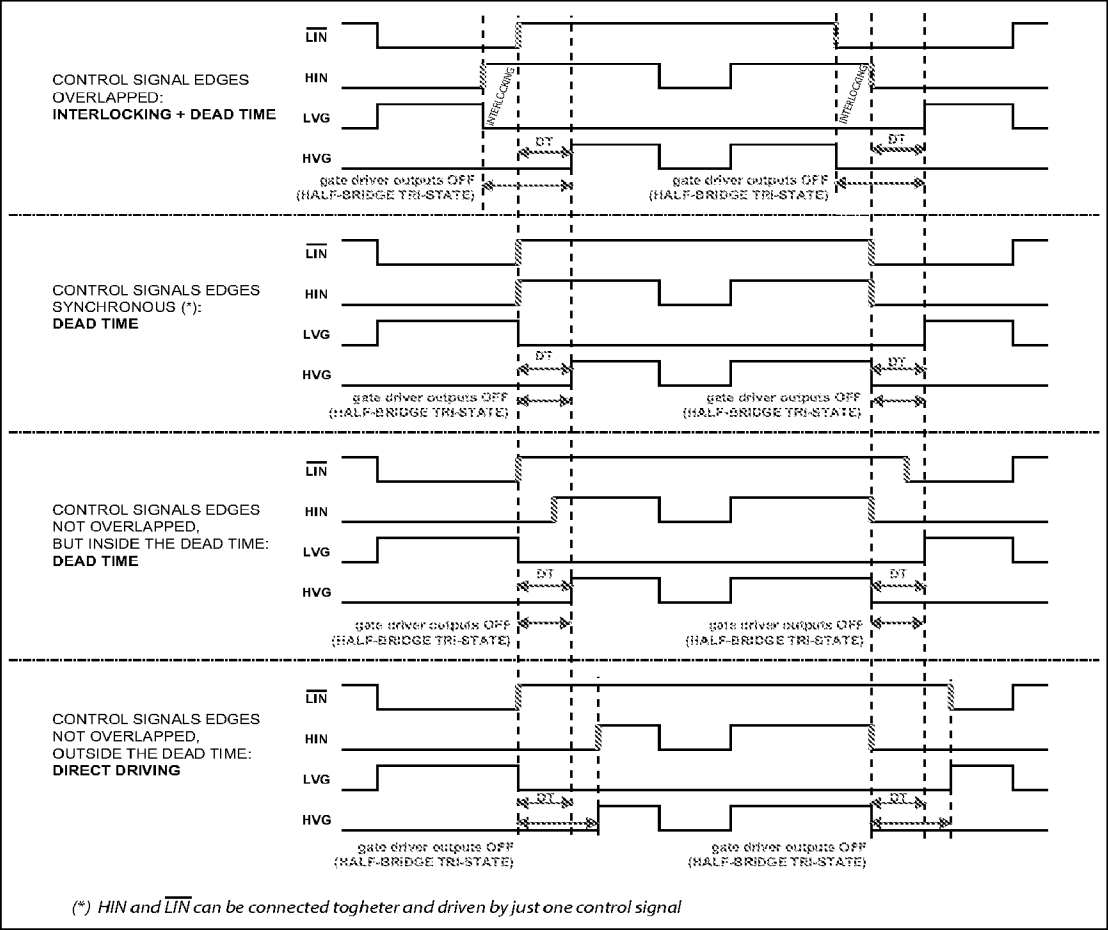
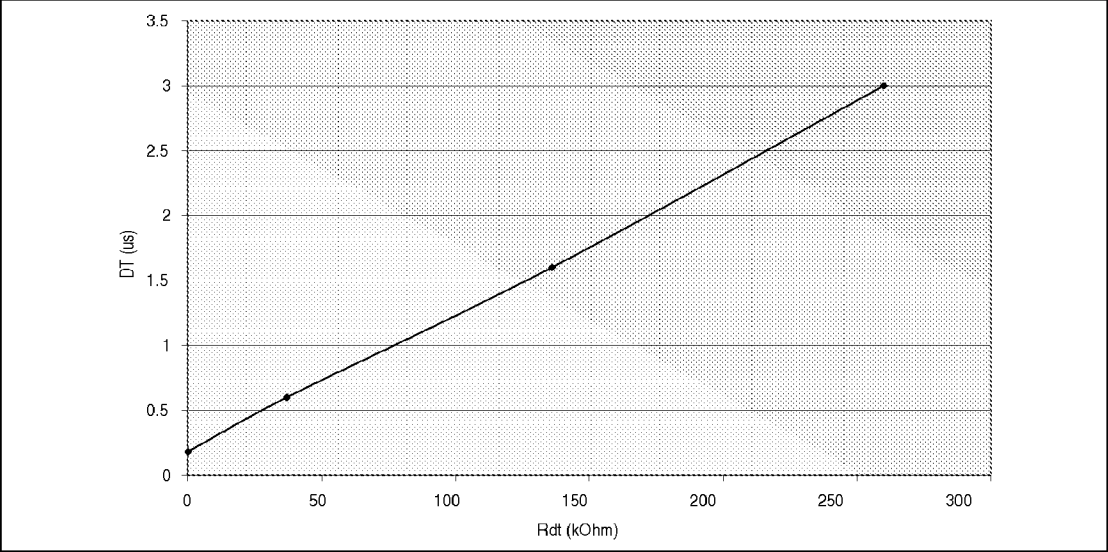


Figure 5. Typical dead time vs. DT resistor value

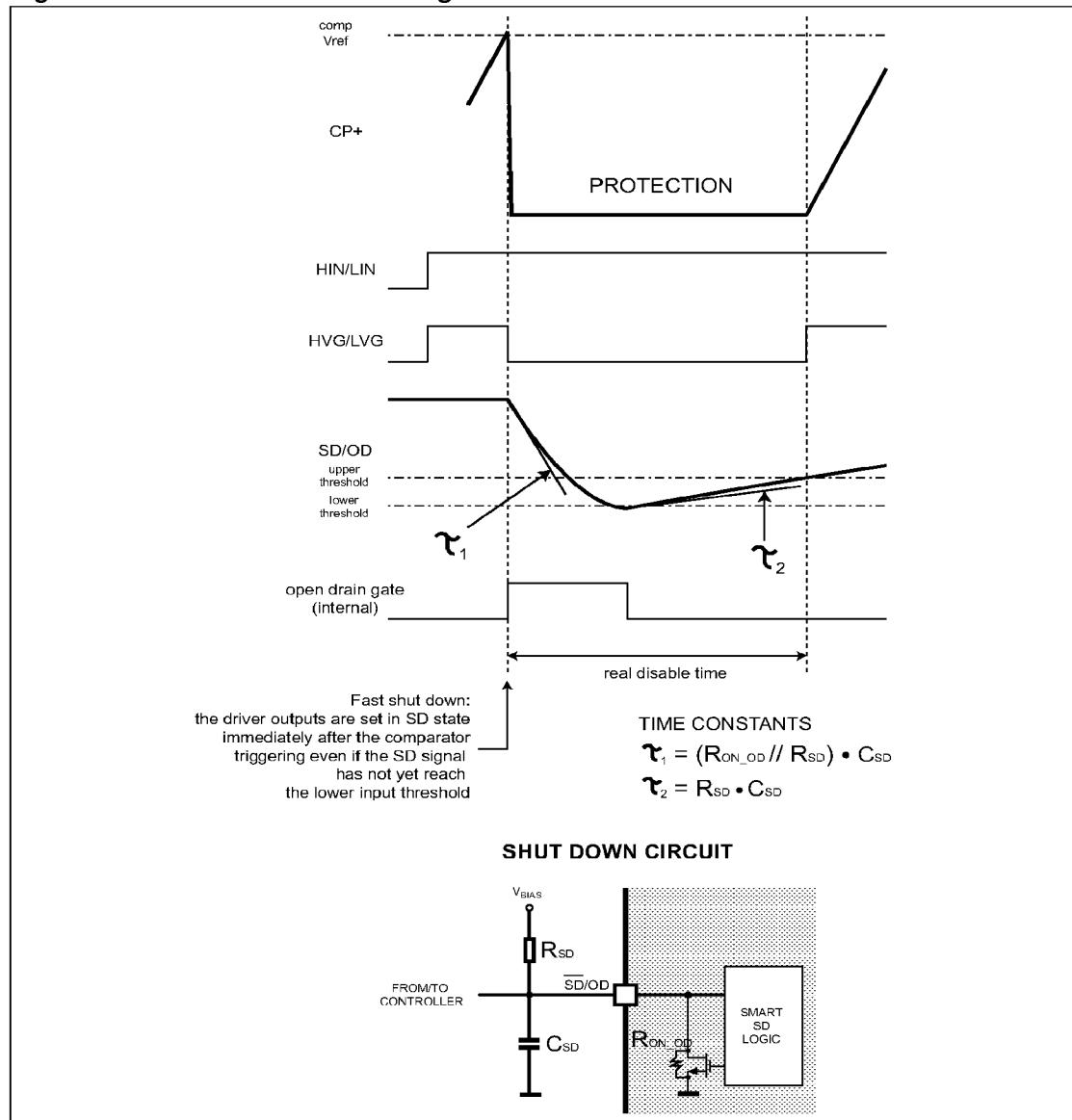


5 Smart shutdown function

The L6390 integrates a comparator for fault sensing purposes. The comparator has an internal reference voltage V_{ref} on its inverting input (see L6390 datasheet), while the non-inverting input is available on pin 10. The comparator input can be connected to an external shunt resistor in order to implement a simple over-current detection function. The output signal of the comparator is fed to an integrated MOSFET with the open drain available on pin 2, shared with the $\overline{SD}$ input.

When the comparator triggers, the device is set in shutdown state and both its outputs are set to low level leading the half-bridge in tri-state.

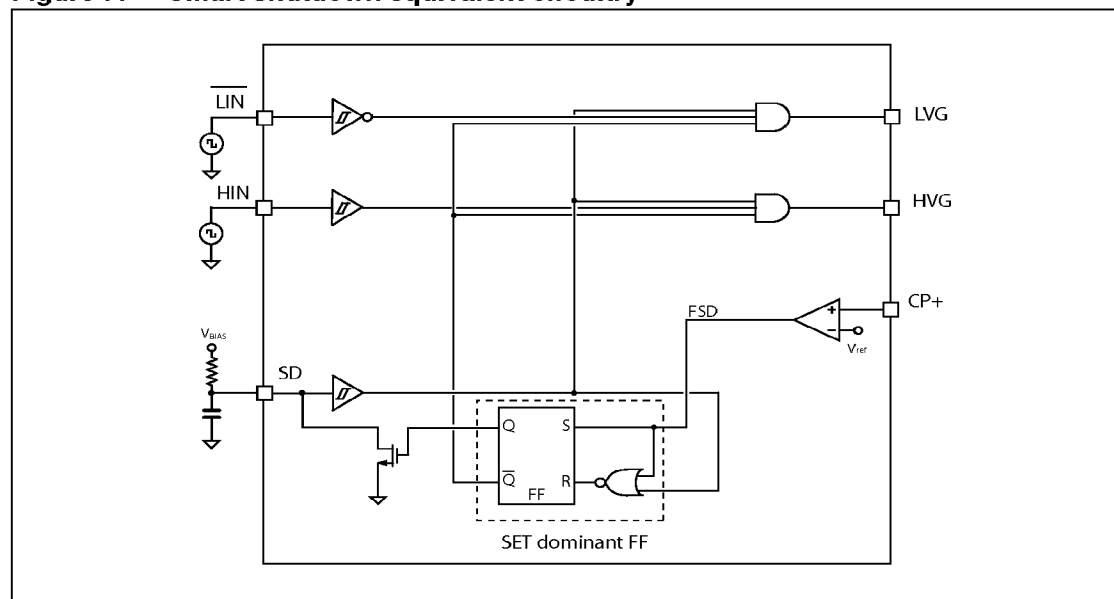
Figure 6. Smart shutdown timing waveforms



In common over-current protection architectures the comparator output is usually connected to the $\overline{SD}$ input and an RC network is connected to this $\overline{SD}/OD$ line in order to provide a mono-stable circuit, which implements a protection time that follows the fault condition. Unlike common fault detection systems, the L6390 smart shutdown architecture allows to immediately turn-off the output gate driver in case of fault, by minimizing the propagation delay between the fault detection event and the actual outputs switch-off. In fact the time delay between the fault and the outputs turn-off is no more dependent on the RC value of the external network connected to the pin. In the smart shutdown circuitry, the fault signal has a preferential path which directly switches off the outputs after the comparator triggering. At the same time the internal logic turns on the open drain output and holds it on until the SD voltage goes below the SD logic input lower threshold. The smart SD system provides the possibility to increase the time constant of the external RC network (that is the disable time after the fault event) without increasing the delay time of the protection. Any external signal provided to the SD pin is not latched and can be used as control signal in order to perform, for instance, PWM chopping through this pin. In fact when a PWM signal is applied to the SD input and the logic inputs of the gate driver are stable, the outputs switch from the low level to the state defined by the logic inputs and vice-versa.

A block diagram of the smart shutdown architecture is depicted in *Figure 7*.

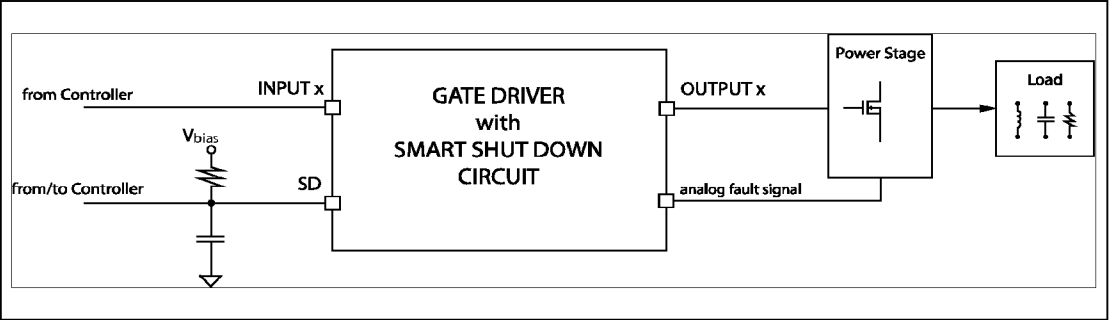
Figure 7. Smart shutdown equivalent circuitry



In normal operation the outputs follow the commands received from respective input signals. When a fault detection event occurs the fault signal (FSD) is set to high by the fault detection circuit output (LVG, HVG) and the FF receives a SET input signal. Consequently the FF outputs set output signals to low level and, at the same time, turn-on the open drain MOSFET which works as active pull-down for the SD signal. Note that the gate driver outputs stay at low level until the $\overline{SD}$ pin has experienced both a falling edge and a rising edge, although the fault signal could be returned to low level immediately after the fault sensing. In fact even if the FF is reset by the falling edge of the $\overline{SD}$ input, the SD signal also works as enable for the outputs, thanks to the two AND ports. Moreover once the internal open-drain transistor has been activated, due to the latch, it cannot be turned-off until the SD pin voltage reaches the low logic level. Note that, since the FF is SET dominant,

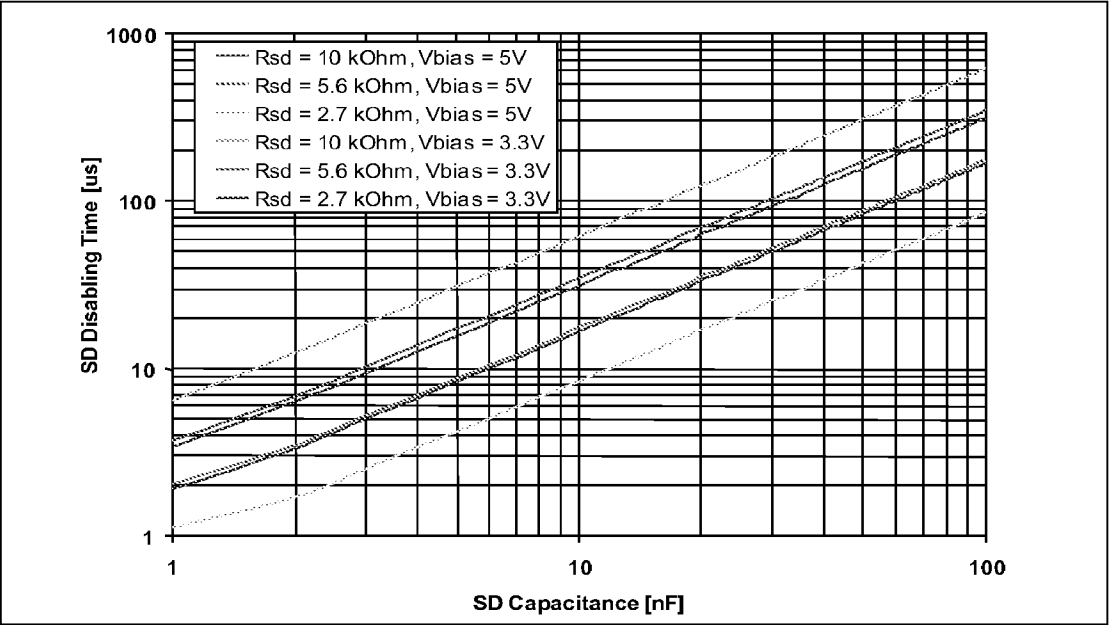
oscillations of the $\overline{\text{SD}}$ pin are avoided if the fault signal remains steady at high level. The block diagram of a power system using the gate driver with the smart shutdown architecture is shown in *Figure 8*. An RC network is used to implement the disable time after a fault detection event.

Figure 8. Protection scheme



In *Figure 9* the typical duration of the disable time vs. the SD capacitance with different R_{sd} values and V_{bias} values is shown.

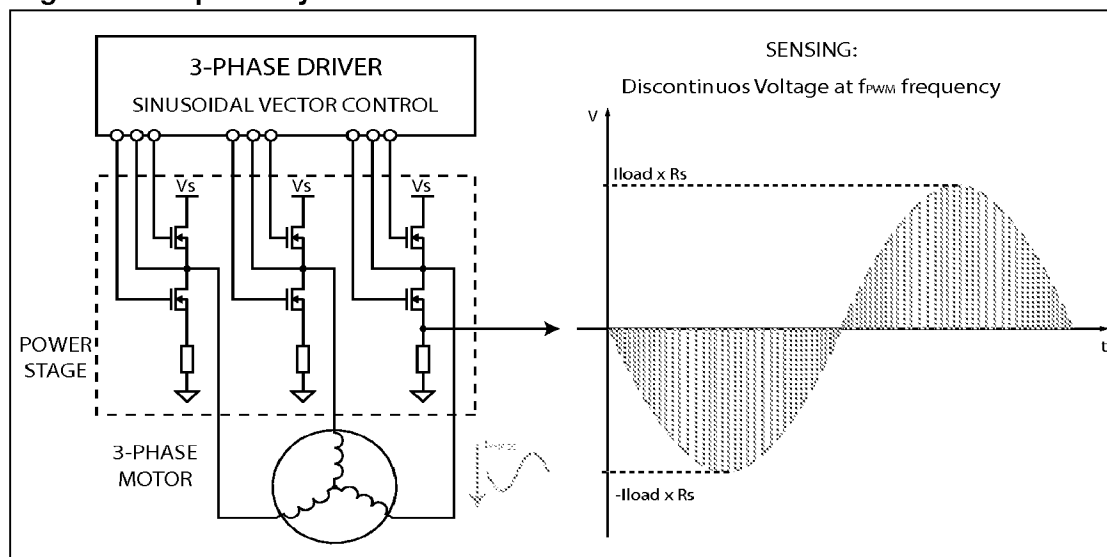
Figure 9. Disable time vs. SD capacitance (typical values)



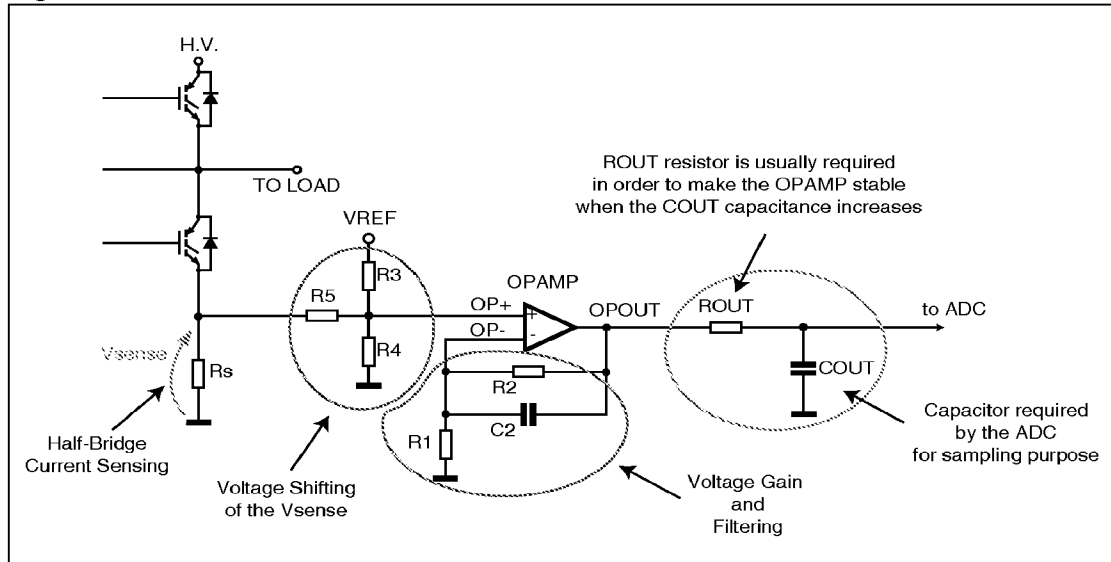
6 L6390 op-amp

The L6390 integrates an operational amplifier optimized for field oriented control (FOC) applications. In a typical FOC application a tri-phase power bridge is used and the currents in the three half-bridges are sensed using a shunt resistor on the source of each low side power switch. The analog current information is transformed in a discontinuous sense voltage signal, available only when the current is flowing in the low side path and having the same frequency of the PWM signal driving the bridge. The sense voltage is a bipolar analog signal, which sign depends on the direction of the current (see *Figure 10*):

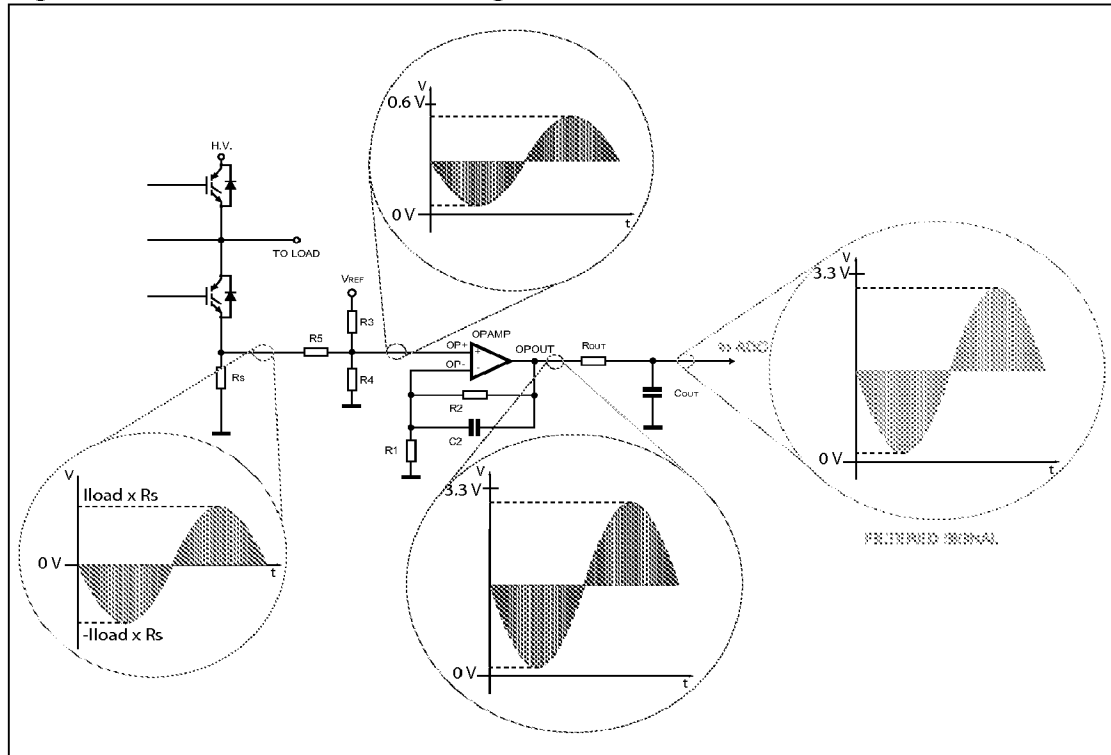
Figure 10. 3-phase system



PWM frequency in a typical home appliance motor driver application is in the range of 5-20 kHz. The sense voltage signals must be provided to an A/D converter in order to perform the matrix calculation related to a certain control technique. Those sense signals are usually shifted and amplified by dedicated op-amps in order to exploit the full range of the A/D converter. The typical scheme shown in *Figure 11* is used.

Figure 11. General advanced current sense scheme

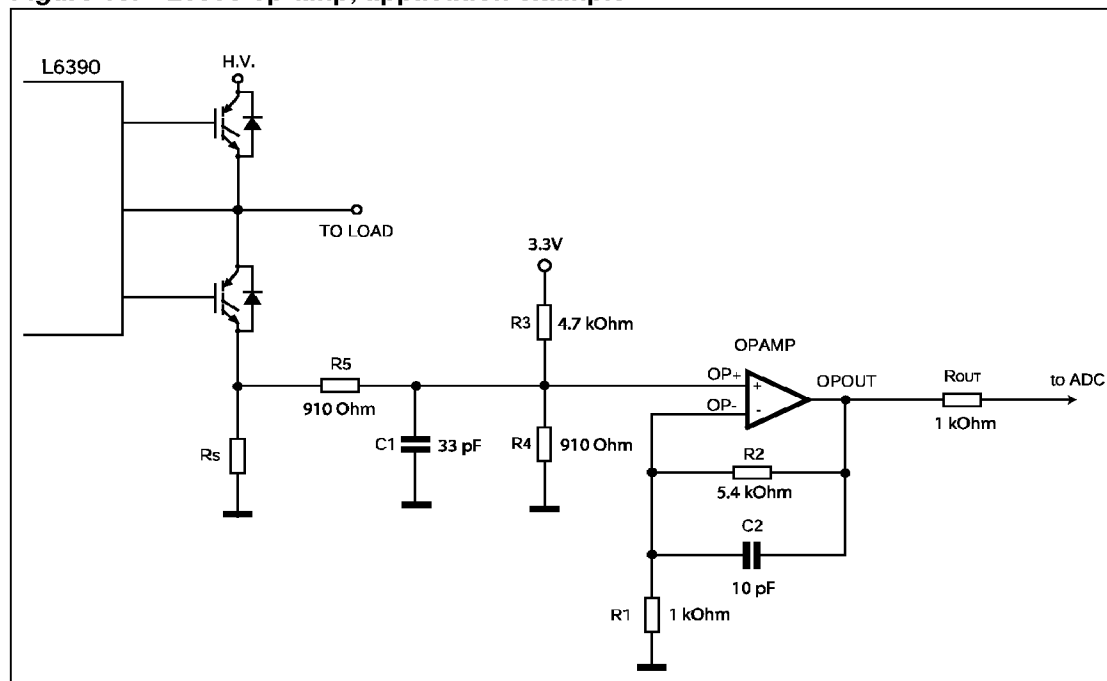
Principle waveforms corresponding to the above scheme are provided in *Figure 12*:

Figure 12. Advanced current sensing waveforms

ADCs used in vector control applications have a typical FSR of about 3.3 V. The sense signals have to be shifted and centered on FSR/2 voltage (about 1.65 V) and amplified with a gain which provides the matching between the maximum value of the sensed signal and

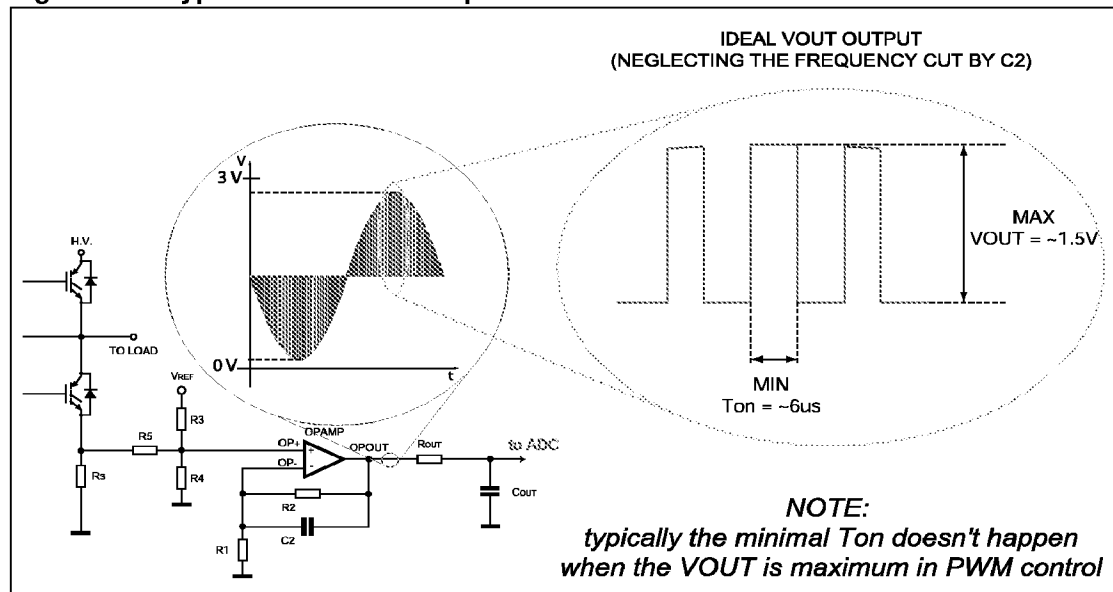
the FSR of the ADC. In *Figure 13* an application circuit of the op-amp with typical passive components values is shown.

Figure 13. L6390 op-amp, application example



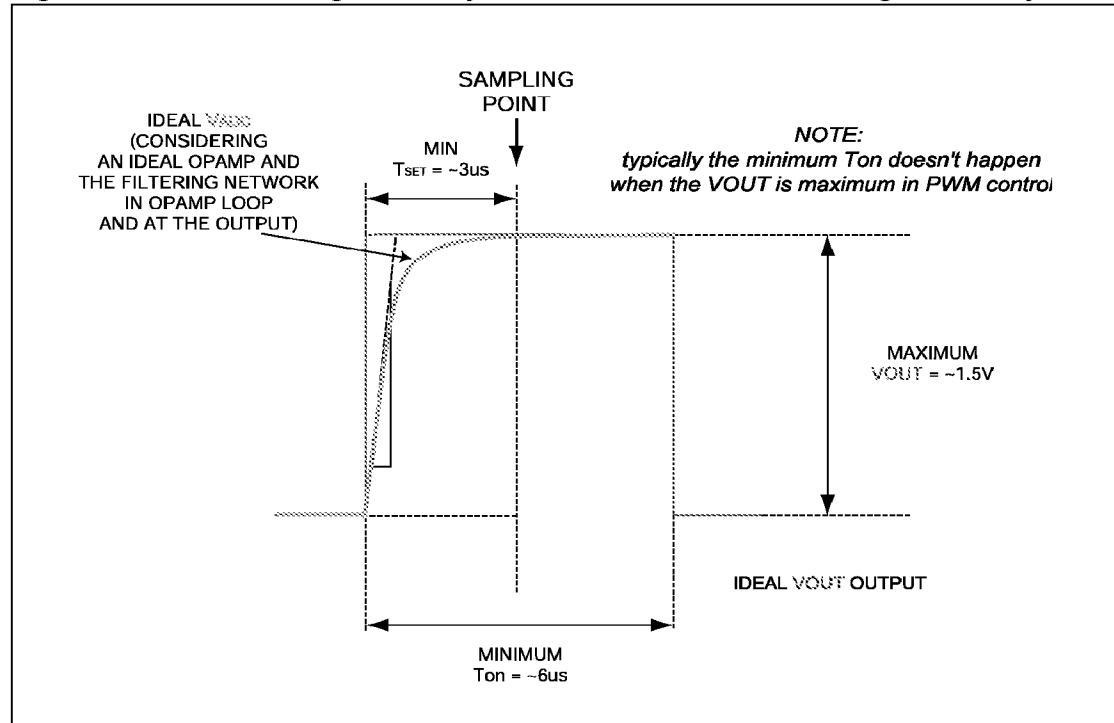
Following *Figure 14* shows the output voltage waveform of the L6390 op-amp considering the voltage level shifting and the gain amplification provided by the amplifier configuration of the op-amp, neglecting the low-pass filtering action of the feedback network. In particular the waveform is zoomed on the maximum amplitude of PWM sinusoidal voltage signal which represents the extreme condition for the external slew rate of the op-amp. The required external slew rate increases proportionally to the voltage excursion and reverse proportionally to the ON time of PWM output.

Figure 14. Typical L6390 ideal output



Typically the maximum voltage variation on op-amp output is $FSR/2$ (about 1.65 V) and the ON time usually does not show its minimum value where the PWM voltage amplitude is higher. In fact the duty cycle of each half-bridge of the 3-phase power stage is proportional to the average voltage applied to each 3-phase terminal which normally is not in phase with the output current of the same bridge, due to the load angle and the BEMF of the running motor. The ADC front-end usually samples the op-amp output voltage in the middle of the ON time, so the output must approach its final value as much as possible within one half of the ON-time. Considering a minimum ON time of about 6 μs , the op-amp output settling time must be lower than 3 μs . If the maximum output voltage variation is 1.5 V and the maximum time needed to reach the desired value is 3 μs , the minimum SR requirement for proper operation would be $1.5 \text{ V} / 3 \mu s = 0.5 \text{ V}/\mu s$ (see Figure 15).

Figure 15. Detail on single PWM cycle in advanced current sensing for FOC systems



7 Bootstrap driver

A bootstrap circuitry is needed to supply the high voltage section. This function is normally accomplished by a high voltage fast recovery diode (*Figure 16 a*). In the L6390 a patented integrated structure replaces the external diode. It is realized with a high voltage DMOS driven synchronously with the low side driver (LVG), with a diode in series, as shown in *Figure 16 b*. An internal charge pump (*Figure 16.b*) provides the DMOS driving voltage.

7.1 C_{BOOT} selection and charging

To choose the proper C_{BOOT} value the external MOSFET can be seen as an equivalent capacitor. This capacitor C_{EXT} is related to the MOSFET total gate charge:

Equation 1

$$C_{EXT} = \frac{Q_{GATE}}{V_{GATE}}$$

The ratio between the capacitors C_{EXT} and C_{BOOT} is proportional to the cyclical voltage loss. It has to be:

Equation 2

$$C_{BOOT} \gg C_{EXT}$$

For example: if Q_{GATE} is 30 nC and V_{GATE} is 10 V, C_{EXT} is 3 nF. With C_{BOOT} = 100 nF the drop would be 300 mV.

If HVG has to be supplied for a long time, the C_{BOOT} selection has to take into account also the leakage and quiescent losses.

For example: since the HVG steady state consumption is lower than 200 μA, if HVG T_{ON} is 5 ms then C_{BOOT} has to supply 1 μC to C_{EXT}. This charge on a 1 μF capacitor means a voltage drop of 1 V.

The internal bootstrap driver gives a great advantage: the external fast recovery high voltage diode can be avoided (it usually has great leakage current). This structure works if V_{OUT} is close to GND (or lower) and in the meanwhile the LVG is on. The charging time (T_{charge}) of C_{BOOT} is the time in which both conditions are fulfilled and it has to be long enough to charge the capacitor.

The bootstrap driver introduces a voltage drop due to the DMOS R_{DS(on)}. At low frequency of operation this drop can be neglected, but if the frequency is increased the drop must be taken into account.

The following equation is useful to compute the drop on the bootstrap DMOS:

Equation 3

$$V_{DROP} = I_{CHARGE} \cdot R_{DS(on)} \rightarrow V_{DROP} = \frac{Q_{GATE}}{T_{CHARGE}} \cdot R_{DS(on)}$$

where Q_{GATE} is the gate charge of the external power MOSFET, $R_{DS(on)}$ is the on resistance of the bootstrap DMOS and T_{CHARGE} is the charging time of the bootstrap capacitor.

For example: given the typical value of $120\ \Omega$ for $R_{DS(on)}$ and using a power MOS with a total gate charge of $30\ nC$, the drop on the bootstrap DMOS is about $1\ V$ if the T_{CHARGE} is $5\ \mu s$. In fact:

Equation 4

$$V_{DROP} = \frac{30nC}{5\mu s} \cdot 120\Omega \approx 0.7V$$

V_{DROP} has to be considered when the voltage drop on C_{BOOT} is calculated: if this drop is too high, or the circuit topology doesn't allow a sufficient charging time, an external diode can be used.

Working at very low frequencies the high side driver on-time can be very long. So C_{BOOT} voltage can drop because of HVG steady state consumption. To avoid extremely large capacitor ($> 1-2\ \mu F$) an external charge pump can be added (see *Figure 17* as example). It is mandatory for the diodes to have a low parasitic capacitance, because C1 and C2 should be greater than diodes capacitance. The oscillator has to work in order to balance the high voltage side consumption, and the minimum frequency is fixed by C1 and C2 values (with $C1 = C2 = 33\ pF$ then $f > 250 - 300\ kHz$).

Figure 16. Bootstrap driver

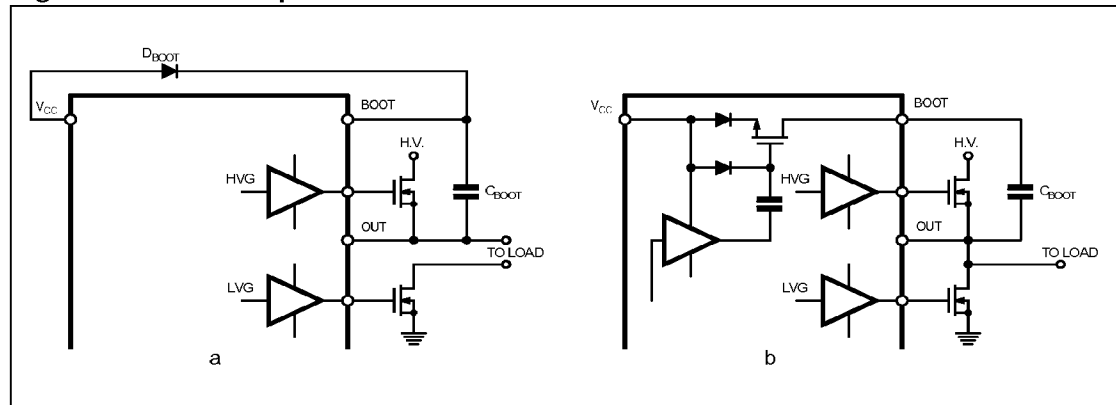
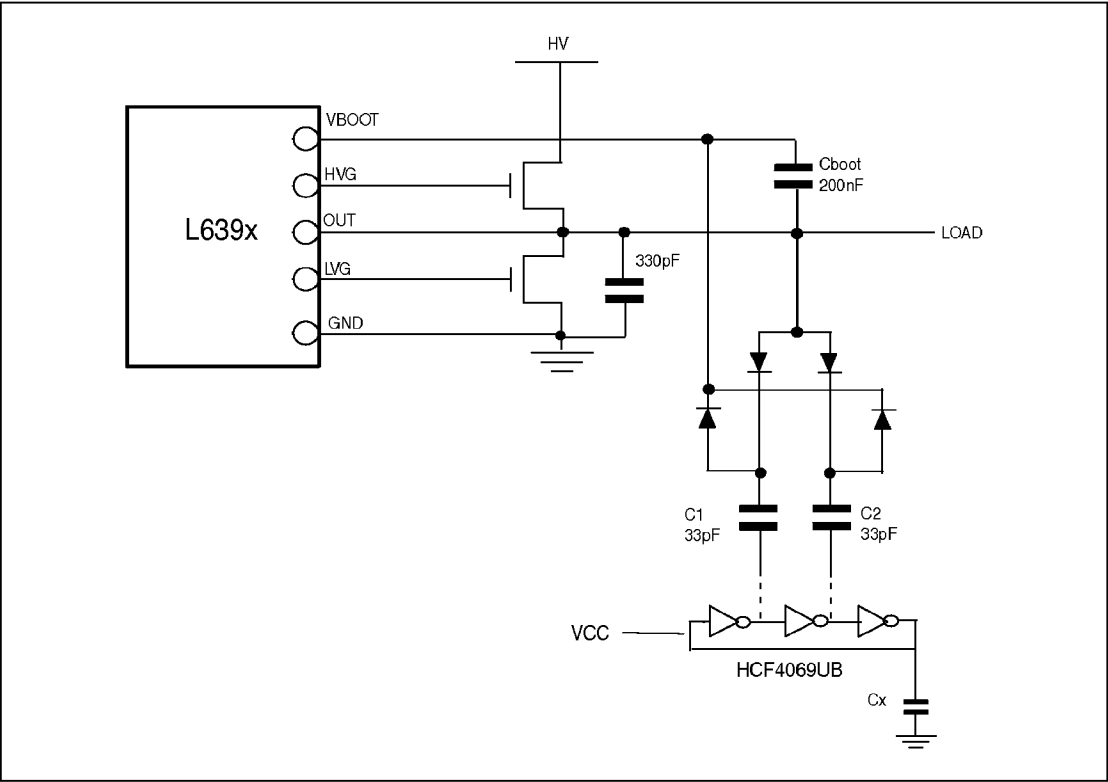


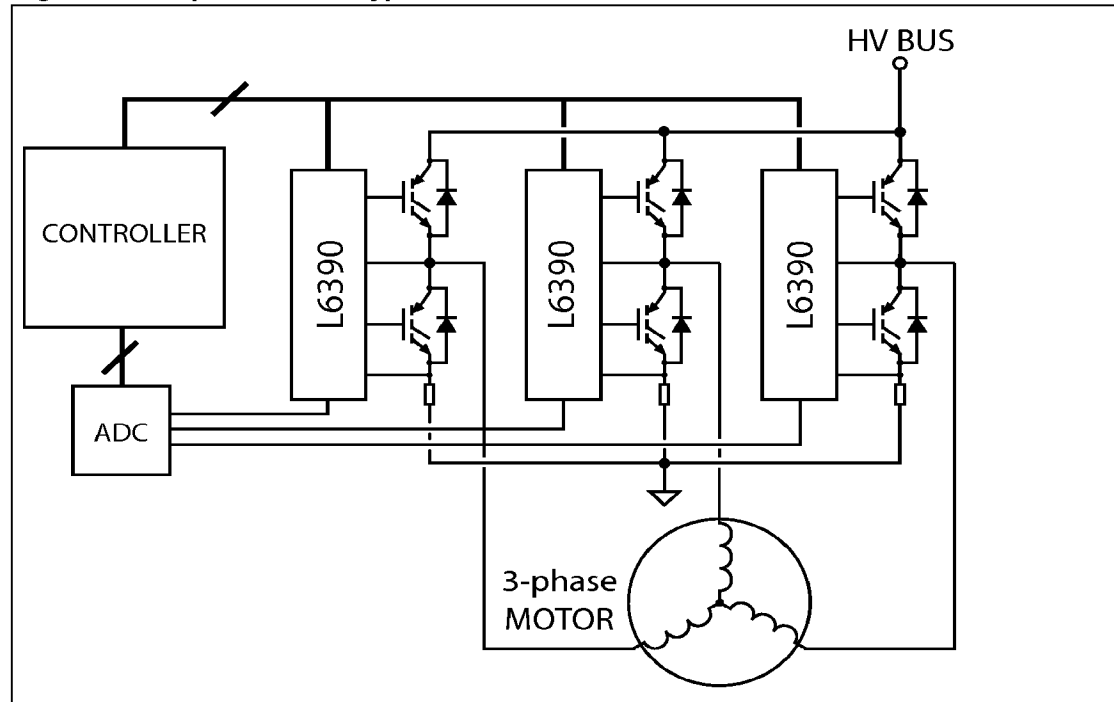
Figure 17. External charge pump



8 Application example

A typical scheme for a 3-phase inverter application is shown in *Figure 18*. In order to drive a 3-phase load as for example a BLDC or Induction AC motor, three L6390 ICs can be used.

Figure 18. 3-phase drive- typical scheme

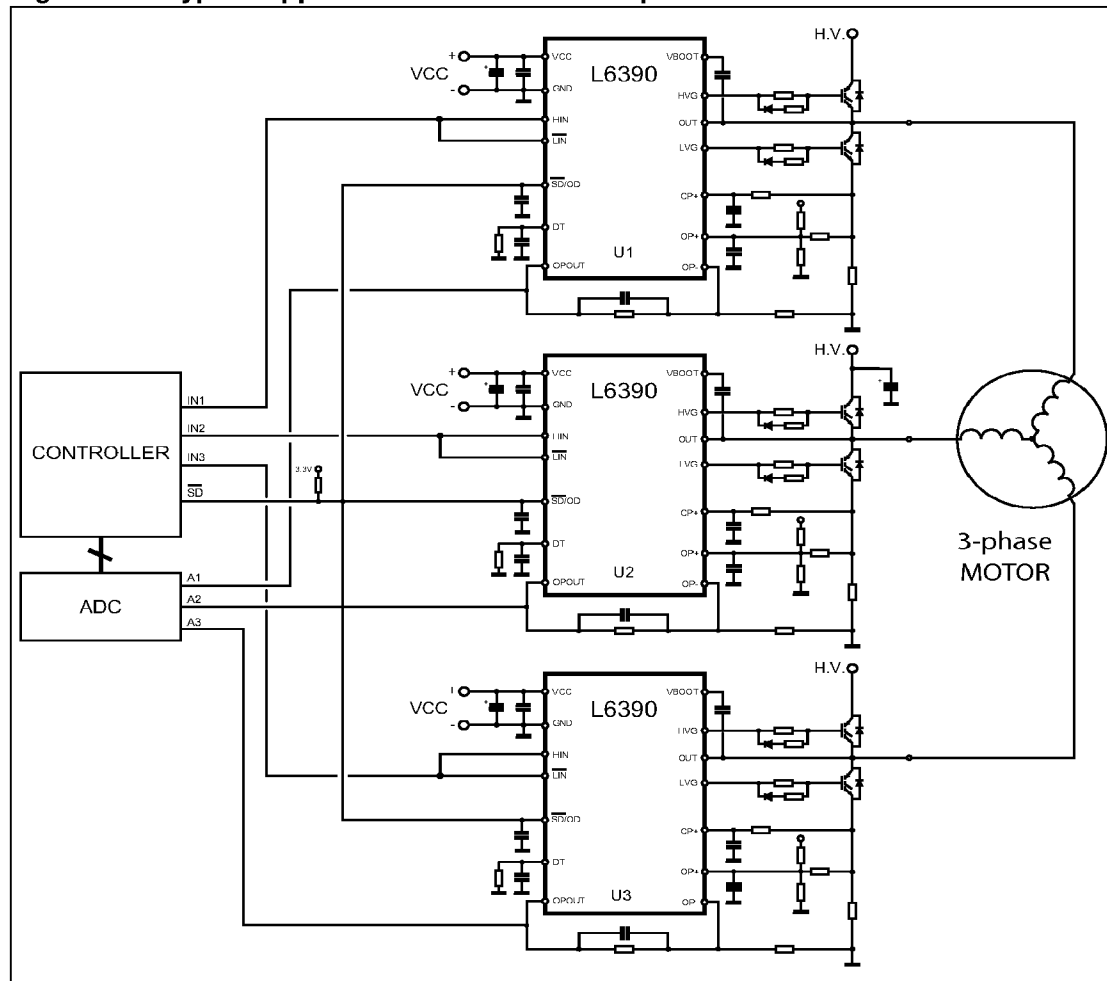


Each L6390 IC drives one half-bridge of the 3-phase power stage. The three gate drivers are driven by the controller and the sensing signals coming from the power stage are managed directly by the L6390 analog blocks (comparators and op-amps) which provide the proper feedback signals to the ADC and the system controller. Each half-bridge comprises two power switches such as IGBTs (or MOSFETs), one high side and one low side. When the high side switch is ON, it brings the output voltage of the half-bridge to the HV bus voltage, which can be a high DC voltage power supply with large power availability, while the low side switch shorts the output to power ground voltage when it is ON. Thanks to the internal floating structure for the high side switch driving, n-type IGBTs (or N-channel MOSFETs) can be used. The gate drivers work exactly as digital/analog-power interface between the controller and the power stage. A shunt resistor can be placed between each low side switch and the power ground, to sense the current on each low side. The information contained in each current sense signal can be conditioned by the L6390 op-amp as described in *Section 6*.

In *Figure 19* a more detailed schematic of a 3-phase power stage topology is shown. Note that usually is recommended (not mandatory) to have some gate resistances between each power switch gate and the correspondent gate driver output, in order to limit the current during the gate charge. Then some filtering and/or level shifting RC networks should be added between the shunt resistors and the correspondent comparator and op-amp input pins of each gate driver. A unique RC network can be used for the $\overline{SD}$ pin even if it is strongly recommended to split the capacitor in three equal components in parallel in order to

place each one very close to the $\overline{SD}$ input pin of each IC. This capacitance and, in general, each capacitance indicated in the schematic must be placed as close as possible to the IC in order to guarantee a good noise filtering action.

Figure 19. Typical application schematic of a 3-phase FOC



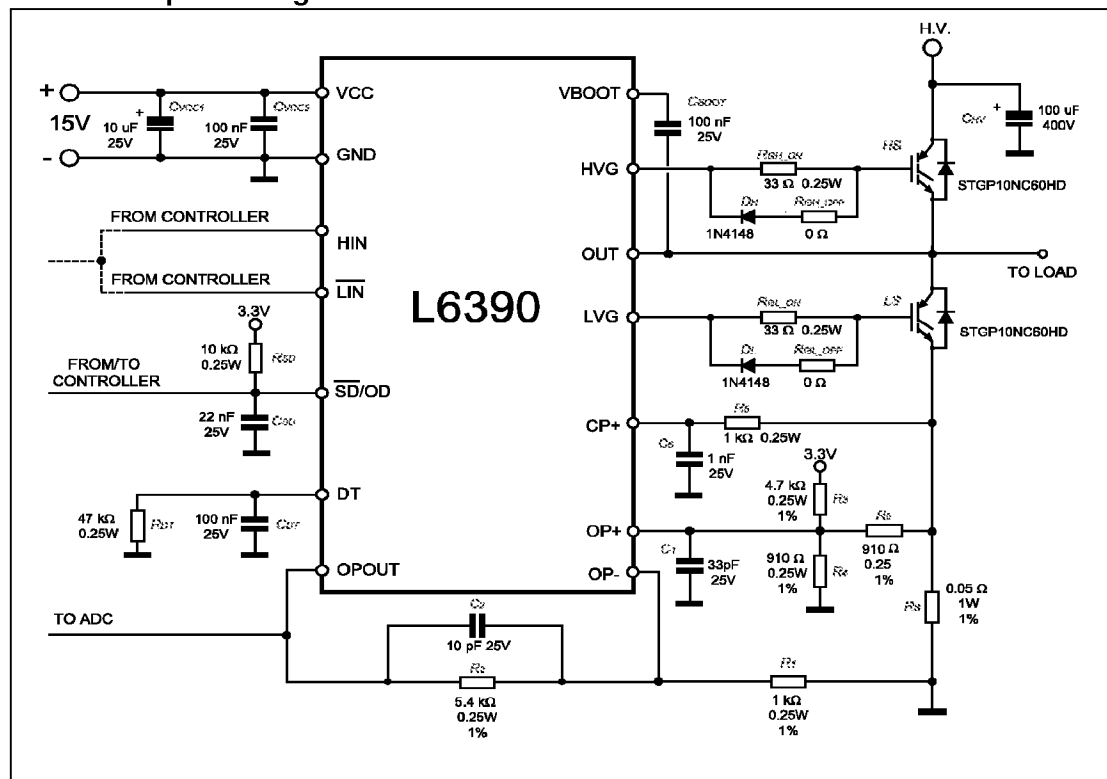
In the example of *Figure 19* the logic inputs HIN and $\overline{LIN}$ are connected together, using just one single signal to drive each L6390.

In *Figure 20* a special focus on a single gate driver circuit of the 3-phase L6390 scheme is provided.

8.1 VCC supply pin

Regarding the VCC pin, a local filtering of the supply voltage very close to the L6390 IC is recommended. Generally the suggestion is to use two capacitors, one electrolytic with greater value ($C_{VCC1} = 10 \mu\text{F}$, for example) which has a great energy capability but also a not negligible ESR (so is quite slow in providing the current) and a second smaller ceramic capacitor ($C_{VCC2} = 100 \text{ nF}$) which has a lower ESR value but a lower energy capability. The first capacitor works mainly as bulk energy storage while the second one is able to supply the dynamic current spikes required by the commutations of the device, so is better for high frequency decoupling of the IC supply voltage. On the other hand, the selection of the proper value for the CBOOT capacitor is already described in *Section 7*.

Figure 20. Ex. of an application circuit for one of the three half-bridges of a 3-phase power stage



In the example a VCC supply voltage of 15 V is used. This voltage is approximately the same voltage provided to the gate of the power switches.

8.2 BOOT (floating) supply pin

The bootstrap capacitor to supply the high-side floating section of the gate driver must be placed between the BOOT pin 16 and the OUT pin 14. For a deep description of the proper dimensioning of the bootstrap capacitor please refer to *Section 7 on page 18*. The capacitor must be placed as close as possible to the related IC pins. The bootstrap diode required for the charge of the bootstrap capacitor is integrated inside the L6390 device.

8.3 Logic input pins

The logic input pins can be connected directly to the controller with the suggestions provided in *Section 2 on page 6*. If the application environment is very noisy and the logic input voltage is low (e.g. 3.3 V), it can be useful to place some small RC network (not showed in *Figure 20*) in series with the logic input lines, in order to avoid false input triggering due external noise.

8.4 Shutdown pin

Dimensioning of the SD network (R_{SD} and C_{SD}) is provided in *Section 5*.

8.5 Dead time pin

The resistance value on the DT pin must be selected following the indications in *Section 4* and in *Figure 5*. It is recommended to connect between DT and GND pins a capacitor with a value of at least 100 nF, as close as possible to the IC and with short PCB tracks.

8.6 Op-amp

As explained in the previous paragraphs, the L6390 op-amp is completely uncommitted so a large amount of amplifier configurations can be implemented by the application designer. An example of amplifying network is provided in *Section 6*. Typically low tolerance resistors are used, if high accuracy is required in the conditioning of analog signals.

8.7 Comparator input

No particular external circuits are required apart some noise filtering network useful to avoid false triggering of the comparator due to voltage spikes on sense resistor. In the example of *Figure 20* a common RC network is implemented (time constant $\sim 1 \mu s$).

8.8 Sense resistor

The value of the sense resistor must be chosen considering mainly the current rating of the application and the amplitude of the sense voltage desired. The power rating of the sense resistor must be large enough to withstand the maximum current of the application. More resistors in parallel are often used in order to withstand higher power requirements and to obtain low resistance values with low parasitic inductance, which must be as low as possible in order to reduce the dynamic below-ground voltage on the OUT pin of the gate driver.

8.9 Gate driver outputs: gate lines

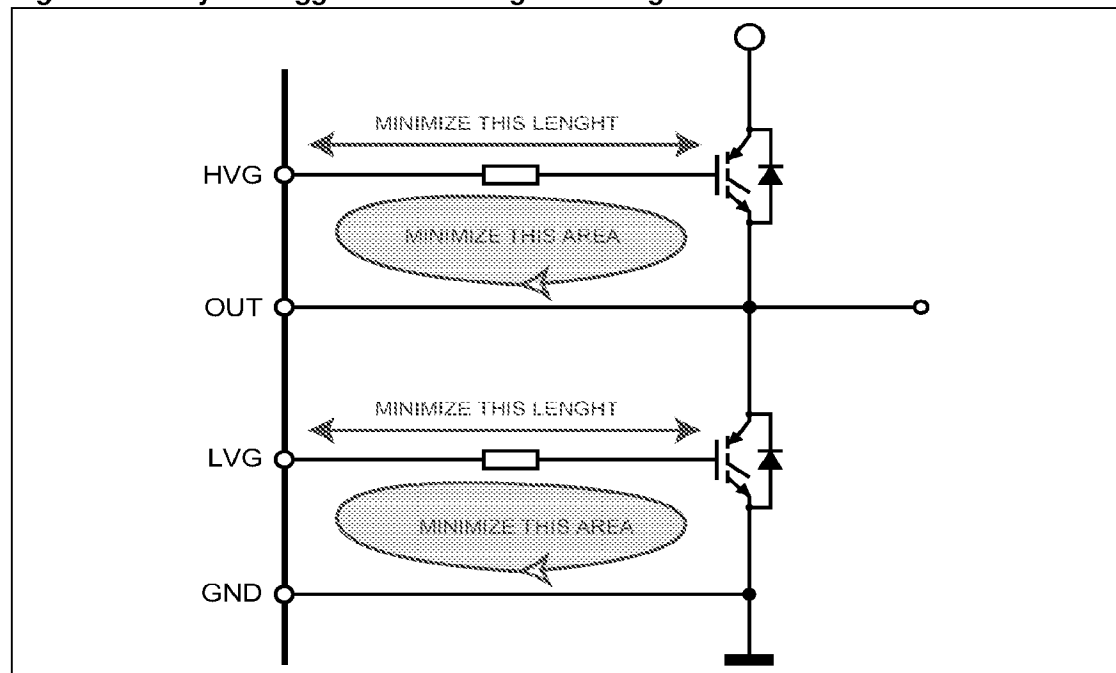
The gates of the power switches and the gate driver outputs can be connected directly, but usually some gate resistors are placed in series on the gate lines in order to limit the gate current during commutations. The final target is to control the dV_{OUT}/dt of each half-bridge output and then reducing the EMI. A more detailed explanation of the mechanisms behind the dV_{OUT}/dt control through the gate resistors is provided in the next sub-paragraph.

Consider that all the following considerations should be considered as approximated analyses of the gate charge phenomenon, so for a proper sizing of the gate resistor is always strongly recommended to evaluate the resulting power bridge transitions in bench analyses.

As shown in *Figure 20*, the gate line is split into separated paths, one for the turn-on (with, in the example, a gate resistor of $33\ \Omega$) and the other one for the turn-off, using a small signal diode as path selector the equivalent turn OFF resistance is in a first approximation the parallel of the turn OFF and the turn ON resistances (neglecting the diode drop). In the example the turn-off resistance is set to $0\ \Omega$ to provide the lowest resistance path for the turn-off of the IGBT. In fact, as explained in the two following paragraphs, low impedance on the gate driver turn-off helps in reducing the induced turn-on phenomena.

Regarding the layout of such gate lines, it is always strongly recommended to place the power IGBT (or MOSFETs) very close to the gate driver. It is important to reduce as much as possible the lengths of such line paths as well the areas included in the gate circuits, because these can act as weak antennas and could catch noise from the surrounding environment. The larger these areas, the higher the gain of such undesired antenna circuits.

Figure 21. Layout suggestion for the gate driving circuits



8.10 Gate driving: principle of working with inductive load

In this paragraph a deeper description of the power IGBT (or MOSFET) gate driving, in case of inductive load, will be provided. The following explanations and calculations are just intended to provide a general understanding of physical principles lying behind the phenomenon of the dV_{OUT}/dt control through the gate current limitation performed by the gate resistors. The target is to help the application designer which uses L6390 ICs to be aware of the various contributes that different parameters of the gate driver system have on the power transitions and also to distinguish the main effects on which he should focus on.

Calculations and formulae should be considered as qualitative indications and not for an accurate quantitative use since, as explained in the above paragraph, they are the result of a first approximation study. The experimental verification of the design choices is always recommended.

The description distinguishes and explores the two main actors of the gate driver system: by one side the inner structure of the gate driver output buffers is described and on the other side the switching mechanism of a generic IGBT (or MOSFET) is analyzed in detail. In *Figure 22* a simplified view of the L6390 gate driver output buffers is provided. Each one can be considered as a CMOS push-pull stage where a P-channel MOSFET works as source driver while an N-channel MOSFET works as sink driver. The structure is similar both for the low side and for the high side and the behavior can be considered the same. In fact the high side driver can be thought as a floating buffer having as supply the VBOOT voltage and as reference the OUT pin. The C_{BOOT} capacitor represents the floating supply voltage source of the high side driver. During the charge of the power switch gate, each source/sink MOSFET can be considered (in first approximation, for simplicity) as if it would be in the ohmic region, so it can be represented as an equivalent resistor with a value equal to its $R_{DS(on)}$. Thanks to this approximation it is possible to use a simplified equivalent circuit for the turn-on and the turn-off commutation (see *Figure 22* and *23*). Regarding the turn-on, this gate charge circuit has two resistors in series (R_{GATE_ON} and $R_{DS(on)_SOURCE}$) and a supply voltage which is VCC for the low side and VBOOT-OUT for the high side. Regarding the turn-off the equivalent circuit is composed just by two resistors (R_{GATE_OFF} and $R_{DS(on)_SINK}$) connected to the source of the power switch.

Gate driver output buffers: equivalent circuit

Figure 22. Gate driver output: equivalent circuit for turn-on

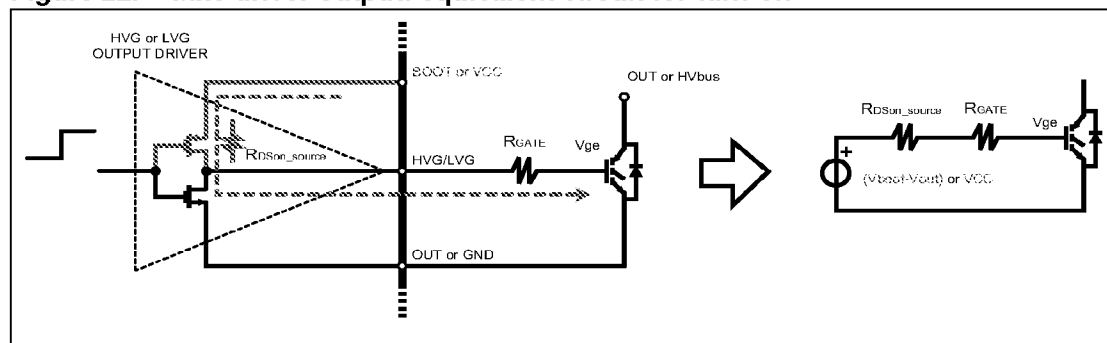
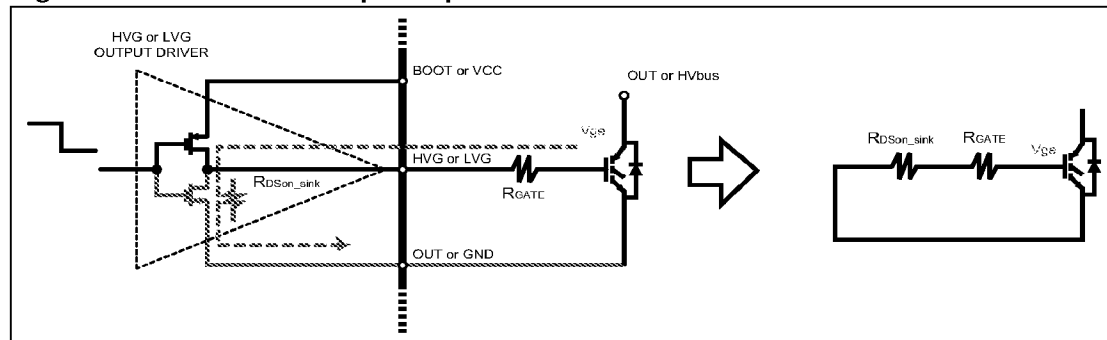


Figure 23. Gate driver output: equivalent circuit for turn-off



This first simplification is related to the inner structure of the gate driver circuitry. Let now focus for a while on the side of the power bridge and its commutations. In a common half-bridge composed by a high side and a low side switch the transitions of the power IGBTs (or MOSFETs) are not all similar, but they can be distinguished in two main types: soft switching and hard switching (see *Figure 24* and *25*). The main element playing a key role in the dynamic of transitions is the direction of the current flowing in the power IGBT (or MOSFET) under evaluation which depends on the direction of the load current respect to the half-bridge. Mainly if the current has the same direction of the power switch the commutation is hard. If it is in opposite direction the commutation is soft.

Figure 24. Hard switching

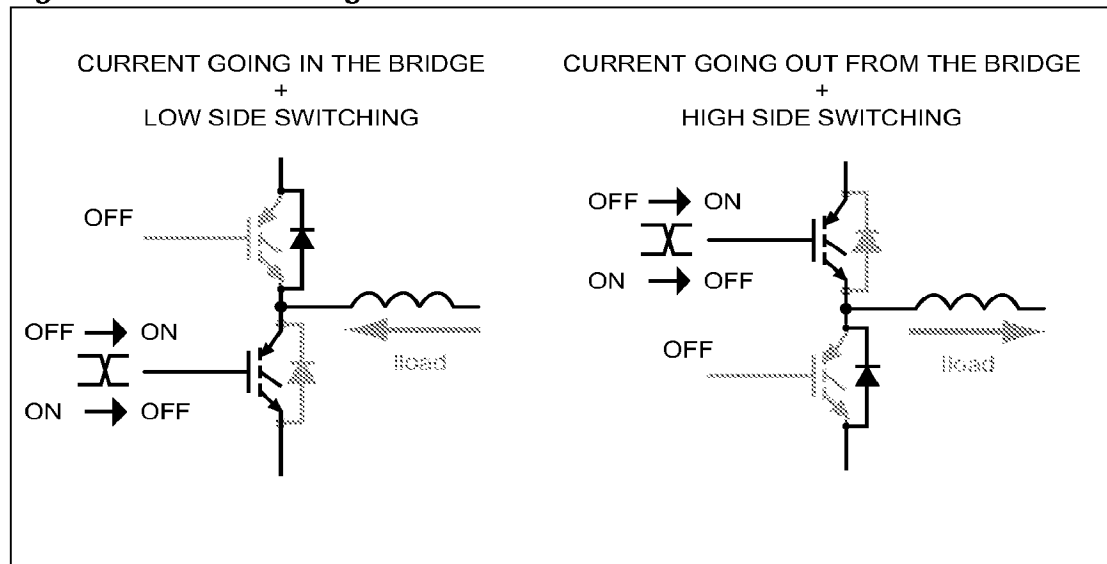
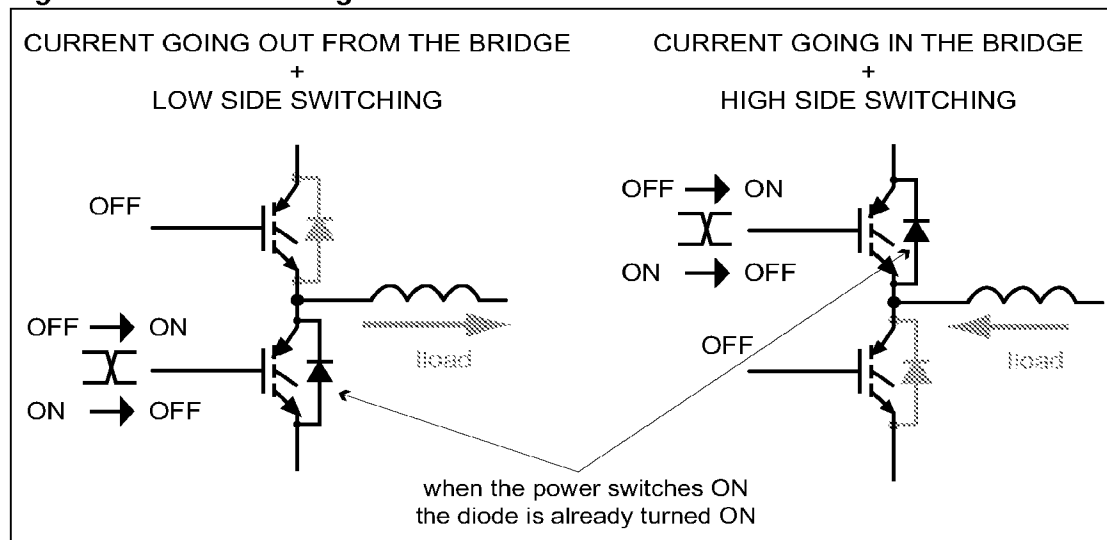


Figure 25. Soft switching



As shown in *Figure 24* and *25*, there is hard switching when the high side is commutating and the load current is exiting the bridge or when the low side is commutating and the load

current is entering the bridge. This type of transition is called “hard” because the power switch turns on when the related V_{CE} (V_{DS}) is at maximum; V_{CE} (V_{DS}) goes back to maximum during turn-off (V_{CE} is close to zero while the switch is steadily ON). The transition is completely managed by the switch that, during the commutation, dissipates some energy. On the other hand, if the switch turns on when its V_{CE} is already closed to zero or V_{CE} remains close to zero after turn-off, the transition is soft and, during the commutation, the switch dissipates almost no energy. This last condition happens when the free-wheeling diode of the switch is bringing the current because the diode is in the same direction of the load current. As a general rule, consider that the dynamic of the transition (dV_{OUT}/dt , V_{CE} rise/fall time etc.) is always managed and controlled by the power switch in hard switching, while its companion switch is necessarily in soft switching because it is in the opposite direction of the load current. Considering this, in order to understand the power IGBT (MOSFET) transition, the hard switching transition will be further investigated. Note that all of the descriptions that follow consider an inductive load connected to the half-bridge stage output.

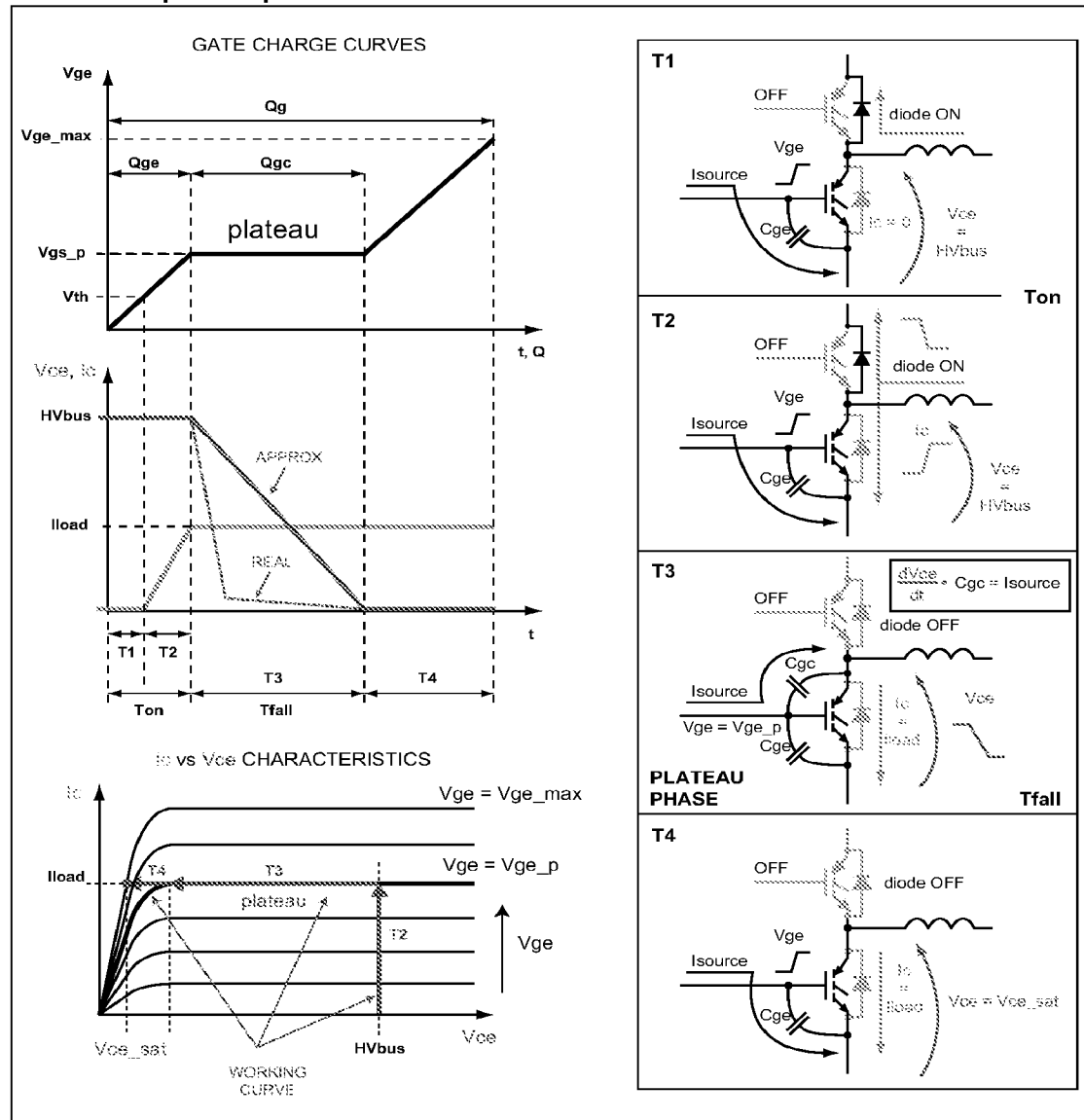
In *Figure 26*, the dynamics of the hard switching for the turn-on transition are described. The first graph is related to the gate charge curves of the power IGBTs (or MOSFETs). This is a typical graph which is also available in the datasheets for the IGBT or MOSFET and describes the dependence of the amount of charge required by the IGBT gate on the voltage drop between the gate and the emitter (or source). In the second graph the collector current (I_C) and V_{CE} voltage versus time are shown. The third graph shows the working point of the IGBT traced on the various I_C vs. V_{CE} characteristic curves for different V_{GE} voltages. For a single turn-on transition, four main phases related to the power IGBT (MOSFET) commutation can be distinguished. On the left side of *Figure 26*, the IGBT conditions for each phase are described. During phase T1 the gate begins to be charged, but the power IGBT still is not conducting because V_{GE} is below the threshold of the IGBT. In this state, the IGBT current is zero and its V_{CE} is at maximum, while V_{GE} is gradually increasing. In phase T2 the V_{GE} voltage goes above to the IGBT threshold and the IGBT starts to bring part of the load current. V_{CE} remains fixed at maximum level because the free-wheeling diode of the companion IGBT is still bringing the rest of the load current and is still conducting, thereby clamping the V_{CE} voltage. Both in T1 and T2 phases, the gate current contributes to charge the equivalent V_{GE} parasitic capacitance of the IGBT. In the I_C vs. V_{CE} characteristic plot the T2 phase is the vertical section of the working curve trace, because the IGBT is moving on its own characteristic with a constant V_{CE} and an increasing current, while its V_{GE} is increasing also. When the amount of current flowing in the IGBT is equal to the load current, the diode turns off and the V_{CE} voltage starts to decrease because it is no longer clamped by the free-wheeling diode of the other IGBT. The working point on I_C vs. V_{CE} curve reaches the load value and starts to move horizontally on the I_C constant curve in the direction of decreasing V_{CE} voltages. This is the T3 phase, usually called the “plateau” phase. This name derives from the fact that the gate charge curve is horizontal for the entire T3 phase, until V_{CE} reaches the V_{CE_sat} value corresponding to the load current. Note that the V_{GE} voltage is constant, although the gate current flowing in the IGBT gate is not zero. The reason for this is that the whole gate current is used to charge the C_{GC} (Miller) parasitic capacitance which then experiences a dV/dt on its terminals because the V_{CE} voltage is decreasing after the diode turn-off. As a first approximation, if C_{GC} was constant (currently it is not), the dV_{OUT}/dt could be calculated as follows:

Equation 5

$$\left(\frac{dV_{OUT}}{dt}\right)_{FALL} = \frac{I_{SOURCE}}{C_{GC}}$$

The formula above shows how the source current coming from the gate driver can directly control the dV_{OUT}/dt of the power half-bridge. This calculation is just a first approximation, since typically the C_{GC} Miller capacitance value is not constant but depends on the V_{CE} voltage, C_{GC} is not linear. The main result of the abrupt variation of the C_{GC} is that the actual slope of the OUT transition is composed of two different dV_{OUT}/dt slopes, one faster and the other slower compared to the value obtained by the above formula (see *Figure 26*). In this document only the first approximation approach will be used.

Figure 26. Turn-on hard switching details with induction load: gate charge and plateau phase



Due to C_{GC} non-linearity, the IGBT (or MOSFET) datasheet also reports the equivalent total amount of charge required during the different gate charge phases: the total gate charge Q_g required for turning on the IGBT (or MOSFET) completely, the Q_{GE} required for increasing

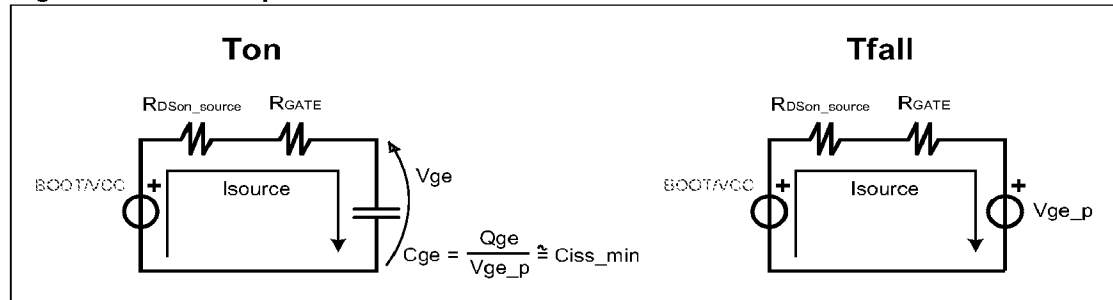
the V_{GE} up to the plateau voltage and the Q_{GC} required during the whole plateau phase. This last amount of charge is also called the plateau charge. The time required to provide the complete plateau charge is the T_{fall} time, and this is the same time necessary for completing the V_{CE} transition. For T_{on} time is intended the time delay between the beginning of the gate charge and the full conduction of the IGBT (or MOSFET), when the power switch current equals the full load current. So T_{fall} can be calculated as follows:

Equation 6

$$T_{fall} = \frac{Q_{GC}}{I_{SOURCE}}$$

It is now possible to merge the equivalent gate driver output circuit reported in figures 24 and 25 and the considerations regarding the different gate charge phases in the total equivalent circuit reported in Figure 27.

Figure 27. Total equivalent circuit for the turn-on



From the above circuit the value of the transition times could be calculated (the following formulae are related to the low side transition, but the same are suitable also for the high side by exchanging "VCC" with "VBOOT-VOUT"):

Equation 7

$$T_{ON} = (R_{DS(on)} + R_{GATE}) \cdot C_{ISSmin} \cdot \ln\left(\frac{VCC}{VCC - V_{GEP}}\right)$$

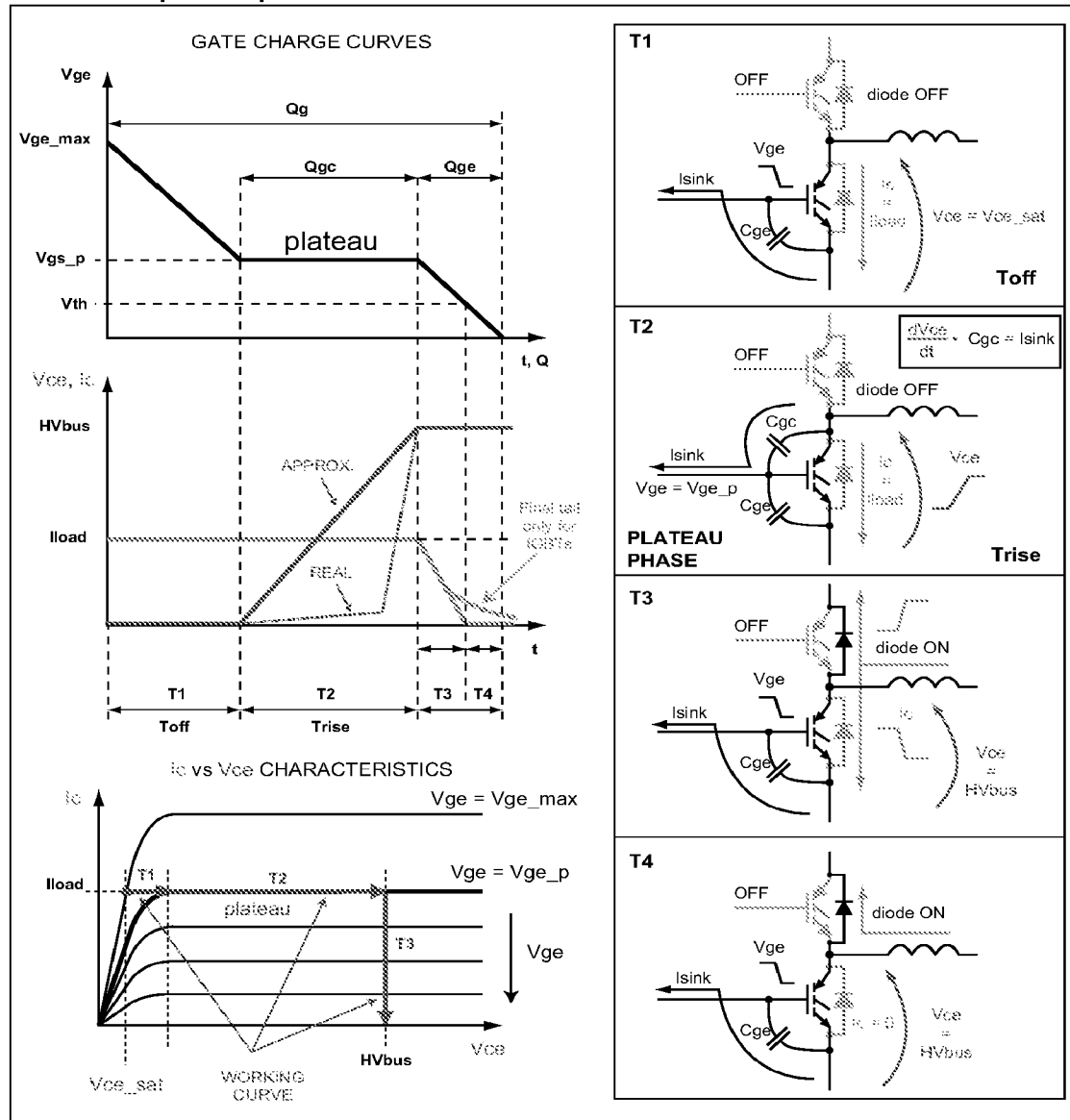
Equation 8

$$T_{FALL} = \frac{Q_{GC}}{I_{SOURCE}} = Q_{GC} \cdot \frac{(R_{DS(on)} + R_{GATE})}{VCC - V_{GEP}}$$

C_{ISSmin} is used because when the V_{CE} (or V_{DS}) of the power IGBT (or MOSFET) is maximum (equal to HV_{bus}), C_{ISS} (which depends on the V_{CE}) shows its minimum value.

During the turn-off of the power IGBT (or MOSFET), the behavior is the same as the turn-on, but in reversed time order. In Figure 28, the gate charge characteristics and V_{CE} vs. I_C curves are provided. The time required to sink the complete plateau charge is the T_{rise} time, while for T_{off} time is intended the time delay between the beginning of the gate charge and the increase of V_{ce} (or V_{ds}) voltage.

Figure 28. Turn-off hard switching details with induction load: gate charge and plateau phase



The rising dV_{OUT}/dt and T_{rise} can be calculated as follows:

Equation 9

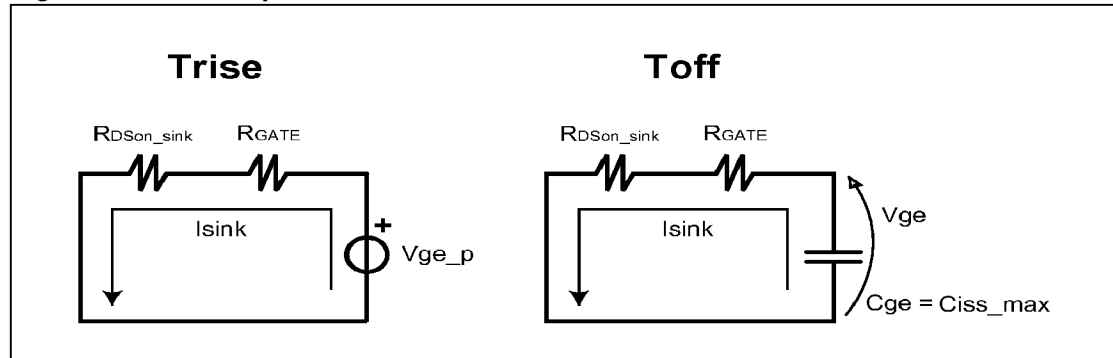
$$\left(\frac{dV_{OUT}}{dt}\right)_{RISE} = \frac{I_{SINK}}{C_{GC}}$$

Equation 10

$$T_{RISE} = \frac{Q_{GC}}{I_{SINK}}$$

As discussed above, the equivalent gate driver output circuit can be represented as in *Figure 29*:

Figure 29. Total equivalent circuit for the turn-off



Given the equivalent circuits in *Figure 29* and the approximations of the power IGBT (MOSFET) switching behavior, the timing equations are as follows:

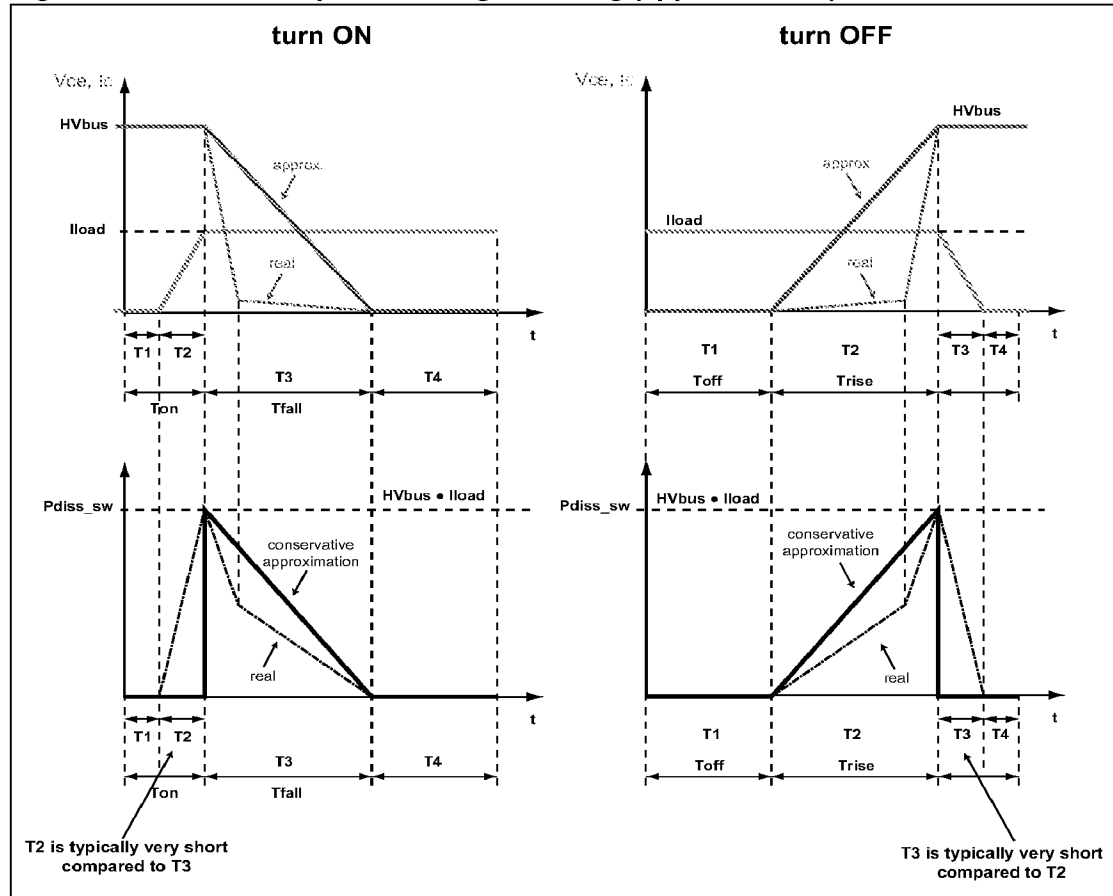
Equation 11

$$T_{OFF} = (R_{DS(on)} + R_{GATE}) \cdot C_{ISSmax} \cdot \ln \cdot \left(\frac{V_{GE_MAX}}{V_{GEP}} \right)$$

Equation 12

$$T_{RISE} = \frac{Q_{GC}}{I_{SINK}} = Q_{GC} \cdot \frac{(R_{DS(on)SINK} + R_{GATE})}{V_{GEP}}$$

As above, C_{ISS_max} is used because when the V_{CE} (or V_{DS}) of the power IGBT (or MOSFET) is minimum (equal to V_{CE_sat}) the C_{ISS} (which depends on the V_{CE}) shows its maximum value. Even if they are just approximations, the T_{fall} and T_{rise} are very important values to be estimated because they provide an indication of the power dissipation during the switching of the power IGBT (or MOSFET), which can be approximated as indicated in *Figure 30*.

Figure 30. Power dissipation during switching (approximation)

Since the T_2 time for the turn-on and T_3 time for the turn-OFF are much shorter compared to T_{fall} and T_{rise} , respectively, the approximated instantaneous amount of energy dissipated during the commutation can be calculated as the triangular area having as its base the T_{fall} , or T_{rise} time and as its height the product of the maximum V_{CE} voltage (equal to HV_{bus}) and the load current value:

Equation 13

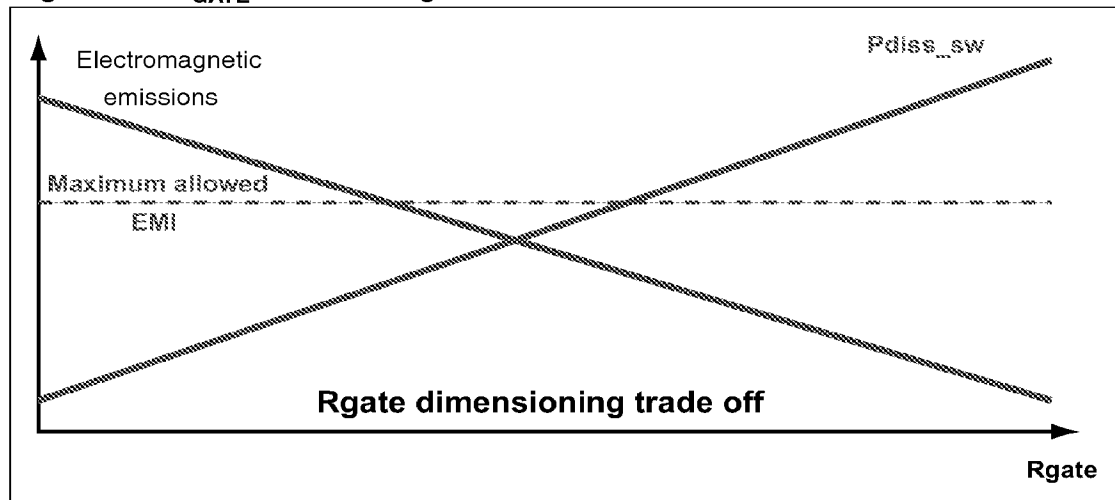
$$E_{DISS-SW} \cong \frac{(HV_{bus} \cdot I_{LOAD}) \cdot (T_{fall} + T_{rise})}{2}$$

Equation 14

$$P_{DISS-SW} \cong \frac{(HV_{bus} \cdot I_{LOAD}) \cdot (T_{fall} + T_{rise}) \cdot f_{SW}}{2}$$

The term f_{SW} represents the switching frequency of the power IGBT (or MOSFET).

Based the results obtained by these approximate calculations, it should be clear that the main consideration when dimensioning the gate resistor values is the trade-off between electromagnetic emission and the power dissipated during power IGBT (or MOSFET) switching, as shown in *Figure 31*.

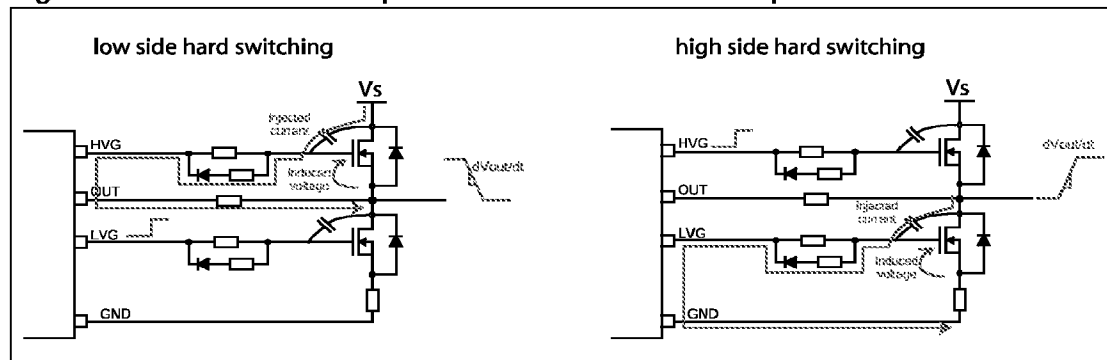
Figure 31. R_{GATE} dimensioning criteria

R_{GATE} influences the power dissipation on one side and the EMI effects on the other. Therefore, the best trade-off must be determined during the application design-in phase.

9 Induced turn-on phenomenon

One possible phenomenon to be analyzed is the induced turn-on that could occur on a turned-off power IGBT (or MOSFET) when its companion on the same half-bridge is switching on. This phenomenon is due to the current injected on the gate by the Miller parasitic capacitance (C_{GC}) which causes an undesired voltage increase on the gate of the power IGBT (or MOSFET) which should be kept well turned off. Induced gate voltage depends on the absolute value of the parasitic capacitance C_{GC} , its relative ratio with C_{GE} , the value of the dV_{OUT}/dt of the half-bridge and the value of the equivalent (turn-off) resistance between the emitter (or the source) and the gate. *Figure 32* illustrates this phenomenon.

Figure 32. Induced turn-on phenomenon - circuitual description



Note: Typically the induced turn-on phenomenon does not cause a complete turn-on of the power MOSFET, so it is rare to have destructive cross-conduction on the half-bridge during commutations. Nevertheless, weak conduction of the opened power switch could increase the power dissipation of the power stage, increasing the overall temperature of the power MOSFETs and reducing the efficiency. This is why this phenomenon deserves particular attention also in terms of thermal performance of the power application.

Some strategies to reduce this phenomenon are:

- a) **Reducing resistance path between gate and emitter (source).** Reducing as much as possible the gate resistance in which the injected current flows, the voltage drop on the gate becomes lower. The drawback is a possible increase of the dV_{OUT}/dt during turn-off of the power IGBT (or MOSFET) which commutates in hard switching, and then an increase of electrical noise and EMI issues (as explained in the previous paragraph, the power IGBT or MOSFET in hard-switching is always the one which has the channel in the same direction as the current). On the other hand, usually the dV_{OUT}/dt during turn-off is lower than during turn-on, because the plateau voltage is closer to the source voltage than the supply voltage of the driver. This results in a lower gate current for the discharge of the gate charge.

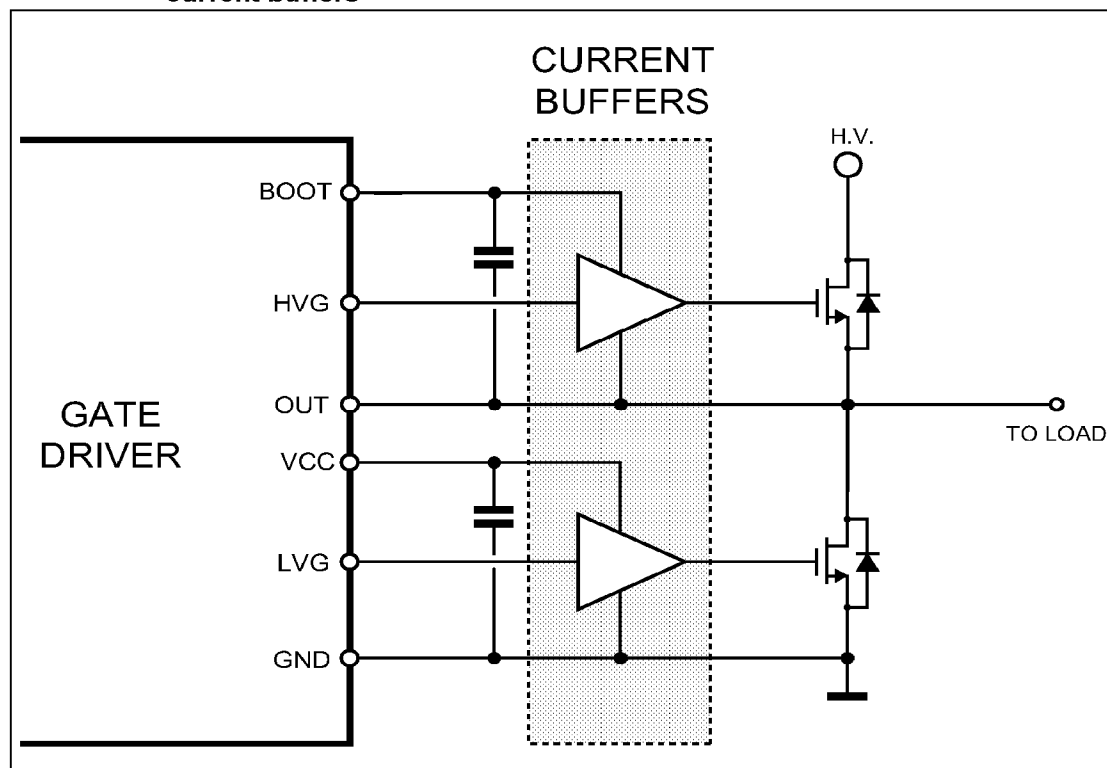
- b) **Reducing the maximum dV_{OUT}/dt .** This reduction can be achieved by increasing the gate resistor value which limits the gate current for the turn-on of the power MOSFET. The drawback is a consequent increase of the switching time which means dissipating more power during commutations.
- c) **Using a MOSFET with lower C_{GD}/C_{GS} (or C_{GC}/C_{GE} for IGBTs) ratio.** This solution is not always the simplest, but it could be the best strategy where the induced turn-on phenomenon is dominant.

10 How to increase the gate driver output current capability

In some cases, certain applications may require more gate driver output current capability. This requirement could be found in systems having a power rating higher than about 1 kW. In those applications, in fact, typically the power MOSFET/IGBTs have a large gate charge which contributes to a slowing down of the power switch transition (the dV_{OUT}/dt), increasing the power dissipation during each commutation. Note that also in applications with power rating higher than 1 kW, the output current capability of the L6390 could be enough if the power dissipation for commutation is acceptable (the power dissipation due to $R_{DS(on)}$ or V_{CESAT} is not dependent on the gate current). But if the limitation of this power contribution is a constraint, the dV_{OUT}/dt of each transition must be increased by enhancing the current capability of the gate driver.

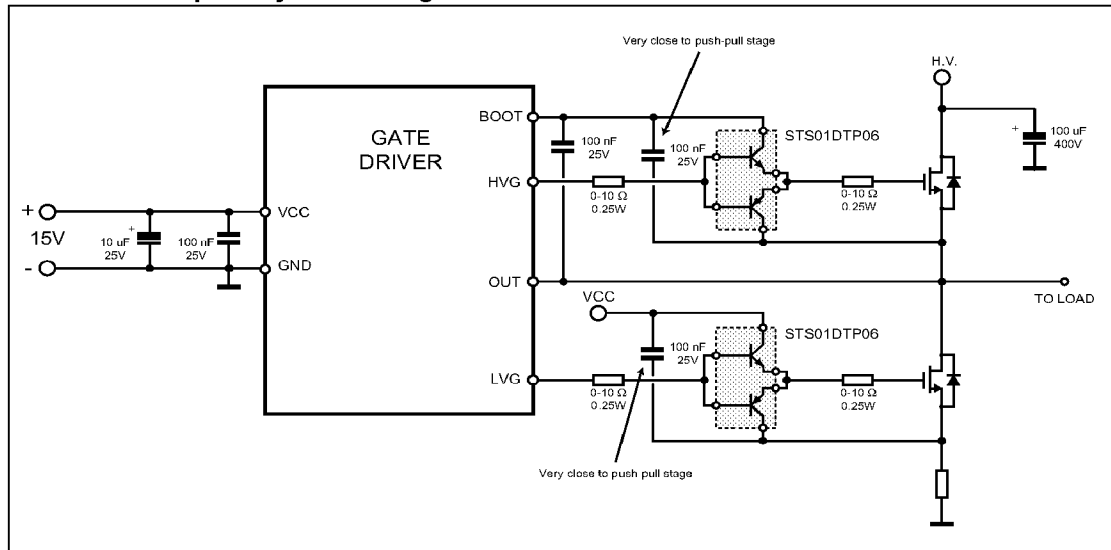
A simple way to increase the current capability of the gate driver outputs is to insert, in series with the two gate lines of one half-bridge gate driver IC, two external current buffers (Figure 33).

Figure 33. Block diagram of output current capability enhancement using external current buffers



Typically the current buffers are implemented using bipolar NPN-PNP push-pull non-inverting (emitter follower configuration) structures. In Figure 34, a typical circuit using a gate driver with bipolar push-pull current buffers is shown.

Figure 34. Example of a gate driving circuit with current buffers for current capability increasing



In the example above, the STMicroelectronics devices STS01DTP06 are dual NPN-PNP complementary bipolar transistors rated for 1 A of current and available in the small SO-8 package. The two push-pull structures are placed on each gate driving path and must be provided with a decoupling capacitor very close to device pins (see *Figure 34*).

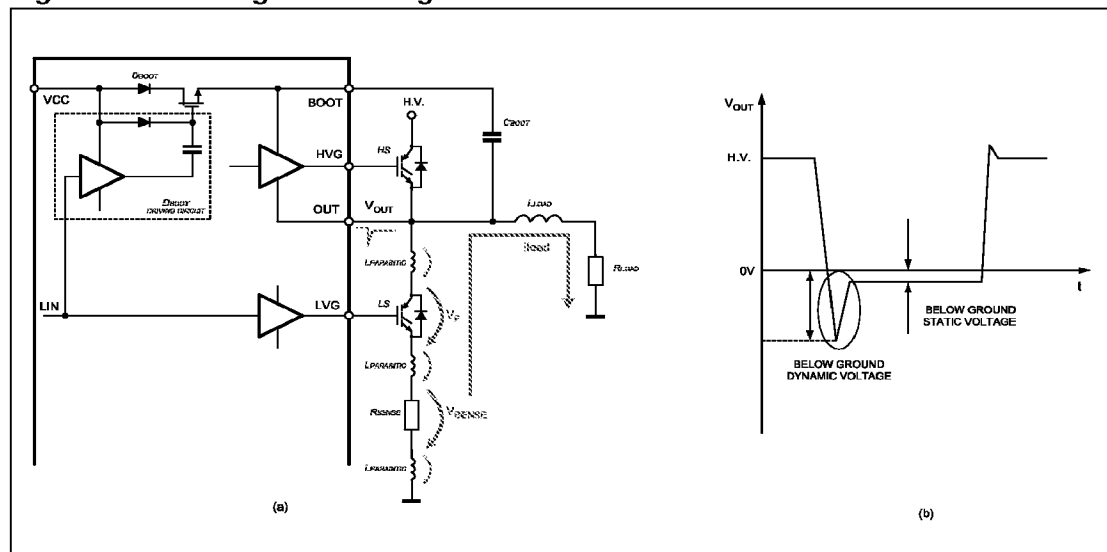
11 The below-ground voltage on the OUT pin

A typical phenomenon found in a great number of power applications is the below-ground voltage experienced by the OUT pin, which sometimes can be very high. Contrary to common opinion, the real problem of the below-ground voltage is not in the maximum absolute value of the OUT pin, but rather the voltage on BOOT pin and the over-charging of the bootstrap capacitance. In the paragraph that follows the root causes of the below-ground voltages will be explained in detail, and the description of the real issues which could result from the phenomenon will be provided.

11.1 The below-ground voltage phenomenon

In power applications using half-bridge topologies and typically driving loads with a significant inductive component, the output of the power half-bridge experiences, systematically, a below-ground voltage transition, which can be distinct in a dynamic contribution consisting of a greater undershoot spike and in a static contribution, which is a below-ground static voltage with lower absolute value (*Figure 35b*). This phenomenon occurs when the bridge carries out a so-called hard switching transition towards a low voltage level and the load current is outgoing (from the bridge to the load). When the high side switch turns off, the output current tends to remain quite constant due to the inductive component of the load, and then has to flow through the low side freewheeling diode, which turns on going from a high voltage reverse condition to a forward condition. It is evident that until the output bridge voltage has reached the "zero" value, the diode is turned off, so the output transition is dominated by the high side turn-off commutation. After the output voltage reaches the zero-voltage level, the diode can turn on and it begins to bring the entire load current in a very brief time, so the high di/dt causes the well-known forward peak voltage, which is the main contribution to the undershoot spikes. Other contributions to dynamic below-ground voltage are the spikes due to the high di/dt experienced by the parasitic inductances in series with the free-wheeling diode located along the turn-off current path of the half-bridge (*Figure 35*).

Figure 35. below-ground voltages in L6390



The overall below-ground voltage on the OUT pin can be calculated as follows:

Equation 15

$$V_{\text{OUTmin_static}} = -(R_{\text{SENSE}} \cdot I_{\text{LOAD}} + V_F)$$

Equation 16

$$V_{\text{BGV_spike}} = \underbrace{V_{\text{FPK}} + L_{\text{PARASITIC}} \cdot \frac{dI_F}{dt}}_{\text{dynamic contribution}} + \underbrace{V_{\text{OUTmin_static}}}_{\text{static contribution}}$$

where:

- V_{FPK} is the free-wheeling diode transient peak forward voltage, which depends mainly on the device technology and on the dI_F/dt of the current in the diode. Typical values could be from a few volts to more than 10 V. In *Figure 36*, VFP vs. dI_F/dt of the STTH1L06 diode is shown.
- the dI_F/dt is the current slope in the low side IGBT/MOSFET and could have a value from a few tens to several hundreds of A/ μ s. The value depends mainly on the power switch characteristics and in part on the driving current.
- $L_{\text{PARASITIC}}$ represents the sum of all parasitic inductances on the current path and mainly depends on the PCB layout. Generally, during the design of the power application it is important to pay attention to the layout of the power bridges in order to limit this parameter. Typical values of a good layout are in the order of some tens of nH. Note that it is useful also to use R_{SENSE} resistors with low parasitic inductances for the same reason.
- $R_{\text{SENSE}} \cdot I_{\text{LOAD}}$ product is the value of the V_{SENSE} voltage and typically is less than 1 V, also for thermal dissipation issues on the same resistor.
- V_F is the forward voltage of the free-wheeling diode and it is usually less than 2 V.

11.2 How to reduce the below ground spike voltage

In order to reduce the below ground spike, the following action should be taken:

- Reduce the parasitic inductances (see *Figure 35*).
- Reduce the dI_F/dt by slowing down the turn-off of the high side IGBT/MOSFET.

In most of application the two previous strategies result to be enough to reduce properly the below ground spike voltage, increasing the robustness of power system and then the margin for safe operation of the application. On the other hand in some cases, where the below ground spike voltage is significantly higher, the above suggestions may be not sufficient to limit that value and then it could be useful to add some external components to improve further the noise robustness of the power stage section:

- Add a small resistor (2÷10 Ω typically) in series to the OUT line (see *Figure 37*). The series resistor has the positive effect of limiting the spike voltage on the BOOT pin, thanks to the filtering effect of such resistor coupled with the bootstrap capacitor. Note that, in order to obtain an effective filtering effect, the OUT resistor must be placed between the minus terminal of bootstrap capacitor and the output of the power stage, as indicated in *Figure 37*.

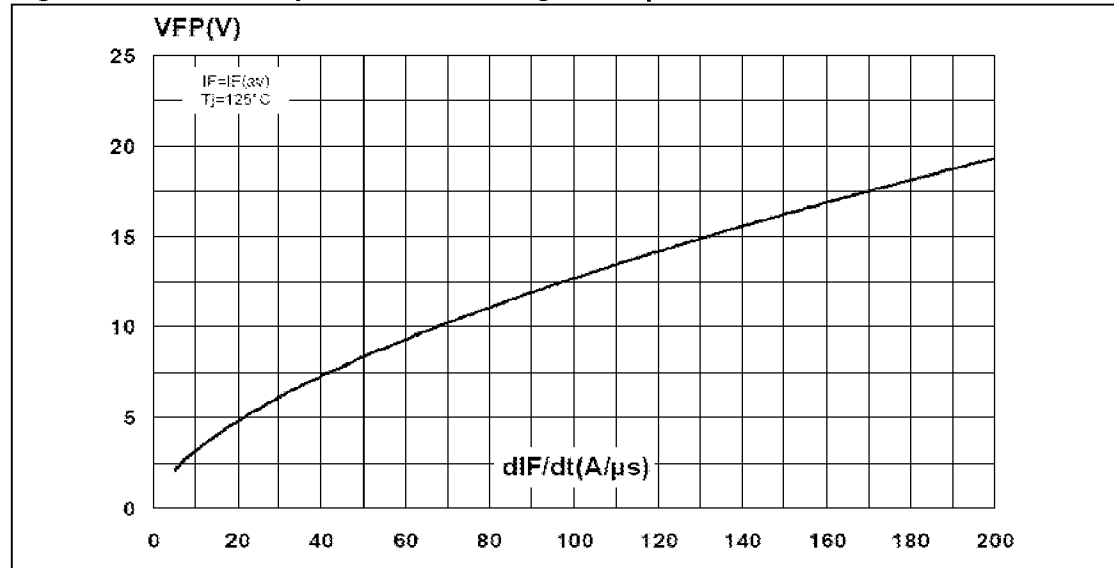
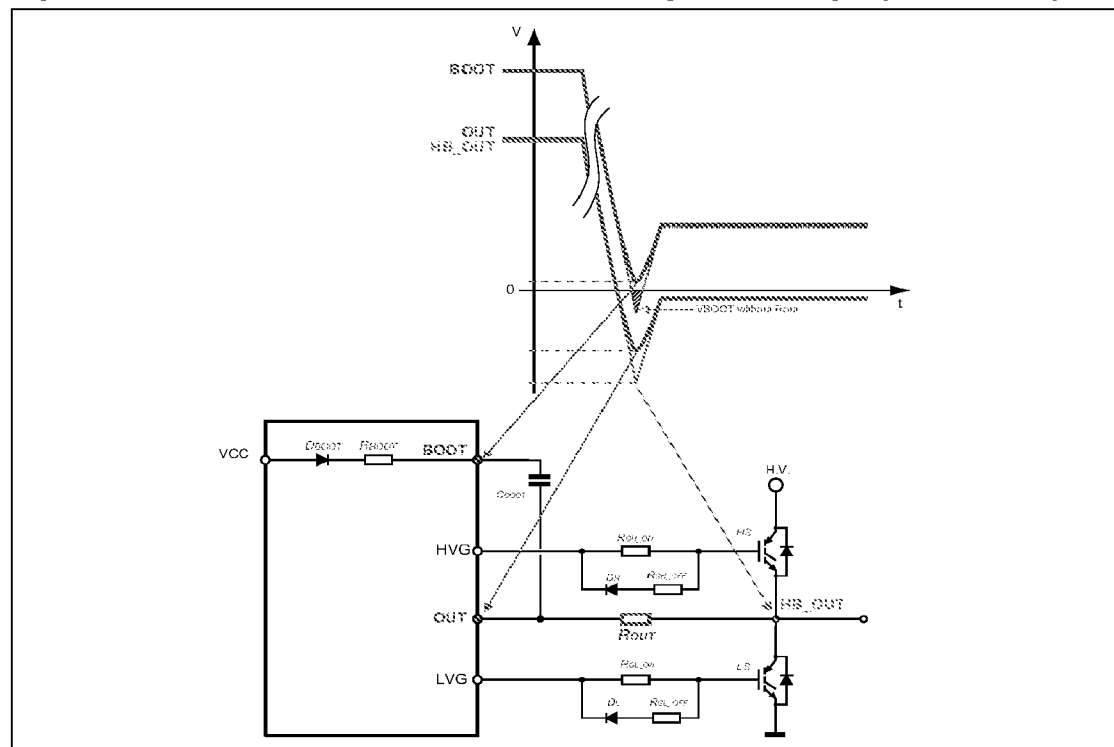
Figure 36. Transient peak forward voltage vs. di_F/dt of STTH1L06 diode

Figure 37. Use of OUT resistor to limit the below ground voltage spike on OUT pin

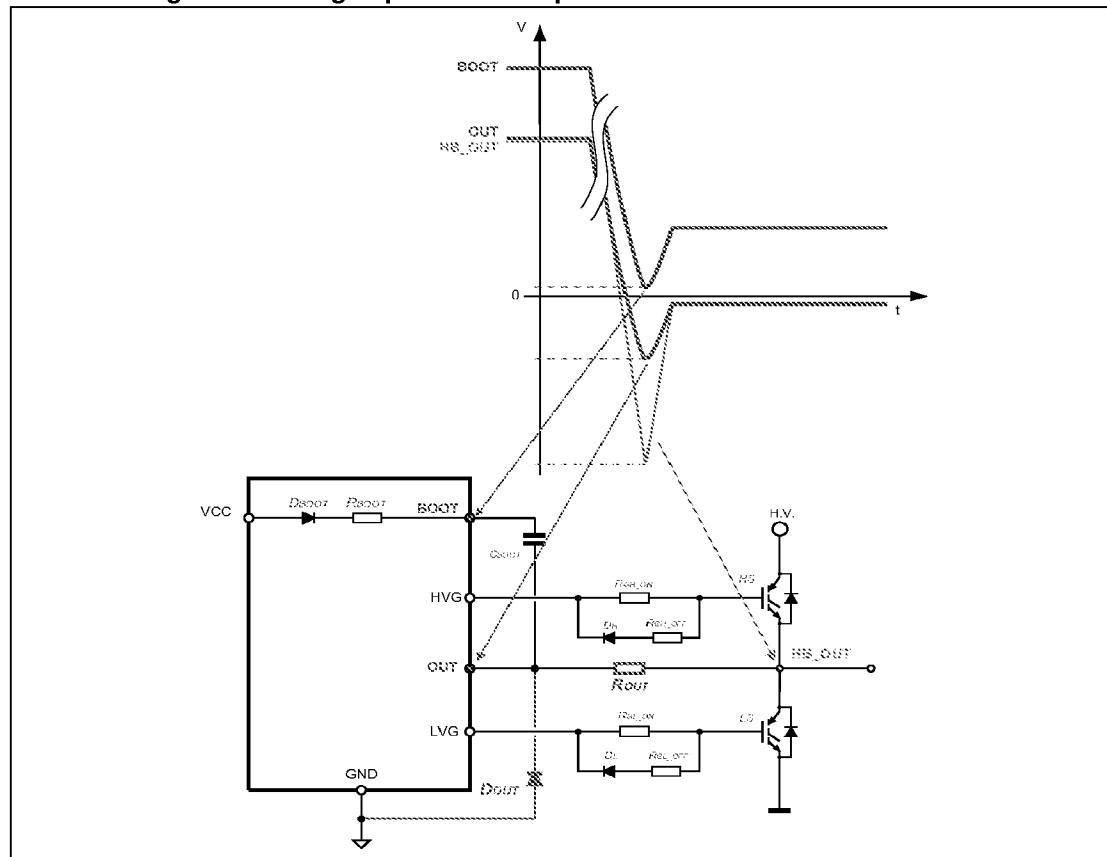


Note that this resistor is in series with both the turn ON and the turn OFF path, so it must be considered in the sizing of overall turn-ON and turn-OFF resistance.

In case neither the OUT resistor would not be enough to limit the below ground voltage, the following final resolving action could be taken:

- Add a high voltage fast diode (e.g. STTH1L06) between the GND pin and the OUT pin (very close to the device pins) in order to clamp directly on the gate driver OUT pin the below ground voltage spike. Note that, in any case, this diode must be used together with the OUT resistance suggested in the previous tip, because it must be avoided that the diode brings the very large load current during low side recirculation, in order to increase the clamping action of the diode itself (see following *Figure 38*).

Figure 38. Use of combination of OUT resistor and OUT diode to limit the below ground voltage spike on OUT pin



Note that R_{OUT} resistor is also in series with the charging path of the bootstrap capacitor. Its effects may be more evident during the first charge of the bootstrap capacitor, when it is completely discharged and a significant charging current may flow into the R_{OUT} resistor, producing an additional voltage drop between ground and OUT pin. Because during the bootstrap charging the HVG pin is set to low level, the OUT pin and the HVG pin are shorted together and have the same voltage, so the voltage drop due to R_{OUT} results directly transferred to the V_{GE} (or V_{GS}) of the high side IGBT (or MOSFET). Then the risk is that a weak turn ON of the high side power switch, when the low side power switch is already ON, could cause a cross-conduction in the power half bridge. Actually in most of cases this doesn't represent an issue for the proper working of the application, because the typical intrinsic resistance R_{BOOT} (~120 Ω) in series to the internal bootstrap diode is much higher than the R_{OUT} commonly used and the voltage drop on R_{OUT} is negligible.

11.3 Issues related to the below-ground voltage phenomenon

Contrary to common belief, the issues resulting from significant below-ground voltages on the OUT pin are not related to the maximum absolute voltage values that the OUT pin is able to withstand, but are instead related mainly to the voltage value of BOOT pin, which is indirectly bound to the OUT pin voltage. There are two main issues to be taken into consideration, all related to the absolute maximum ratings of the IC:

- VBOOT absolute minimum voltage
- VBOOT-VOUT absolute maximum voltage

11.3.1 V_{BOOT} voltage safe operating condition

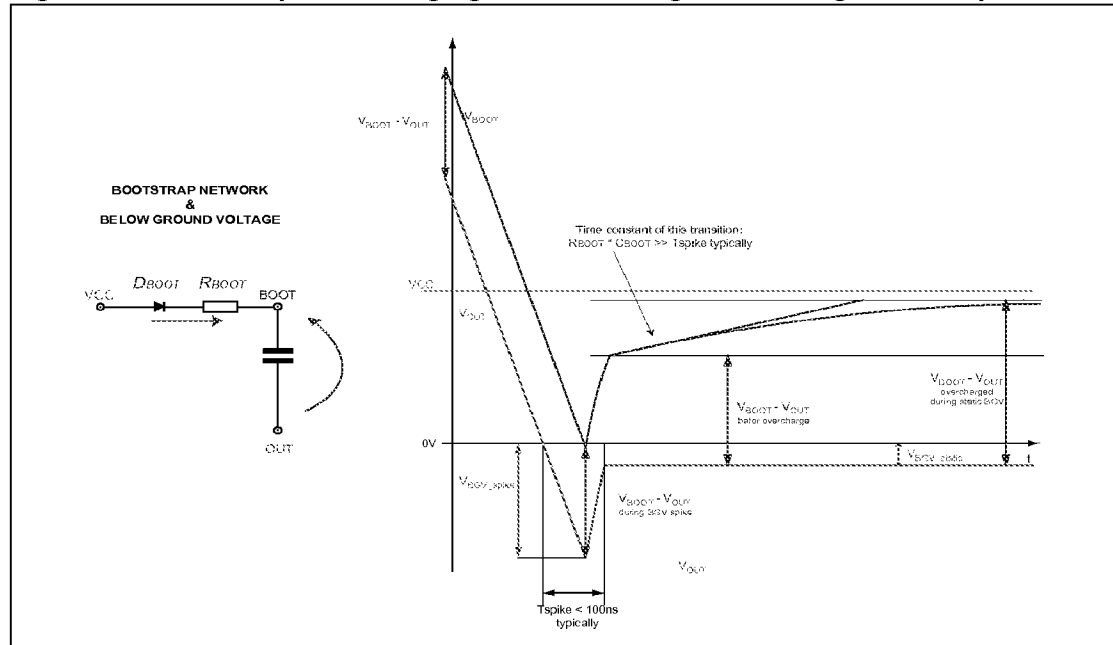
Electrically, the OUT pin could safely tolerate even many volts below ground without problems, but the BOOT pin cannot. The absolute value of the VBOOT voltage must not go steadily below -0.3 V, in order to avoid the turn-on of the built-in junction between BOOT pin and the IC gate driver substrate (connected to GND), which is normally in reverse condition. The turn-on of this junction could, in fact, cause a current large enough to damage the device.

11.3.2 Bootstrap capacitor over-charging

Another very important consideration is bootstrap capacitor over-charging. Actually, even before the OUT pin approaches the zero voltage value, the bootstrap diode (internal or external) tends to tie V_{BOOT} close to the VCC supply voltage. Since the OUT pin is below ground, the bootstrap capacitor is overcharged through the current coming from the bootstrap diode, and the V_{BOOT} - V_{OUT} voltage increases. It is very important that the bootstrap over-charge does not exceed the recommended maximum value for the V_{BOOT} - V_{OUT} voltage (see the relevant datasheet), as the high side floating section of the gate driver could be damaged.

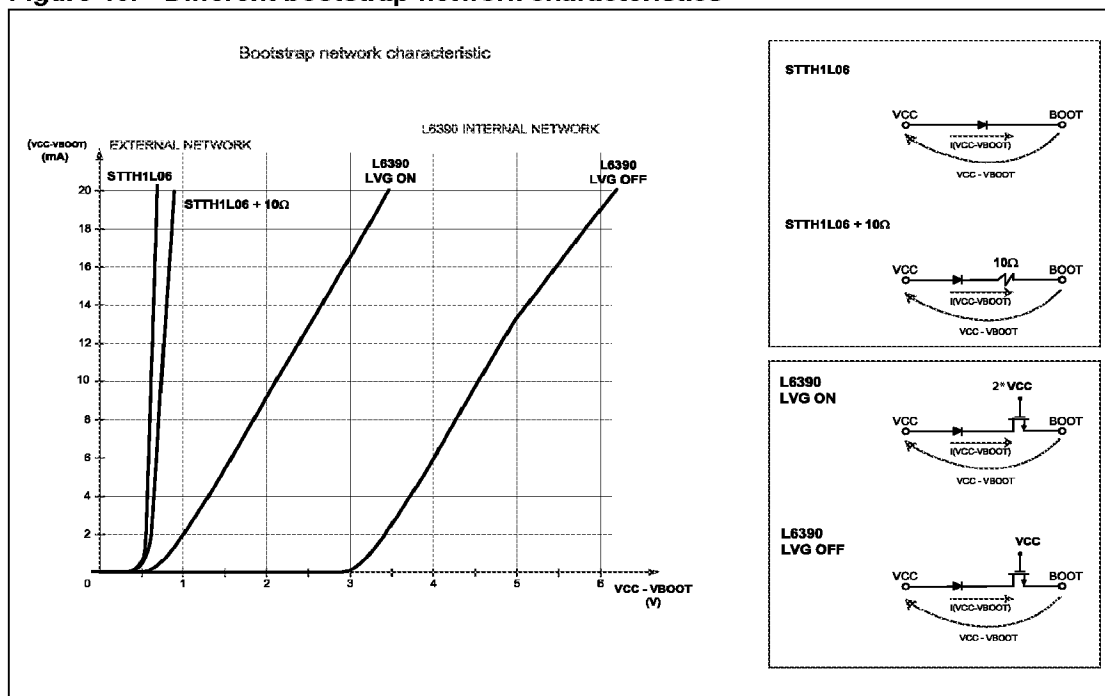
Note that a significant over-charging of the bootstrap capacitor is only possible through the static contribution of the below-ground voltage, because the dynamic below-ground voltage is very brief and the over-charging transition is limited by the time constant RC of the bootstrap capacitor and by the overall resistance in series with the bootstrap diode.

In *Figure 39*, the over-charging phenomenon is illustrated in detail.

Figure 39. Bootstrap over-charging due to below-ground voltage on OUT pin

Note that in the L6390 IC gate driver the internal DMOS in series with the integrated bootstrap diode is fully turned on only when LVG is ON. In fact, when the LVG is OFF, the gate of the bootstrap DMOS is biased at V_{CC} voltage. This means that if the V_{BOOT} is pulled down by the bootstrap capacitor (due to below-ground voltage on OUT pin) at about 3 V (typ) below V_{CC} voltage, the DMOS turns on again. Figure 40 shows some characteristics of different (internal and external) bootstrap networks.

Figure 40. Different bootstrap network characteristics

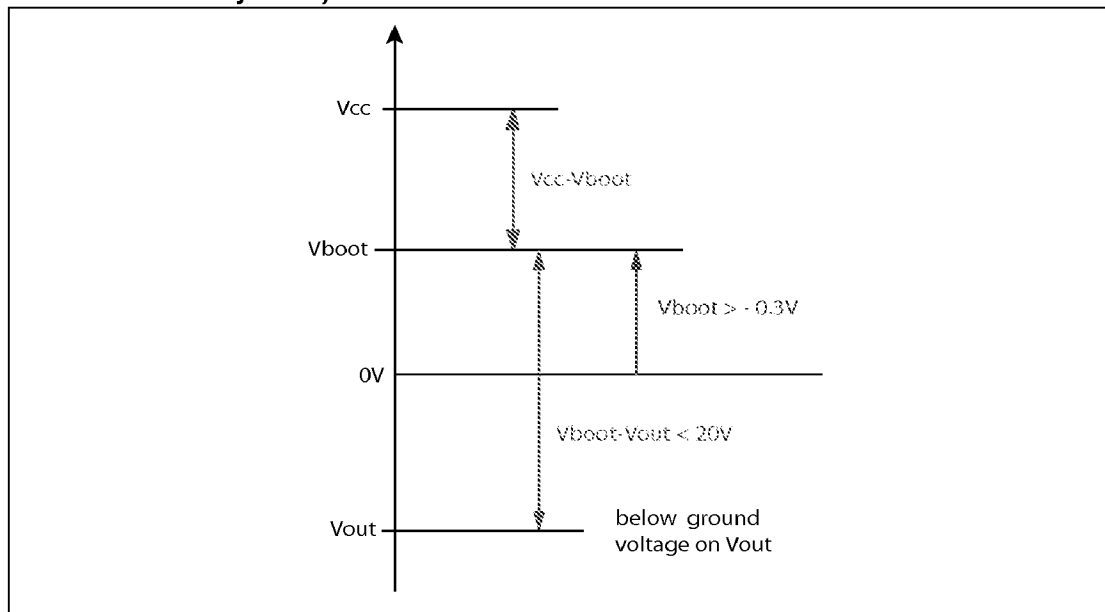


Regarding the below-ground voltage spike, typically this undershoot voltage has a very brief duration (less than 100 ns) and is not enough to further over-charge the bootstrap capacitor, because the charging time constant of the bootstrap is equal to the product of the C_{BOOT} and the R_{BOOT} resistance in series with the diode. Moreover, the higher this resistance, the lower the risk of bootstrap over-charging during BVG spikes.

For example, with a R_{BOOT} of 120 Ω and a C_{BOOT} of 100 nF, the associated time constant would be about 12 μ s, much higher than typical below-ground voltage spike duration. Using an internal bootstrap diode it is very difficult to over-charge the floating section up to dangerous voltage levels with the very short duration of the undershoot spike. More attention should be given to the below-ground voltage of the V_{BOOT} during this spike because, as explained previously, the internal junction V_{BOOT} -to-substrate could turn on; typically for very short time (a few tenths of nanoseconds), this is not a problem.

In Figure 41, the main conditions to avoid issues related to the below-ground voltage phenomenon in steady-state are resumed. As explained in paragraph 11.1, typically the static below-ground voltage is hardly higher than 2 V, so these constraints are usually not a limitation for most applications.

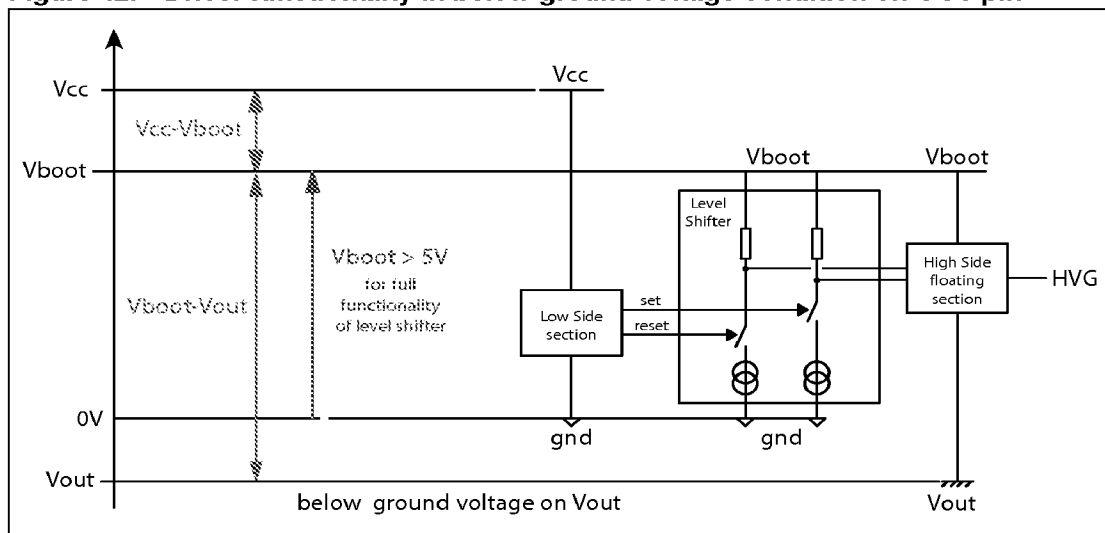
Figure 41. L6390 safe operating range when the OUT pin is below ground voltage (in steady state)



11.4 Functionality of L6390 outputs in below-ground condition

The L6390 IC gate driver makes use of a level shifter to send the set-reset information to the high side floating section. The level shifter, shown in *Figure 42*, is mainly composed of two high voltage switches, for set/reset signals, driven by the low voltage section and linked to two pull-up resistors connected to the BOOT pin.

Figure 42. Driver functionality in below-ground voltage condition on OUT pin



The two level-shifted signals are then fed into a logic latch in the high side floating section, providing the logic state for the high side gate driver (HVG output). The L6390 IC provides

full functionality of the switching operation of the high side section until the level shifter is operating. Because the two pull-up resistors are connected to the floating supply V_{BOOT} of the high side floating section, the driver still functions properly if the OUT voltage begins to go below ground. This means that the operating limit for the level shifter structure must be referred to the absolute value of the BOOT voltage relative to ground. In fact, the BOOT voltage can be considered the supply of the level shifter block. The minimum operating value of the level shifter supply is 5 V, so the V_{BOOT} value must be at least 5 V. The $V_{BOOT} - V_{OUT}$ voltage can be any value in the 12.4 V to 20 V range. If the BOOT voltage is between 0 V and 5 V, the functionality of the level shifter is not guaranteed, but internal structures should not be damaged. Some examples are given in the following section.

11.4.1 Steady state (DC) conditions

When the LVG is off, the bootstrap diode turns on only when the BOOT voltage is about 2 V below the VCC (worst case: minimum voltage drop value), so it could over-charge the bootstrap capacitor up to a maximum voltage of about $VCC - 2 V - V_{OUT}$. To avoid exceeding the value of 20 V on $V_{BOOT} - V_{OUT}$ voltage, the OUT pin could be forced permanently to the value of $VCC - 2 V - 20 V$ with the L6390 IC still operating safely, making the HVG output switching as the HIN logic input (see the table below).

Table 3. Minimum V_{OUT} in DC condition providing safe and full operation of the high side section

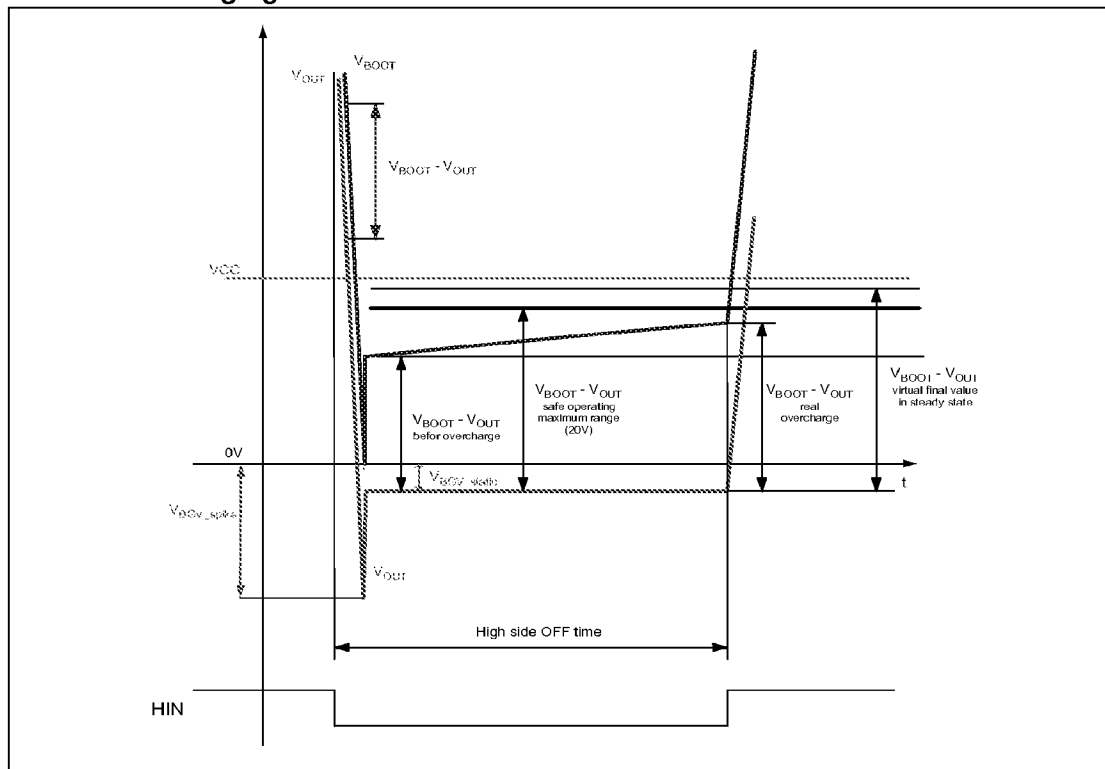
	Example 1	Example 2	Example 3
VCC	12.5	15	17
V_{BOOT}	10.5	13	15
$V_{OUT} \text{ (min)}$	-9.5	-7	-5
$V_{BOOT} - V_{OUT} \text{ (max)}$	20	20	20

It must be emphasized, as explained in the previous paragraph, how in most applications the static below-ground voltage of the OUT pin is very rarely lower than about - 2 V.

11.4.2 Transient conditions

If the time that the OUT pin goes below ground is limited, the bootstrap over-charge probably will not exceed the safe operating range (20 V) of $V_{BOOT} - V_{OUT}$, even if the V_{OUT} voltage is lower than the limit of $VCC - 2 V - 20 V$. In this case the gate driver and the high side section are fully operational if the V_{BOOT} voltage remains above 5 V, as explained previously, for proper functioning of the high side level shifter. The logic state is anyway held if the V_{BOOT} remains above ground.

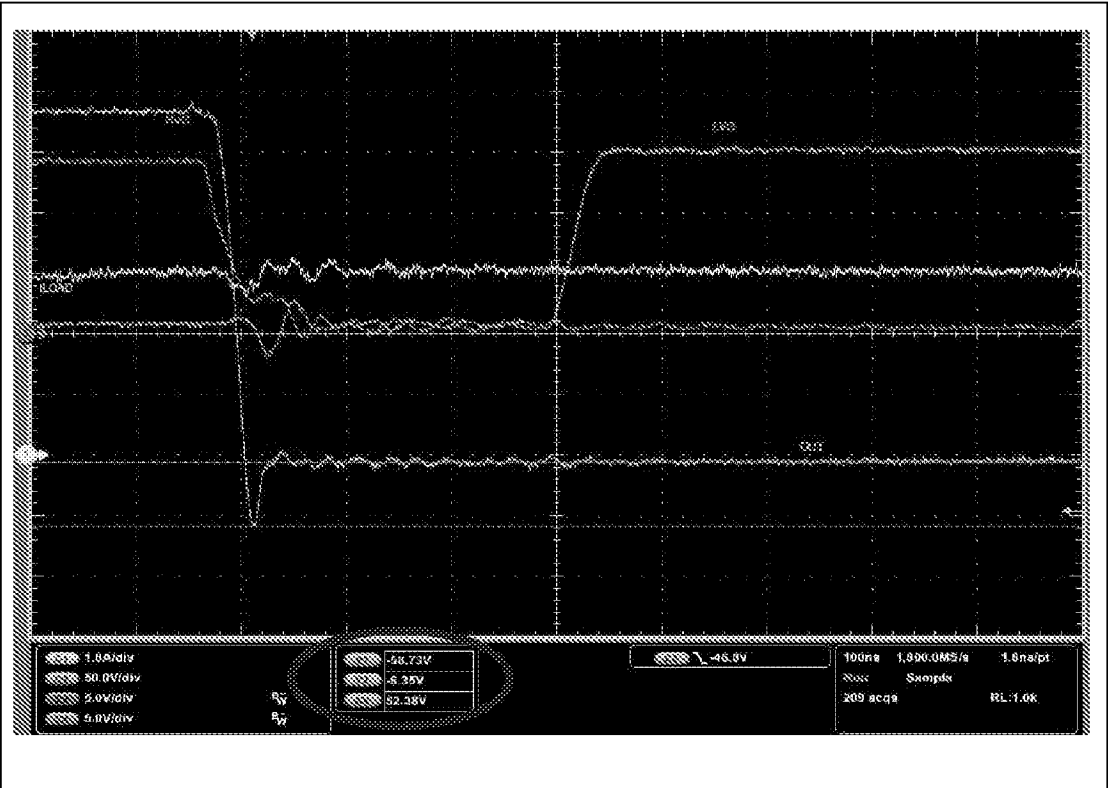
Figure 43. OUT below-ground voltage in transient conditions: limited boot over-charging



11.4.3 Below-ground voltage spikes

As previously stated, the duration of the dynamic contribution of the below-ground voltage on the OUT pin is not enough to over-charge the bootstrap capacitor. This fact usually removes the risk of exceeding the recommended maximum value of 20 V for $V_{BOOT} - V_{OUT}$ voltage, but it increases the danger of pulling down the BOOT pin below ground, thus violating its absolute minimum rating. As mentioned above, this phenomenon should be avoided for safe IC gate driver operation. However, note that applicative bench tests have shown the L6390 proper operation even with below-ground spikes on the OUT pin well above - 50 V (see *Figure 44*).

Figure 44. Example of below-ground voltage spike



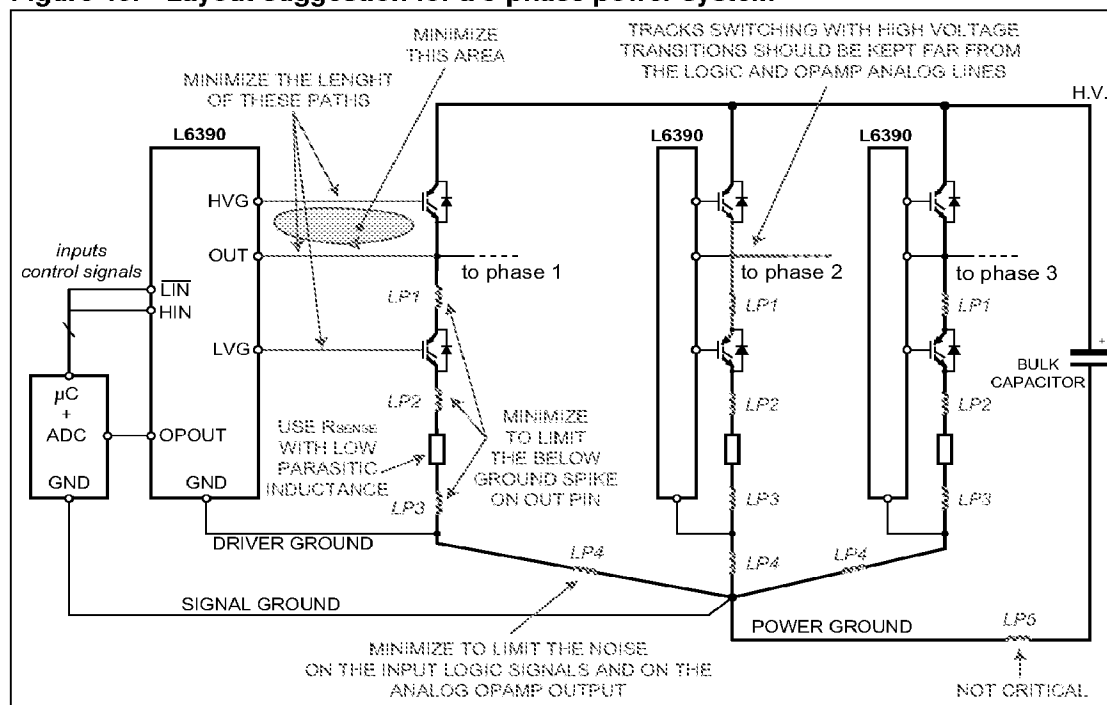
12 Layout suggestions

Typically, for power applications using high voltages and large load currents, the board layout of all circuits related to the power stage is important. Board layout includes several aspects, such as track dimensions (length and width), circuit areas, but also the proper routing of the traces and the optimized reciprocal arrangement of the various system elements and power sources in the PCB area.

Reasons to give particular attention to the PCB layout include EMI issues (both induced and perceived by the application) and over-voltage spikes due to parasitic inductances along the PCB traces, the proper connection of the sense blocks, the logic inputs and the analog outputs of the L6390 device. In fact, the L6390 IC not only has the function of driving the power stage, but also embeds analog sensing blocks such as comparator and op-amps. For example, especially regarding multi-phase power stages, it is important to keep the current reading, performed through the integrated op-amp, from ground noise.

In Figure 45 some layout guidelines and suggestions for a 3-phase application are provided.

Figure 45. Layout suggestion for a 3-phase power system



As explained in Section 8.9, the gate driving PCB traces should be designed to be as short as possible and the area of the circuits should be minimized to avoid the sensitivity of such structures to the surrounding noise. Typically, a good power system layout keeps the power IGBTs (or MOSFETs) of each half-bridge as close as possible to the related gate driver.

In Figure 45 a set of parasitic inductances related to the different circuit tracks is shown. The various groups of inductances may have undesired effects which should be limited as much as possible. Moreover, note that Figure 45 emphasizes parasitic inductances located on the lines usually managing high voltages and fast current transitions, which are very noisy.

The group of LP1, LP2 and LP3 parasitic inductances is located along the low side path of each half-bridge, between the OUT pin and the ground of the related driver and provides an undesired contribution to the issue of below-ground voltage spike on each OUT pin of the L6390 device (as described in *Section 11.1*). In fact, at the beginning of the current recirculation on the low side switches, the current may experience a high di/dt which may produce, on those parasitic inductances, significant voltage spikes. These spikes are summed with the voltage drop of the low side diode and with the R_{SENSE} and have a negative sign, so the overall voltage drop between OUT and GND may be significant. The recommendation is to limit as much as possible each contribution to this phenomenon by limiting the length of tracks LP1, LP2 and LP3 and using an R_{SENSE} resistor with low intrinsic inductance. More specifically, LP1 may be reduced by connecting the OUT line directly to the collector (or drain) of the low side IGBT (or MOSFET). LP2 may be reduced by placing the R_{SENSE} resistor as close as possible to the emitter (or source) of the low side IGBT (or MOSFET). The LP3 may be minimized by connecting the ground line (also called driver ground) of the related gate driver directly on the R_{SENSE} resistor.

LP4 represents the parasitic inductance located between the ground connections of each gate driver (driver ground) and the ground connection of the application controller (also called signal ground). Due to its location, this parasitic inductance introduces noise which is experienced by the input logic signals and the op-amp output analog signals. In fact, each phase of the bridge causes high currents (with high di/dt) to flow on these paths, resulting in voltage noise which drops between the gate driver ground and the controller ground. This noise between the two grounds is directly added to all logic and analog voltage signals between the gate driver and the microcontroller included the input logic signals and the analog output of the related gate driver op-amp. It is recommended to minimize this noise by reducing the distance between the signal ground and the driver ground (for each gate driver in the system) as much as possible. Generally, it is recommended to connect the signal ground to the three driver grounds through a star connection, in order to improve the balancing and symmetry of the 3-phase driving topology.

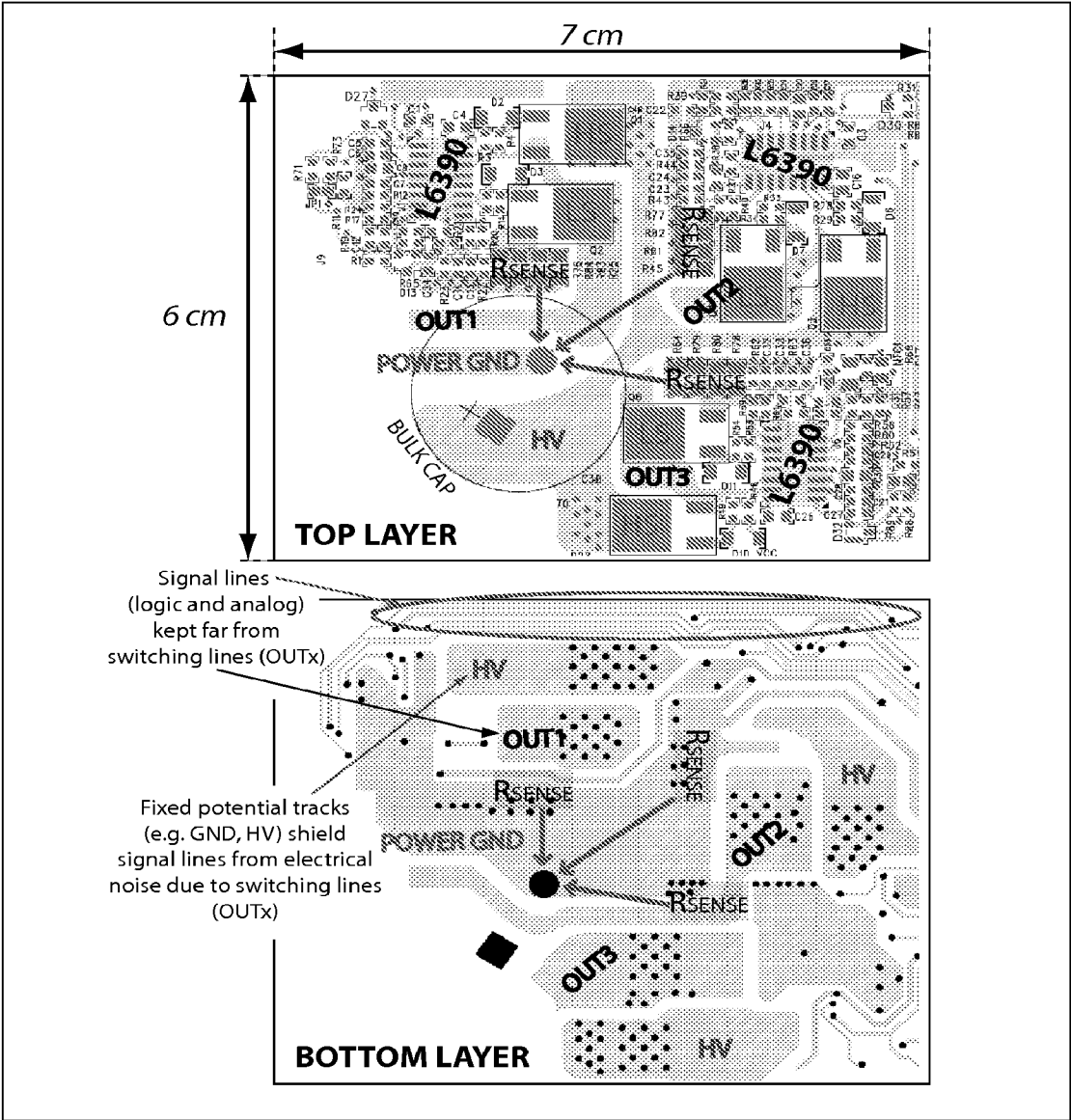
Note: It is very important to avoid any ground loop; only a single path must connect two different ground nodes.

The LP5 parasitic inductance is usually not critical, because it stands between the negative terminal of the bulk capacitor and the signal/power ground. The spikes on this parasitic element minimally influence other nodes of the system.

Another useful suggestion is to ensure some distance between the lines switching with high voltage transitions, and the signal lines sensitive to electrical noise. Specifically, the tracks of each OUT phase bringing significant currents and high voltages should be separated from the logic lines and analog sensing circuits of op-amps and comparators.

In *Figure 46* a practical example of the layout is provided. The example represents the 3-phase power stage section of the STEVAL-IHM021V1 demonstration board. It includes the L6390 gate drivers, the six IGBTs (or MOSFETs) in the DPAK package and the bulk capacitor. The layout suggestions described in this paragraph are implemented in the layout example in *Figure 46*. As described in *Figure 46*, the fixed voltage tracks, such as GND or HV lines, can be used to shield the logic and analog lines from the electrical noise produced by the switching lines (e.g. OUT1, OUT2 and OUT3). Each half-bridge ground is connected in a star configuration and the three R_{SENSE} resistors are very close to each other and to the power ground. Note also that the suggested 3-phase configuration occupies a modest amount of surface area (7 cm x 6 cm), and thanks to the SMD packages and board vias, no heat sinks are required.

Figure 46. Layout example from the STEVAL-IHM021V1 3-phase board



13 Revision history

Table 4. Document revision history

Date	Revision	Changes
02-Oct-2008	1	Initial release
03-Aug-2009	2	Removed chapter 11 The use of the resistor on the OUT line and added <i>Section 11.2: How to reduce the below ground spike voltage on page 40.</i>

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FIVR – Fully Integrated Voltage Regulators on 4th Generation Intel® Core™ SoCs

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Abstract—Intel’s® 4th generation Core™ microprocessors are powered by Fully Integrated Voltage Regulators (FIVR). These 140 MHz multi-phase buck regulators are integrated into the 22nm processor die, and feature up to 80 MHz unity gain bandwidth, non-magnetic package trace inductors and on-die MIM capacitors. FIVRs are highly configurable, allowing them to power a wide range of products from 3W fanless tablets to 300W servers. FIVR helps enable 50% or more battery life improvements for mobile products and more than doubles the peak power available for burst workloads.

I. INTRODUCTION

Intel’s® 4th generation Core™ microprocessors (code name Haswell) are powered by Fully Integrated Voltage Regulators (FIVR), the industry’s first large scale deployment of high current switching regulators integrated into a VLSI die and package. An overview of the scheme is given in Fig. 1(a). A first stage VR, which is on the motherboard, converts from the PSU or battery voltage (12-20V) to approximately 1.8V, which is distributed across the microprocessor die. The second conversion stage is comprised of between 8 and 31 (depending on the product) FIVRs, which are 140MHz synchronous multiphase buck converters with up to 16 phases. A simplified schematic for a two phase FIVR domain is shown in Fig. 1(b). The power FETs, control circuitry and high frequency decoupling are on the die, while the inductors and mid-frequency input decoupling capacitors are placed on the package. Each FIVR is independently programmable to achieve optimal operation given the requirements of the domain it is powering. The settings are optimized by the Power Control Unit (PCU), which specifies the input voltage, output voltage, number of operating phases, and a variety of other settings to minimize the total power consumption of the die.

FIVR is the enabling technology behind key improvements for Intel’s® 4th generation Core™ microprocessors including a 50% or more increase in battery life for mobile products, and a 2-3x increase in peak available power (which converts into burst performance). The motherboard voltage regulators eliminated by FIVR free up space that can be used to add platform features or reduce platform dimensions. Details are discussed in Section V.

A. Background

Intel’s® 2008 microprocessor microarchitecture introduced the Power Control Unit (PCU) [1], a microcontroller that monitored conditions across the die in real time, and dynamically adjusted a variety of settings to optimally manage power consumption and performance. One of the most important features controlled by the PCU were newly added high current power gates, which provided a significant improvement in CPU energy efficiency by eliminating large leakage losses on idle compute domains. The power gates on high current domains were introduced by adding gate transistors into thin ‘cracks’ between major functional blocks and represented a very small percentage of the total die area. The die bumps required to support the high current domains required a much larger area than the gate transistors themselves. The large “bump area” posed a hard barrier to productization until a scheme was devised to “borrow” bumps from surrounding circuitry using a thick, low loss routing layer. An extension of this scheme makes FIVR affordable.

A limitation of power gate is that all active domains still operate at the highest voltage required by any individual domain. To create separate voltage domains an entirely new regulator must be added to the motherboard, which adds cost, increases area, and requires extra package pins. An improvement suggested by recent research is the integration of high frequency buck regulators directly on the microprocessor package, or in the die itself [2] [3] [4] [5]. This allows a much larger number of independent power domains, each managed dynamically to match the local computational demand. For example, this would allow one CPU core to run at an elevated voltage and frequency to satisfy a heavy computational load, while other cores execute lower priority code at a much lower voltage and frequency to save power.

Each of the previous works cited had at least one issue making it poorly suited for broad, high volume deployment. The multi-chip approach taken by [2] resulted in an effective current density of 1.3A/mm² that would make it expensive to implement because of the silicon area required to support a full power microprocessor product. In [3] another

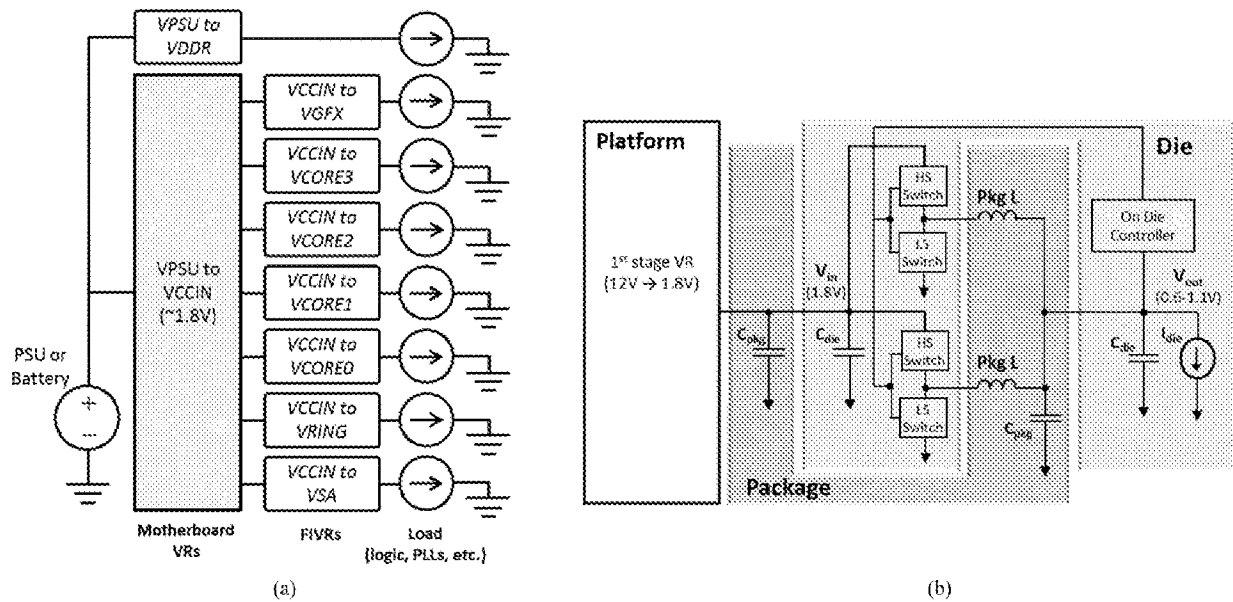


Figure 1. (a) Representative partitioning of the separate high current power domains on a 4th generation Core™ Microprocessor. (b) Simplified schematic of a single FIVR domain, showing the partitioning of the components between the die and the package.

example of the multi-chip approach is demonstrated. This work, which used a 90nm process and inductors integrated onto the die, achieved a higher current density (8A/mm²) but reported a relatively low efficiency of 76% (compared to 85% for 3.3V to 1.0V conversion in [2]). Instead of the multi-chip approach, the authors of [4] integrated the regulator directly into the die on a 45nm process, but still suffered from relatively low current density (1.7A/mm²). The authors also report an efficiency of 83% for 1.5V to 1.0V conversion due in part to the quality of the discrete inductors that were used.

FIVR builds on the VR designs presented in [2], while the implementation strategy that makes FIVR affordable, is an extension of the bump “borrowing” scheme developed for the high current power gates [1].

B. Motivation

This paper will show that FIVR addresses the issues in prior work that prevented broad deployment of integrated switching regulators in high volume products. Extending the earlier bump borrowing scheme yields the same current density increase, and corresponding cost decrease, that first made power gates affordable. Improvements in the inductors and transistors yield efficiency in the 90% range for typical high power workloads. The high unity gain frequency (up to 80MHz) allows FIVR to work with just on die MIM for output capacitance.

While these advancements are necessary, they’re insufficient to make a reasonable business case for FIVR. To pay the costs of developing and fabricating FIVR, it’s important to quantify the actual customer-visible benefits provided in a real implementation. At the start of the design FIVR’s expected benefits fell into half a dozen categories. FIVR delivered material benefits in every category, and some benefits were far larger than expected. The benefit categories were: battery life increase, increased available power (for increased burst performance), decreased power required for a given level of

performance (or almost equivalently, increased performance for a given power consumed), decreased platform cost and size, improved product flexibility and scalability. See section V for the detailed FIVR impact.

II. IMPLEMENTATION, DESIGN, AND SIMULATION

A. Circuitry

A block diagram representing the circuitry for a single FIVR domain is shown in Fig. 2. The buck regulator bridges are formed by replacing the power gates in previous products with NMOS and PMOS cascode power switches. The cascode configuration allows the power switches to be implemented with standard 22nm logic devices while still handling an input voltage of up to 1.8VDC [2]. This avoids the cost of extra processing steps for high voltage devices, while achieving excellent switching characteristics. The bridge drivers are controlled thru high-voltage level-shifters and support ZVS (zero-voltage switching) and ZCS (zero-current-switching) soft-switching operation. The gates of the cascode devices are connected to the “half-rail”, V_{ccdrv} , regulated to $V_m/2$. This is also the negative supply of the PMOS bridge driver as well as the positive supply of the NMOS bridge driver.

The area occupied by the power switches and drivers is small, so they are distributed across the die, immediately above the connection to their associated package inductor which minimizes routing losses. This is illustrated in Fig. 3(a), which shows the location of the package inductors under the die for a four core LGA part. The driver circuitry is interleaved with the power switches in an array which minimizes parasitics to allow for very high switching frequencies. This also allows the size of the bridge to be easily scaled based on the current requirement and optimization points for each power domain.

Each FIVR domain is controlled by a FIVR Control Module (FCM). The FCM contains the circuitry for generating the PWM signals using double-edge modulation, as indicated in

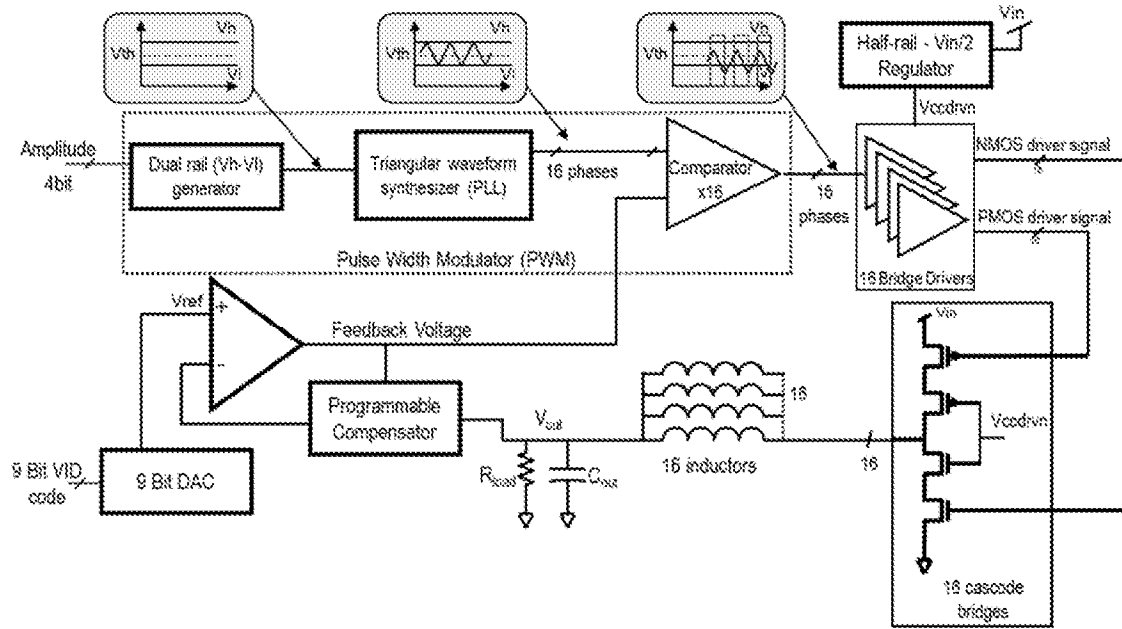


Figure 2. Simplified block diagram of the circuitry for a single representative FIVR domain

Fig. 2 by the dashed box. Separate circuitry not shown in Fig. 2 manages phase current balancing, and the resulting digital PWM signals are distributed from the FCM to individual bridges. The PWM frequency, PWM gain, phase activation, and the angle of each phase are all programmable in fine increments to enable optimal efficiency and minimum voltage ripple across a span of different operating points. Spread-spectrum is used for EMI and RFI (Radio Frequency Interference) control.

The FCM module also contains the feedback control circuitry (compensator). A high-precision 9-bit DAC generates a reference voltage for a programmable, high bandwidth analog fully differential type-3 compensator. Sense lines feed the output voltage back to the compensator. The endpoint of these sense lines is strategically placed to achieve minimum DC error and optimal transient response at an important circuit location in the domain. The compensator is programmed individually for each voltage domain based on its output filter, and can be reprogrammed while the domain is active to maintain optimal transient response as phase shedding occurs.

The key to making FIVR affordable was integrating the power devices directly into the microprocessor die. As with the power gate circuitry discussed in the introduction, the power switching circuitry for FIVR can be placed in small areas between major circuit blocks. The lower current handling of the die bumps makes the die bump area requirements for FIVR much larger than the actual die area required. Since FIVR is integrated into the microprocessor die, routing on the thick metal die layer allows extra bumps to be ‘borrowed’ from areas over other circuits, which avoids wasting any excess die area due to bump current limits. This makes the effective current density of FIVR 31A/mm², a 24x increase over the bump-limited 1.3A/mm² reported in [2].

B. Passives

In order to keep the buck output filter small enough to fit on the die and package it is necessary for FIVR to switch at a high frequency – 140 MHz in most cases. This allows the buck output filter inductors to be implemented using only the bottom metal layers of a standard flip-chip package. Power routing is constrained to the top layers of the package as a result, but the proximity of the inductors to the load ensures that minimal power is dissipated on these layers. The inductors are non-magnetic, i.e. Air Core Inductors (ACI). A representative ACI from an 8-phase domain of a product with an LGA package is shown in Fig. 3(b), including the connection points to the power switches, the DC current path through the inductor, and the connection of the inductor to the output plane. Package design rules allow the ACIs to be placed in to close proximity with one another. On a representative LGA package with four CPU cores, this allowed 59 inductors on 10 different voltage rails to be implemented in a 20mm x 8mm area. The package implementation also allows inductor designs to be customized on a per rail basis to meet efficiency, ripple, and transient response requirements.

Decoupling for the input rail is provided by a combination of ceramic package capacitors and on-die MIM capacitors [6]. The on package ceramic capacitors keep the output impedance of the input rail low from approximately 1 MHz to their self-resonance around 20 MHz. The MIM capacitors are on the die along with the power circuitry and provide high frequency decoupling, including at the switching frequency and its harmonics. Decoupling for the output rails is provided primarily by the MIM capacitors, which are sufficient to provide good transient response if wide bandwidth feedback control is used (see the results section). In some cases the MIM capacitors are supplemented with extra package ceramic capacitors. The comparatively low self-resonant frequency of the ceramic capacitors complicates the control loop design and

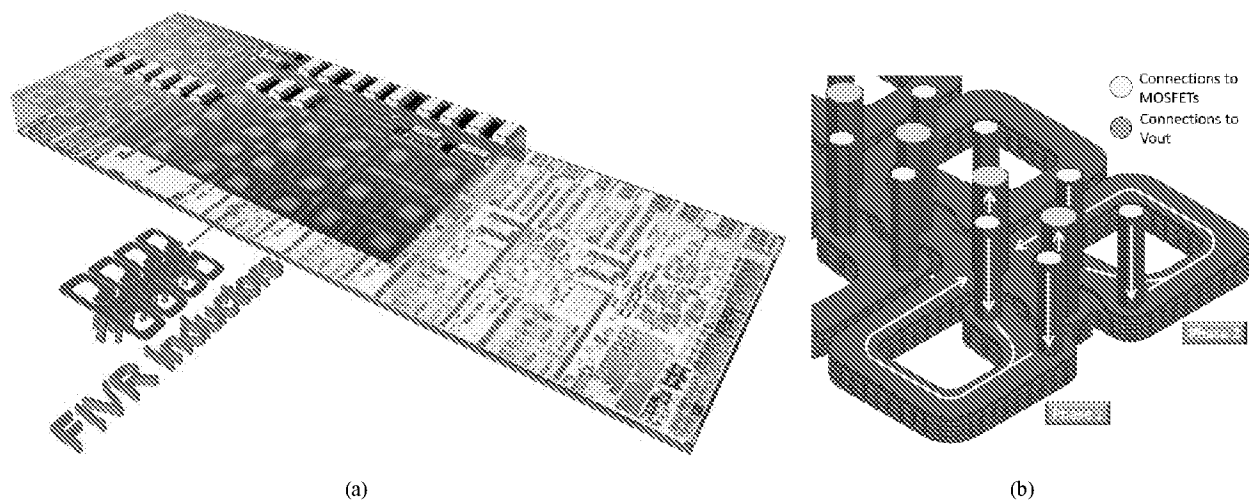


Figure 3. (a) The bottom of an Intel® 4th generation Core™ microprocessor LGA package is shown along with a picture of the corresponding die. A group of eight FIVR inductors is pulled off to the side. (b) An enlarged 3D view of two FIVR inductors is shown with current flow arrows.

does little to attenuate voltage ripple, but the ceramic capacitors can provide a net transient response benefit if they are robustly connected to the output power plane.

C. System Control

In order to minimize losses from FIVR, a modified version of the PCU [1] dynamically configures each FCM based on the current activity level of the domain. The PCU turns each rail on or off based on activity, and specifies an output voltage target to support the desired frequency. It also optimizes the settings discussed in section II.A for the anticipated operating conditions. These settings include the number of active phases (i.e. phase shedding to improve light load performance), the compensator settings (to maintain optimal transient response as the number of phases changes), and the timing of switch drivers (to ensure zero voltage switching at light loads). This allows each FIVR domain to operate at near peak efficiency across a wide range of load conditions from retention to Turbo. An example of the benefit this provides is shown in Section IV.A.

III. CHARACTERIZATION AND PERFORMANCE TESTING

Validating and optimizing a voltage regulator requires the measurement of key parameters such as voltage ripple, efficiency, power supply rejection ratio, transient response, and control loop stability margins. Due to FIVR's high level of integration and fast switching frequency many of these standard measurements are difficult or impossible to perform using off the shelf test equipment. For example, an IA Core voltage domain powered by a FIVR capable of supplying over 30A occupies less than 15mm² on the package, which is completely covered by the microprocessor die. This renders the attachment of an external load for a full current efficiency measurement impossible. This section describes some key Design For Test (DFT) features that are included in FIVR to allow accurate characterization.

A. Control Loop Transfer Function

To enable characterization of the control, a high frequency programmable signal generator is placed in the feedback network on every FIVR domain. The signal generator is activated in a test mode to inject a known, synchronized signal

into the on-die feedback loop. By controlling the test feature and monitoring the output voltage on the package with an oscilloscope, the response of the control loop is directly measured. Repeated measurements are used to tune the compensator to achieve fast response and good stability margins.

B. Load Transient Response and Rejection Ratio

Microprocessors require a nearly constant DC voltage in the presence of large load transients. For characterization purposes, however, it is difficult to create a well behaved step load using only the execution of code in normal operation. The authors in [7] instead use a scheme called Integrated Frequency Domain Impedance Meter (IFDIM), which gates the clock network on and off at a fixed frequency creating a large load transient. The frequency is programmable, the magnitude of the transient can be precisely calibrated using DC measurements, and the load step is known to occur within one clock cycle, so the microprocessor itself is effectively turned into an alternating current load. This feature is included on every FIVR domain, allowing the transient response to a known load to be measured. FIVR domains are characterized across a wide frequency range at multiple operating points for both output impedance and output load coupling across domains.

C. Efficiency

As was previously mentioned, the level of integration makes it impossible to connect a high current load directly to the output of a FIVR rail. An additional complication is that the circuitry on the die cannot be disconnected from the FIVR output, so whenever FIVR is powered some extra current draw due to leakage results. This required the development of a new technique for accurately measuring efficiency. A brief summary of the method is given here. First, a procedure using an external low current load and FIVR operating in a test mode is used to calibrate the leakage. The completely ungated clock tree is then operated at varying frequencies to create a large effective adjustable DC output current. An iterative series of measurements is then used to precisely calibrate the total current drawn by the clocks and the leakage, which allows the efficiency to be measured, when combined with conventional

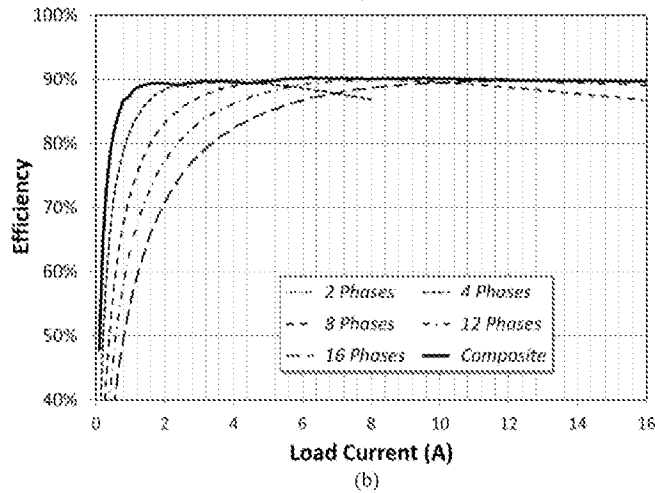
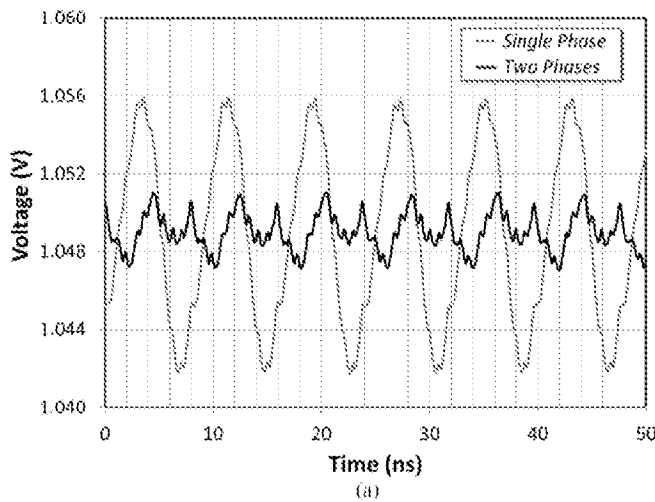


Figure 4. (a) Measured voltage ripple for a low noise domain for single phase and two phase operation (128 averages) (b) Measured efficiency for a voltage domain as a function of the number of active phases

measurements of the voltage and the current at the output of the first stage regulator.

IV. RESULTS

Due to the high switching frequency used, the performance of FIVR is sensitive to the layout of the die and the package which includes the inductors. Each combination of die and package is individually optimized and validated. The following section contains some key validation results from an Intel® 4th generation Core™ microprocessor with four microprocessor cores on an LGA package.

A. Measurements

Fig. 4(a) shows the voltage ripple for a low noise domain measured under the die near the connection of the ACI to the power plane. The measurement was averaged 128 times against the PWM clock (with spread spectrum clocking turned off). To achieve an accurate measurement, a controlled impedance differential sense line was routed on the package from the measurement location to a probe connection point with a matched termination. An active differential oscilloscope probe was then connected to the probe point. This ensures an accurate, wide bandwidth measurement is achieved, as opposed to

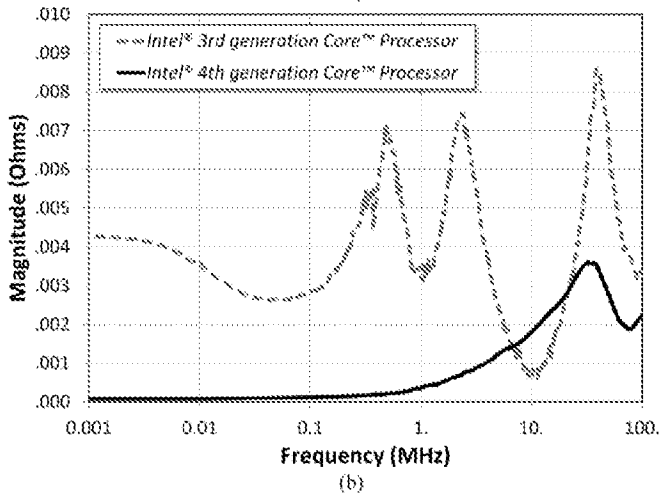
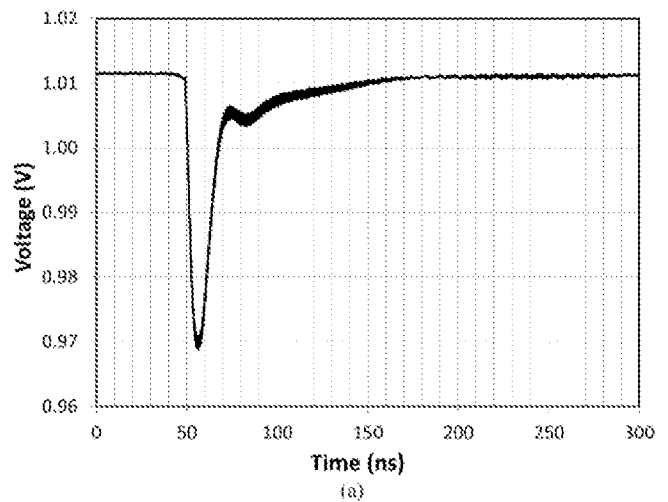


Figure 5. (a) Measured voltage droop on a graphics domain in response to an 8.5A step load (b) Comparison of the effective impedance profile for the graphics voltage domain on a 3rd generation Core™ microprocessor versus a 4th generation Core™ microprocessor

probing the package power planes directly or using a single ended sense line, which can substantially attenuate the measurement over a distance as short as a few millimeters. In two phase operation less than 4mV (less than 1% of the voltage set point) of ripple is achieved with a rail driven by air core inductors well under 2mm² in area.

Fig. 4(b) shows the efficiency as measured using the procedure in section III.C for 1.70V to 1.05V conversion with the bridges configured for hard switching. The efficiency measurement has been repeated for varying numbers of phases, in each case showing a peak efficiency of approximately 90% at 0.75A/phase. By employing a phase shedding scheme it is possible to keep the efficiency of the domain within a few percent of the peak efficiency of the domain from 1A to 15A. This is managed by the PCU which can phase shed when the efficiency can be improved, but also has the intelligence to avoid phase shedding when it could be problematic, for example when a large load transient is possible.

The measured output voltage (averaged 128 times) during an 8.5A load step generated by the IFDIM feature on the graphics voltage rail is shown in Fig 5(a). The measurement

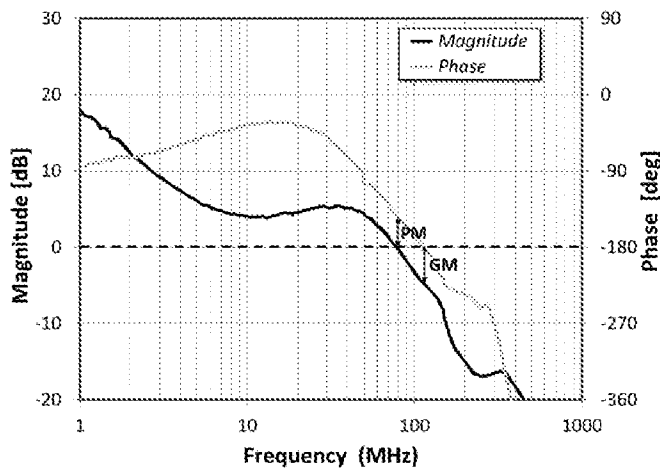


Figure 6. The measured open loop gain and phase of a FIVR showing 78MHz bandwidth with 40° phase margin

was performed with a similar probing configuration to that used for the voltage ripple measurement. The combination of a high bandwidth feedback loop and on die decoupling keep the voltage droop under 50mV despite a rise time for current step of under 1ns (orders of magnitude faster than normal graphics circuit behavior). The main droop event lasts under 30ns, and the DC voltage is restored within 100ns. A nonlinear control feature saturates the duty cycle when a large transient is detected (not active in Fig. 5(a)). The feature was found to provide up to a 25% reduction in voltage droop for step loads, but the benefit is significantly reduced for certain aperiodic load patterns. The effective output impedance profile for the same rail is shown in Fig. 5(b). The peak impedance demonstrates the fast bandwidth of the compensator. Because the inductors are located immediately below the actual area of the die that consumes current, the DC and low frequency load line is virtually zero. The figure also shows the impedance profile for an Intel® 3rd generation Core™ microprocessor graphics rail, which is powered by a platform VR. For this rail, a DC load line is required due to the distance between the VR and the die. Several resonant peaks occur from the various stages of decoupling capacitors on the motherboard and package, and the parasitic inductance between them and the actual point of current consumption on the die.

Fig. 6 shows the open loop transfer function for a FIVR domain, measured using the signal generator DFT. This rail demonstrates a unity gain bandwidth of 78 MHz while still maintaining 40° phase margin. Robust compensator circuit design and very small propagation delays were necessary to achieve this bandwidth, which, in turn, was required to maintain good transient response on rails with limited output capacitance. The high bandwidth also enables fast voltage transitions. A FIVR rail turning on and turning off are shown in Fig. 7. Both transitions are programmed to about half a microsecond for a full range transition – two orders of magnitude faster than a typical platform-based solution. The fast ramp rate translates into power savings for the system, as the voltage rails can be turned on, used, and turned off again almost instantly.

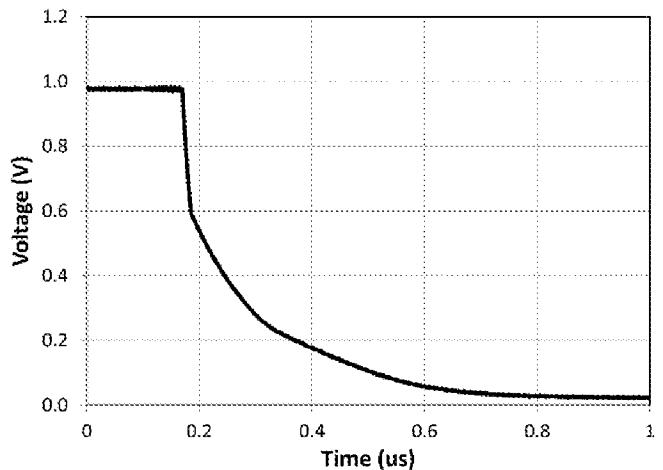
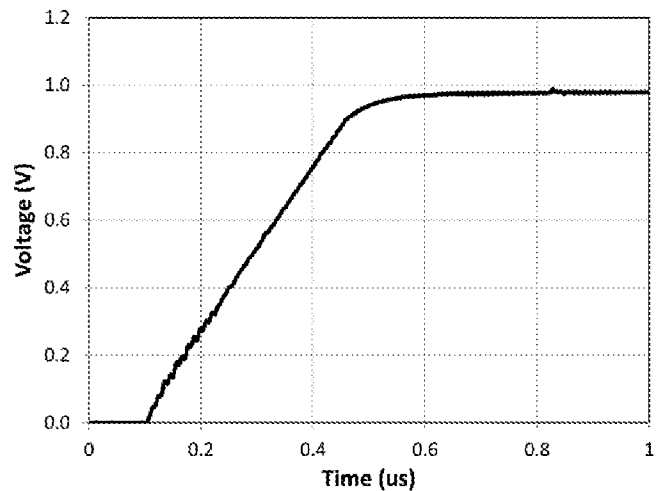


Figure 7. A FIVR rail ramping to its voltage set point from fully off, and then turning off again. Voltage transition times are programmable, but typically set for half a microsecond for a 1V transition.

A large number of additional measurements are taken for validation purposes that are not shown here due to space constraints. These include the output impedance of the V_m rail, audio susceptibility measurements, and the coupling noise due to load transients from one rail to another (particularly from very high current domains such as core and graphics to low current system agent domains). EMI/RFI characterization is also performed.

B. Comparison to Previous Work

Table I contains a comparison to previous works discussed in the introduction. FIVR operates at a higher switching frequency than previous works, which is enabled in part by very good gate charge characteristics for the MOSFETs. This allows up to 90% efficiency at a common conversion ratio.

V. FIVR IMPACT TO PRODUCTS

Battery life improvement: Sufficient battery life for a complete work day has long been desired from mobile products. FIVR, combined with power management architecture improvements, has enabled this for Intel® 4th generation Core™ products. Increases of well over 50% have been widely reported (for example, [8] and [9]). FIVR's battery life benefit comes by several means:

TABLE I. COMPARISON OF FIVR TO PREVIOUSLY REPORTED INTEGRATED VOLTAGE REGULATORS

Parameter	G. Schrom et al., 2010 [2]	T. DiBene et al., 2010 [3]	N. Sturcken et al., 2012 [4]	This Work
Process node	130 nm	90 nm	45 nm	22 nm
Switching Frequency	60 MHz	100 MHz	80 MHz	140 MHz
Unity Gain Freq	5 MHz	Not Published	Not Published	80MHz
Efficiency	85-88%, 3.3V:1.0V	76%	83%, 1.5V:1.0V	90%, 1.7V:1.05V
Total Output I _{max} capability	50 A	Limited by first stage and thermals (Up to 400 A)	1.2 A	Limited by first stage and thermals (Up to 700 A)
I _{max} /VR die area	1.3 A/mm ²	8 A/mm ²	1.7 A/mm ²	31 A/mm ²
Voltage rail count	4	20	1	8 to 31
Phase count	16	320	4	49 to 360
Integration level	MCM ^a	MCM ^a	Integrated into network die	Integrated into CPU die
Inductor technology	Package trace, & magnetic discrete	Magnetic thin-film on VR die	Discrete wire-wound air core	2D array of package trace
Capacitor type	Ceramic package caps	Ceramic package caps	Die Cap	Die Cap - MIM
Cout per Max Amp	2000 nF/A	not published	15 nF/A	7 nF/A

^a MCM – Multi Chip Module – the active circuitry is on a separate die assembled on the same package

- Standby current historically consumes a large fraction of the battery's stored energy. FIVR's fast bandwidth allows low frequency supply noise to be rejected, resulting in up to a 90% reduction in decoupling requirements. This allows both the first and second stages of regulation to be power cycled much faster than on previous products, enabling new deep sleep states with up to 20x lower standby power. With the lowered capacitance, power expended, and time wasted entering and exiting the states is similarly reduced. Reduced sleep-state entry/exit time also saves power by increasing sleep-state usage.
- FIVR's fast control loop and integration into the package result in one tenth the peak impedance of prior solutions (see Fig. 5(b)) in the sub-MHz stimulus range most relevant to the graphics architecture. The resulting low frequency supply noise reduction improves power at a given performance by up to 30%.
- FIVR increases the number of voltage rails, allowing each domain to be set at the minimum possible voltage that supports error-free operation, reducing both leakage and dynamic power.
- Replacing multiple high current voltage regulators on the motherboard with a single first stage regulator reduces the PCB footprint of the power delivery solution. This extra space can be used for a larger battery, with some examples demonstrating up to 10% growth.
- Trimming FIVR together with the microprocessor removes manufacturing guard-bands normally required to ensure that every VR will work with every CPU.

Increased available peak power: An illustrative example shows how FIVR can increase the peak power available to the microprocessor. A typical mobile processor platform using the prior generation power delivery scheme had two 30A, 1.1V VRs providing 33W for cores and 33W for graphics. Using the same power FETs and inductors for the FIVR's 1.8V input VR, the part has 108W power rail (30A/phase * 2 phases * 1.8V),

which can be dynamically allocated to a combination of FIVRs by the PCU. For core-only workloads, nearly the entire 108W can be allocated for the cores, increasing the available power ceiling by 3x. For graphics workloads, 36W can be partitioned to the cores with the remaining 72W going to graphics – more than double the power available from the 33W platform VR. Because power consumption scales as CV^2F and frequency scales with voltage, the increase in available power could be used to operate the graphics at up to 26% higher frequency than possible with the platform VR. A similar calculation yields a 44% higher core frequency in the core-only scenario. The duration of these scenarios is limited by the thermal capabilities of the platform, but translates into improved speed in many real scenarios.

Decreased power at a given performance: Intel's® Iris™ Pro graphics uses FIVR's higher available power to deliver high end graphics. FIVR's high unity gain bandwidth presents less than a tenth the peak output impedance provided by the prior generation's platform VR in the sub-MHz range important to the graphics load (see fig. 5b). Because FIVR has doubled the graphics power ceiling, few (if any) of our shipping parts would fit within the bounds of the older generation's platform. The higher currents typically imply hundreds of millivolt droops on the older platforms. The combination of high currents with high impedance peaks yields a hypothetical power tax in the 20-30% range (assuming one could, and actually would, ship these high current levels into the old platforms). FIVR avoids that tax.

Improved product flexibility and scalability: FIVR's ability to add voltage rails onto a common shared input rail without package growth or even platform changes brings significant flexibility and modularity into the design space that was not available before. New voltage rails can be added as needed, without any platform change. This ability allowed us to introduce the Iris™ Pro graphics into standard platforms even though new rails were needed for the EDRAM and its high speed OPIO link.

Platform size and cost reduction: Since four platform VR controllers are eliminated, along with the associated decoupling caps, power FETs and inductors, there's a clear platform size and cost advantage. FIVR's total platform BOM cost reduction is expected to be several billion dollars over the product lifetime. The power inductors feeding power to the CPU often show up in the critical thickness cross-section of small form-factor laptops and tablets, and trading FIVR's total component count reduction for a thickness reduction is straightforward. A platform phase count increase results in lower current per phase, and the lower phase current can be satisfied with a lower profile set of inductors.

In the prior generation platforms, some dual-sided PCBs have tall components like ICs and inductors on the primary side and lower profile discrete components on the secondary side. Often the secondary-side components are mostly high frequency decoupling, located immediately underneath front-side ICs, with sparsely populated areas between. In such cases, FIVR eliminates most of the secondary-side components, and frees up space on the primary side to accommodate the rest. The resulting populated PCB thickness is reduced by the height of the tallest removed backside components.

In small systems, the platform size tends to limit the feature set, leading to fewer connectivity options, smaller storage space, etc. FIVR's platform size reductions can provide more space to implement these features.

VI. CONCLUSION

Consumers expect every generation of mobile computer products to have more compute power, thinner and lighter form factors, and longer battery life than the last. The 4th generation Intel® Core™ power architecture using FIVR provides improvements in all three of these areas. To the author's knowledge, this is the first consumer product to make use of integrated switching regulators on this scale. Furthermore, FIVR's performance is improved versus previously reported prototypes. The authors therefore feel that FIVR is an important advancement in the field of power electronics.

ACKNOWLEDGMENT

We would like to acknowledge the following FIVR team members who were not already listed as authors: the FIVR silicon design team including George Geannopoulos, Keith Hodgson, Narayanan Raghuraman, Alex Lyakhov, Michael W Rogers, Ravi Vunnam, Lan D Vu, Mark S Milshtein, Chiu Keung Tang, Hong Yun Tan, Seh Leon Goh, Samie Samaan, Narayanan Natarajan, Rajan Vijayaraghavan, Ashish Khanna and Munish Chauhan; top-level integration by Pankaj Aswal; modeling and implementation development by Doug Huard and Alex Waizman; layout studies, inductor development and package designs by John Smith, Brad Larson, and Huong Do; mask design by Neal Tanksley and Galyna Burenkova; test and manufacturing support from RJ Hayes and Arun Krishnamoorthy; modeling support by Alex Levin and Anne Augustine.

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Dual Synchronous Rectified MOSFET Drivers

The ISL6210 integrates two ISL6208A drivers and is optimized to drive two independent power channels in a synchronous-rectified buck converter topology. These drivers combined with an Intersil ISL62xx multiphase PWM controller forms a complete single-stage core-voltage regulator solution with high efficiency performance at high switching frequency for advanced microprocessors.

The IC is biased by a single low voltage supply (5V), minimizing driver switching losses in high MOSFET gate capacitance and high switching frequency applications. Each driver is capable of driving a 3nF load with less than 10ns rise/fall time. Bootstrapping of the upper gate driver is implemented via an internal low forward drop diode, reducing implementation cost, complexity, and allowing the use of higher performance, cost effective N-Channel MOSFETs. Adaptive shoot-through protection is integrated to prevent both MOSFETs from conducting simultaneously.

The ISL6210 features 4A typical sink current for the lower gate driver, enhancing the lower MOSFET gate hold-down capability during PHASE node rising edge, preventing power loss caused by the self turn-on of the lower MOSFET due to the high dV/dt of the switching node.

The ISL6210 also features an input that recognizes a high impedance state, working together with Intersil multiphase PWM controllers to prevent negative transients on the controlled output voltage when operation is suspended. This feature eliminates the need for the Schottky diode that may be utilized in a power system to protect the load from negative output voltage damage.

Ordering Information

PARTNUMBER (Note)	PART MARKING	TEMP. RANGE (°C)	PACKAGE (Pb-Free)	PKG. DWG. #
ISL6210CRZ	62 10CRZ	-10 to +100	16 Ld 4x4 QFN	L16.4x4
ISL6210CRZ-T*	62 10CRZ	-10 to +100	16 Ld 4x4 QFN	L16.4x4

*Please refer to TB347 for details on reel specifications.

NOTE: These Intersil Pb-free plastic packaged products employ special Pb-free material sets, molding compounds/die attach materials, and 100% matte tin plate plus anneal (e3 termination finish, which is RoHS compliant and compatible with both SnPb and Pb-free soldering operations). Intersil Pb-free products are MSL classified at Pb-free peak reflow temperatures that meet or exceed the Pb-free requirements of IPC/JEDEC J STD-020.

Features

- 5V Quad N-Channel MOSFET Drives for Two Synchronous Rectified Bridges
- Adaptive Shoot-Through Protection
 - Active Gate Threshold Monitoring
 - Programmable Dead-Time
- 0.4Ω ON-Resistance and 4A Sink Current Capability
- Supports High Switching Frequency
 - Fast Output Rise and Fall
 - Ultra Low Three-State Hold-Off Time (20ns)
- Low V_F Internal Bootstrap Diode
- Low Bias Supply Current
- Power-On Reset
- QFN Package
 - Compliant to JEDEC PUB95 MO-220 QFN-Quad Flat No Leads-Product Outline
 - Near Chip-Scale Package Footprint; Improves PCB Efficiency and Thinner in Profile
- Pb-Free Available (RoHS Compliant)

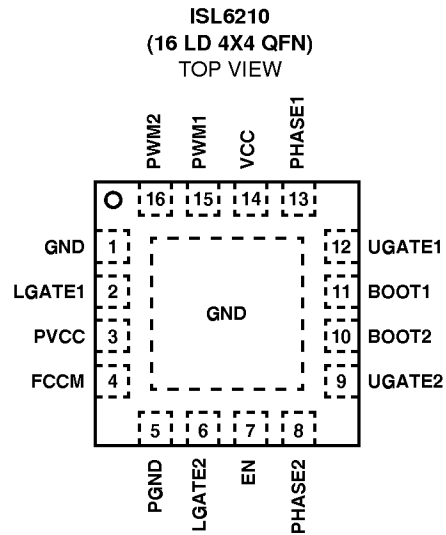
Applications

- Core Voltage Supplies for Intel® and AMD® Microprocessors
- High Frequency Low Profile High Efficiency DC/DC Converters
- High Current Low Voltage DC/DC Converters
- Synchronous Rectification for Isolated Power Supplies

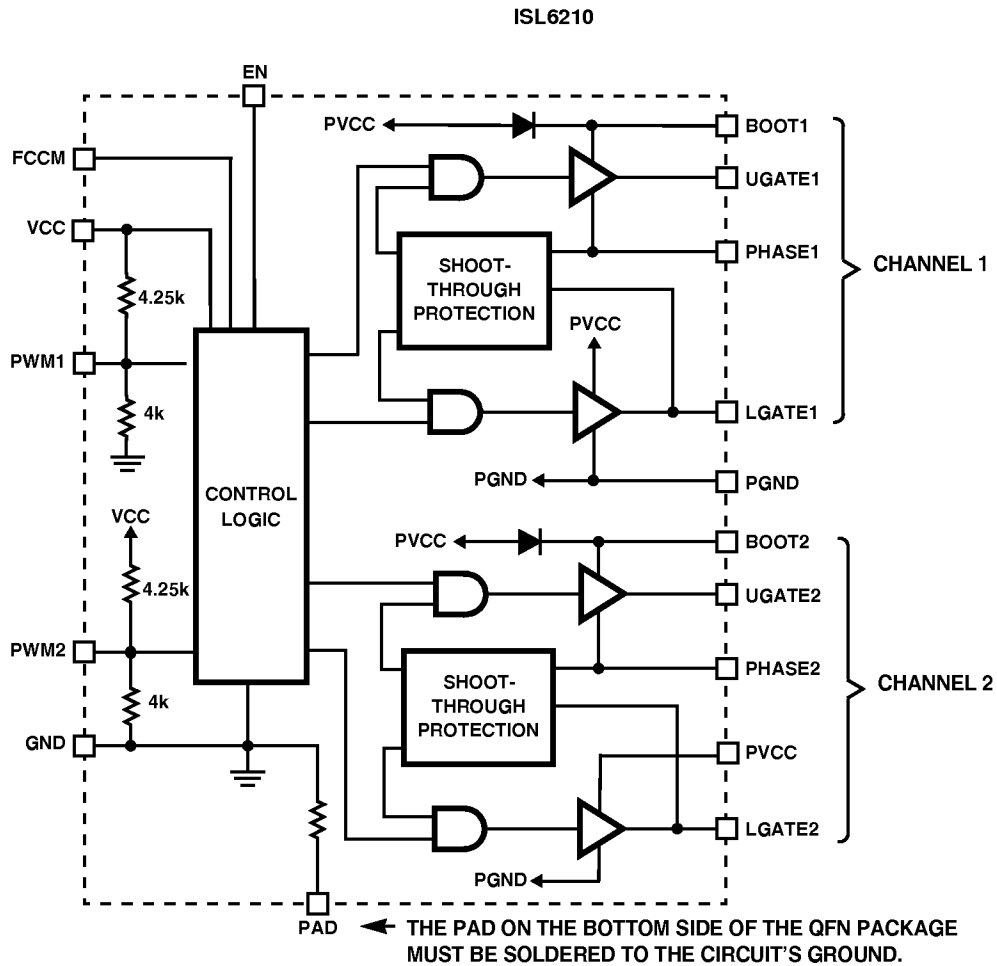
Related Literature

- Technical Brief TB363 "Guidelines for Handling and Processing Moisture Sensitive Surface Mount Devices (SMDs)"
- Technical Brief 400 and Technical Brief 417 for Power Train Design, Layout Guidelines, and Feedback Compensation Design
- Technical Brief 447 "Guidelines for Preventing Boot-to-Phase Stress on Half-Bridge MOSFET Driver ICs"

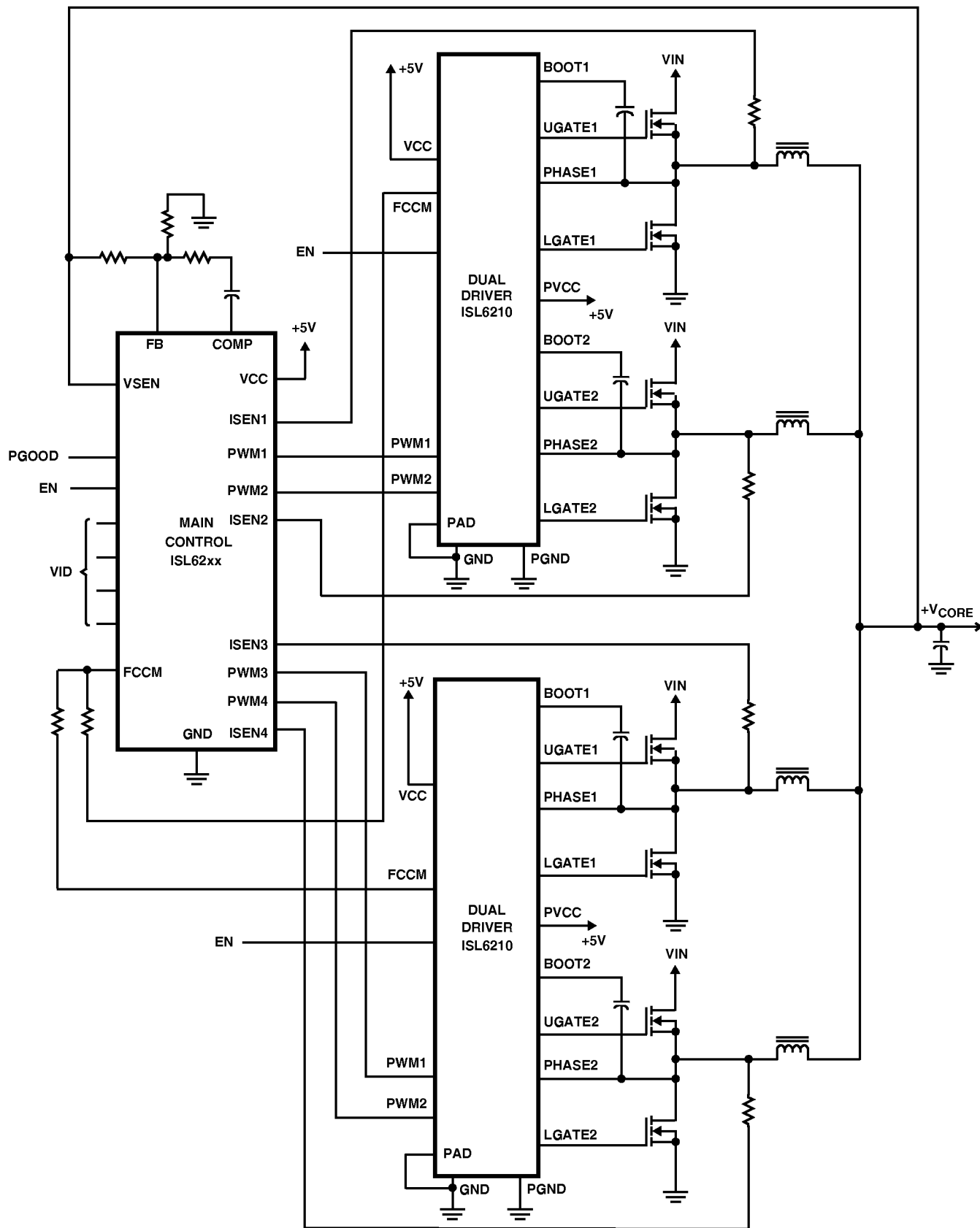
Pinout



Block Diagram



Typical Application - Multiphase Converter Using ISL6210 Gate Drivers



Absolute Maximum Ratings

Supply Voltage (PVCC, VCC)	-0.3V to 7V
Input Voltage (V _{EN} , V _{PWM})	-0.3V to VCC + 0.3V
BOOT Voltage (V _{BOOT-GND})	-0.3V to 33V (DC) or 36V (<200ns)
BOOT To PHASE Voltage (V _{BOOT-PHASE})	-0.3V to 7V (DC) -0.3V to 9V (<10ns)
PHASE Voltage	GND - 0.3V to 30V (DC) GND - 8V (<20ns Pulse Width, 10μJ)
UGATE Voltage	V _{PHASE} - 0.3V (DC) to V _{BOOT} V _{PHASE} - 5V (<20ns Pulse Width, 10μJ) to V _{BOOT}
LGATE Voltage	GND - 0.3V (DC) to VCC + 0.3V GND - 2.5V (<20ns Pulse Width, 5μJ) to VCC + 0.3V
Ambient Temperature Range	-40°C to +125°C

CAUTION: Do not operate at or near the maximum ratings listed for extended periods of time. Exposure to such conditions may adversely impact product reliability and result in failures not covered by warranty.

NOTES:

1. θ_{JA} is measured in free air with the component mounted on a high effective thermal conductivity test board with “direct attach” features.
2. θ_{JC} , “case temperature” location is at the center of the package underside exposed pad. See Tech Brief TB379 for details.

Thermal Information

Thermal Resistance (Notes 1 and 2)	θ_{JA} (°C/W)	θ_{JC} (°C/W)
QFN Package	46	8.5
Maximum Junction Temperature	+150°C	
Maximum Storage Temperature Range	-65°C to +150°C	
Pb-Free Reflow Profile	see link below http://www.intersil.com/pbfree/Pb-FreeReflow.asp	

Recommended Operating Conditions

Ambient Temperature Range	-10°C to +100°C
Maximum Operating Junction Temperature	+125°C
Supply Voltage, VCC	5V ±10%

Electrical Specifications

These specifications apply for T_A = -10°C to +100°C, Unless Otherwise Noted. Parameters with MIN and/or MAX limits are 100% tested at +25°C, unless otherwise specified. Temperature limits established by characterization and are not production tested.

PARAMETER	SYMBOL	TEST CONDITIONS	MIN	TYP	MAX	UNITS
SUPPLY CURRENT						
Bias Supply Current	I _{VCC}	PWM pin floating, V _{VCC} = 5V	-	170	-	μA
POWER-ON RESET						
POR Rising			-	3.4	4.2	V
POR Falling			2.6	3.0	-	V
Hysteresis			-	400	-	mV
BOOTSTRAP DIODE						
Forward Voltage Drop	V _F	V _{VCC} = 5V, forward bias current = 2mA	0.3	0.60	0.7	V
PWM INPUT						
Sinking Impedance	R _{PWM_SNK}		8.0	10.4	15	kΩ
Source Impedance	R _{PWM_SRC}		8.3	10.6	25	kΩ
Three-State Rising Threshold		V _{VCC} = 5V	1.08	1.3	1.5	V
Three-State Falling Threshold		V _{VCC} = 5V	3.4	3.65	3.98	V
Three-State Shutdown Holdoff Time	t _{TSSHD}	t _{PDLU} or t _{PDLL} + Gate Falling Time	-	80	-	ns
Three-state to UG/LG Rising Propagation Delay	t _{PTS}		-	20	-	ns
SWITCHING TIME (See Figure 1)						
UGATE Rise Time (Note 4)	t _{RU}	V _{VCC} = 5V, 3nF Load	-	8.0	-	ns
LGATE Rise Time (Note 4)	t _{RL}	V _{VCC} = 5V, 3nF Load	-	8.0	-	ns
UGATE Fall Time (Note 4)	t _{FU}	V _{VCC} = 5V, 3nF Load	-	8.0	-	ns
LGATE Fall Time (Note 4)	t _{FL}	V _{VCC} = 5V, 3nF Load	-	4.0	-	ns
UGATE Turn-Off Propagation Delay	t _{PDLU}	V _{VCC} = 5V, Outputs Unloaded	-	20	-	ns
LGATE Turn-Off Propagation Delay	t _{PDLL}	V _{VCC} = 5V, Outputs Unloaded	-	27	-	ns
UGATE Turn-On Propagation Delay	t _{PDHU}	V _{VCC} = 5V, Outputs Unloaded; R _{SET} = 0Ω	-	26	-	ns
LGATE Turn-On Propagation Delay	t _{PDHL}	V _{VCC} = 5V, Outputs Unloaded; R _{SET} = 0Ω	-	26	-	ns
UGATE Turn-On Propagation Delay	t _{PDHU}	V _{VCC} = 5V, Outputs Unloaded; R _{SET} = 80kΩ	-	41	-	ns
LGATE Turn-On Propagation Delay	t _{PDHL}	V _{VCC} = 5V, Outputs Unloaded; R _{SET} = 80kΩ	-	33	-	ns
Minimum LGATE On Time in DCM (Note 4)	t _{LGMIN}		-	400	-	ns

Electrical Specifications

These specifications apply for $T_A = -10^{\circ}\text{C}$ to $+100^{\circ}\text{C}$, Unless Otherwise Noted. **(Continued)** Parameters with MIN and/or MAX limits are 100% tested at $+25^{\circ}\text{C}$, unless otherwise specified. Temperature limits established by characterization and are not production tested.

PARAMETER	SYMBOL	TEST CONDITIONS	MIN	TYP	MAX	UNITS
OUTPUT						
Upper Drive Source Resistance	R_{UG_SRC}	250mA Source Current	-	1.0	2.5	Ω
Upper Drive Source Current (Note 4)	I_{UG_SCR}	$V_{UGATE-Phase} = 2.5V$	-	2.00	-	A
Upper Drive Sink Resistance	R_{UG_SNK}	250mA Sink Current	-	1.0	2.5	Ω
Upper Drive Sink Current (Note 4)	I_{UG_SNK}	$V_{UGATE-Phase} = 2.5V$		2.00	-	A
Lower Drive Source Resistance	R_{LG_SRC}	250mA Source Current	-	1.0	2.5	Ω
Lower Drive Source Current (Note 4)	I_{LG_SCR}	$V_{LGATE} = 2.5V$	-	2.00	-	A
Lower Drive Sink Resistance	R_{LG_SNK}	250mA Sink Current	-	0.4	1.0	Ω
Lower Drive Sink Current (Note 4)	I_{LG_SNK}	$V_{LGATE} = 2.5V$	-	4.00	-	A

NOTE:

- Parameters with MIN and/or MAX limits are 100% tested at $+25^{\circ}\text{C}$, unless otherwise specified. Temperature limits established by characterization and are not production tested.
- Limits established by characterization and are not production tested.

Functional Pin Description

NUMBER	NAME	FUNCTION
1	GND	Bias and reference ground. All signals are referenced to this node.
2	LGATE1	Lower gate drive output of Channel 1. Connect to gate of the low-side power N-Channel MOSFET.
3	PVCC	This pin supplies power to both the lower and higher gate drives in ISL6210. Connect to a +5V supply. Place a high quality low ESR ceramic capacitor from this pin to GND.
4	FCCM	Logic control input that will force continuous conduction mode (HIGH state) or allow discontinuous conduction mode (LOW state). Placing a series resistor in this input will allow the switching dead-time to be programmed.
5	PGND	It is the power ground return of both low gate drivers.
6	LGATE2	Lower gate drive output of Channel 2. Connect to gate of the low-side power N-Channel MOSFET.
7	EN	Logic control input that will enable (HIGH state) or disable (LOW state) the IC. Shutdown current is $<1\mu\text{A}$.
8	PHASE2	Connect this pin to the SOURCE of the upper MOSFET and the DRAIN of the lower MOSFET in Channel 2. This pin provides a return path for the upper gate drive.
9	UGATE2	Upper gate drive output of Channel 2. Connect to gate of high-side power N-Channel MOSFET.
10	BOOT2	Floating bootstrap supply pin for the upper gate drive of Channel 2. Connect the bootstrap capacitor between this pin and the PHASE2 pin. The bootstrap capacitor provides the charge to turn on the upper MOSFET. See "Internal Bootstrap Diode" on page 7 for guidance in choosing the capacitor value.
11	BOOT1	Floating bootstrap supply pin for the upper gate drive of Channel 1. Connect the bootstrap capacitor between this pin and the PHASE1 pin. The bootstrap capacitor provides the charge to turn on the upper MOSFET. See "Internal Bootstrap Diode" on page 7 for guidance in choosing the capacitor value.
12	UGATE1	Upper gate drive output of Channel 1. Connect to gate of high-side power N-Channel MOSFET.
13	PHASE1	Connect this pin to the SOURCE of the upper MOSFET and the DRAIN of the lower MOSFET in Channel 1. This pin provides a return path for the upper gate drive.
14	VCC	Connect a +5V bias supply to this pin. It supplies the internal analog circuits. Place a high quality, low ESR ceramic capacitor from this pin to GND. This should be a separate capacitor than the one used for PVCC (Pin 3).
15	PWM1	The PWM signal is the control input for the Channel 1 driver. The PWM signal can enter three distinct states during operation. See "Three-State PWM Input" on page 6 for further details. Connect this pin to the PWM output of the controller.
16	PWM2	The PWM signal is the control input for the Channel 2 driver. The PWM signal can enter three distinct states during operation. See "Three-State PWM Input" on page 6 for further details. Connect this pin to the PWM output of the controller.
N/A	PAD	Connect this pad to the power ground plane (GND) via thermally enhanced connection.

Timing Diagram

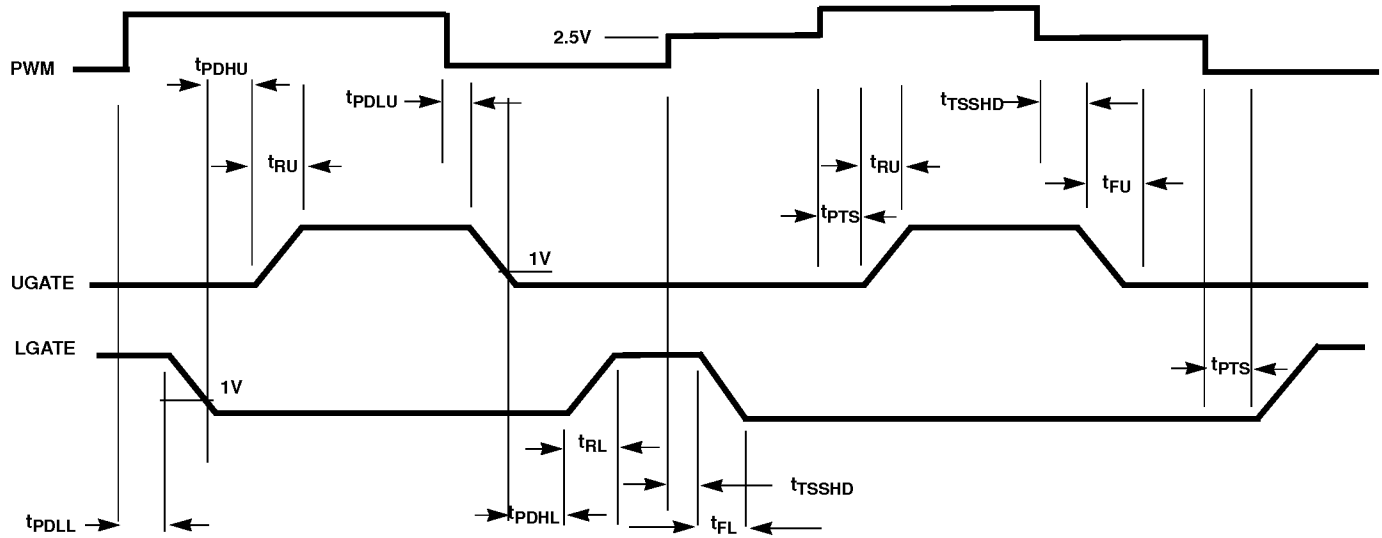


FIGURE 1. TIMING DIAGRAM

Description

Theory of Operation

Designed for speed, the ISL6210 dual MOSFET driver controls both high-side and low-side N-Channel FETs for two separate channels of a Multiphase PWM system from two independent PWM signals.

A rising edge on PWM initiates the turn-off of the lower MOSFET (see "Timing Diagram" on page 6). After a short propagation delay [t_{PDLL}], the lower gate begins to fall. Typical fall times [t_{FL}] are provided in the "Electrical Specifications" table on page 4. Adaptive shoot-through circuitry monitors the LGATE voltage. When LGATE has fallen below 1V, UGATE is allowed to turn ON. This prevents both the lower and upper MOSFETs from conducting simultaneously, or shoot-through.

A falling transition on PWM indicates the turn-off of the upper MOSFET and the turn-on of the lower MOSFET. A short propagation delay [t_{PDLU}] is encountered before the upper gate begins to fall [t_{FU}]. The upper MOSFET gate-to-source voltage is monitored, and the lower gate is allowed to rise after the upper MOSFET gate-to-source voltage drops below 1V. The lower gate then rises [t_{RL}], turning on the lower MOSFET.

This driver is optimized for converters with large step down compared to the upper MOSFET because the lower MOSFET conducts for a much longer time in a switching period. The lower gate driver is therefore sized much larger to meet this application requirement.

The 0.5Ω ON-resistance and 4A sink current capability enable the lower gate driver to absorb the current injected to the lower gate through the drain-to-gate capacitor of the

lower MOSFET and prevent a shoot through caused by the high dv/dt of the phase node.

Diode Emulation

Diode emulation allows for higher converter efficiency under light-load situations. With diode emulation active, the ISL6210 will detect the zero current crossing of the output inductor and turn off LGATE. This ensures that discontinuous conduction mode (DCM) is achieved. Diode emulation is asynchronous to the PWM signal. Therefore, the ISL6210 will respond to the FCCM input immediately after it changes state.

Please note that Intersil does not recommend Diode Emulation use with $r_{DS(ON)}$ current sensing topologies. The turn-OFF of the low side MOSFET can cause gross current measurement inaccuracies.

Three-State PWM Input

A unique feature of the ISL6210 and other Intersil drivers is the addition of a shutdown window to the PWM input. If the PWM signal enters and remains within the shutdown window for a set holdoff time, the output drivers are disabled and both MOSFET gates are pulled and held low. The shutdown state is removed when the PWM signal moves outside the shutdown window. Otherwise, the PWM rising and falling thresholds outlined in the "Electrical Specifications" table on page 4 determine when the lower and upper gates are enabled.

Adaptive Shoot-Through Protection

Both drivers incorporate adaptive shoot-through protection to prevent upper and lower MOSFETs from conducting simultaneously and shorting the input supply. This is accomplished by ensuring the falling gate has turned off one MOSFET before the other is allowed to turn on.

During turn-off of the lower MOSFET, the LGATE voltage is monitored until it reaches a 1V threshold, at which time the UGATE is released to rise. Adaptive shoot-through circuitry monitors the upper MOSFET gate-to-source voltage during UGATE turn-off. Once the upper MOSFET gate-to-source voltage has dropped below a threshold of 1V, the LGATE is allowed to rise.

In addition to gate threshold monitoring, a programmable delay between MOSFET switching can be accomplished by placing a resistor in series with the FCCM input. This delay allows for maximum design flexibility over MOSFET selection. The delay can be programmed from 5ns to 50ns and is obtained from the absolute value of the current flowing into the FCCM pin. If no resistor is used, the minimum 5ns delay is selected. Gate threshold monitoring is not affected by the addition or removal of the additional dead-time. Refer to Figure 2 and Figure 3 for more detail.

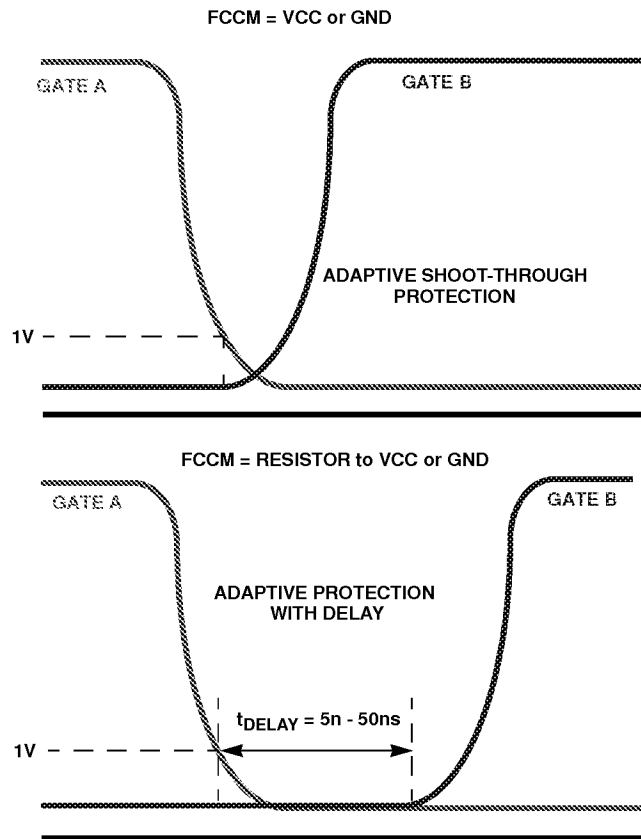


FIGURE 2. PROGRAMMABLE DEAD-TIME

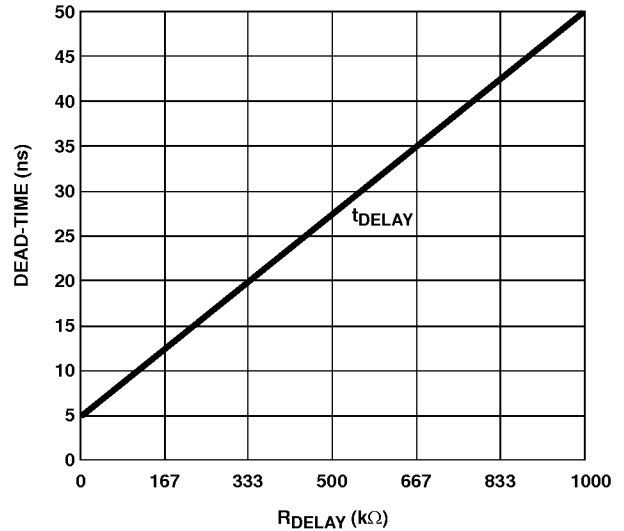


FIGURE 3. ISL6210 PROGRAMMABLE DEAD-TIME vs DELAY RESISTOR

The equation governing the dead-time seen in Figure 3 is expressed in Equation 1:

$$t_{DELAY(ns)} = [0.045 \times R_{DELAY(k\Omega)}] + 5ns \quad (EQ. 1)$$

The equation can be rewritten to solve for R_{DELAY} as follows:

$$R_{DELAY(k\Omega)} = \frac{(t_{DELAY(ns)} - 5ns)}{0.045} \quad (EQ. 2)$$

Internal Bootstrap Diode

This driver features an internal bootstrap diode. Simply adding an external capacitor across the BOOT and PHASE pins completes the bootstrap circuit.

Equation 3 helps select a proper bootstrap capacitor size:

$$C_{BOOT_CAP} \geq \frac{Q_{GATE}}{\Delta V_{BOOT_CAP}} \quad (EQ. 3)$$

$$Q_{GATE} = \frac{Q_{G1} \cdot PVCC}{V_{GS1}} \cdot N_{Q1}$$

where Q_{G1} is the amount of gate charge per upper MOSFET at V_{GS1} gate-source voltage and N_{Q1} is the number of control MOSFETs. The ΔV_{BOOT_CAP} term is defined as the allowable droop in the rail of the upper gate drive.

As an example, suppose two IRLR7821 FETs are chosen as the upper MOSFETs. The gate charge, Q_G , from the data sheet is 10nC at 4.5V (V_{GS}) gate-source voltage. Then the Q_{GATE} is calculated to be 22nC at PVCC level. We will assume a 200mV droop in drive voltage over the PWM cycle. We find that a bootstrap capacitance of at least 0.110μF is required. The next larger standard value capacitance is 0.22μF. A good quality ceramic capacitor is recommended.

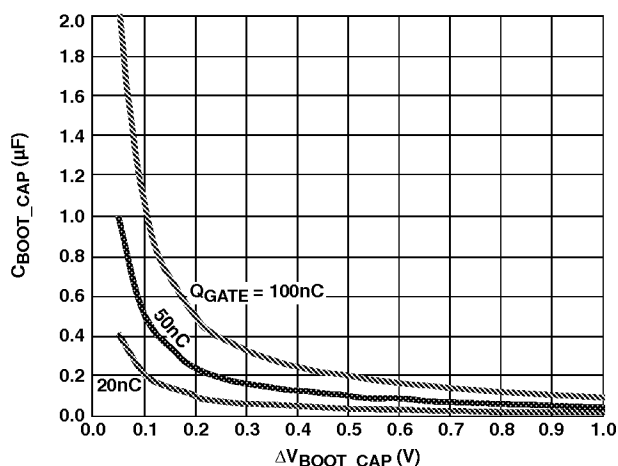


FIGURE 4. BOOTSTRAP CAPACITANCE vs BOOT RIPPLE VOLTAGE

Power Dissipation

Package power dissipation is mainly a function of the switching frequency (F_{SW}), the output drive impedance, the external gate resistance, and the selected MOSFET's internal gate resistance and total gate charge. Calculating the power dissipation in the driver for a desired application is critical to ensure safe operation. Exceeding the maximum allowable power dissipation level will push the IC beyond the maximum recommended operating junction temperature of +125°C. The maximum allowable IC power dissipation for the SO14 package is approximately 1W at room temperature, while the power dissipation capacity in the QFN packages, with an exposed heat escape pad, is around 2W. See "Layout Considerations" on page 9 for thermal transfer improvement suggestions. When designing the driver into an application, it is recommended that the following calculation is used to ensure safe operation at the desired frequency for the selected MOSFETs. The total gate drive power losses due to the gate charge of MOSFETs and the driver's internal circuitry and their corresponding average driver current can be estimated with Equations 4 and 5, respectively,

$$P_{Qg_TOT} = P_{Qg_Q1} + P_{Qg_Q2} + I_Q \cdot V_{CC} \quad (\text{EQ. 4})$$

$$P_{Qg_Q1} = \frac{Q_{G1} \cdot PV_{CC}^2}{V_{GS1}} \cdot F_{SW} \cdot N_{Q1}$$

$$P_{Qg_Q2} = \frac{Q_{G2} \cdot PV_{CC}^2}{V_{GS2}} \cdot F_{SW} \cdot N_{Q2}$$

$$I_{DR} = \left(\frac{Q_{G1} \cdot N_{Q1}}{V_{GS1}} + \frac{Q_{G2} \cdot N_{Q2}}{V_{GS2}} \right) \cdot F_{SW} + I_Q \quad (\text{EQ. 5})$$

where the gate charge (Q_{G1} and Q_{G2}) is defined at a particular gate to source voltage (V_{GS1} and V_{GS2}) in the corresponding MOSFET data sheet; I_Q is the driver's total quiescent current with no load at both drive outputs; N_{Q1} and N_{Q2} are number of upper and lower MOSFETs, respectively. The $I_Q V_{CC}$ product is the quiescent power of the driver without capacitive load and is typically negligible.

The total gate drive power losses are dissipated among the resistive components along the transition path. The drive resistance dissipates a portion of the total gate drive power losses, the rest will be dissipated by the external gate resistors (R_{G1} and R_{G2} , should be a short to avoid interfering with the operation shoot-through protection circuitry) and the internal gate resistors (R_{G11} and R_{G12}) of MOSFETs. Figures 5 and 6 show the typical upper and lower gate drives turn-on transition path. The power dissipation on the driver can be roughly estimated as follows:

$$P_{DR} = P_{DR_UP} + P_{DR_LOW} + I_Q \cdot V_{CC} \quad (\text{EQ. 6})$$

$$P_{DR_UP} = \left(\frac{R_{HI1}}{R_{HI1} + R_{EXT1}} + \frac{R_{LO1}}{R_{LO1} + R_{EXT1}} \right) \cdot \frac{P_{Qg_Q1}}{2}$$

$$P_{DR_LOW} = \left(\frac{R_{HI2}}{R_{HI2} + R_{EXT2}} + \frac{R_{LO2}}{R_{LO2} + R_{EXT2}} \right) \cdot \frac{P_{Qg_Q2}}{2}$$

$$R_{EXT2} = R_{G1} + \frac{R_{GI1}}{N_{Q1}} \quad R_{EXT2} = R_{G2} + \frac{R_{GI2}}{N_{Q2}}$$

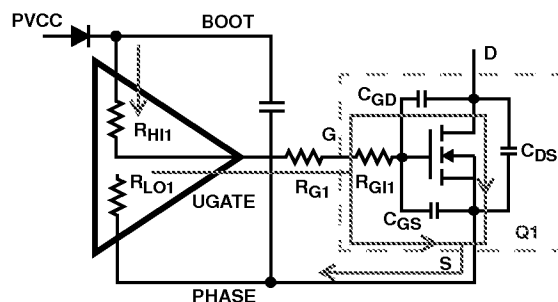


FIGURE 5. TYPICAL UPPER-GATE DRIVE TURN-ON PATH

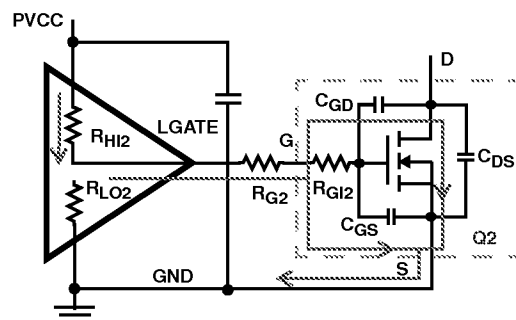


FIGURE 6. TYPICAL LOWER-GATE DRIVE TURN-ON PATH

Layout Considerations

Reducing Phase Ring

The parasitic inductances of the PCB and the power devices (both upper and lower FETs) could cause serious ringing, exceeding absolute maximum rating of the devices. The negative ringing at the edges of the PHASE node could add charges to the bootstrap capacitor through the internal bootstrap diode, in some cases, it could cause over stress across BOOT and PHASE pins. Therefore, user should do a careful layout and select proper MOSFETs and drivers. The D²PAK and DPAK package MOSFETs have high parasitic lead inductance, which can exacerbate this issue. FET selection plays an important role in reducing PHASE ring. If higher inductance FETs must be used, a Schottky diode is recommended across the lower MOSFET to clamp negative PHASE ring.

A good layout would help reduce the ringing on the phase and gate nodes significantly:

- Avoid uses via for decoupling components across BOOT and PHASE pins and in between VCC and GND pins. The decoupling loop should be short.
- All power traces (UGATE, PHASE, LGATE, GND, VCC) should be short and wide, and avoid using via; otherwise, use two vias for interconnection when possible.
- Keep SOURCE of upper FET and DRAIN of lower FET as close as thermally possible.
- Keep connection in between SOURCE of lower FET and power ground wide and short.
- Input capacitors should be placed as close to the DRAIN of upper FET and SOURCE of lower FETs as thermally possible.

NOTE: Refer to Intersil Tech Brief TB447 for more information.

Thermal Management

For maximum thermal performance in high current, high switching frequency applications, connecting the thermal pad of the QFN part to the power ground with multiple vias is recommended. This heat spreading allows the part to achieve its full thermal potential.

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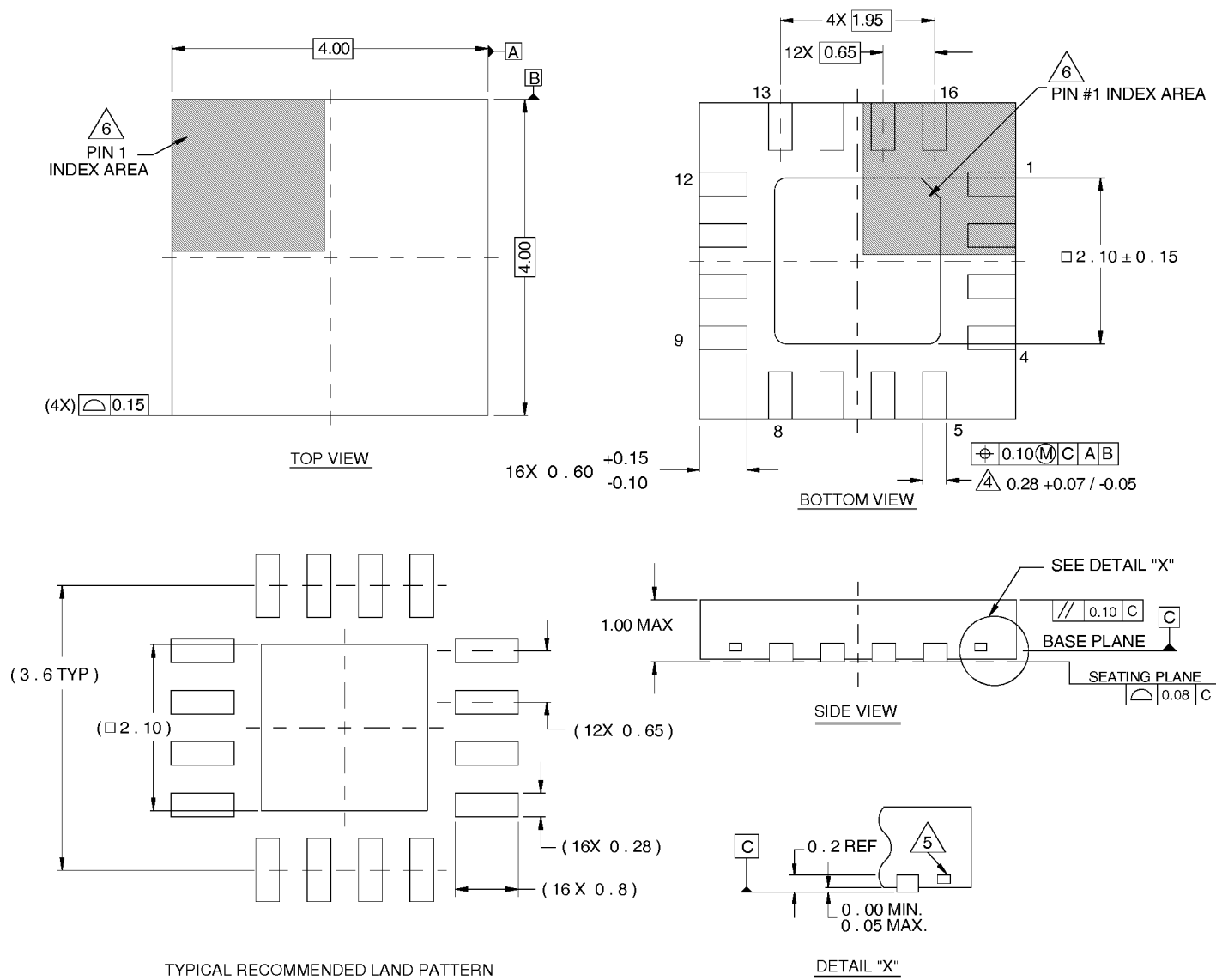
For information regarding Intersil Corporation and its products, see www.intersil.com

Package Outline Drawing

L16.4x4

16 LEAD QUAD FLAT NO-LEAD PLASTIC PACKAGE

Rev 6, 02/08



NOTES:

- Dimensions are in millimeters.
Dimensions in () for Reference Only.
- Dimensioning and tolerancing conform to AMSE Y14.5m-1994.
- Unless otherwise specified, tolerance : Decimal ± 0.05
- Dimension b applies to the metallized terminal and is measured between 0.15mm and 0.30mm from the terminal tip.
- Tiebar shown (if present) is a non-functional feature.
- The configuration of the pin #1 identifier is optional, but must be located within the zone indicated. The pin #1 identifier may be either a mold or mark feature.



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NOTICE OF ALLOWANCE AND FEE(S) DUE

26214 7590 01/17/2017
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

LEE, JYE-JUNE

ART UNIT

PAPER NUMBER

2838

DATE MAILED: 01/17/2017

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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14/682,187

04/09/2015

Patrizio Vinciarelli

00614-0166004

3198

TITLE OF INVENTION: A Fault Tolerant Power Converter having Converter Operating Cycles

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	04/17/2017

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

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Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/682,187	04/09/2015	Patrizio Vinciarelli	00614-0166004	3198

TITLE OF INVENTION: A Fault Tolerant Power Converter having Converter Operating Cycles

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nonprovisional	UNDISCOUNTED	\$960	\$0	\$0	\$960	04/17/2017

EXAMINER	ART UNIT	CLASS-SUBCLASS
LEE, JYE-JUNE	2838	363-017000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
- ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____
- (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____
- 3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
- ☐ Publication Fee (No small entity discount permitted)
- ☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (**Please first reapply any previously paid issue fee shown above**)

- ☐ A check is enclosed.
- ☐ Payment by credit card. Form PTO-2038 is attached.
- ☐ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. **Change in Entity Status** (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
- ☐ Applicant asserting small entity status. See 37 CFR 1.27
- ☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

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EXAMINER

LEE, JYE-JUNE

ART UNIT	PAPER NUMBER
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2838

DATE MAILED: 01/17/2017

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 14/682,187	Applicant(s) VINCIARELLI, PATRIZIO	
	Examiner JYE-JUNE LEE	Art Unit 2838	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to the amendment filed on 11/10/2016.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 61-98. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying Indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|--|--|
| 1. ☐ Notice of References Cited (PTO-892) | 5. ☒ Examiner's Amendment/Comment |
| 2. ☒ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date _____ | 6. ☒ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | 7. ☐ Other _____. |
| 4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date _____. | |

/ADOLF BERHANE/
Primary Examiner, Art Unit 2838

Examiner's Amendment

1. An examiner's amendment to the record appears below. Should the changes and/or additions be unacceptable to applicant, an amendment may be filed as provided by 37 CFR 1.312. To ensure consideration of such an amendment, it MUST be submitted no later than the payment of the issue fee.

Authorization for this examiner's amendment was given via telephone with Applicants representative Rex Huang on 9/26/2016.

2. The application has been amended as follows:

Regarding **claims 72 and 81**, the claims are amended so the claims would read as follows.

72. A method of converting power received from an input source at an input voltage via a power transformer for supplying a load at a DC output voltage, comprising

providing a driver circuit having an input for receiving power from the source and a driver output,

providing an output circuit for deployment near the load, the output circuit having an AC input and an output for supplying the DC output voltage to the load;

providing the power transformer having a first winding for electrical connection to the driver output and a second winding for electrical connection to the AC input of the output circuit;

forming a resonant circuit including a winding of the power transformer, the resonant circuit being characterized by a characteristic resonant period;

providing a power bus between the driver circuit and the power transformer;
operating the driver circuit to drive the power transformer in a series of converter
operating cycles, each converter operating cycle characterized by

(a) two power transfer phases of essentially equal duration each having a
duration less than the characteristic resonant period, during which one or more switches
in the driver circuit are ON and power is transferred from the input to the output via the
power transformer and

(b) a clamp phase during which a winding of the power transformer is clamped,
the clamp phase being characterized by (i) essentially zero voltage across the clamped
winding and (ii) an average value of current flowing in the clamped winding, the average
value of current having a polarity, wherein the average is taken over the duration of the
clamp phase, and the polarity is the same as a polarity of an average value of current
flowing in the clamped winding during the power transfer phase immediately preceding
the clamp phase, and (iii) essentially zero current is conducted between the secondary
winding and the output; and

deploying the driver circuit as a self-contained unit at a first location separated
from the output circuit.

81. An apparatus comprising:

a first package comprising a driver circuit having an input for receiving power
from an input source and a driver output;

an output circuit having an AC input and an output for supplying a DC output

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voltage to a load, the output circuit being disposed outside of the first package;

a power transformer having a first winding for electrical connection to the driver output of the driver circuit and a second winding for electrical connection to the AC input of the output circuit;

a resonant circuit formed in part by one of the windings of the power transformer, the resonant circuit having a characteristic resonant period; and

a power bus for electrical connection to the driver circuit and the power transformer;

wherein the driver circuit is configured to drive the power transformer in a series of converter operating cycles, each converter operating cycle comprising

(a) two power transfer phases of essentially equal duration each having a duration less than the characteristic resonant period, during which one or more switches in the driver circuit are ON and power is transferred from the input of the driver circuit to the output of the output circuit via the power transformer and

(b) a clamp phase during which a winding of the power transformer is clamped, the clamp phase being characterized by (i) essentially zero voltage across the clamped winding and (ii) an average value of current flowing in the clamped winding, the average value of current having a polarity, wherein the average is taken over the duration of the clamp phase, and the polarity is the same as a polarity of an average value of current flowing in the clamped winding during the power transfer phase immediately preceding the clamp phase, and (iii) essentially zero current is conducted between the secondary winding and the output.

Reasons For Allowance

3. **Claims 61-98** are allowed.

The following is an examiner's statement of reasons for allowance:

Regarding **claim 61**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which "...operating the primary switches in a series of converter operating cycles, each converter operating cycle characterized by (a) two power transfer phases of essentially equal duration during which (i) one or more of the primary switches are ON, (ii) a primary winding of the transformer is driven with power received from the input, (iii) a magnetizing current flows through the primary winding and increases in magnitude, and (iv) power is transferred between the input and the output via the transformer; and (b) a clamp phase during which a winding of the power transformer is clamped, the clamp phase being characterized by (i) essentially zero voltage across the clamped winding and (ii) an average value of current flowing in the clamped winding, the average value of current having a polarity, wherein the average is taken over the duration of the clamp phase, and the polarity is the same as a polarity of an average value of current flowing in the clamped winding during the power transfer phase immediately preceding the clamp phase, and (iii) essentially zero current is conducted between the secondary winding and the output."

Regarding **claim 72**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which "...providing an output circuit for deployment near the load, the output circuit having an AC input and an output for supplying the DC

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output voltage to the load; providing the power transformer having a first winding for electrical connection to the driver output and a second winding for electrical connection to the AC input of the output circuit; forming a resonant circuit including a winding of the power transformer, the resonant circuit being characterized by a characteristic resonant period; providing a power bus between the driver circuit and the power transformer; operating the driver circuit to drive the power transformer in a series of converter operating cycles, each converter operating cycle characterized by (a) two power transfer phases of essentially equal duration each having a duration less than the characteristic resonant period, during which one or more switches in the driver circuit are ON and power is transferred from the input to the output via the power transformer and (b) a clamp phase during which a winding of the power transformer is clamped, the clamp phase being characterized by (i) essentially zero voltage across the clamped winding and (ii) an average value of current flowing in the clamped winding, the average value of current having a polarity, wherein the average is taken over the duration of the clamp phase, and the polarity is the same as a polarity of an average value of current flowing in the clamped winding during the power transfer phase immediately preceding the clamp phase, and (iii) essentially zero current is conducted between the secondary winding and the output; and deploying the driver circuit as a self-contained unit at a first location separated from the output circuit.”.

Regarding **claim 81**, none of the cited prior art alone or in combination disclose or teach the claimed inventions in which “...a first package comprising a driver circuit having an input for receiving power from an input source and a driver output; an output

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circuit having an AC input and an output for supplying a DC output voltage to a load, the output circuit being disposed outside of the first package; a power transformer having a first winding for electrical connection to the driver output of the driver circuit and a second winding for electrical connection to the AC input of the output circuit; a resonant circuit formed in part by one of the windings of the power transformer, the resonant circuit having a characteristic resonant period; and a power bus for electrical connection to the driver circuit and the power transformer; wherein the driver circuit is configured to drive the power transformer in a series of converter operating cycles, each converter operating cycle comprising (a) two power transfer phases of essentially equal duration each having a duration less than the characteristic resonant period, during which one or more switches in the driver circuit are ON and power is transferred from the input of the driver circuit to the output of the output circuit via the power transformer and (b) a clamp phase during which a winding of the power transformer is clamped, the clamp phase being characterized by (i) essentially zero voltage across the clamped winding and (ii) an average value of current flowing in the clamped winding, the average value of current having a polarity, wherein the average is taken over the duration of the clamp phase, and the polarity is the same as a polarity of an average value of current flowing in the clamped winding during the power transfer phase immediately preceding the clamp phase, and (iii) essentially zero current is conducted between the secondary winding and the output.”

The aforementioned limitations in combination with all remaining limitations of the

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respective claims are believed to render the aforementioned indicated claim and any dependent claims thereof patentable over the art of record.

4. Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to JYE-JUNE LEE whose telephone number is (571)270-7726. The examiner can normally be reached on Monday to Thursday 9:00 AM to 7:30 PM EST.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Monica Lewis can be reached on 571-272-1838. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Art Unit: 2838

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/JYE-JUNE LEE/

Examiner, Art Unit 2838

/ADOLF BERHANE/

Primary Examiner, Art Unit 2838

Sheet 1 of 1

Substitute Disclosure Form Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	U.S. Department of Commerce Patent and Trademark Office		Attorney Docket No. 00614-0166004	Application No. 14/682,187
	Applicant VLT, Inc.			
	Filing Date April 9, 2015		Group Art Unit	

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1	6,031,743	02/29/2000	Carpenter et al.			
	2	6,035,261	03/07/2000	Carpenter et al.			
	3	6,275,958	08/14/2001	Carpenter et al.			
	4	6,466,458	10/15/2002	Zhang et al.			

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
	5						Yes	No

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	6	

Examiner Signature /JYE-JUNE LEE/	Date Considered 12/27/2016
---	--------------------------------------

EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

Substitute Disclosure Form

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /J.L/

Sheet 1 of 1

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0166004	Application No. 14/682,187
	Applicant VLT, Inc.	
	Filing Date April 9, 2015	Group Art Unit

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1.						

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No
	2.							

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	3.	Office Action; U.S. Appln. No. 13/044,063; Examiner Nguyen; May 23, 2014
	4.	Reply to Action of May 23, 2014; U.S. 13/044,063; Examiner Nguyen; Oct. 24, 2014
	5.	Notice of Allowance; U.S. Appln. No. 13/044,063; Examiner Nguyen; Feb. 20, 2015
	6.	Amendment after Allowance Pursuant to 37 C.F.R. §1.312; U.S. Appln. No. 13/044,063; Examiner Nguyen; Mar. 3, 2015
	7.	Supplemental Notice of Allowability; U.S. Appln. No. 13/044,063; Examiner Nguyen; Apr. 20, 2015
	8.	Notice of Allowance; U.S. Appln. No. 13/044,063; Examiner Nguyen; Jun. 5, 2015

Examiner Signature /JYE-JUNE LEE/	Date Considered 12/27/2016
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

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Sheet 1 of 1

Substitute Disclosure Form Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0166004	Application No.
	Applicant VLT, Inc.		
	Filing Date	Group Art Unit	

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1	6,788,033	09/07/2004	Vinciarelli			
	2	6,911,848	06/28/2005	Vinciarelli			
	3	6,930,893	08/16/2005	Vinciarelli			
	4	6,934,166	08/23/2005	Vinciarelli			
	5	6,969,909	11/29/2005	Briere			
	6	6,975,098	12/13/2005	Vinciarelli			
	7	6,984,965	01/10/2006	Vinciarelli			
	8	6,985,341	01/10/2006	Vinciarelli, et al.			
	9	7,145,786	12/05/2005	Vinciarelli			
	10	RE40,072	02/19/2008	Prager, et al.			
	11	7,561,446	07/14/2009	Vinciarelli			
	12	7,768,807	08/2010	Chen, et al.			

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No
	13							

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	14	AN2738 (Application Note AN2738, "L6390 half-bridge gate driver," STMicroelectronics, August 2009)
	15	ISL6210 (Datasheet ISL6210, "Dual Synchronous Rectified MOSFET Drivers," Intersil Americas Inc., December 2008)

Examiner Signature /JYE-JUNE LEE/	Date Considered 12/27/2016
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

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Sheet 1 of 1

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0166004	Application No. 14/682,187
	Applicant VLT, Inc.	
	Filing Date April 9, 2015	Group Art Unit 2838

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1.	2013/0069608	03-2013	Gakhar et al.			
	2.	6,300,749	10-2001	Castelli et al.			
	3.	6,603,292	08-2003	Schouten et al.			
	4.	6,700,361	03-2004	Gregorius, Peter			
	5.	7,521,909	04-2009	Dow et al.			
	6.	8,232,783	07-2012	Yanagawa, Kenji			

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document

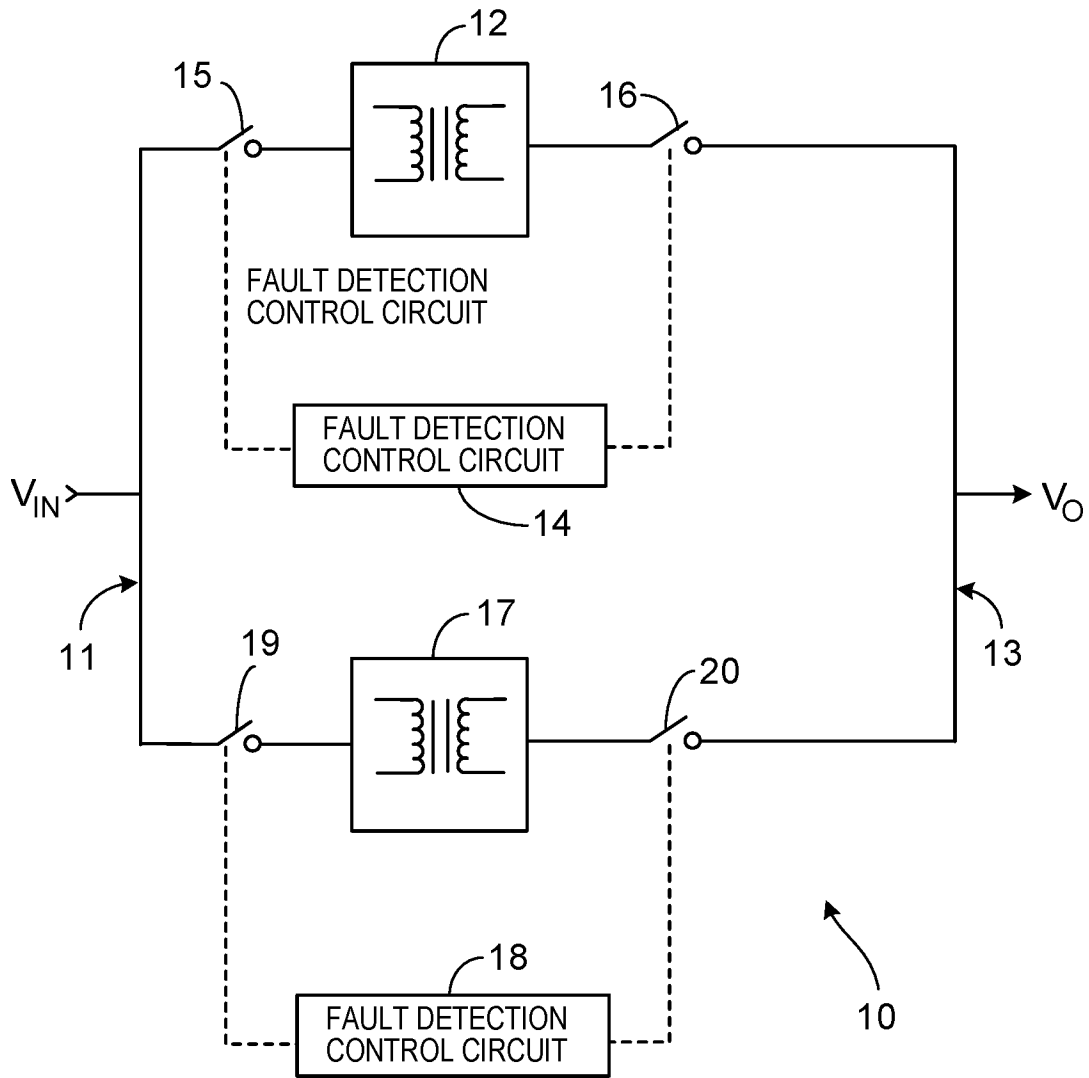
Examiner Signature /JYE-JUNE LEE/	Date Considered 01/09/2017
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

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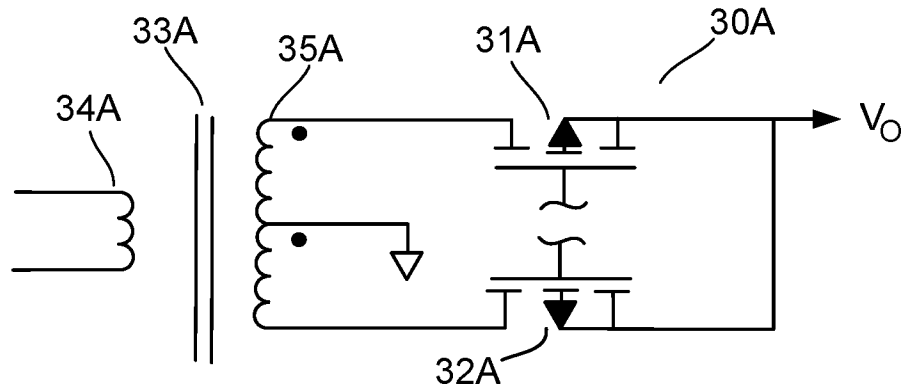
OK TO ENTER: /J.L./

12/27/2016

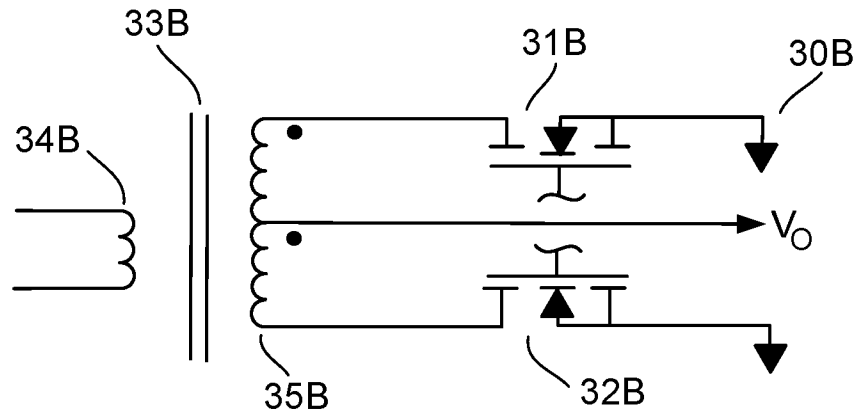


(PRIOR ART)

FIG. 1



(PRIOR ART)

FIG. 2A

(PRIOR ART)

FIG. 2B

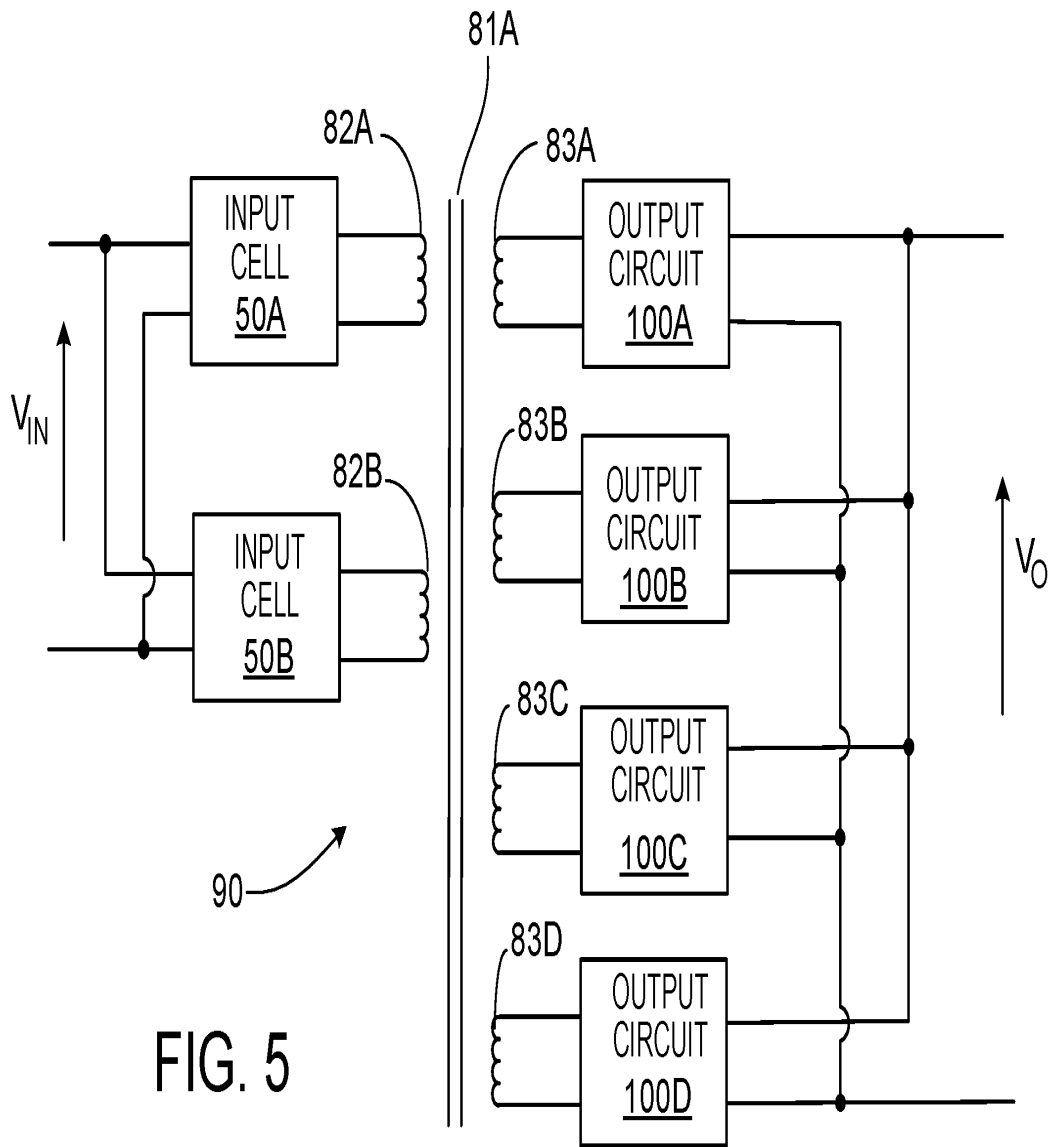


FIG. 5

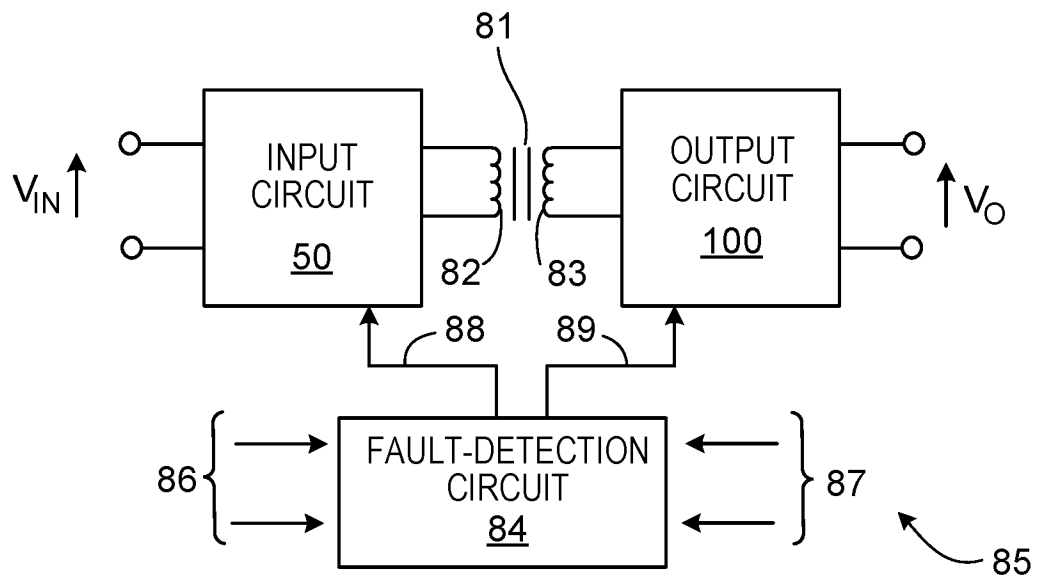
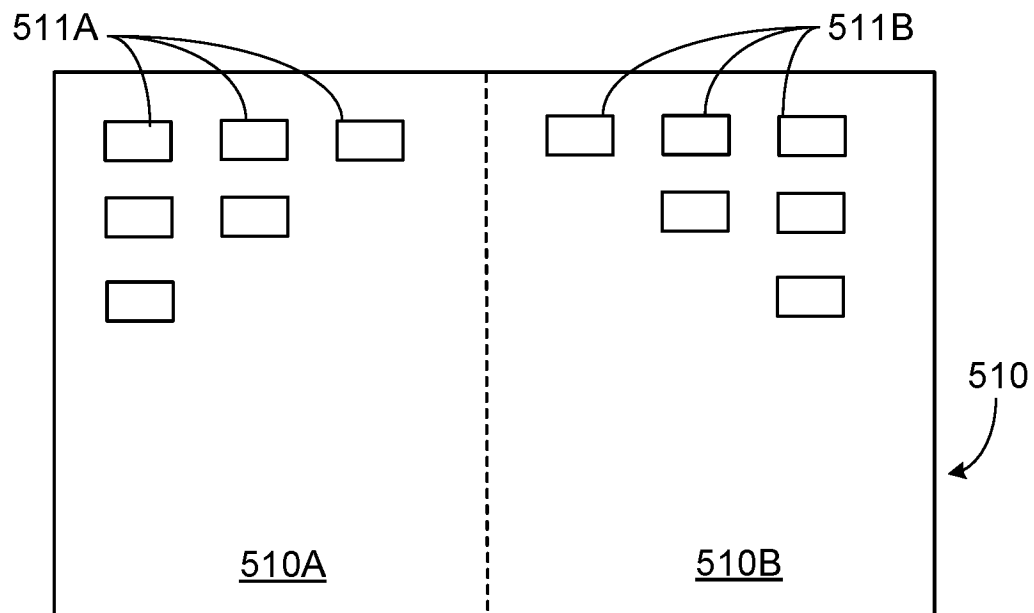


FIG. 6



(PRIOR ART)

FIG. 9

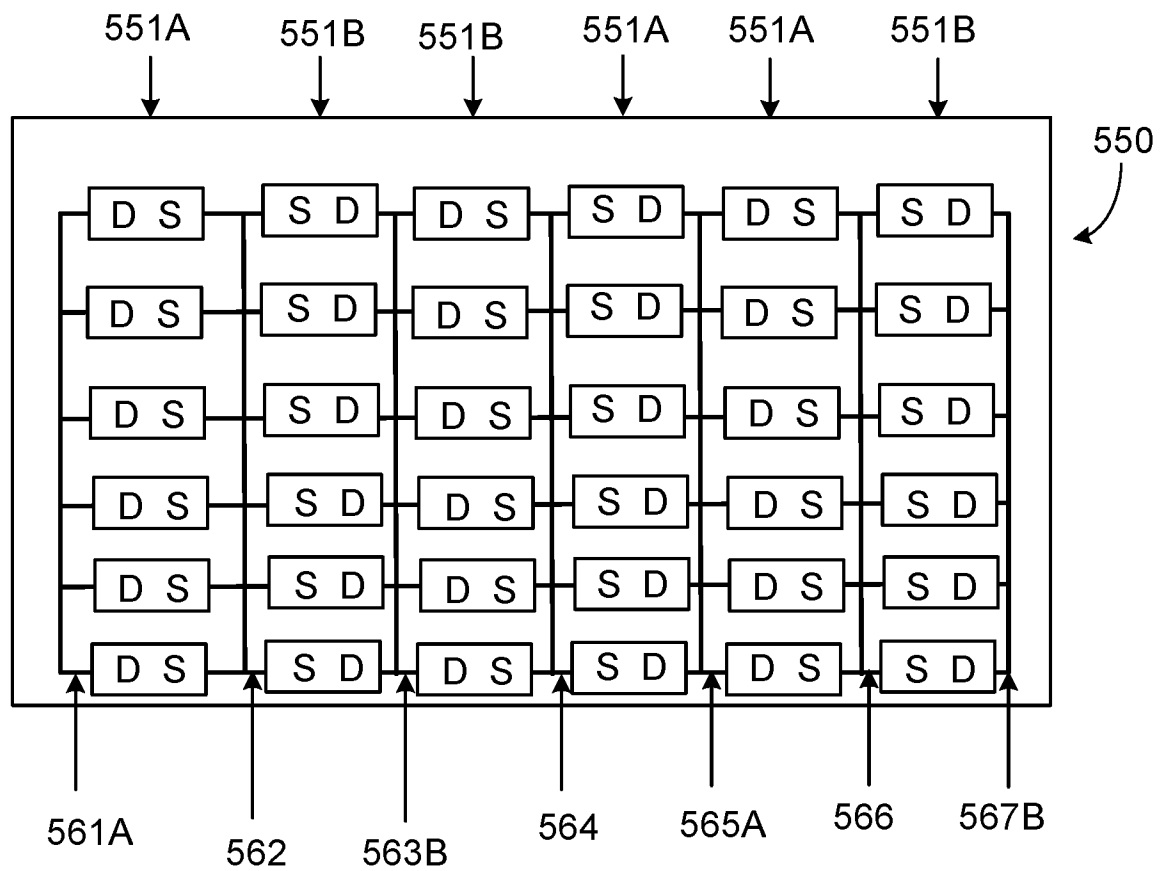


FIG. 10

Electronic Acknowledgement Receipt

EFS ID:	29949597
Application Number:	15616288
International Application Number:	
Confirmation Number:	7438
Title of Invention:	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Anastasia Renard
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0180001
Receipt Date:	01-AUG-2017
Filing Date:	07-JUN-2017
Time Stamp:	15:10:35
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Information Disclosure Statement (IDS) Form (SB08)	IDS_Transmittal.pdf	70188	no	1
			044d9f4b0c77f8b81bf2f3adf83d4e7e2d4b0e09		

Warnings:

Information:					
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2	Information Disclosure Statement (IDS) Form (SB08)	PTO1449.pdf	132125	no	2
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Warnings:					
Information:					
This is not an USPTO supplied IDS fillable form					
3	Non Patent Literature	0166001_USPTO_Amendment _312_030315.pdf	107851	no	17
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Information:					
4	Non Patent Literature	0166001_USPTO_Notice_of_All owance_022015.pdf	876112	no	15
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Information:					
5	Non Patent Literature	0166001_USPTO_Notice_of_All owance_060515.pdf	113764	no	11
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6	Non Patent Literature	0166001_USPTO_Office_Action _052314.pdf	1159461	no	24
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Information:					
7	Non Patent Literature	0166001_USPTO_Response_to _Office_Action_102414.pdf	685490	no	39
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9	Non Patent Literature	AN2738_2009_08.pdf	1998086	no	54
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10	Non Patent Literature	Burton.pdf	1494724	no	8
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Warnings:					
Information:					
12	Non Patent Literature	NOA_14682187_2017_01_17.pdf	187816	no	22
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Warnings:					
The page size in the PDF is too large. The pages should be 8.5 x 11 or A4. If this PDF is submitted, the pages will be resized upon entry into the Image File Wrapper and may affect subsequent processing					
Information:					
			Total Files Size (in bytes):	7367076	
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	2842
Serial No. :	15/616,288	Examiner :	Unknown
Filed :	June 7, 2017	Conf. No. :	7438
Title :	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS		

MAIL STOP AMENDMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Please consider the references listed on the enclosed PTO-1449 Form. Cited U.S. patents will be provided on request.

This statement is being filed before the receipt of a first action on the merits.

Apply any necessary charges or credits to deposit account 06-1050, referencing the above attorney docket number.

Please contact the undersigned if there are any questions regarding this statement.

Respectfully submitted,

Date: February 27, 2018_____

/Rex I. Huang/_____

Rex I. Huang
Reg. No. 57,661

Customer Number 26214
Fish & Richardson P.C.
Telephone: (617) 542-5070
Facsimile: (877) 769-7945

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0180001	Application No. 15/616,288
	Applicant VLT, Inc.	
	Filing Date June 7, 2017	Group Art Unit 2842

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1.	8,582,333	11/12/2013	Oraw et al.			
	2.	9,660,537	05/23/2017	Vinciarelli			

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document

Examiner Signature	Date Considered
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

Electronic Acknowledgement Receipt

EFS ID:	31919017
Application Number:	15616288
International Application Number:	
Confirmation Number:	7438
Title of Invention:	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Michelle Leidall
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0180001
Receipt Date:	28-FEB-2018
Filing Date:	07-JUN-2017
Time Stamp:	15:34:53
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Information Disclosure Statement (IDS) Form (SB08)	00614-0180001_Transmittal.pdf	69913	no	1
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Warnings:

Information:					
This is not an USPTO supplied IDS fillable form					
2	Information Disclosure Statement (IDS) Form (SB08)	00614-0180001_1449.pdf	120007	no	1
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Warnings:					
Information:					
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Total Files Size (in bytes):				189920	
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Apply any necessary charges or credits to deposit account 06-1050, referencing the above attorney docket number.

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Respectfully submitted,

Date: May 24, 2018_____

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		Filing Date June 7, 2017	Group Art Unit 2842
(37 CFR §1.98(b))			

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Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	1.	Cao et al., "A Family of Zero Current Switching Switched-Capacitor DC-DC converters," Applied Power Electronics Conference and Exposition (APEC), 2010 Twenty-Fifth Annual IEEE, Mar. 18, 2010
	2.	Cao et al., "Zero-Current-Switching Multilevel Modular Switched-Capacitor DC-DC Converter" IEEE Transactions on Industry Applications, 46(6):2536-2544, Sep. 2010
	3.	Oraw et al., "Load Adaptive, High Efficiency, Switched Capacitor Intermediate Bus Converter," Telecommunications Energy Conference, IEEE 2007, pp. 628-635 Sept. 30 – Oct. 4, 2007
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A Family of Zero Current Switching Switched-Capacitor DC-DC Converters

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Abstract— This paper presents a new zero current switching (ZCS) technique for a family of switched-capacitor dc-dc converters. Compared to the traditional ZCS switched-capacitor dc-dc converters by inserting a magnetic core in the circuit, these new ZCS switched-capacitor dc-dc converters employ the stray inductance present in the circuit as the resonant inductor and provide soft switching for the devices. These ZCS switched-capacitor dc-dc converters do not utilize any additional components to minimize switching loss and reduce the current and voltage spike, thus leading to high efficiency and reliable benefit over traditional switched-capacitor dc-dc converters. Moreover, the bulky capacitor bank existing in traditional switched-capacitor circuits for high power high current application to achieve high efficiency was reduced significantly. Small size, low capacitance, low ESR, high current rating and high temperature rating ceramic capacitors can be employed. Therefore, by using proposed ZCS technique, small size, high power density, high efficiency, high temperature rating, and high current rating switched-capacitor dc-dc converter could be built. Simulation and experimental results are given to demonstrate the validity and features of the soft switching switched-capacitor dc-dc converters.

I. INTRODUCTION

With the technological development of silicon carbide (SiC) and ceramic materials, very high temperature (250 °C) SiC switching devices and ceramic capacitors will be available. But the magnetic cores become dysfunctional at very high temperature. Therefore, magnetic-less switched-capacitor dc-dc converters operating at very high temperatures become possible and attractive with the adoption of natural air cooling by eliminating bulky heat sinks and magnetic cores. There are three main basic structures of traditional switched-capacitor dc-dc converters that is most popular, Marx generator type switched-capacitor dc-dc voltage multiplier shown in Fig. 1 [1-3], charge pump type multi-level modular switched-capacitor dc-dc converter shown in Fig. 2 [4, 5], and generalized multi-level type switched-capacitor dc-dc converter shown in Fig. 3 [6-9]. The derivation circuits of generalized multi-level type switched-capacitor dc-dc converter are also popular in automotive applications [10-14].

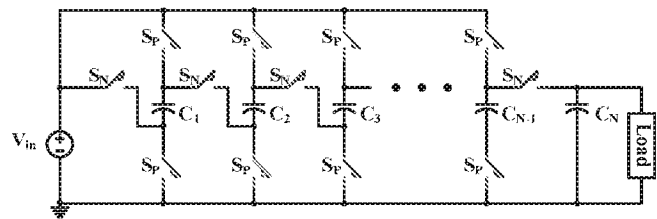


Fig. 1. Switched-capacitor dc-dc voltage multiplier.

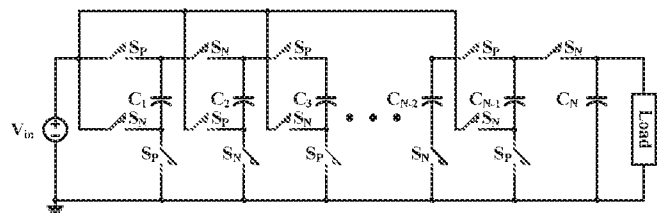


Fig. 2. Multilevel modular switched-capacitor dc-dc converter.

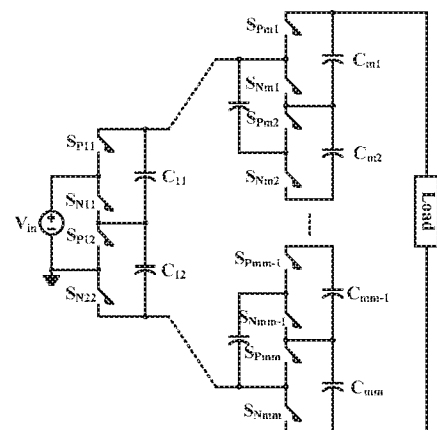


Fig. 3. Generalized multi-level switched-capacitor dc-dc converter.

However, there are many common drawbacks in traditional switched-capacitor dc-dc converters for high power and high current automotive application. A huge capacitor bank with high capacitance has to be utilized in order to

reduce the voltage ripple of the capacitors and achieve high efficiency, which will undoubtedly increase the size of the converter [7, 10, 13, 15]. Besides, with the increase of current rating, huge turn off current leads to unneglectable switching loss of the device. High voltage overshoot is also caused by large turn off current. Moreover, EMI problems caused by the di/dt and dv/dt generated during the switching transient are undesirable in automotive applications.

In order to reduce the switching loss, voltage spikes, and EMI, several ZCS methods have been proposed by inserting an inductor in series with the capacitor [16-21]. But these methods require a relatively big resonant inductor (larger than 1 μ H) which is not viable to be achieved by the circuit stray inductance as claimed. By inserting a magnetic core in the switched-capacitor circuit to achieve ZCS is a contradiction by itself. The switched-capacitor circuit with a magnetic core is not a switched-capacitor circuit anymore. Many good features of switched-capacitor will lose including good integration capability, small size and high temperature operation. Also, to utilize the parasitic inductance in the circuit, the switching frequency has to be pushed to tens of megahertz by using these methods, which is not technically feasible [22-24].

This paper presents a new ZCS strategy for a family of switched-capacitor dc-dc converters that is able to overcome the aforementioned drawbacks of traditional switched-capacitor dc-dc converters without inserting big magnetic cores or increasing switching frequency to megahertz level. This new strategy can achieve the ZCS for all the switches without losing any the good features of switched-capacitor circuit compared to the traditional ZCS strategy. This ZCS strategy employs the distributed parasitic inductance in the circuit resonating with the capacitors to provide zero current transition to the switching devices. So, the switching loss is minimized, voltage spike is reduced and EMI is limited. In case the stray inductance is not sufficiently large or equalized as expected, small air core inductors (less than 100n) or layout PCB wires can be used to meet the requirement of the stray inductance. By considering the influence of stray inductance, the power loss is only related to the conduction loss of capacitors and switching devices without considering gate drive and control loss, which is different from the traditional switched-capacitor circuits. Therefore, bulky capacitor banks with high capacitance to reduce voltage ripple and achieve high efficiency are no longer needed, while small size MLCC capacitors with low ESR and low capacitance can be adopted to increase power density and reduce the converter size. No extra components are added in the presented soft switching strategy, thus making it reliable and low in cost. A 160 W ZCS switched-capacitor dc-dc multiplier is analyzed and simulated to confirm the proposed ZCS strategy. A 12 V input 480 W ZCS voltage doubler prototype is also designed and built to confirm the operation, simulation and experiment results are provided.

II. PROPOSED ZCS SWITCHED-CAPACITOR DC-DC CONVERTER FAMILY

Fig. 4, Fig. 5, and Fig. 6 show the proposed ZCS switched-capacitor family utilizing the distributed parasitic inductance existing in the circuit. Fig. 4 shows the N-level ZCS switched-

capacitor dc-dc voltage multiplier. Fig. 5 shows the N-level ZCS multi-level modular switched-capacitor dc-dc converter (ZCS-MMSCC) which has already been discussed in detail in [25]. Fig. 6 shows the N-level ZCS generalized multi-level switched capacitor dc-dc converter. The parasitic inductance mainly includes the stray inductance due to the circuit layout, MOSFETs package parasitic inductance and the capacitor ESL. The stray inductance caused by circuit layout is usually the dominant part. By designing the circuit layout properly, the inductance needed for resonant can be achieved. In case the circuit layout is not well designed or equalized as expected, an air core inductor could be inserted to achieve the prerequisite distributed stray inductance. Because resonant inductors are distributed in the circuit, only small resonant inductance is needed for resonant. Therefore, ZCS of all the switching devices can be achieved by the resonant of the capacitor and the distributed stray inductance in the circuit.

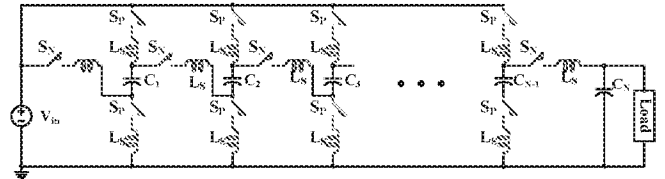


Fig. 4. ZCS switched-capacitor dc-dc voltage multiplier.

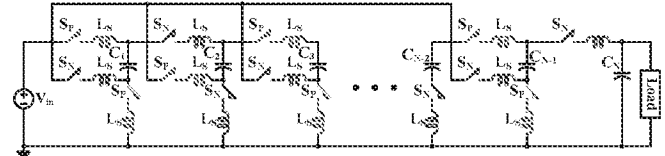


Fig. 5. ZCS multilevel modular switched-capacitor dc-dc converter.

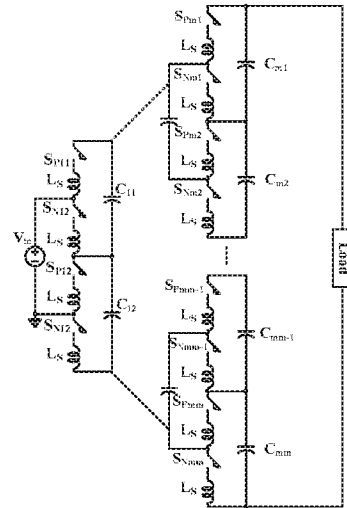


Fig. 6. ZCS generalized multi-level switched-capacitor dc-dc converter.

III. OPERATION PRINCIPLE

A. ZCS Switched-Capacitor DC-DC Voltage Multiplier

Fig. 7 shows the four-level ZCS switched-capacitor dc-dc voltage multiplier as an example. Similar to the four-level ZCS-MMSCC, this circuit also works as a four times step-up dc-dc converter. V_{in} represents the ideal input voltage source.

$L_{S1} - L_{S7}$ represents the equivalent stray inductance present in the circuit. L_{S2} , L_{S4} and L_{S6} are the equivalent stray inductance when the capacitor is charged in parallel. L_{S1} , L_{S3} , L_{S5} and L_{S7} are the equivalent stray inductance between each capacitor when the capacitor discharges in series. S_P and S_N are the same switching devices controlled complementary at 50% duty cycle. C_1 to C_3 are the capacitors with the average voltage of input voltage, while C_4 has the average voltage of four times of input voltage. L_S does not have to be in the position drawn in Fig. 7, it could be distributed anywhere in series with switches or capacitors in the circuit. Similar to the ZCS-MMSCC, the total equivalent stray inductance is the sum of the connection wire parasitic inductance, capacitor parasitic inductance and the MOSFETs package parasitic inductance. Usually the connection wire parasitic inductance is the major part of the stray inductance. For the analysis convenience, only one equivalent L_S is used to represent the total stray inductance present in each parallel resonant loop. And one equivalent L_S is used between two capacitors in the series resonant loop.

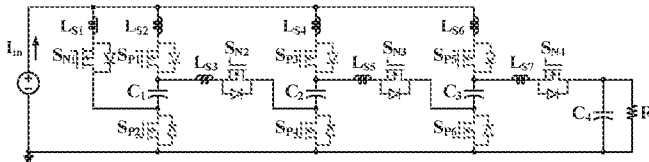


Fig. 7. Four-level ZCS switched-capacitor voltage multiplier main circuit.

Fig. 8 shows the idealized waveform of proposed ZCS switched-capacitor dc-dc voltage multiplier under steady-state conditions. V_{GS_SP} and V_{GS_SN} are the gate signal of switch S_P and S_N with 50% duty cycle. V_{DS_SP} and V_{DS_SN} are the drain source voltage of the switch S_P and S_N . Actually, the drain source voltage of different S_P and S_N are different in value but similar in shape due to the characteristics of voltage multiplier circuit. I_{SP} and I_{SN} are the drain source current of the switch S_P and S_N . All the switches of S_P or S_N have the same current waveforms. Assume input voltage is an ideal voltage source. By considering the stray inductance present in the circuit, when the switches are turned on, the current through the stray inductance, capacitors and the switch will begin to resonate from zero. By adjusting switching frequency to the resonant frequency, the current through switches S_P , S_N and stray inductance will decrease to zero when the switches are turned off. The half switching period is a half sinusoidal waveform. Therefore, the ZCS of all the switches is achieved in both turn on and turn off. The capacitor C_1 , C_2 and C_3 are charged in parallel in the half-period with the sinusoidal current waveform when S_P is on. And they are discharged in series with the input voltage source in another half-period also in the sinusoidal shape when S_N is on. $I_{C1,2,3}$ are the current through capacitor C_1 , C_2 , and C_3 , which is the sum of the current through the switch with the sinusoidal shape. When the capacitor is charged, the current is the positive part; when the capacitor discharges, the current is negative. The capacitor C_4 is charged when other capacitors are in series with input voltage when S_N is on. And it is discharged to the load when S_P is on. I_{C4} is the current through capacitor C_4 , which is a sinusoid waveform when S_N is on. When S_P is on, the current through C_4 is the negative dc load current with small ripple.

$V_{C1,2,3}$ is the voltage across the capacitor C_1 , C_2 and C_3 with the input voltage dc offset and a sinusoidal ripple. The voltage ripple is determined by the capacitor current and capacitance. V_{C4} is the voltage across the capacitor C_4 , with the dc offset four times input voltage and a voltage ripple also determined by the capacitor current and capacitance. The operation of this circuit can be described in two states shown in Fig. 9 and Fig. 10 with different switches turned on.

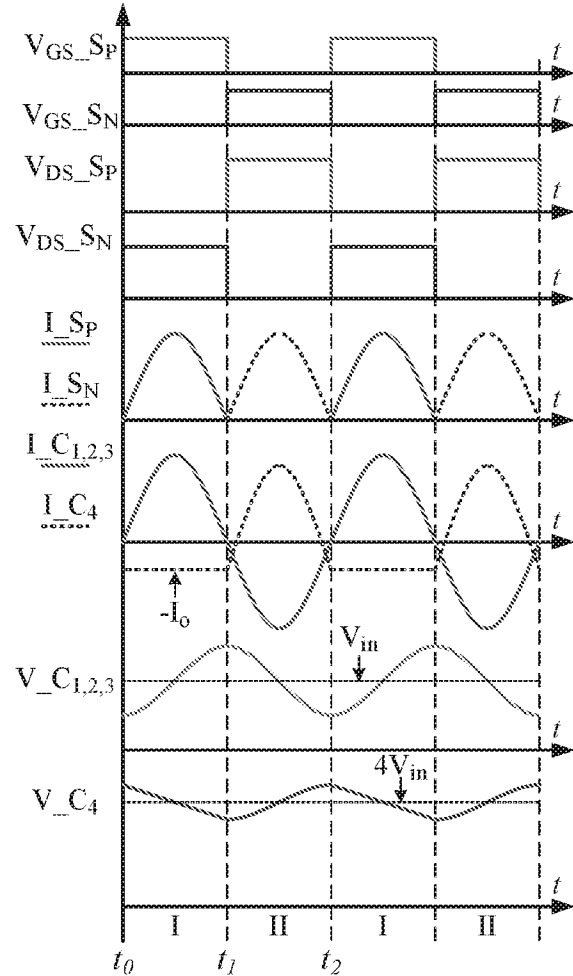


Fig. 8. Ideal waveforms of four-level ZCS voltage multiplier.

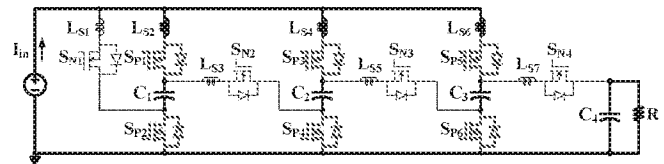


Fig. 9. Operation modes of state I when S_P is on.

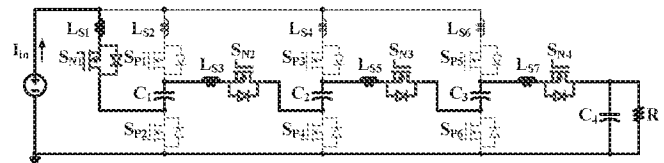


Fig. 10. Operation modes of state II when S_N is on.

1) State I [t_0, t_1]

Fig. 9 shows the state when S_P is turned on at $t = t_0$ while S_N is off. During this state C_1 , C_2 and C_3 are charged by V_{in} . Fig. 11 show the three simplified equivalent circuits of state I. Fig. 11(a) shows the situation when V_{in} , L_{S2} , S_{P1} , C_1 and S_{P2} form a resonant loop. Fig. 11(b) shows the situation when V_{in} , L_{S4} , S_{P3} , C_2 and S_{P4} form a resonant loop. Fig. 11(c) shows the situation when V_{in} , L_{S6} , S_{P5} , C_3 and S_{P6} form a resonant loop. Because of the presence of the stray inductance L_S , before the switch is turned on, the current through L_S already decreases to zero. The current through S_P will increase from zero when the switch is turned on, so S_P is turned on at zero current. For the case shown in Fig. 11(a), after L_{S2} and C_1 resonate for half cycle, the current through S_{P1} and S_{P2} falls to zero. Therefore S_{P1} and S_{P2} turn off at zero current. Similarly, for the case shown in Fig. 11(b) and Fig. 11(c) the current through S_{P3} , S_{P4} , S_{P5} and S_{P6} will realize zero current turn on and turn off too. So, ZCS is achieved on all the switches. The required stray inductance value is the same for the three circuit and is easy to be achieved by the circuit layout due to the symmetry of the circuit, assuming the capacitance are the same.

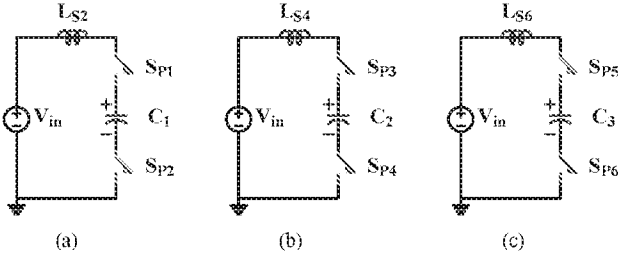


Fig. 11. Simplified equivalent circuits of state I when S_P is on.

Without loss of generality, the following assumptions have been made for the analysis: all the switches are ideal, i.e. no conduction resistance is considered; input voltage source is ideal, i.e. constant and no internal impedance; the capacitor ESR is zero. Assuming $C_1 = C_2 = C_3 = C_4$, so in order to have the same resonant frequency, $L_{S2} = L_{S4} = L_{S6}$. Assume $L_{S1} = L_{S3} = L_{S5} = L_{S7}$. The state equations of Fig. 11(a) are:

$$V_{in} = L_{S2} \frac{di_{S2}}{dt} + v_{C1} \quad (1)$$

$$i_{L_{S2}} = C_1 \frac{dv_{C1}}{dt} \quad (2)$$

The solutions are:

$$i_{S2}(t) = \frac{\pi P_o}{4V_{in}} \sin \omega_r t \quad (3)$$

$$v_{C1}(t) = V_{in} - \frac{\pi P_o}{4V_{in} C_1 \omega_r} \cos \omega_r t \quad (4)$$

Where V_{in} is the value input voltage, L_{S2} is the value of stray inductance, ω_r is the resonant frequency equals to $1/\sqrt{L_{S2} C_1}$, and P_o is the output power. After half cycle, the capacitor voltage is charged to:

$$v_{C1}(t_1) = V_{in} + \frac{\pi P_o}{4V_{in} C_1 \omega_r} \quad (5)$$

The capacitor voltage ripple is:

$$\Delta v_{C1} = \frac{\pi P_o}{4V_{in} C_1 \omega_r} \quad (6)$$

The state equations of Fig. 11(b) and Fig. 11(c) are similar to the state equations of Fig. 11(a). The voltage across C_2 and C_3 are the same as the voltage across C_1 . The voltage ripples of C_2 and C_3 are also the same as Δv_{C1} .

2) State II [t_1, t_2]

Fig. 10 shows the state when S_N is turned on at $t = t_1$ while S_P is off. During this state, C_4 is charged by the V_{in} , C_2 and C_3 in series. Assume the load is zero. Fig. 12 shows the simplified equivalent circuits of state II. It shows the situation when V_{in} , L_{S1} , S_{N1} , C_1 , L_{S3} , S_{N2} , C_2 , L_{S5} , S_{N3} , C_3 , L_{S7} , S_{N4} , and C_4 form a resonant loop. Similar as the last state, because of the presence of the L_S , the current through S_N will also increase its value from zero in a resonant manner. The zero current turn-on of S_N can be achieved. After stray inductance L_{S1} , L_{S3} , L_{S5} and L_{S7} in series resonate with capacitor C_1 , C_2 , C_3 and C_4 in series for half cycle at $t = t_2$, S_{N1} , S_{N2} , S_{N3} and S_{N4} will have a zero current turn-off when the current through them decreases to zero. Hence, zero current switching can be achieved on these switches.

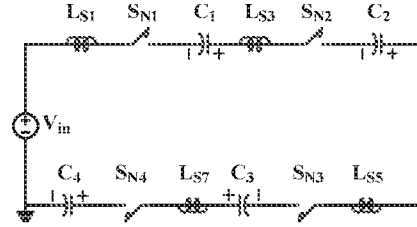


Fig. 12. Simplified equivalent circuits of state II S_N is on.

The state equations of Fig. 12 are:

$$V_{in} + v_{C1} + v_{C2} + v_{C3} = 4L_{S1} \frac{di_{L_{S1}}}{dt} + v_{C4} \quad (7)$$

$$i_{L_{S1}} = C_4 \frac{dv_{C4}}{dt} \quad (8)$$

The solutions are:

$$v_{C4}(t) = 4V_{in} - \frac{\pi P_o}{4V_{in} C_4 \omega_r} \cos \omega_r t \quad (9)$$

$$i_{L_{S1}}(t) = \frac{\pi P_o}{4V_{in}} \sin \omega_r t \quad (10)$$

After half cycle, the capacitor voltage is charged to

$$v_{C4}(t_2) = 4V_{in} + \frac{\pi P_o}{4V_{in} C_4 \omega_r} \quad (11)$$

The capacitor voltage ripple is:

$$\Delta v_{C4} = \frac{\pi P_o}{4V_{in} C_4 \omega_r} \quad (12)$$

If zero load is considered, the required stray inductance in the series resonant loop is the same as the parallel resonant loop shown in Fig. 11 which means $L_{S1} = L_{S3} = L_{S5} = L_{S7} = L_{S2} = L_{S4} = L_{S6}$. After considering the load effect with a negative dc offset in the capacitor C_4 current. The required stray inductance of series resonant loop will be a little bit smaller than the parallel resonant loop. Because the resonant frequency of a pure LC network is different with the LCR network.

B. ZCS Voltage Doubler

Fig. 13(a) shows the two-level ZCS generalized multi-level switched-capacitor dc-dc converter as an example. This circuit is called ZCS voltage doubler in short because the output voltage is twice as much as the input. V_{in} represents the ideal input voltage source. L_{S1} , L_{S2} , L_{S3} and L_{S4} can be considered as the distributed stray inductance in series with the switches in the circuit layout. S_P and S_N are the same switching devices controlled complementary at 50% duty cycle. C_1 and C_2 are the capacitors with the average voltage the same as the input voltage. Because of the symmetry of the circuit, the distributed stray inductance L_{S1} , L_{S2} , L_{S3} and L_{S4} can be replaced with an equivalent stray inductance L_S in the input side shown in Fig. 13(b) for the analysis convenience. Only one equivalent stray inductance L_S is needed to represent the total stray inductance in two resonant loop.

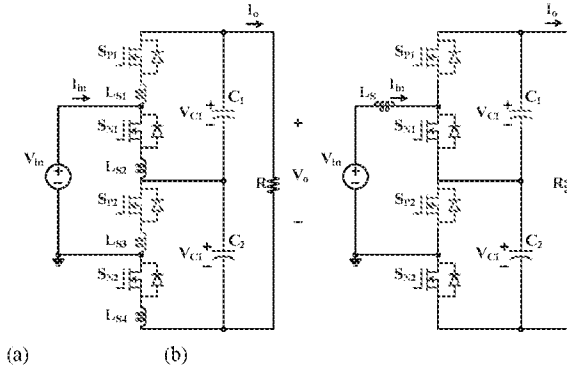


Fig. 13. ZCS voltage doubler and its topology simplification.

Fig. 14 shows the idealized waveform of proposed ZCS voltage doubler under steady-state conditions. V_{GS_SP} and V_{GS_SN} are the gate drive signal of switch S_P and S_N complementary with 50% duty cycle. V_{DS_SP1} to V_{DS_SN2} are the drain source voltage of switch S_P and S_N . I_{SP} and I_{SN} are the drain source current of the switch S_P and S_N . All the switches of S_P or S_N have the same current waveforms. Assume input voltage is an ideal voltage source. By considering the equivalent stray inductance in the input side, when the switches are turned on, the current through the stray inductance, capacitors and the switch will begin to resonate from zero. By adjusting switching frequency to the resonant frequency, the current through switches S_P , S_N and stray inductance will decrease to zero when the switches are turned off, which is the half period of the sinusoidal waveform. Therefore, the ZCS of all the switches is achieved in both turn

on and turn off. I_{C1} and I_{C2} are the current through capacitor C_1 and C_2 . The capacitor C_1 and C_2 are charged in each half-period with the sinusoidal current waveform when S_P or S_N is on. And they are always discharged in series with load current. So, the current through the capacitor C_1 , and C_2 , is the current through the switch subtracts the output current. V_{C1} and V_{C2} are the voltage across the capacitor C_1 and C_2 . V_o is the output voltage with the voltage ripple half of the voltage ripple of the capacitor. Because of the 180 degree phase shift of the capacitor voltage ripple, the capacitor voltage ripple will cancel out with each other. The operation of the circuit can be described in two states as shown in Fig. 15(a) and Fig. 15(b) with different switches turned on.

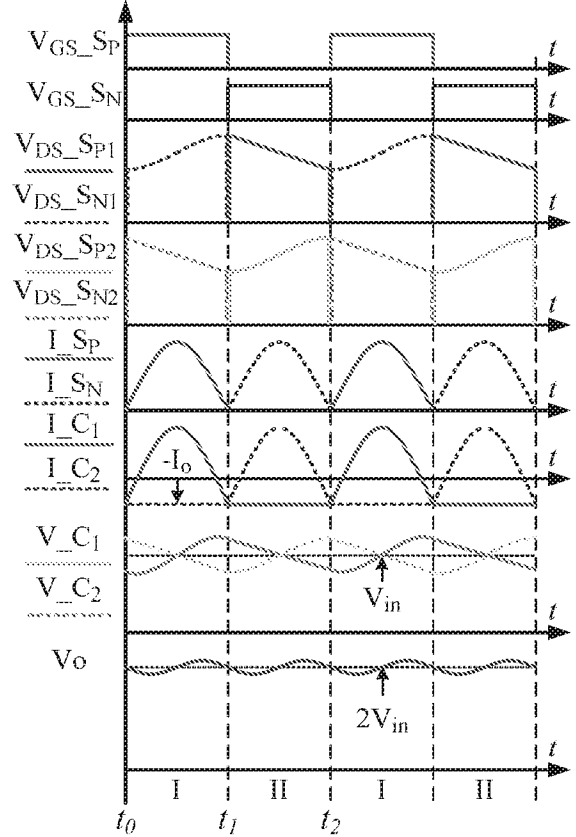


Fig. 14. Ideal waveforms of ZCS voltage doubler.

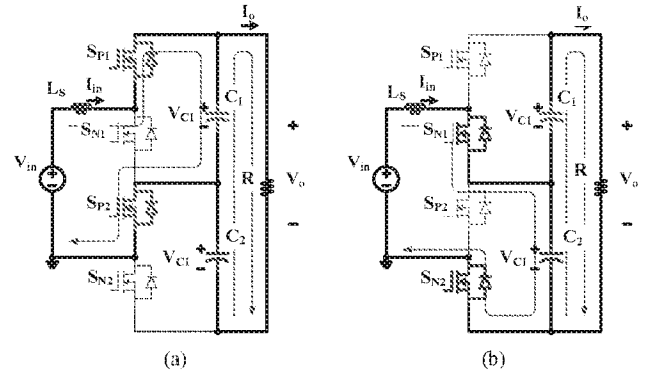


Fig. 15. (a) Operation modes of state I when S_P is on. (b) Operation modes of state II when S_N is on.

IV. DESIGN GUIDELINES

The 480 W ZCS voltage doubler design procedure is shown here as an example. The specification of the prototype converter is $V_{in}=12$ V, $V_o=24$ V, $P_o=480$ W, $f_s=44.25$ kHz.

A. Capacitance

Capacitance value should be chosen according to the voltage ripple of the capacitor using the following equation.

$$C = \frac{\pi P_o}{2V_{in}\Delta v_C \omega_r} \quad (13)$$

Because of the special structure of the circuit, the voltage ripple across all the capacitors are the same. And the value of the voltage ripple is determined by the output power, input voltage, capacitance and the resonant frequency. The capacitor voltage ripple should be chosen smaller than the input voltage to prevent the voltage across the capacitor from resonating to the negative region and lose the zero current switching. The output voltage ripple is about five times smaller than the capacitor voltage ripple because of the 180 degrees phase shift operation of the switches. And the capacitor voltage ripple will cancel out and the output voltage is two times bigger than one capacitor voltage. The output ripple is chosen around 8% of output voltage. So the capacitor voltage ripple is 40% of the capacitor voltage. According to the (13), the capacitance is chosen 47 μ F in the simulation.

B. Stray inductance

The circuit layout should be designed carefully, in order to make the stray inductance of each equivalent circuit equal. Assume all the stray inductance in the circuit is equalized. And the required stray inductance can be satisfied by using proper input inductance. Required stray inductance for ZCS can be determined by the resonant LCR network.

C. Switching frequency

Switching frequency should be set the same as resonant frequency in order to promise the zero current switching. Usually, the stray inductance of the circuit is already determined when the circuit layout is finished. The capacitance is also determined by the required voltage ripple. Assume the air core inductor is used for the input inductance, and the switching frequency can be determined by the input inductance value. If the input inductance is determined, the switching frequency can be twisted by monitoring the power device current when the switch is turned off. The switching frequency should be set exactly at the point when the switch current resonates to zero when the switch is turned off.

V. SIMULATION AND EXPERIMENT RESULTS

A. ZCS Voltage Multiplier

Fig. 16 shows saber® simulation waveforms of a 160 W four-level ZCS voltage multiplier, where V_p and V_n are the switch gate-source control voltage, V_{ds_Sp} and V_{ds_Sn} are the switch drain-source voltage, I_{Sp} and I_{Sn} are the switch drain-source current, $I_{C1,2,3}$ is the current through capacitor C_1 , C_2 and C_3 . I_{C4} is the current through capacitor C_4 . $V_{C1,2,3}$ is the voltage across the capacitor C_1 , C_2 and C_3 . V_{in} is the input voltage. V_o is the output voltage or the

voltage across the capacitor C_4 . The input voltage is 5 V. Switching frequency is about 70 kHz. Capacitance is 47 μ F and the value of the stray inductance $L_{S2} = L_{S4} = L_{S6} = 108$ nH, $L_{S1} = L_{S3} = L_{S5} = L_{S7} = 98$ nH. The series resonant stray inductance is smaller caused by the load effect. The simulation results is consistent with the theoretical analysis, which verifies the above analysis.

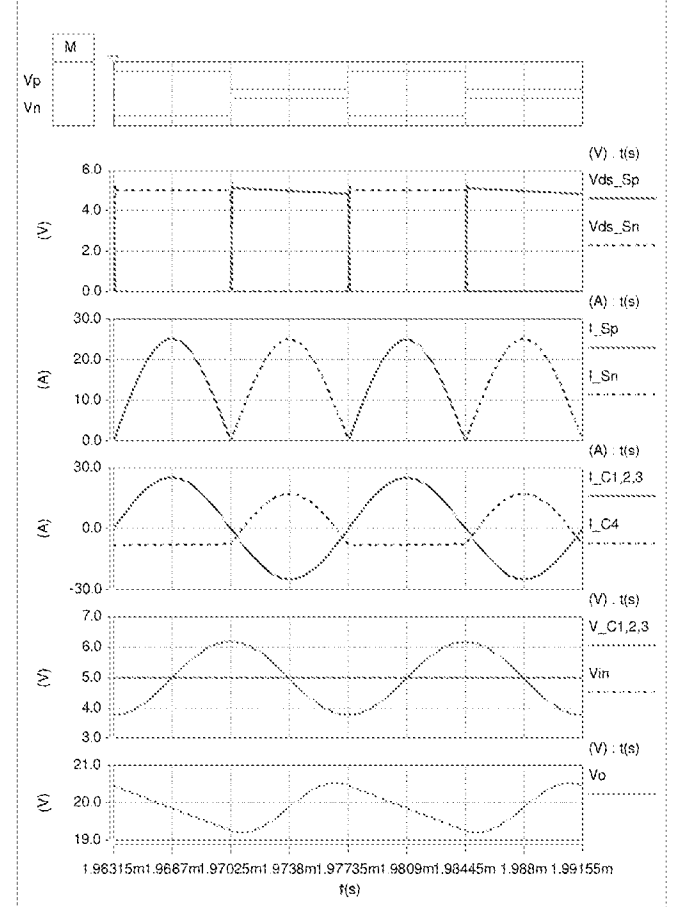


Fig. 16. Simulation results of 160 W ZCS voltage multiplier.

B. ZCS Voltage Doubler

Fig. 17 and Fig. 18 shows saber® simulation waveforms of a 480 W ZCS voltage doubler, where S_p and S_n are the switch gate-source control voltage, V_{ds_Sp1} , V_{ds_Sn1} , V_{ds_Sp2} , V_{ds_Sn2} are the switch drain-source voltage of switch S_{P1} , S_{N1} , S_{P2} and S_{N2} . I_{Sp1} and I_{Sn1} are the switch drain-source current of switches S_{P1} and S_{N1} , which is omitted here. The current through S_{P2} is the same as S_{P1} , the current through S_{N2} is the same as S_{N1} . I_{in} is the current through the input stray inductance L_S . I_{C1} and I_{C2} are the current through capacitor C_1 and C_2 . V_{C1} and V_{C2} are the voltage across the capacitor C_1 and C_2 . V_{in} is the input voltage. V_o is the output voltage. The input voltage is 12 V. Switching frequency is about 44 kHz. Capacitance is 47 μ F and stray inductance $L_S = 165$ nH. The simulation results is consistent with the theoretical analysis, which verifies the analysis.

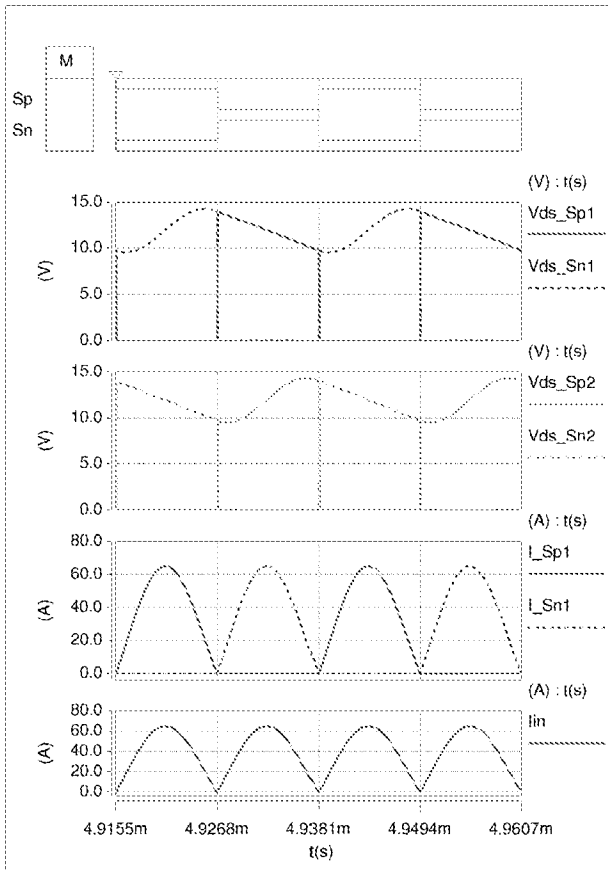


Fig. 17. Simulation results of 480 W ZCS voltage doubler.

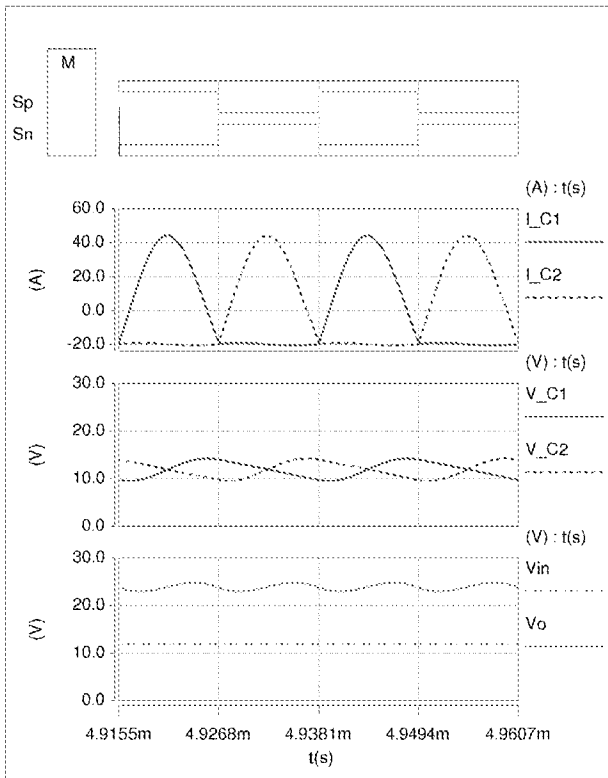


Fig. 18. Simulation results of 480 W ZCS voltage doubler.

In the 480 W ZCS voltage doubler prototype, the switching devices are two 30 V 180 A MOSFETs IPB009N03L from infineon connected in parallel. Resonant capacitor are ten 100 V 4.7 μ F MLCC capacitors C5750X7R2A475K from TDK connected in parallel. Input capacitor are twelve 16 V 470 μ F conductive polymer aluminum solid electrolytic capacitors PLG1C471MDO1 from nichicon connected in parallel. Gate drive voltage is 7 V. The switching frequency is about 40 kHz. The input stray inductance is around 200 nH.

Fig. 19 and Fig. 20 show the experiment waveforms of the 480 W ZCS voltage doubler prototype with the parts mentioned above. Fig. 19 shows the complementary gate drive signals of S_P and S_N with duty cycle about 49% due to some dead time. V_{GS_SP} and V_{GS_SN} are the complementary gate-source control voltage.

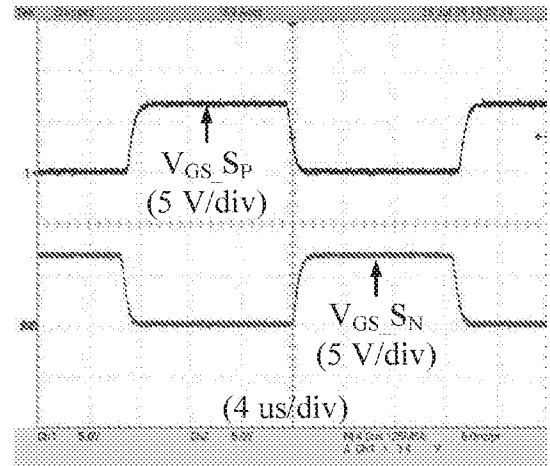


Fig. 19. Gate drive signal of ZCS voltage doubler.

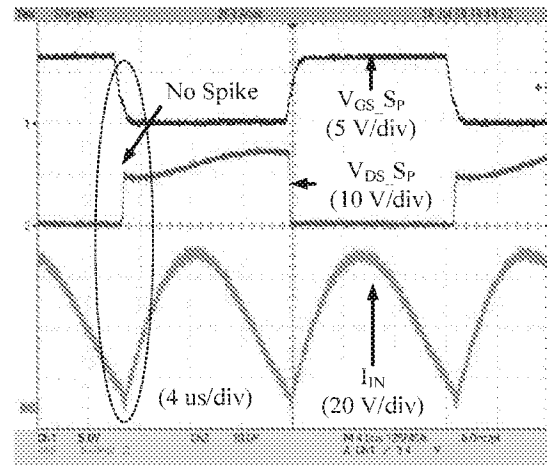


Fig. 20. Switch voltage and input current waveform of ZCS voltage doubler.

VI. CONCLUSION

In this paper, a family of zero current switching switched-capacitor dc-dc converters is proposed. By eliminating the bulky, lossy inductive component with a magnetic core, the

converter is able to operate at very high temperatures with high efficiency. By using the proposed soft-switching strategy, switching loss has been minimized and EMI has been restricted; the size of capacitors has been reduced, thus making the converter small and light. Owing to the utilization of distributed stray inductance or distributed air core inserting technique, the big inductor is avoided to achieve ZCS. Hence, the proposed ZCS switched-capacitor converter family shows great potential in high temperature, high power future automotive applications.

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Zero-Current-Switching Multilevel Modular Switched-Capacitor DC–DC Converter

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Abstract—This paper presents a quasi-resonant technique for multilevel modular switched-capacitor dc–dc converter (MMSCC) to achieve zero-current switching (ZCS) without increasing cost and sacrificing reliability. This ZCS-MMSCC employs stray inductance distributed in the circuit as the resonant inductor to resonate with the capacitor and provide low dv/dt and di/dt switching transition for the device. The ZCS-MMSCC does not utilize any additional components to achieve ZCS and meanwhile solves the current and voltage spike problem during the switching transition, thus leading to reliable and high-efficiency advantages over traditional MMSCC. Furthermore, the ZCS-MMSCC reduces the capacitance needed in the circuit to attain high efficiency. In this case, the bulky capacitor bank with high capacitance in a traditional MMSCC to reduce voltage difference and achieve high efficiency is no longer necessary. A 150-W four-level ZCS-MMSCC prototype has been built. Simulation and experimental results are given to demonstrate the validity and features of the proposed soft-switching switched-capacitor circuit.

Index Terms—Multilevel, switched capacitor, zero-current switching (ZCS).

I. INTRODUCTION

IN ORDER to obtain lightweight, small-sized, high-efficiency, high-power-density, and integrated dc–dc converters, switched-capacitor circuits have been investigated since 1970s [1], [2]. Conventional switched-capacitor dc–dc converters have some common drawbacks: 1) weak output regulation ability and structurally determinate voltage conversion ratio; 2) pulsating input current and high current spike; 3) high voltage spike across the switching device; 4) high electromagnetic interference (EMI); and 5) unidirectional power conversion ability. Many approaches have been proposed to achieve voltage regulation by using duty cycle control [2]–[4]. However, all these methods have to sacrifice efficiency to achieve the voltage regulation, which is only acceptable in low-power-conversion field. Several zero-current-switching (ZCS) techniques by inserting an inductor in the switched-capacitor circuit have been proposed to solve the voltage spike, current spike, and EMI problem [5]–[8]. However, adding a relatively big resonant

inductor into the switched-capacitor circuit to achieve ZCS is a contradiction by itself. A switched-capacitor circuit with a magnetic core is not a switched-capacitor circuit any more, and the good features such as high-temperature operation, good integration, and small size will be lost [9], [10].

Another type of switched-capacitor circuit, the flying capacitor multilevel dc–dc converter (FCMDC), has become more and more popular recently because of the potential automotive application [11]–[15]. However, there are several limitations of FCMDC, such as nonmodular structure, complex control strategy, and high-conversion-ratio unreachability. When the conversion ratio is larger than five or more than five levels have to be used, the control of FCMDC is becoming very complex. The high current stress of the switching device of FCMDC also prevents it from a high-current application.

In order to solve the problems of FCMDC in a high-voltage-conversion-ratio high-current automotive application, a multilevel modular switched-capacitor dc–dc converter (MMSCC) was proposed recently, as shown in Fig. 1 [16]. MMSCC has a modular structure, a complementary control scheme, capability of reaching high voltage gain, and reduced switch current stress. It shows good potential in automotive application [17]. Nevertheless, because MMSCC still uses hard switching, there still exist several problems limiting its power rating, such as high current spike, high voltage spike, high switching loss, and severe EMI noise. With the increasing of power rating or current rating of the MMSCC, the voltage spike and switching loss problems become more and more serious. In addition, in order to increase the total system efficiency by reducing the voltage difference between capacitors, higher capacitance electrolytic capacitors have to be used [13], [16]–[18]. The current rating of the electrolytic capacitor is small to some extent. In order to meet the high-current-ripple requirement, a huge electrolytic capacitor bank has to be used, which will increase the size of the converter significantly. Multilayer chip-type ceramic (MLCC) capacitor has much better performance than the electrolytic capacitor, such as equivalent series resistance (ESR), life time, and size in terms of current rating. However, the capacitance of MLCC capacitor is very small, so it is not feasible to use MLCC capacitor for the traditional MMSCC.

This paper presents a quasi-resonant technique for MMSCC that is able to achieve ZCS without increasing cost and sacrificing reliability. The proposed ZCS strategy employs distributed stray inductances present in each module of the circuit resonating with the capacitors to provide zero current turn-on and turn-off to the devices. Because MMSCC is particularly suitable for high-voltage-gain with high-input-current applications [16]–[21], the ZCS version also inherits all the benefits

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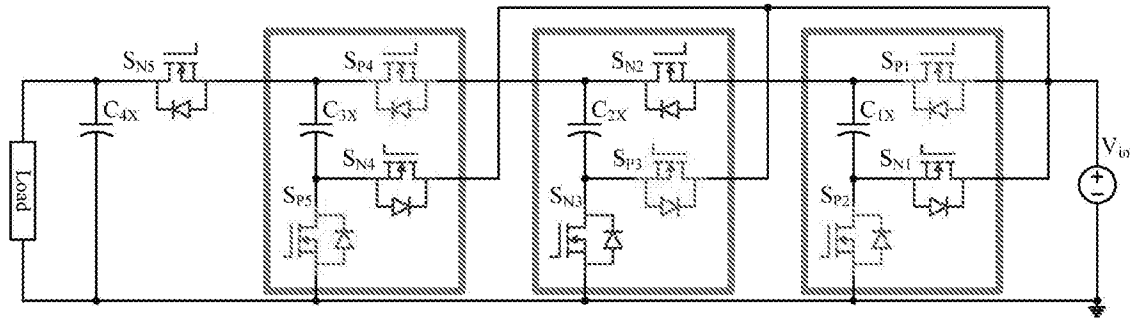


Fig. 1. Four-level MMSCC with three modular blocks.

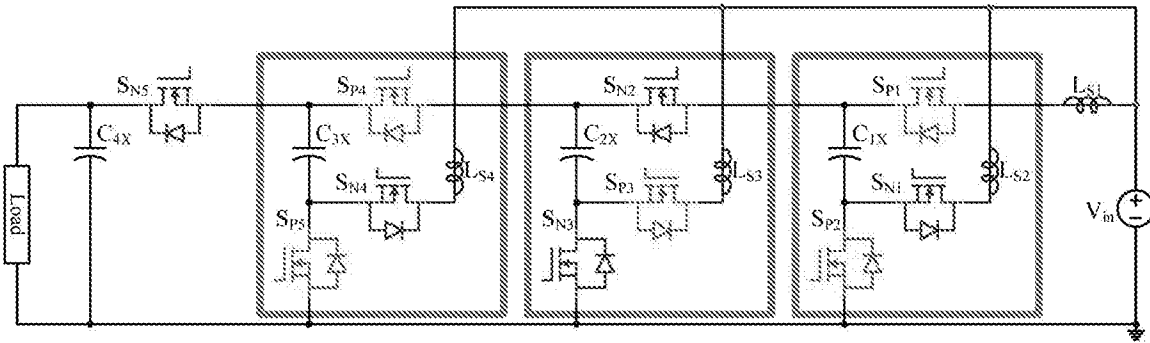


Fig. 2. Four-level ZCS-MMSCC with three modular blocks.

from it. By using the ZCS, the switching loss of the traditional MMSCC is minimized, and the current spike, voltage spike, and EMI are reduced. Aside from the benefits gained from ZCS and MMSCC, the proposed ZCS-MMSCC has other special features.

- 1) Different from other soft-switching techniques, this ZCS strategy does not use additional active devices or passive components, thus making it reliable and cost efficient.
- 2) Because of the module structure, the required resonant inductance is quite small for each module, thus making it feasible to be realized by the distributed stray inductance existing in the circuits.
- 3) The ZCS-MMSCC does not need high capacitance to achieve high efficiency like traditional MMSCC. In this case, a small-capacitance, high-current-rating, low-ESR, and small-size MLCC capacitor could be used. The total size of the converter is reduced significantly, and the power density and the converter current rating are increased correspondingly.

Hence, the proposed ZCS-MMSCC solves the aforementioned problems of traditional MMSCC and achieves ZCS without inserting big magnetic cores. For the same current rating and voltage gain, the ZCS version of MMSCC is smaller and has higher power density and higher efficiency than the traditional MMSCC. Therefore, the proposed ZCS-MMSCC is particularly suitable to the high-current high-voltage-gain automotive application. A 150-W four-level ZCS-MMSCC prototype was built to confirm the operation. Simulation and experimental results are given to demonstrate the validity and features of the proposed ZCS-MMSCC.

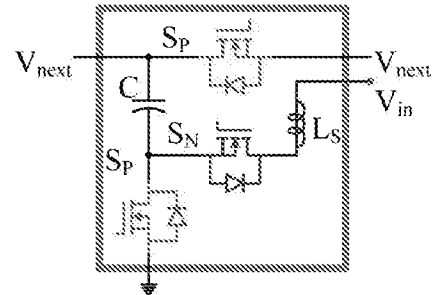


Fig. 3. Proposed ZCS-MMSCC modular block.

II. PROPOSED CIRCUIT TOPOLOGY

Fig. 2 shows the proposed four-level ZCS-MMSCC as an example. The circuit works as a four times step-up converter. Fig. 3 shows the proposed ZCS-MMSCC converter modular block. By using $n - 1$ modular blocks, a switch, and a capacitor, n -level ZCS-MMSCC with n times voltage gain can be generated. V_{in} represents the ideal input voltage source. L_S represents the equivalent stray inductance present in each module of the circuit. S_P and S_N are the same switching devices controlled complementary at 50% duty cycle. C_{1X} to C_{4X} are the capacitors with the average voltage from one time to four times of input voltage. L_S does not have to be in the position shown in Fig. 2; it could be distributed anywhere in series with switches or capacitors in the circuit shown in Fig. 4. L_S is the sum of the connection wire parasitic inductance L_{SW} , capacitor parasitic inductance L_{ESL} , and the MOSFETs package parasitic inductance L_{SP} . Usually, the connection wire parasitic inductance L_{SW} is the major part of the stray inductance, because all the stray inductances are in series when the current flows through. In each simplified equivalent circuit

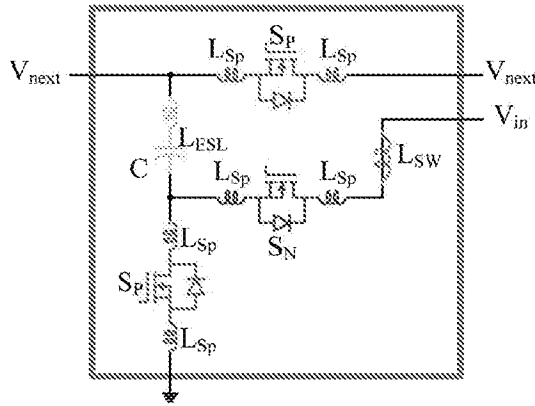


Fig. 4. ZCS-MMSCC modular block with all the stray inductances.

discussed in the next section, only one equivalent L_S is needed to represent the total stray inductance present in the circuit.

Due to the modular structure of the circuit, the stray inductance is distributed equally in each module. The required stray inductance for resonant is more than 100 times smaller (2–13.5 nH) than other ZCS strategies using the same switching frequency [6]–[8], [22], [23]. In case the stray inductance is not sufficiently large or equally distributed in each module, a small air core could be utilized to promise that each module will have the same stray inductance for resonant.

III. OPERATION PRINCIPLE

Fig. 5 shows the idealized waveform of the proposed ZCS-MMSCC under steady-state conditions. The gate signal of switches S_P and S_N is complementary, and duty cycle is 50%. Assume that the input voltage is an ideal voltage source. By considering the stray inductance present in the circuit, when the switch is turned on, the current through the stray inductance, capacitors, and the switch will begin to resonate from zero. By adjusting the switching frequency to resonant frequency, the current through switches S_P and S_N and the stray inductance will decrease to zero when the switch is turned off, which is the half period of the sinusoidal waveform. Therefore, the ZCS of the switch is achieved at both turn-on and turn-off. The capacitor is charged in half period with the sinusoidal current waveform and discharged in another half period also in the sinusoidal shape. Therefore, the current through the capacitor is the sum of the current through the switch, which is a sinusoidal waveform. When the capacitor is charged, the current is positive; when the capacitor discharges, the current is negative. The voltage across the capacitor has a dc offset with a sinusoidal ripple. The dc voltage offset is determined by the capacitor position in the circuit from one time input voltage to n times input voltage. The voltage ripple is determined by the capacitor current and capacitance.

The operation of the circuit can be described in two states, as shown in Figs. 6 and 7, with different switch turned on.

A. State I [t_0, t_1]

Fig. 6 shows the state when S_P is turned on at $t = t_0$ while S_N is off. During this state, C_{1X} is charged by V_{in} . C_{3X} is

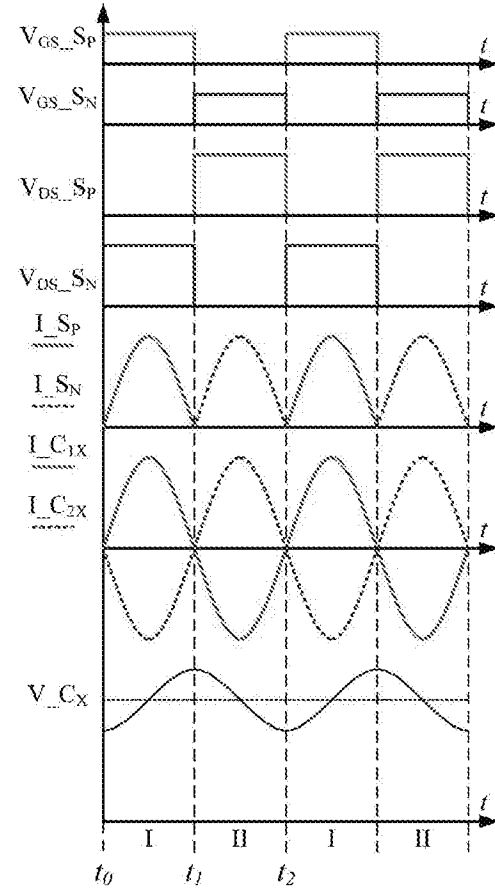


Fig. 5. Typical waveforms of the proposed ZCS-MMSCC.

charged by V_{in} and C_{2X} in series. Figs. 8 and 9 show the two simplified equivalent circuits of state I. Fig. 8 shows the situation when V_{in} , L_{S1} , S_{P1} , C_{1X} , and S_{P2} form a resonant loop. Fig. 9 shows the situation when V_{in} , L_{S3} , S_{P3} , C_{2X} , S_{P4} , C_{3X} , and S_{P5} form a resonant loop. Because of the presence of the equivalent stray inductance L_S , before the switch is turned on, the current through L_S already decreases to zero. The current through S_P will increase from zero when the switch is turned on, so S_P is turned on at zero current. For the case shown in Fig. 8, after L_{S1} and C_{1X} resonate for half cycle, the current through S_{P1} and S_{P2} falls to zero. Therefore, S_{P1} and S_{P2} should be turned off at this point so that the switches are turned off at zero current. For the case shown in Fig. 9, after L_{S3} resonates with C_{2X} and C_{3X} in series for half cycle, the current through S_{P3} , S_{P4} , and S_{P5} will decrease to zero, too. Therefore, zero-current turn-off is also realized on these switches. For the situation shown in Fig. 8, there is only one capacitor in the equivalent circuit other than two capacitors in series in other equivalent circuits. The required stray inductance value is two times smaller than that of the other circuits assuming the capacitances are the same. This will not be a problem in the real circuit design and layout; one only needs to pay attention to make sure that the stray inductance of all the other loops is twice as large as that of the first resonant loop.

Without loss of generality, the following assumptions have been made for the analysis: All the switches are ideal, i.e.,

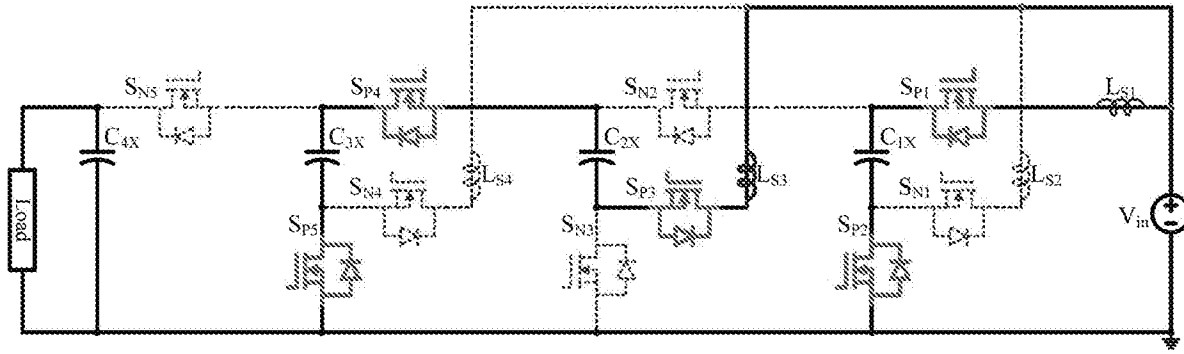
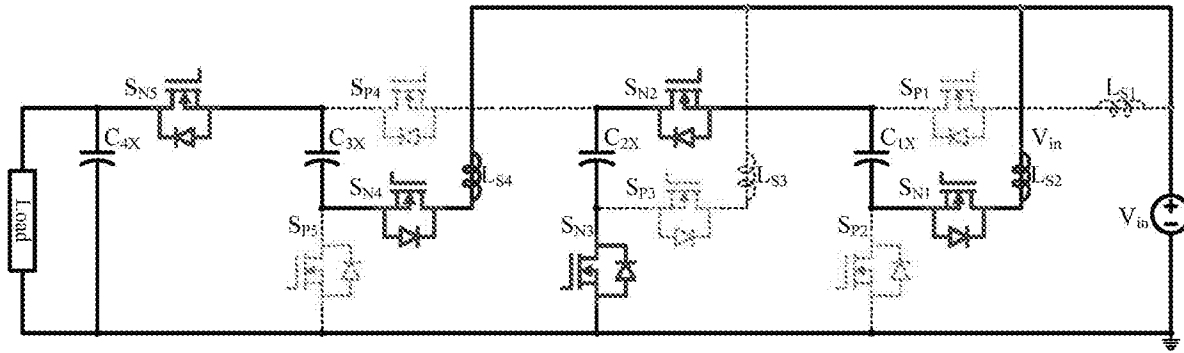
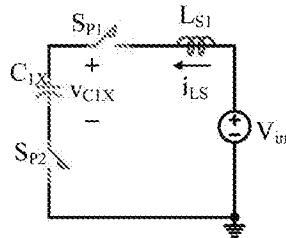
Fig. 6. Operation modes: state I- S_P is on.Fig. 7. Operation modes: state II- S_N is on.

Fig. 8. Simplified equivalent circuits of state I (a).

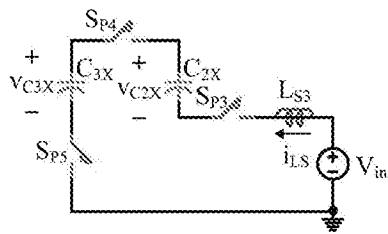


Fig. 9. Simplified equivalent circuits of state I (b).

no conduction resistance is considered; input voltage source is ideal, i.e., constant and no internal impedance; and the capacitor ESR is zero. Assume that $C_{1X} = C_{2X} = C_{3X} = C_{4X}$, so in order to have the same resonant frequency, $L_{S2} = L_{S3} = L_{S4} = 2L_{S1}$. The state equations of the simplified equivalent circuit shown in Fig. 8 are

$$V_{in} = L_{S1} \frac{di_{L_{S1}}}{dt} + v_{C_{1X}} \quad (1)$$

$$i_{L_{S1}} = C_{1X} \frac{dv_{C_{1X}}}{dt} \quad (2)$$

The solutions are

$$i_{L_{S1}}(t) = \frac{\pi P_o}{4V_{in}} \sin \omega_r t \quad (3)$$

$$v_{C_{1X}}(t) = V_{in} - \frac{\pi P_o}{4V_{in} C_{1X} \omega_r} \cos \omega_r t \quad (4)$$

where V_{in} is the value input voltage, L_{S1} is the value of stray inductance, ω_r is the resonant frequency equal to $1/\sqrt{L_{S1} C_{1X}}$, and P_o is the output power. After half cycle, the capacitor voltage is charged to

$$v_{C_{1X}}(t_1) = V_{in} + \frac{\pi P_o}{4V_{in} C_{1X} \omega_r} \quad (5)$$

The capacitor C_{1X} voltage ripple is

$$\Delta v_{C_{1X}} = \frac{\pi P_o}{4V_{in} C_{1X} \omega_r} \quad (6)$$

The state equations of the simplified equivalent circuit shown in Fig. 9 are

$$V_{in} = L_{S3} \frac{di_{L_{S3}}}{dt} + v_{C_{3X}} - v_{C_{2X}} \quad (7)$$

$$i_{L_{S3}} = C_{3X} \frac{dv_{C_{3X}}}{dt} \quad (8)$$

$$i_{L_{S3}} = -C_{2X} \frac{dv_{C_{2X}}}{dt} \quad (9)$$

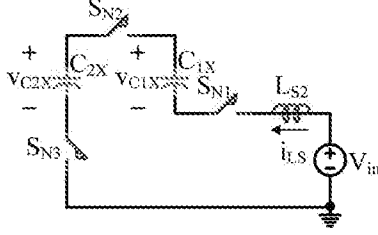


Fig. 10. Simplified equivalent circuits of state II (a).

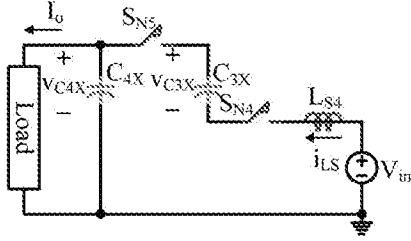


Fig. 11. Simplified equivalent circuits of state II (b).

Plug (17) into (7) and (9) to solve the following equations:

$$i_{LS3}(t) = \frac{\pi P_o}{4V_{in}} \sin \omega_r t \quad (10)$$

$$v_{C3X}(t) = 3V_{in} - \frac{\pi P_o}{4V_{in} C_{1X} \omega_r} \cos \omega_r t. \quad (11)$$

The capacitor \$C_{3X}\$ voltage ripple is

$$\Delta v_{C3X} = \frac{\pi P_o}{4V_{in} C_{1X} \omega_r}. \quad (12)$$

B. State II [\$t_1, t_2\$]

Fig. 7 shows the state when \$S_N\$ is turned on at \$t = t_1\$ while \$S_P\$ is off. During this state, \$C_{2X}\$ is charged by \$V_{in}\$ and \$C_{1X}\$ in series; \$C_{4X}\$ is charged by \$V_{in}\$ and \$C_{3X}\$ in series. Figs. 10 and 11 show the simplified equivalent circuits of state II. Fig. 10 shows the situation when \$V_{in}\$, \$L_{S2}\$, \$S_{N1}\$, \$C_{1X}\$, \$S_{N2}\$, \$C_{2X}\$, and \$S_{N3}\$ form a resonant loop. Fig. 11 shows the situation when \$V_{in}\$, \$L_{S4}\$, \$S_{N4}\$, \$C_{3X}\$, \$S_{N5}\$, and \$C_{4X}\$ form a resonant loop. Because of the presence of \$L_S\$, the current through \$S_N\$ will also increase its value from zero in a resonant manner. The zero-current turn-on of \$S_N\$ is achieved. For the case shown in Fig. 10, after \$L_{S2}\$, \$C_{1X}\$, and \$C_{2X}\$ resonate for half cycle at \$t = t_2\$, \$S_{N1}\$, \$S_{N2}\$, and \$S_{N3}\$ will have a zero-current turn-off when the current through them decreases to zero. For the case shown in Fig. 11, after \$L_{S4}\$ resonates with \$C_{3X}\$ and \$C_{4X}\$ in series for a half cycle, the current through \$S_{N3}\$, \$S_{N4}\$, and \$S_{N5}\$ will also decrease to zero. Hence, zero-current turn-off is also realized on these switches.

The state equations of the simplified equivalent circuit shown in Fig. 10 are

$$V_{in} = L_{S2} \frac{di_{LS2}}{dt} + v_{C2X} - v_{C1X} \quad (13)$$

$$i_{LS2} = C_{2X} \frac{dv_{C2X}}{dt} \quad (14)$$

$$i_{LS2} = -C_{1X} \frac{dv_{C1X}}{dt}. \quad (15)$$

Plug (4) into (13) and (15) to solve the following equations:

$$i_{LS2}(t) = -\frac{\pi P_o}{4V_{in}} \sin \omega_r t \quad (16)$$

$$v_{C2X}(t) = 2V_{in} + \frac{\pi P_o}{4V_{in} C_{1X} \omega_r} \cos \omega_r t. \quad (17)$$

The capacitor \$C_{2X}\$ voltage ripple is

$$\Delta v_{C2X} = \frac{\pi P_o}{4V_{in} C_{1X} \omega_r}. \quad (18)$$

The state equations of the simplified equivalent circuit shown in Fig. 11 are

$$V_{in} = L_{S4} \frac{di_{LS4}}{dt} + v_{C4X} - v_{C3X} \quad (19)$$

$$i_{LS4} = C_{4X} \frac{dv_{C4X}}{dt} + I_o \quad (20)$$

$$i_{LS4} = -C_{3X} \frac{dv_{C3X}}{dt} \quad (21)$$

where \$I_o\$ is the output current. Since this converter is a four times boost circuit, \$V_o = 4V_{in}\$, so \$I_{in} = 4I_o\$. Plug (11) into (19) and (21) to solve the following equations:

$$i_{LS4}(t) = -\frac{\pi P_o}{4V_{in}} \sin \omega_r t \quad (22)$$

$$v_{C4X}(t) = 4V_{in} + \frac{\pi P_o}{4V_{in} C_{1X} \omega_r} \cos \omega_r t. \quad (23)$$

The capacitor \$C_{4X}\$ voltage ripple is

$$\Delta v_{C4X} = \frac{\pi P_o}{4V_{in} C_{1X} \omega_r}. \quad (24)$$

The soft-switching condition in both states is \$f_s = f_r\$, where \$f_s\$ is the switching frequency and \$f_r\$ is the resonant frequency.

IV. DESIGN GUIDELINES

The 150-W four-level ZCS-MMSCC design procedure is shown here as an example. The specifications of the prototype converter are as follows: \$V_{in} = 5\$ V, \$V_o = 20\$ V, \$P_o = 150\$ W, and \$f_s = 45\$ kHz.

A. Capacitance

Capacitance value should be chosen according to the voltage ripple

$$C = \frac{\pi P_o}{4V_{in} \Delta v_C \omega_r}. \quad (25)$$

Because of the special structure of the circuit, the voltage ripple across all the capacitors is the same, and the value of the voltage ripple is determined by the input current, capacitance, and the resonant frequency. The capacitor voltage ripple should be chosen to be smaller than the input voltage to prevent the voltage across the first capacitor from resonating to the negative region and losing the ZCS. That means, the first capacitor should have a positive dc voltage offset. In fact, because of the special structure of the converter, the output voltage is \$n\$ times

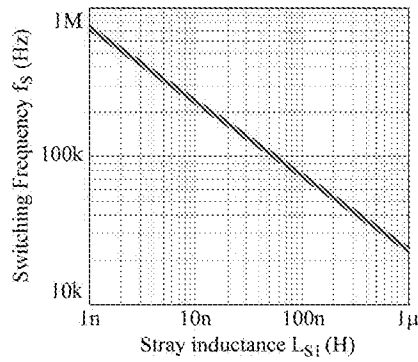


Fig. 12. ZCS stray inductance and switching frequency.

of the input voltage; even when the voltage ripple is as large as the input voltage, the voltage ripple on the last capacitor is still $1/n$. When the conversion ratio is large, the ripple will be small enough in the output side, and high capacitance is not needed to reduce the output voltage ripple.

The output ripple is chosen to be around 9% of the output voltage. According to (25), the capacitance is chosen to be $47 \mu\text{F}$ in the prototype. Another parameter to determine the capacitance value is the current rating of the capacitor; if the current rating of the capacitor is too small to meet the current requirements, then more capacitors in parallel have to be used. In this prototype, high-current-rating MLCC capacitors are used. Ten 100-V $4.7\text{-}\mu\text{F}$ C5750X7R2A475K capacitors from TDK are used in parallel. The current rating of each capacitor is around 3 A each.

B. Stray Inductance

The circuit layout should be designed carefully, in order to make the stray inductance of each equivalent circuit equal. In case the stray inductance is not equal or large enough, a small air core can be inserted to equalize the stray inductance of each module. The switching frequency should be adjusted according to the stray inductance present in the circuit if the circuit layout is already fixed. Required stray inductance for ZCS can be determined by the following equation, using the equivalent circuit shown in Fig. 8:

$$L_{S1} = \frac{1}{C_{1X} (2\pi f_s)^2}. \quad (26)$$

When the resonant capacitance is $47 \mu\text{F}$, Fig. 12 shows the curve of switching frequency versus stray inductance.

Table I shows the relevant required stray inductance according to the switching frequency, when the capacitance is $47 \mu\text{F}$. When the switching frequency is larger than 200 kHz, the required stray inductance is already smaller than the regular circuit layout stray inductance.

Obviously, for other equivalent circuits shown in Figs. 9–11, the stray inductance values of L_{S2} , L_{S3} , and L_{S4} for resonant need to be doubled. The capacitance value becomes half when two capacitors are connected in series in these equivalent circuits. The switching frequency can be easily determined by the actual stray inductance in the circuit. Around 100 and 200 kHz, the most often used switching frequencies in dc-dc converters,

TABLE I
REQUIRED STRAY INDUCTANCE ACCORDING TO SWITCHING FREQUENCY

Switching Frequency f_s	Stray inductance L_{S1}
500kHz	2.15 nH
400 kHz	3.37 nH
300 kHz	6 nH
200 kHz	13.5 nH
100 kHz	54 nH
50 kHz	215 nH

the needed stray inductance for resonating is less than 55 nH. This value is small enough to be realized by the stray inductance in the circuit layout or by a small air core. The inductance of a 10-cm-long wire or air core inductor is already about 50 nH. Only about 2-nH stray inductances are needed when the switching frequency is increased to 500 kHz. However, the stray inductance inside a normal MOSFET package such as TO-220 is already about 1–2 nH; when several MOSFETs are connected in series, this value could be higher considering the stray inductance of the MOSFET leg which means that the switching frequency does not need to be higher than 500 kHz to achieve ZCS.

C. Switching Frequency

Switching frequency should be set the same as the resonant frequency in order to promise the ZCS. Usually, the stray inductance of the circuit is already determined when the circuit layout is finished. The capacitance is also determined by the required voltage ripple. Assume that the stray inductance of each module is equalized. The switching frequency can be twisted by monitoring the power device current when the switch is turned off. The switching frequency should be set exactly at the point when the switch current resonates to zero when the switch is turned off.

V. POWER LOSS ANALYSIS

A. Power Device Loss

Conduction Loss: Assume that the input power is P_{in} , the power device turn-on resistance is R_{on} , the rms value of switch current is I_{S_rms} , the average value of input current is I_{in_ave} , the conversion ratio of the converter is N_T , and the conduction loss of power device is P_{S_con} . Thus, the power device conduction loss equation can be derived as follows:

$$I_{in_ave} = \frac{P_{in}}{V_{in}} \quad (27)$$

$$I_{S_rms} = \frac{I_{in_ave}}{N_T} \frac{\pi}{2} \quad (28)$$

$$P_{S_con} = I_{S_rms}^2 R_{on}. \quad (29)$$

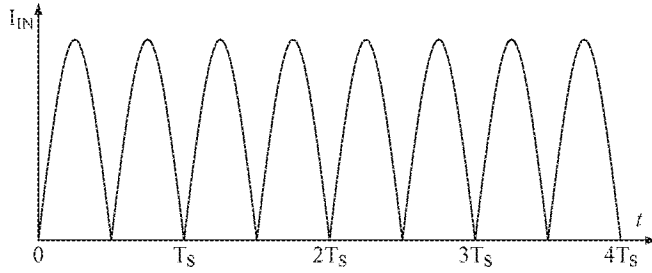


Fig. 13. Ideal input current waveform.

Plugging (27) and (28) into (29), the following conduction loss can be derived:

$$P_{S_con} = \frac{\pi^2 P_{in}^2}{4N_T^2 V_{in}^2} R_{on}. \quad (30)$$

Gate Drive Loss: Assume that the gate drive voltage is V_{GS} , the gate total charge is Q_g , and the gate drive loss of power device is P_{S_drv} . Thus, the power device gate drive loss can be derived using the following equation:

$$P_{S_drv} = V_{GS} Q_g f_s. \quad (31)$$

B. Capacitor Conduction Loss

Resonant Capacitor Conduction Loss: Assume that the capacitor rms current is I_{C_rms} , the capacitor ESR is R_{ESR} , and the capacitor conduction loss is P_{C_con} . Thus, the capacitor conduction loss can be derived using the following equation:

$$I_{C_rms} = \frac{\pi I_{in_ave}}{\sqrt{2} N_T} \quad (32)$$

$$P_{C_con} = I_{C_rms}^2 R_{ESR}. \quad (33)$$

Plugging (32) into (33), one can obtain

$$P_{C_con} = \frac{\pi^2 I_{in_ave}^2}{2N_T^2} R_{ESR}. \quad (34)$$

Plugging (27) into (34), one can obtain

$$P_{C_con} = \frac{\pi^2 P_{in}^2}{2N_T^2 V_{in}^2} R_{ESR}. \quad (35)$$

Input Capacitor Conduction Loss: Assume that the rms value of input current is I_{in_rms} , the capacitor ESR is R_{ESR_in} , and the conduction loss of input capacitor is $P_{C_in_con}$. The input current is shown in Fig. 13.

$$I_{in_rms} = \frac{\pi I_{in_ave}}{2\sqrt{2}} \quad (36)$$

$$P_{C_in_con} = I_{in_rms}^2 R_{ESR_in}. \quad (37)$$

Plugging (36) into (37), one can obtain

$$P_{C_in_con} = \frac{\pi^2 I_{in_ave}^2}{8} R_{ESR_in}. \quad (38)$$

Plugging (27) into (38), one can obtain

$$P_{C_in_con} = \frac{\pi^2 P_{in}^2}{8V_{in}^2} R_{ESR_in}. \quad (39)$$

C. Total Power Loss

There are $3N_T - 2$ switching devices and N_T capacitors in a N_T -level ZCS-MMSCC, so the total power loss P_{loss} is

$$P_{loss} = (3N_T - 2)(P_{S_con} + P_{S_drv}) + N_T P_{C_con} + P_{C_in_con}. \quad (40)$$

Plugging (30), (31), and (34) into (40), one can obtain

$$P_{loss} = (3N_T - 2) \left(\frac{\pi^2 P_{in}^2 R_{on}}{4N_T^2 V_{in}^2} + V_{GS} Q_g f_s \right) + \frac{\pi^2 P_{in}^2 R_{ESR}}{2N_T V_{in}^2} + \frac{\pi^2 P_{in}^2 R_{ESR_in}}{8V_{in}^2}. \quad (41)$$

Assume that η is the converter efficiency

$$\eta = \frac{P_{in} - P_{loss}}{P_{in}}. \quad (42)$$

Plugging (41) into (42), one can obtain (43), shown at the bottom of the page.

VI. SIMULATION AND EXPERIMENT RESULTS

A. 150-W Four-Level ZCS-MMSCC Prototype

Fig. 14 shows Saber simulation waveforms of a 150-W four-level ZCS-MMSCC, where V_{gs_Sp} and V_{gs_Sn} are the switch gate–source control voltages, V_{ds_Sp1} and V_{ds_Sn2} are the switch drain–source voltages, and I_{d_Sp1} and I_{d_Sn2} are the switch drain–source currents. The input voltage is 5 V. The switching frequency is about 45 kHz. Capacitance is 47 μ F, and stray inductance L_{S1} is about 250 and 500 nH for others. The simulation results are consistent with the theoretical analysis, which verifies the aforementioned analysis.

In the 150-W four-level ZCS-MMSCC prototype, the switching devices are two 30-V 180-A MOSFETs (IPB009N03L, from Infineon) connected in parallel. Resonant capacitors are ten 100-V 4.7- μ F MLCC capacitors (C5750X7R2A475K, from TDK) connected in parallel. Input capacitors are 14 6.3-V 470- μ F conductive polymer aluminum solid electrolytic capacitors (PLE0J471MDO1, from Nichicon) connected in parallel. Gate drive voltage is 15 V.

$$\eta = \frac{P_{in} - (3N_T - 2) \left(\frac{\pi^2 P_{in}^2 R_{on}}{4N_T^2 V_{in}^2} + V_{GS} Q_g f_s \right) - \frac{\pi^2 P_{in}^2 R_{ESR}}{2N_T V_{in}^2} - \frac{\pi^2 P_{in}^2 R_{ESR_in}}{8V_{in}^2}}{P_{in}} \quad (43)$$

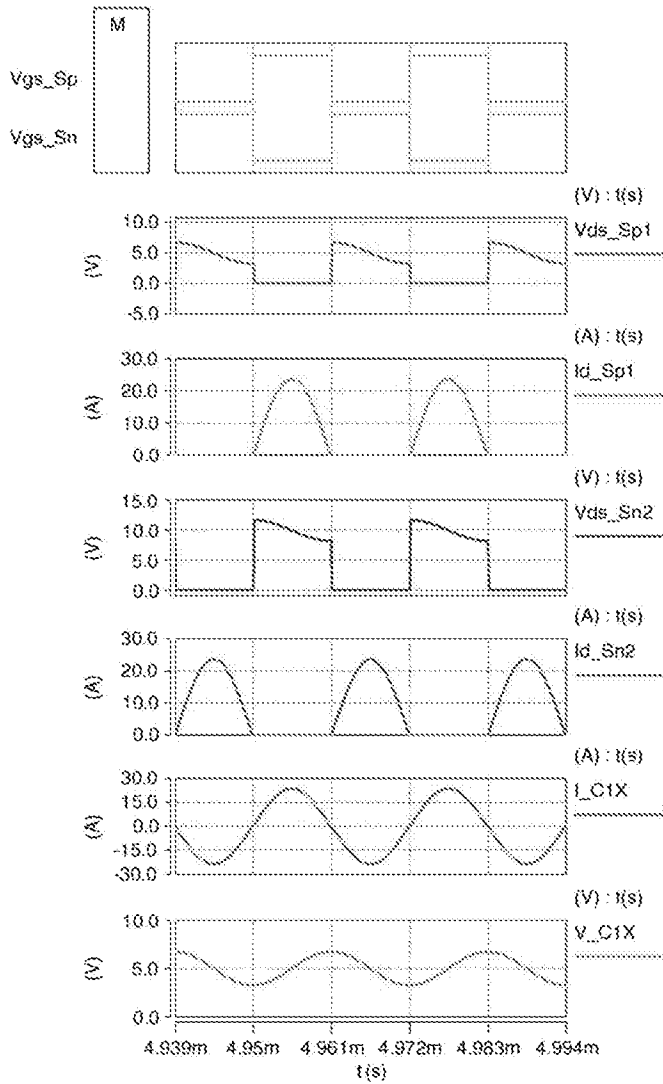
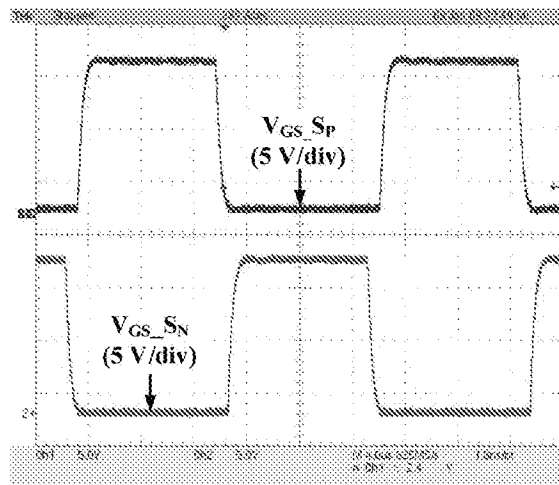


Fig. 14. Simulation waveforms of a 150-W four-level ZCS-MMSCC.

Fig. 15. Gate drive signals of S_P and S_N .

Figs. 15–18 show the experiment waveforms of the 150-W four-level ZCS-MMSCC prototype with the parts mentioned earlier. Fig. 15 shows the complementary gate drive signals of

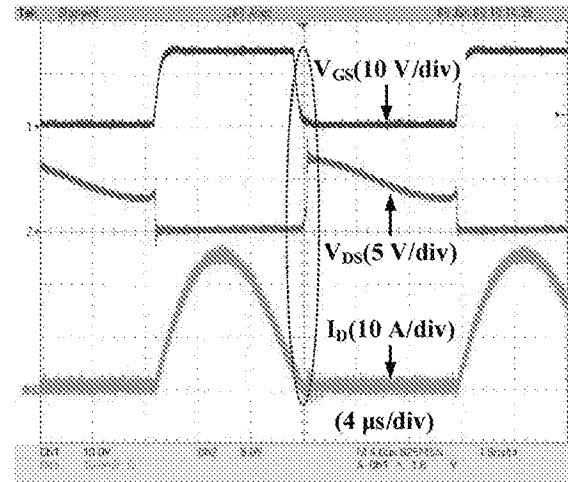
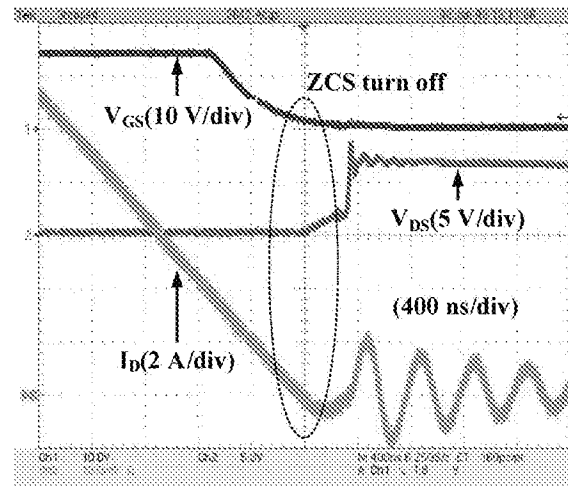
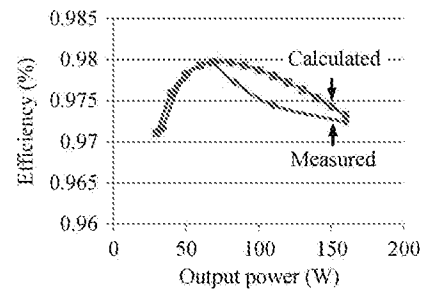
Fig. 16. Voltage and current waveforms of S_{P1} .Fig. 17. Zoomed-in voltage and current waveforms of S_{P1} .

Fig. 18. Measured efficiency and calculated efficiency.

S_P and S_N with duty cycle of about 48% due to some dead time. V_{GS_SP} and V_{GS_SN} are the complementary gate–source control voltages.

Fig. 16 shows the voltage and current waveforms of S_{P1} . V_{GS} is the gate–source voltage of S_{P1} , V_{DS} is the drain–source voltage of S_{P1} , and I_D is the drain–source current of S_{P1} . The resonant current waveform realizing ZCS is shown. Fig. 17 shows the zoomed-in waveforms of the switch S_{P1} turn-off transient, which indicates that the switch achieves ZCS during turn-off. From the experiment results, we can derive that the switch S_{P1} can achieve ZCS in both turn-on and turn-off. Other

switches have the voltage and current waveforms similar to that of S_{P1} which will not be shown here.

Fig. 18 shows the measured efficiency curve using Yokogawa WT1600 digital power meter of the proposed ZCS-MMSCC and the calculated efficiency using (43). The efficiency is above 97%, which is a big improvement than traditional MMSCC.

VII. CONCLUSION

The presented ZCS-MMSCC has all the advantages of ZCS and MMSCC, and it has some special features besides the features got from ZCS and MMSCC. All the features can be concluded as follows.

- 1) It reduces dv/dt , di/dt , and EMI because of ZCS.
- 2) It reduces the switching loss which improves efficiency.
- 3) It uses simple control strategy and possesses modular structure and bidirectional power conversion.
- 4) It increases the potential power rating.
- 5) It reduces the capacitance and capacitor size and lowers the cost.
- 6) It utilizes distributed stray inductance to achieve ZCS and improves reliability.

A new soft-switching operation strategy to the existing MMSCC has been proposed and validated by both simulation and experiment results. A 150-W 5-V input four-level converter prototype for automotive application has been built and tested based on the proposed circuit, and the efficiency of the prototype is higher than 97%. The ZCS-MMSCC overcomes the current and voltage spike problems of traditional MMSCC and provides a technically viable soft-switching strategy with circuit layout stray inductance. This technique could also be applied to the switched-capacitor circuits on a semiconductor chip with very small stray inductance in the circuit without pushing the switching frequency to megahertz. The proposed ZCS-MMSCC also shows great potential to high-voltage-gain and high-input-current applications.

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Analysis and Optimization of Switched-Capacitor DC–DC Converters

Michael D. Seeman, *Student Member, IEEE*, and Seth R. Sanders, *Member, IEEE*

Abstract—Analysis methods are developed that fully determine a switched-capacitor (SC) dc–dc converter's steady-state performance through evaluation of its output impedance. This analysis method has been verified through simulation and experimentation. The simple formulation developed permits optimization of the capacitor sizes to meet a constraint such as a total capacitance or total energy storage limit, and also permits optimization of the switch sizes subject to constraints on total switch conductances or total switch volt-ampere (V-A) products. These optimizations then permit comparison among several switched-capacitor topologies, and comparisons of SC converters with conventional magnetic-based dc–dc converter circuits, in the context of various application settings. Significantly, the performance (based on conduction loss) of a ladder-type converter is found to be superior to that of a conventional magnetic-based converter for medium to high conversion ratios.

Index Terms—Analysis, dc–dc converter, output impedance, switched-capacitor.

I. INTRODUCTION

THIS paper develops analysis methods that fully determine a switched-capacitor (SC) dc–dc converter's steady-state performance through evaluation of its output impedance. This resistive impedance is a function of frequency and has two asymptotic limits: one where resistive paths dominate the impedance, and another where charge transfers among idealized capacitors dominate the impedance. This work develops a network theoretic analysis of these two asymptotic limits, which can be used to evaluate both the converter efficiency and output regulation as a function of load for a broad class of SC converters. Simulations and experiments have been performed to verify the analysis methods.

The comprehensive analysis and design calculations given here are new, but connect with the analysis framework developed in the pioneering work of [1]. The work in [1], [2] offered a network theoretic formulation for computation of open-circuit dc–dc conversion ratios, and a rather involved method for computation of output impedance. Reference [1] and other previous analysis work [3]–[7] mainly focused on the performance analysis (i.e., output impedance computation) for a single converter topology.

The simple formulation developed permits optimization of the capacitor sizes to meet a constraint such as a total capacitance or total energy storage limit, and also permits optimization

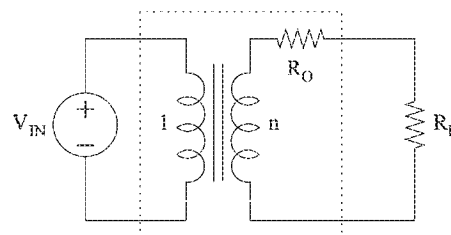


Fig. 1. Model of an idealized switched-capacitor converter.

of the switch sizes subject to constraints on total switch conductances or total switch volt-ampere (V-A) products. These optimizations are carried out for a set of representative switched-capacitor topologies. These optimizations then permit comparison among several switched-capacitor topologies, and comparisons of SC converters with conventional magnetic-based dc–dc converter circuits. The performance (based on conduction loss) of a ladder-type converter is found to be superior to that of a conventional boost converter for medium to high conversion ratios.

II. SWITCHED-CAPACITOR CONVERTER IMPEDANCE ANALYSIS

With the model in Fig. 1, the converter provides an ideal dc voltage conversion ratio under no load conditions, and all conversion losses are manifested by voltage drop associated with non-zero load current through the output impedance [1], [5]. The resistive output impedance accounts for capacitor charging and discharging losses and resistive conduction losses. Additional losses due to short-circuit current and parasitic capacitances, in addition to gate-drive losses, can be incorporated into the model. However, they will not be considered here since these effects are generally application and implementation dependent. Insight gained can be used to model effects of parasitic capacitances [8]. For the present, our aim is to provide a general analysis and design framework.

The low-frequency output impedance R_O in Fig. 1 sets the maximum converter power, constrained by a minimal efficiency objective, and also determines the open-loop load regulation properties. There are two asymptotic limits to output impedance, the slow and fast switching limits, as related to switching frequency. The slow switching limit (SSL) impedance is calculated assuming that the switches and all other conductive interconnects are ideal, and that the currents flowing between input and output sources and capacitors are impulsive, modeled as charge transfers. The SSL impedance is inversely proportional to switching frequency. The fast switching limit (FSL) occurs when the resistances associated with switches, capacitors and interconnect dominate, and the capacitors act

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effectively as fixed voltage sources. In the FSL, current flow occurs in a frequency-independent piecewise constant pattern.

The set of converters considered in this paper is limited to two-phase converters made solely of ideal capacitors, resistive switches, and input and output voltage sources. Two-phase converters switch alternately between two configurations. Multi-phase converters [4] are outside the scope of this paper, but can be considered using similar methods. This paper does not address the more fundamental topological conditions needed to determine whether or not a specific circuit constitutes a well-formulated two-phase converter. Rather, the paper assumes that the circuits under consideration all have well-defined two-phase operation. References [1], [2] begin to address the topological question of what constitutes a well-formulated two-phase SC dc-dc circuit, though the characterization given is not complete.

A. Slow-Switching Limit Impedance

For the slow-switching limit (SSL) impedance analysis, the finite resistances of the switches, capacitors, and interconnect are neglected. A pair of charge multiplier vectors $\mathbf{a}^1$ and $\mathbf{a}^2$ can be derived for any standard non-degenerate two-phase SC converter. The charge multiplier vectors correspond to charge flows that occur immediately after the switches are closed to initiate each respective phase of the SC circuit. Each element of a charge multiplier vector corresponds to a specific capacitor or independent voltage source, and represents the charge flow into that component, normalized with respect to the output charge flow. As outlined in [1], the charge multiplier vectors can be uniquely computed using the KCL constraints in each topological phase and the constraint that the two charge multiplier quantities on each capacitor are equal and opposite.

The charge multiplier vector $\mathbf{a}^1$ is defined as

$$\mathbf{a}^1 = [q_{out}^1 \ q_1^1 \ \cdots \ q_n^1 \ q_{in}^1]^T / q_{out} \quad (1)$$

where each component is the ratio of charge transfer in each element during phase 1 of the switching period to the charge delivered to the output during a full period. If charge flows into the element during phase 1, the corresponding entry in the $\mathbf{a}^1$ vector is positive. Vector $\mathbf{a}^2$ is defined analogously, for phase 2. The charge multiplier vector can be partitioned into output, capacitor and input components, respectively

$$\mathbf{a}^1 = [a_{out}^1 \ \mathbf{a}_c^1 \ a_{in}^1]^T. \quad (2)$$

For the ladder network example of Fig. 2, the charge multiplier vectors can be obtained through network analysis using Kirchhoff's Current Law (KCL) [1]. In this example, and in all other examples encountered by the authors, the charge multiplier vectors can be obtained by inspection (in Fig. 3). The charge from the input source flows into C4 during phase 2. In phase 1, that charge is transferred into C3. By considering alternating phases, the charge flow in each component can be found

$$\mathbf{a}^1 = [2/3 \ -2/3 \ 1/3 \ -1/3 \ 0]^T \quad (3)$$

$$\mathbf{a}^2 = [1/3 \ 2/3 \ -1/3 \ 1/3 \ -1/3]^T. \quad (4)$$

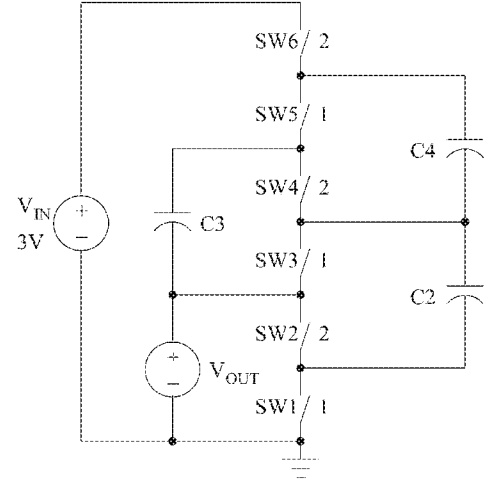


Fig. 2. 3 V to 1 V ladder circuit.

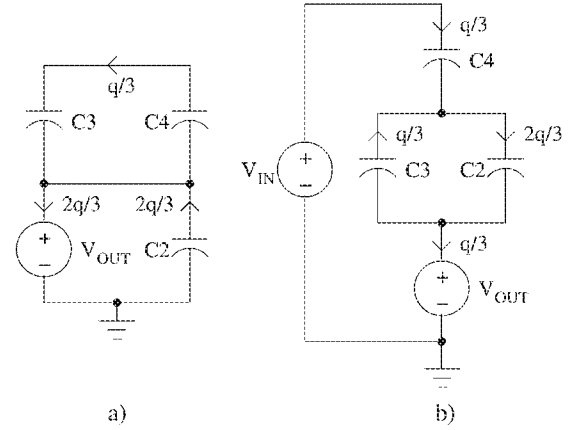


Fig. 3. Charge flow in ladder converter. (a) phase 1 (b) phase 2.

In each of these charge multiplier vectors, the first component corresponds to the output charge flow, thus these two components must sum to one. The last component of each charge multiplier vector corresponds to the charge flow into the input source, and is non-zero during only phase 2 in this example.

The charge multiplier vectors, the capacitor characteristics, and the switching frequency are the only data needed to determine the output impedance under the asymptotic SSL condition. The calculation, developed here, is based on Tellegen's Theorem [9] which states that for any network, any vector of branch voltages that satisfies KVL is orthogonal to any vector of branch currents (or equivalently charge flows) that satisfies KCL. This theorem is applied in each of the two configurations for a two-phase switched capacitor converter operating in periodic steady state, where the input is short-circuited and the output is connected to an independent dc voltage source. The charge flow per period (or average current flow) into the single independent source then defines the output impedance.

Application of Tellegen's theorem to the switched capacitor converter, in each of its two configurations, yields $\mathbf{a}^1 \cdot \mathbf{v}^1 = 0$ and $\mathbf{a}^2 \cdot \mathbf{v}^2 = 0$, where $\mathbf{v}^1$ and $\mathbf{v}^2$ are the respective steady

state network voltage vectors in phases 1 and 2. Additively combining these two applications of Tellegen's theorem, and noting that the input voltage source has value zero, yields

$$v_{\text{out}} (a_{\text{out}}^1 + a_{\text{out}}^2) + \sum_{\text{capacitors}} (a_{c,i}^1 v_{c,i}^1 + a_{c,i}^2 v_{c,i}^2) = 0 \quad (5)$$

where the first term corresponds to the constant output voltage source and the terms under the summation correspond to the capacitor branches. Recall that $a_{\text{out}}^1 + a_{\text{out}}^2 = 1$ and that $a_{c,i}^1 = -a_{c,i}^2$ for each capacitor branch (due to charge conservation in periodic steady-state). By defining $a_{c,i} = a_{c,i}^1 = -a_{c,i}^2$ and $q_i = a_{c,i} q_{\text{out}}$ and multiplying (5) by q_{out} , the net charge delivered to the output in a period, we obtain

$$q_{\text{out}} v_{\text{out}} + \sum_{\text{capacitors}} q_i \Delta v_i = 0 \quad (6)$$

where $\Delta v_i = v_{c,i}^1 - v_{c,i}^2$. In (6), the first term corresponds to the product of the constant output voltage and the total charge flow into this independent voltage source, and each term in the summation corresponds to energy loss associated with a specific capacitor. It is of direct interest here that none of the capacitor voltages need to be explicitly calculated for this analysis. Rather, Δv_i can be computed from

$$\Delta v_i = q_i / C_i \quad (7)$$

where C_i is the capacitance value of the i th capacitor, assuming linear capacitors. Introducing (7) into (6), and then dividing the result by q_{out}^2 yields

$$\frac{v_{\text{out}}}{q_{\text{out}}} + \sum_{\text{capacitors}} \left(\frac{q_i}{q_{\text{out}}} \right)^2 \frac{1}{C_i} = 0. \quad (8)$$

We note that q_i / q_{out} corresponds to the i th entry of the charge multiplier vector $\mathbf{a}_c$, since these entries are for the capacitors. Dividing (8) by the switching frequency then directly yields the average output impedance for the slow-switching asymptotic limit

$$R_{\text{SSL}} = -\frac{v_{\text{out}}}{i_{\text{out}}} = \sum_i \frac{(a_{c,i})^2}{C_i f_{\text{sw}}}. \quad (9)$$

The converter's loss in terms of the series output impedance R_{SSL} can be expressed in terms of capacitor loss. The product $q_i \Delta v_i$ in (6) represents the energy loss by charging and discharging capacitor i in each cycle, and could be used to calculate the converter's loss even with a nonlinear capacitor. In the following discussion, attention is restricted to the case of linear capacitors. The sum of the energy lost through the capacitors is equal to the calculated loss associated with the output impedance for a given load.

This powerful result yields a simple calculation of this asymptotic output impedance and some intuition into the operation of SC converters. The output impedance directly models the losses in the circuit due to capacitor charging and discharging. This

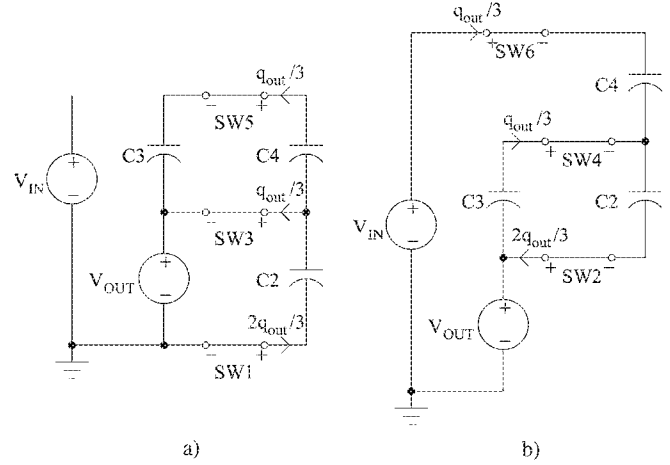


Fig. 4. Switch charge flow in ladder converter. (a) Phase 1, (b) Phase 2.

impedance can be determined by simply examining the charge flow in the converter without simulation or complicated network analysis.

B. Fast Switching Limit Impedance

The other asymptotic limit, the fast switching limit (FSL), is characterized by constant current flows between capacitors. The switch on-state impedances and other resistances are sufficiently large such that during each phase, the capacitors do not approach equilibrium. In the asymptotic limit, the capacitor voltages are modeled as constant. The circuit loss is related only to conduction loss in resistive elements. The concept of the FSL impedance is introduced informally in [5].

The duty cycle of the converter is important when considering the FSL impedance since currents flow during the entirety of each phase. For this analysis, a duty cycle of 50% is assumed for simplicity. Duty cycle differing from 50% can be included in the following analysis without much difficulty if another duty cycle is used. Additionally, only the on-state switch resistance is considered; other parasitic resistance [e.g., capacitor equivalent series resistance (ESR)] can be similarly incorporated into the model if desired.

The $a_{r,i}$ values are defined as the charge flow through each switch during the phase in which the switch is on. Even in the FSL, the charge flows must follow the same pattern as in the SSL, constrained by $\mathbf{a}^1$ and $\mathbf{a}^2$. For the switches that are on during phase 1, the corresponding $a_{r,i}$ values can be determined from the $\mathbf{a}^1$ components. Analogously, corresponding $a_{r,i}$ values for switches that conduct during phase 2 can be determined from the $\mathbf{a}^2$ components. The values of $a_{r,i}$ are independent of duty cycle as they simply represent the charge flow through the switches that ensure charge conservation on the circuit's capacitors. The $a_{r,i}$ values for the switches in the ladder converter in Fig. 2 can be determined directly. The charge flows in the switches during both phases are shown in Fig. 4, resulting in an $\mathbf{a}_r$ vector of

$$\mathbf{a}_r = [-2/3 \quad -2/3 \quad 1/3 \quad 1/3 \quad 1/3 \quad 1/3]. \quad (10)$$

For positive power flow (i.e., from the input to output source), the sign of each component of the $\mathbf{a}_r$ vector indicates the direction of current flow with respect to the blocking voltage of a switch. A positive quantity indicates the switch conducts positive current while on and blocks positive voltage while off. This switch must be implemented using an active transistor. A negative quantity indicates the switch conducts negative current and blocks positive voltage and is suitable for diode implementation (if the forward voltage drop is tolerable). For power flow in the opposite direction, the switch types reverse.

In the FSL, the current through the on-state switches is assumed to be constant. Given the charge flow vector, the current in each switch is easily determined

$$i_{r,i} = 2q_{r,i}f_{sw} \quad (11)$$

where $q_{r,i}$ is the charge flow through switch i during a single period. The factor of two appears because of the 50% duty cycle. Substituting $q_{r,i} = a_{r,i}q_{out}$ and $q_{out} = i_{out}/f_{sw}$ into (11) yields

$$i_{r,i} = 2a_{r,i}i_{out}. \quad (12)$$

The current through the switches is only dependent on the vector $\mathbf{a}_r$, which is obtainable by inspection. The network voltages never need to be found in this analysis, simplifying computation significantly.

The average power loss due to each individual switch is equal to the instantaneous on-state power loss multiplied by its duty cycle. Since the total loss of the SC converter in the FSL is just the sum of the switch losses, the total circuit loss is given by

$$P_{FSL} = \sum_{\text{switches}} \frac{1}{2} R_i (2a_{r,i}i_{out})^2 \quad (13)$$

where R_i is the on-state resistance of switch i .

Since the input and output charge flow in the SC converter is constrained by the conversion ratio n , all the power loss in an ideal SC converter (as analyzed here) is modeled by the output voltage drop. Thus the output impedance can be determined by equating the actual power loss of the circuit with the apparent power loss due to the output impedance. Since this power loss is proportional to the square of the output current, the FSL output impedance can be obtained by inspection

$$R_{FSL} = 2 \sum_i R_i (a_{r,i})^2. \quad (14)$$

Similar to the SSL output impedance in (9), the FSL output impedance is given simply in terms of component parameters and the switch charge multiplier coefficients of each switch. The power loss due to these conduction losses is equal to the equivalent power loss through the output impedance. These two simple forms of the output impedance (given in (9) for the SSL and (14) for the FSL) can be used to provide strong guidance for the design of switched-capacitor power converters.

III. COMPONENT OPTIMIZATION

Given that all converter losses attributed to the capacitors and resistive switches can be reflected in the computation of a

single real output resistance, it is now possible to optimize the components in order to minimize that output impedance. Minimal output impedance corresponds to maximum efficiency for a given power delivered, and dually, corresponds to maximum power delivery for a given loss. This section develops optimality computations for the slow switching limit (SSL) and the fast switching limit (FSL) impedance. When optimizing over capacitances, one should minimize the output impedance that is associated only with the capacitances, namely the SSL impedance. Analogously, when optimizing over switch sizes, one should minimize the FSL output impedance. The final design step is to choose a maximum operating frequency for which the parasitic losses are acceptable. The total capacitance and switch conductance should be adjusted such that the total impedance meets the design goal and the SSL and FSL impedances are balanced. The last step ensures that the total switch and capacitor sizes (and costs) are minimized for the intended power level.

The optimization procedure requires knowledge of the component working voltages, unlike the output impedance analysis. The working voltage for a capacitor is the maximum voltage on the capacitor during steady-state converter operation. For a transistor (switch), the working voltage is the voltage it blocks during steady-state converter operation. For open-circuit operation, these working voltages can be found by inspection in most examples, or by the process outlined in [1]. This analysis is based on combining KVL constraints for the two phase topologies, in combination with a known source voltage. The result is the computation of vectors denoted $\mathbf{v}_c$ and $\mathbf{v}_r$ for the working voltages of the capacitors and switches, respectively, ratioed to the converter output voltage.

The optimization is based on a physical size (or cost) constraint for the devices. When capacitors are optimized, their total energy storage capability is held constant. Or, in the case when all capacitors must be rated for the same voltage, the total capacitance is held constant. Likewise, when the switch sizes are optimized, the total V-A capacity product is held constant. This V-A metric translates to a constraint on the G-V² products summed over the switches (G refers to switch conductance). If all the switches are rated for the same voltage, the constraint reduces to holding the sum of the switch conductances constant.

A. SSL Capacitor Optimization

The capacitor optimization uses a constraint that holds the total energy storage capability, summed over all capacitors, fixed to a constant E_{tot} . This constraint can be mathematically expressed as

$$\sum_i \frac{1}{2} (v_{c,i(\text{rated})})^2 C_i = E_{tot} \quad (15)$$

where C_i represents the value of capacitor i and $v_{c,i(\text{rated})}$ represents the rated voltage of capacitor i . The energy storage capability of a capacitor is related to its rated voltage, as that dictates its size and cost, not the maximum voltage it sees during operation. However, the capacitor's working voltage must be less than the rated voltage to avoid damaging the component, and should be close to the rated voltage to achieve good utilization of the device.

A function $\mathcal{L}$ is defined to perform the constrained optimization

$$\mathcal{L} = \sum_i \frac{(a_{c,i})^2}{C_i} + \lambda \left(\sum_i \frac{1}{2} (v_{c,i(\text{rated})})^2 C_i - E_{\text{tot}} \right) \quad (16)$$

where the first term represents the SSL output impedance (scaled by switching frequency as it does not effect the minimization) and the second term incorporates the constraint in (15). The impedance is minimized by equating the partial derivatives of $\mathcal{L}$ with respect to C_i and λ with zero

$$\frac{\partial \mathcal{L}}{\partial C_i} = -\frac{(a_{c,i})^2}{C_i^2} + \lambda \frac{1}{2} (v_{c,i(\text{rated})})^2 = 0 \quad (17)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = \sum_i \frac{1}{2} (v_{c,i(\text{rated})})^2 C_i - E_{\text{tot}} = 0. \quad (18)$$

Equation (18) simply repeats the constraint in (15).

The relationship in (17) sets up a proportionality between C_i , $a_{c,i}$, and $v_{c,i(\text{rated})}$. The energy constraint can be used to find an expression for the value of each capacitor

$$C_i = \left| \frac{a_{c,i}}{v_{c,i(\text{rated})}} \right| \frac{2E_{\text{tot}}}{\sum_k |a_{c,k} v_{c,k(\text{rated})}|}. \quad (19)$$

The optimal energy storage of each capacitor is proportional to the V - Q product of each capacitor

$$E_i = \frac{|a_{c,i} v_{c,i(\text{rated})}|}{\sum_k |a_{c,k} v_{c,k(\text{rated})}|} E_{\text{tot}}. \quad (20)$$

When the total energy is constrained, the optimal capacitor energies are proportional to the product of their rated voltage and their charge multiplier coefficients. In addition, the ripple voltage on each capacitor is directly proportional to that capacitor's rated voltage.

The optimized output impedance can be calculated by combining (9) and (19)

$$R_{\text{SSL}}^* = \frac{1}{2E_{\text{tot}} f_{\text{sw}}} \left(\sum_i |a_{c,i} v_{c,i(\text{rated})}| \right)^2. \quad (21)$$

By optimizing the capacitors, the output impedance becomes proportional to the square of the sum of the products of voltages and charge flows (V - A product) of each capacitor. The optimization can improve the performance of an SC converter designed in an ad-hoc manner significantly, especially one with a large conversion ratio.

If all capacitors in a SC converter are rated for the same voltage, as in the ladder topology or in applications with integrated capacitors, the optimization results can be simplified. In this case, we constrain total capacitance to a value of C_{tot} , and the value of each individual capacitor is given by

$$C_i^* = \frac{|a_{c,i}|}{\sum_k |a_{c,k}|} C_{\text{tot}}. \quad (22)$$

Each capacitor is sized proportionally to its charge multiplier coefficient. With optimized capacitors, the voltage ripple on each capacitor is set equal in magnitude.

The optimized SSL output impedance (from (21)) thus simplifies to

$$R_{\text{SSL}}^* = \frac{1}{C_{\text{tot}} f_{\text{sw}}} \left(\sum_i |a_{c,i}| \right)^2. \quad (23)$$

These optimization results for the single-voltage technology are very simple to utilize in switched-capacitor converter design.

B. FSL Switch Optimization and Sizing

Like capacitors, the switches in a SC converter can be optimized, yielding dramatic performance increases. This optimization is carried out in the asymptotic fast switching limit where output impedance is directly related to switch conductance. This optimization assumes a duty cycle of 50%.

The switch VA product, summed over all switches, is used as the cost-based metric in this optimization. This V - A metric corresponds to a constraint on the G - V^2 product summed over the switches, for both discrete and integrated switches. Parallel discrete switches increases total conductance, whereas placing switches in series increases voltage blocking while decreasing conductance. To increase voltage blocking without reducing conductance, the number of devices used scales quadratically, motivating the G - V^2 metric.

In an integrated application, the same total G - V^2 constraint applies. The transistor length and nominal voltage scale linearly with process feature size. In addition, switch conductance scales proportionally with transistor width and inversely with transistor length. A cost metric A_{sw} , related to the area (or width multiplied by length) of a specific transistor, can be written as $A_{\text{sw}} = GV^2$ (in units of GV^2 , i.e., S - V^2).

This constraint, applicable to both discrete and integrated transistors, can be expressed as

$$A_{\text{tot}} = \sum_{\text{switches}} G_i (v_{r,i(\text{rated})})^2 \quad (24)$$

where G_i is the conductance of switch i and $v_{r,i(\text{rated})}$ is the rated voltage of switch i . As in the capacitor optimization, $v_{r,i(\text{rated})}$ is the voltage the device can support, not necessarily the voltage it blocks in normal operation. Naturally, the rated voltage must be larger than the nominal blocking voltage.

A Lagrange optimization function $\mathcal{L}$ is formed to minimize the FSL output impedance while satisfying the constraint in (24)

$$\mathcal{L} = \sum_i \frac{(a_{r,i})^2}{G_i} + \lambda \left(\sum_{\text{switches}} G_i (v_{r,i(\text{rated})})^2 - A_{\text{tot}} \right). \quad (25)$$

The first term corresponds to the FSL output impedance (the factor of two in (14) does not affect the optimization) and the second term corresponds to the constraint in (24). The minimization is performed by taking the partial derivative of (25) with respect to G_i and setting it to zero

$$\frac{\partial \mathcal{L}}{\partial G_i} = -\frac{(a_{r,i})^2}{G_i^2} + \lambda (v_{r,i(\text{rated})})^2 = 0. \quad (26)$$

Again, differentiating with respect to λ yields the constraint in (24).

Equation (26) yields a proportionality between G_i and the ratio between the switch's charge multiplier coefficient and its voltage rating. This proportionality, when combined with the G - V^2 constraint in (24), yields an expression for the optimal conductance of each switch

$$G_i^* = \frac{1}{R_i^*} = \left| \frac{a_{r,i}}{v_{r,i}(\text{rated})} \right| \frac{A_{\text{tot}}}{\sum_k |a_{r,k} v_{r,k}(\text{rated})|}. \quad (27)$$

Comparing the optimal conductance G_i to the optimal capacitance in (19) makes it evident that the two optimizations are analogous.

The optimal FSL output impedance is obtained by substituting (27) into (14)

$$R_{\text{FSL}}^* = \frac{2}{A_{\text{tot}}} \left(\sum_i |a_{r,i} v_{r,i}(\text{rated})| \right)^2. \quad (28)$$

Similar to the optimal SSL impedance, the optimal FSL output impedance is related to the square of the sum of the V-A products. This simple form of the optimal output impedance allows the comparison of various SC converter topologies. Several SC converter topologies are compared in Section IV.

Many SC converters use switches with a single voltage rating. For instance, many IC-based converters only use the native NMOS transistors of the process since these transistors perform the best. In addition, topologies such as the ladder converter utilize switches that must all block the same voltage. The switch-cost constraint discussed in the previous section simplifies into a constraint on total switch conductance G_{tot} . The optimal conductance of each switch simplifies to

$$G_i^* = \frac{|a_{r,i}|}{\sum_k |a_{r,k}|} G_{\text{tot}} \quad (29)$$

likewise, when all switches are rated for an identical voltage, the optimal FSL output impedance simplifies to

$$R_{\text{FSL}}^* = \frac{2}{G_{\text{tot}}} \left(\sum_i |a_{r,i}| \right)^2. \quad (30)$$

The performance of a converter is related to the square of the sum of the charge multiplier coefficients. Topologies with a small sum of these coefficients perform better for a given switch conductance than a topology with a large sum of coefficients. In integrated applications or other applications where single-voltage switches must be used, this optimization can be used. A comparison of SC converters based on single-voltage devices is performed in Section IV.

IV. COMPARISON OF SC CONVERTER TOPOLOGIES

A number of SC converter topologies exist in the literature [1], [3]–[5], [10], [11] but the merits of each have never been compared in a methodical way. The optimizations in Sections III-A and III-B can be used to provide a performance comparison among different common SC converter topologies. Fig. 5 shows five converter topologies discussed in the literature.

The first comparison uses the cost metrics in Section III and assumes that devices of every voltage rating are available.

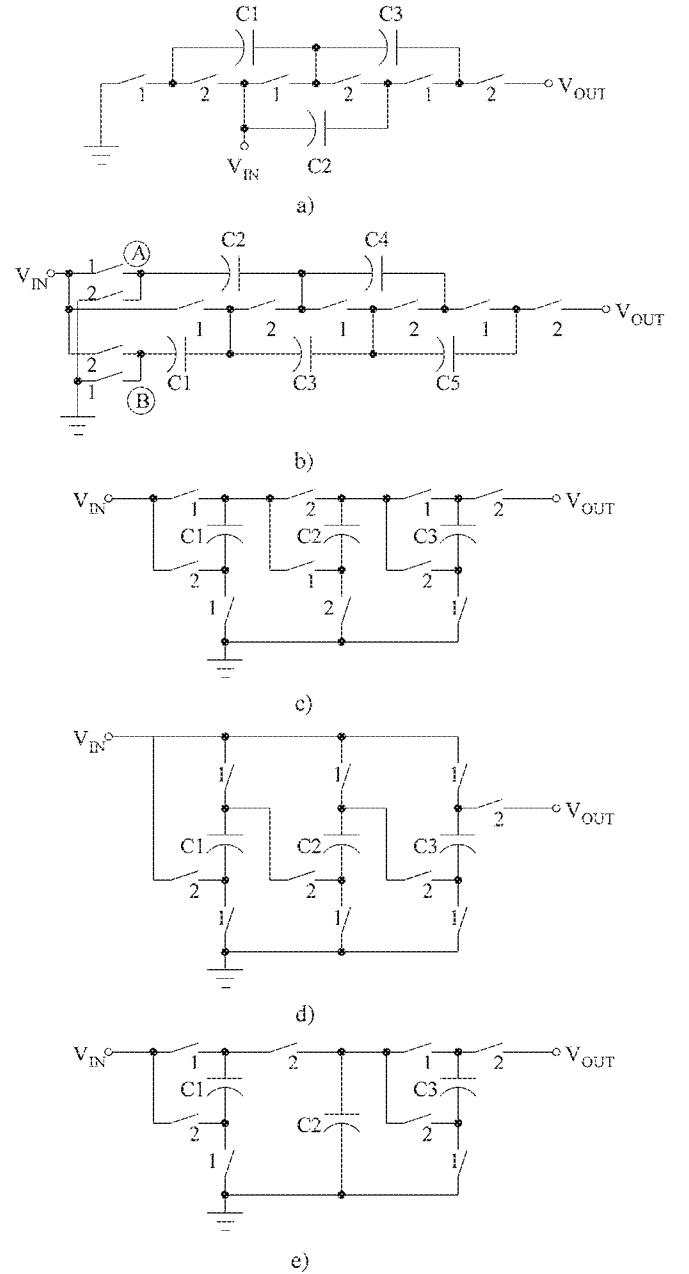


Fig. 5. Five step-up SC converter topologies. (a) Ladder. (b) Cockcroft–Walton Multiplier. (c) Fibonacci. (d) Series-Parallel. (e) Doubler.

Step-up versions of the topologies are considered, as shown in Fig. 5, although step-down versions would yield identical results. The commonly-used Dickson Charge Pump is a simple transformation of the Cockcroft–Walton multiplier, constructed by connecting the negative plate of each capacitor to either node A or B. The capacitors in the Dickson charge pump form two star networks while the capacitors in the Cockcroft–Walton multiplier form two linear strings. The optimal output impedance for all topologies in both asymptotic limits is evaluated for a range of conversion ratios (represented by n).

When evaluating the FSL output impedance, the converters are evaluated on the ratio $(V_{\text{OUT}}^2 / R_{\text{FSL}} A_{\text{tot}})$ (the ratio between the G - V^2 product of the converter and the switch G - V^2 product

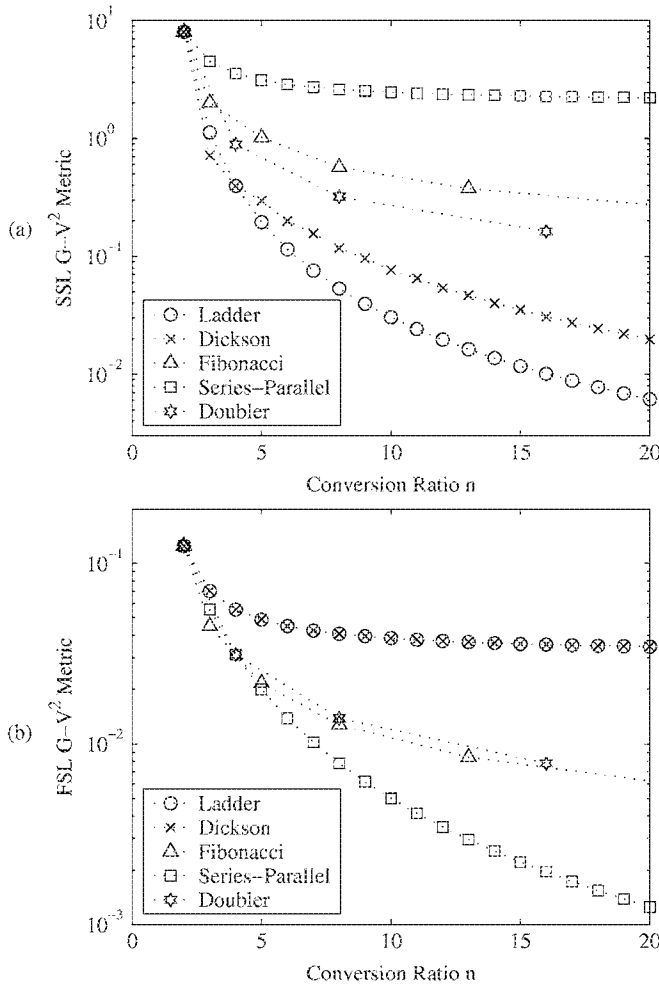


Fig. 6. (a) SSL and (b) FSL performance metrics with optimal-voltage devices.

summed over all switches). For a given cost constraint and conversion ratio, the converter with the highest metric is the one with the lowest FSL output impedance. Likewise, when the SSL output impedance is considered, the converters are evaluated on the ratio $V_{OUT}^2 / (R_{FSL} / E_{tot} f_{sw})$, where a larger metric corresponds to a smaller SSL impedance.

After performing the optimization and comparison, the five topologies are compared in Fig. 6. At a conversion ratio of two, all topologies perform identically. Upon further inspection, for $n = 2$ only, these five topologies are actually identical. Converters that do well in the SSL comparison, such as the series-parallel topology, do poorly in the FSL comparison. Conversely, topologies such as the Cockcroft-Walton multiplier and the Ladder topology that perform well in the FSL comparison typically perform poorly in the SSL comparison. Some converters use capacitors efficiently and others use switches efficiently, but none of these converters are superior in both asymptotes. For converters designed using a capacitor-limited process, a series-parallel topology would work best, while switch-limited designs should use a topology such as the Cockcroft-Walton multiplier or ladder topology.

The exponential converters (where the conversion ratio is exponentially related to the number of capacitors), such as

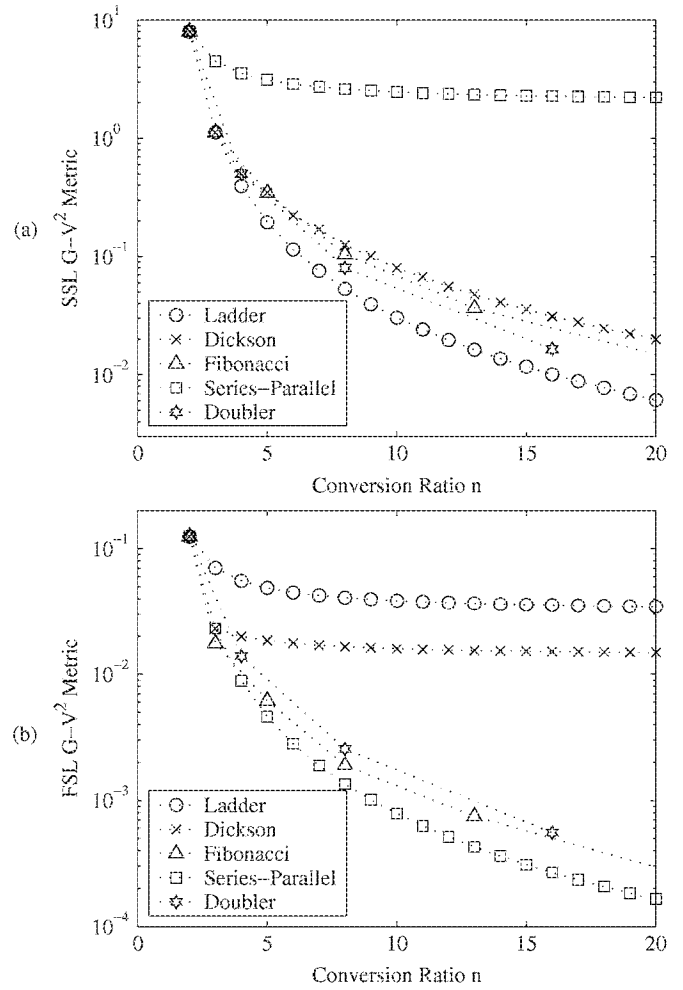


Fig. 7. (a) SSL and (b) FSL performance metrics with single-voltage devices.

the Fibonacci and Doubler topologies, exhibit mediocre performance in both the SSL and FSL comparisons. However, since the switches and capacitors used in their implementations support a range of different voltages and most of the switches are not ground-referenced, practical implementation would be difficult if not impossible.

The second comparison performed assumes that all devices must be of the same voltage rating. In integrated applications using standard CMOS processes, the switches and capacitors are usually all rated for the same voltage. The process is chosen such that this voltage rating corresponds to the maximum voltage seen on any device. However, the switches and capacitors can be rated differently from each other, i.e. if the highest-voltage switch is rated for 1 V, a 1 V process would be used, even if some capacitors support a higher voltage.

The comparative results using identically-rated switches and transistors are shown in Fig. 7. The series-parallel topology is still optimal in the SSL comparison, as all capacitors in that topology also support the same voltage. Likewise, the ladder topology is optimal in the FSL comparison, as all switches in that topology support the same voltage. However, the exponential converters are now relatively poor in both comparisons because they involve a wide range of device stresses, which is impractical in implementation. These comparisons can be used to

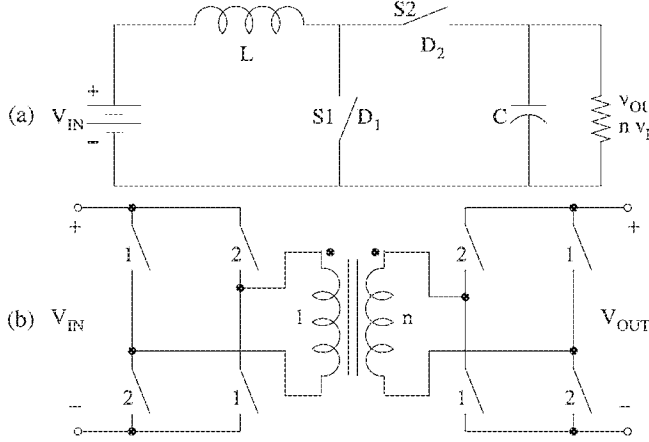


Fig. 8. (a) Standard boost converter. (b) Transformer-bridge converter.

select the best topology for any given application. Reference [8] includes more computational details for these converters.

V. COMPARISON WITH CONVENTIONAL DESIGNS

Switched-capacitor converters have several advantages over conventional inductor-based dc-dc converters. With a switched-capacitor converter, conduction and switching losses are not additional losses, but are already incorporated in the output impedance based losses calculated in Sections II-A and Sections II-B. The only losses that are not included in the output impedance are the gate drive losses, losses associated with parasitic capacitances and control power. Short-circuit (shoot-through) power can be eliminated by the use of sufficiently non-overlapping clocks. Stray capacitances from dynamic nodes must be minimized and their losses incorporated into the efficiency of the converter if the strays are not eliminated. These losses are further considered in [8].

A SC converter and a conventional dc-dc converter can be compared directly when conduction loss is considered. The silicon area (for the switches and control functions) is the dominant cost in many dc-dc converters. A converter with a significantly-lower switch conductance loss may have a cost advantage over a converter with a higher switch loss. For the SC converter, the conduction loss is equal to the loss corresponding to the FSL output impedance. The switch-related loss of an inductor-based converter is made up of conduction losses, due to switch on-state resistance, and switching losses during the switch state transitions. Only the conduction loss will be considered here, using the FSL performance metric $V_{OUT}^2 / (R_{FSL} / A_{tot})$ developed in Section IV.

A ladder-type step-up converter [such as the one in Fig. 5(a)] is considered, as it uses switches most efficiently in the FSL. Two magnetic-based converters are compared, the boost converter and transformer-bridge converter, both shown in Fig. 8. Total switch $G \cdot V^2$ product is held constant for all converters, and the SC converter is assumed to operate in the FSL. Finally, all switches are sized optimally based on the optimization methods presented in this paper. All converters are designed and optimized for a given conversion ratio n , and without loss of generality, an input voltage of 1 V is assumed.

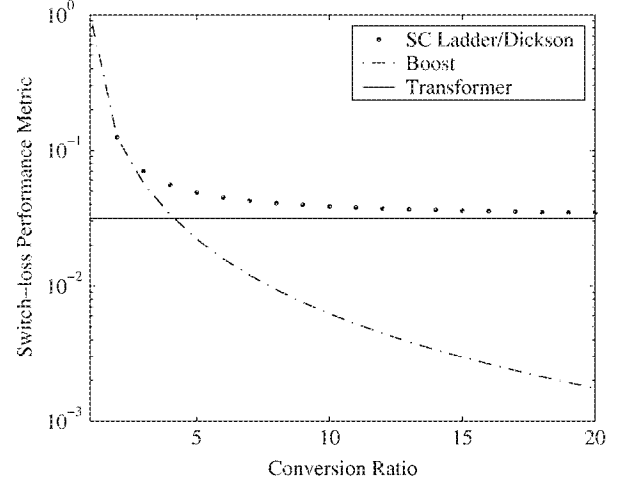


Fig. 9. Performance metric comparison.

The step-up ladder-type SC converter is considered first. All switches in the ladder topology must be rated for 1 V. The lowest two switches in the ladder structure have an a_r component of $(n - 1)$ while the other $2(n - 1)$ switches simply have an a_r component of 1. Thus, the sum of the a_r components is

$$\sum_i |a_{r,i}| = 2(n - 1) + 2(n - 1) = 4(n - 1). \quad (31)$$

The optimal FSL output impedance of this converter (constrained such that $\sum_{\text{switches}} G_i V_i^2 = A_{tot} = 1$) is thus

$$R_{out} = 2 \left(\sum_i |a_{r,i} v_{r,i(\text{rated})}| \right)^2 = 32(n - 1)^2. \quad (32)$$

Computing the ratio of this output resistance to the square of the output voltage yields the performance metric of the ladder circuit, $(n^2) / (32(n - 1)^2)$. This metric is also plotted in Fig. 9.

The boost converter in Fig. 8(a) is operated at duty cycle $D = 1/n$ to achieve a step-up ratio of n . The duty cycle of switch S1 is $D_1 = D = 1/n$ and the duty cycle of switch S2 is $D_2 = 1 - D = (n - 1)/n$. The conduction loss in this circuit is directly computed as

$$P_{cond} = \left(\frac{D}{G_1} + \frac{1 - D}{G_2} \right) I_{in}^2 = R_{out} I_{out}^2. \quad (33)$$

The equivalent loss impedance R_{out} can be directly compared to the output impedance of the SC converter. Optimizing the ratio of the two switch conductances for a given duty cycle, the following constraint can be derived

$$\left(\frac{G_1}{G_2} \right)^2 = \frac{D}{1 - D}. \quad (34)$$

Since the total $G \cdot V^2$ product of the switches is again constrained at one and each switch in the boost converter must be rated for the output voltage of n , the total conductance is restricted to $G_{tot} = 1/n^2$. From this constraint, the equivalent

loss impedance can be determined (note that $D = 1/n$ to achieve the correct conversion ratio):

$$R_{\text{out}} = n^4 \left(\frac{D}{G_1} + \frac{1-D}{G_2} \right) = n^4 \left(1 + 2\sqrt{\frac{n-1}{n^2}} \right). \quad (35)$$

Computing the ratio of the square of the output voltage to this optimal output resistance yields the performance metric of the boost circuit, $1/(n^2(1 + 2\sqrt{(n-1)/(n^2)}))$. This metric is plotted in Fig. 9.

Finally, the transformer-based direct converter in Fig. 8(b) is considered. The transformer is assumed to be ideal and to have an up-conversion ratio of n . The output switches are all identical and must be rated for the output voltage of n V. The on-current of these switches is equal to the output current I_{out} . Likewise, the input switches must be rated for 1 V and conduct a current of nI_{out} . To constrain the total G-V² product equal to one, the output switches must have conductances of $1/8n^2$ and the input switches must have conductances of $1/8$. The conduction loss can then be calculated as

$$P_{\text{cond}} = 2(8n^2)I_{\text{out}}^2 + 2(8)(nI_{\text{out}})^2 = 32n^2I_{\text{out}}^2. \quad (36)$$

The resulting performance metric for this converter is then constant at $1/32$ for any conversion ratio. This makes intuitive sense as only the transformer turns ratio is changed to achieve different conversion ratios.

The conduction losses of the three converters, represented by the performance metric $V_{\text{OUT}}^2/(R_{\text{FSL}}/A_{\text{tot}})$ developed in Section IV, are compared in Fig. 9. The SC and boost converters' performance metrics decrease as the conversion ratio increases, but the SC converter approaches an asymptotic limit at $1/32$ (the same as the transformer-based converter), but the boost converter's performance continues to decrease.

At large conversion ratios, the step-up ladder-type SC converter is significantly superior to the boost converter as the switches in the ladder topology block only the input voltage and most switches carry less than the input current. However, the boost converter's switches carry the full input current and block the full output voltage. Even though the SC converter has many more switches, the low working V-A product of these switches yields a lower conduction loss than that of the boost converter, with its much higher working V-A product switches.

In an application where switches are the limiting factor in performance or cost, switched-capacitor converters are evidently advantageous over conventional magnetics-based dc-dc converters at high or moderate conversion ratios.

VI. VERIFICATION BY SIMULATION

As this analysis method is based on idealized devices, circuit-level simulation, through SPICE or spectre is appropriate for verification of this analysis. Ideal capacitor and voltage-controlled resistances are used in the simulation. Parasitics, while an important consideration in real-world implementations, are not considered in this paper, and are not considered in the verification simulation.

Five step-up switched-capacitor converters have been simulated over a range of switching frequencies. By varying the

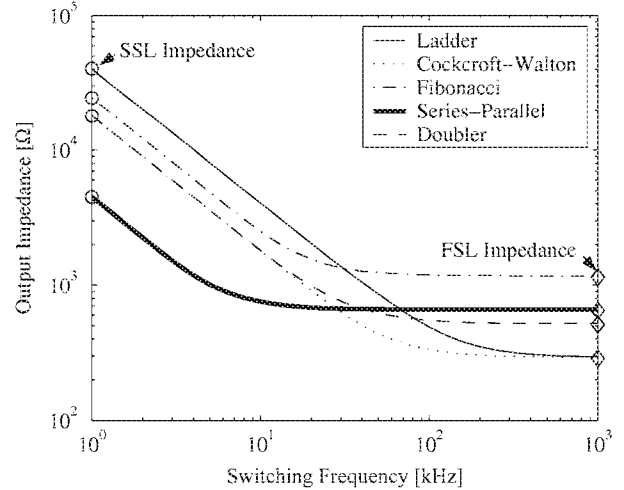


Fig. 10. Simulated output impedance versus switching frequency.

TABLE I
SSL AND FSL VA-PRODUCTS AND IMPEDANCES
FOR FIVE CONVERTERS (R_{SSL} AT 1 KHz)

	$\sum a_c v_c $	$\sum a_r v_r $	R_{SSL}	R_{FSL}
1:4 Ladder	9	12	40.5 kΩ	288 Ω
1:4 Cockcroft-Walton	6	12	18 kΩ	288 Ω
1:5 Fibonacci	7	24	24.5 kΩ	1152 Ω
1:4 Series-Parallel	3	18	4.5 kΩ	648 Ω
1:4 Doubler	6	16	18 kΩ	512 Ω

switching frequency, both the SSL and FSL output impedances can be determined and compared with those from the mathematical analysis. The converters simulated are the 1:4 (i.e., 1 V input to 4 V output) ladder converter, 1:4 Cockcroft-Walton multiplier, 1:4 doubler, 1:5 Fibonacci converter, and a 1:4 series-parallel converter, all shown in Fig. 5. The capacitors and switches in all converters are optimized using the methods in this paper using a switch GV² product of 1 VA and a total capacitor energy of 1 μJ. The output impedance was determined by measuring the current transfer between the input and output voltage sources in a transient simulation.

The charge-multiplier-voltage products and output impedances of the five converters found via the methods in Section II are shown in Table I. These calculations and detailed converter analysis are given in [8]. Fig. 10 shows the simulated impedance of the converters between 100 Hz and 1 MHz. The symbols in the plots indicate the calculated FSL and SSL impedances, showing a match between the theoretical and simulated values.

Five very different switched-capacitor converters were simulated for a range of frequencies encompassing both the SSL asymptote and the FSL asymptote. The simulation results verify that the analysis methods developed in Section II determine the correct output impedance in both asymptotes for all five converters. This simulation verifies that the methods developed in this paper accurately predict the performance of any two-phase SC converter.

VII. EXPERIMENTAL VERIFICATION

An ultra-low-power switched-capacitor power conversion integrated circuit (IC) has been fabricated and tested with design

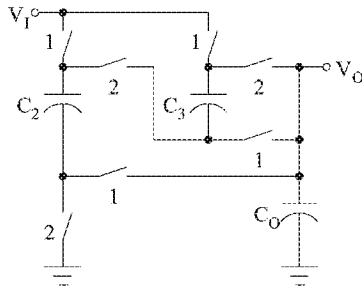


Fig. 11. 3:2 Series-parallel converter topology.

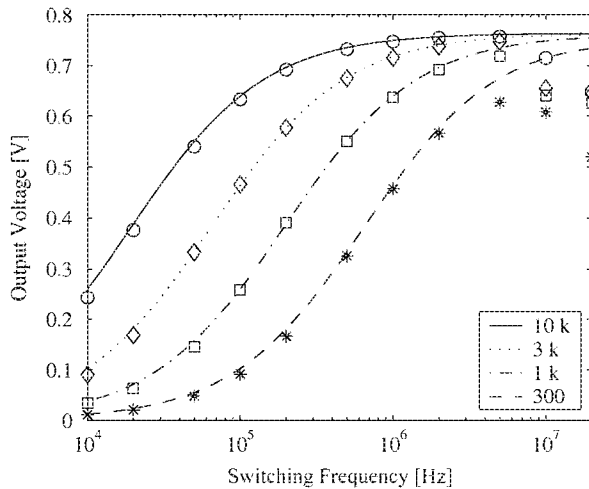


Fig. 12. Measured and calculated output voltage versus load and switching frequency.

guidance from the analysis method developed here. The design and experimental results of this IC are presented in [12]. The results from the 3:2 converter (1.2 V to 0.8 V) is presented here as an example circuit further verifying the calculations for a single converter topology.

A schematic of this converter is shown in Fig. 11. It is similar to the series-parallel circuit in that the capacitors are placed in series in one phase and parallel in the other phase. Capacitors C_1 and C_2 both have a charge multiplier of $1/3$ in this circuit and capacitor C_0 is ignored as it is in parallel with the output. Each of the seven switches also has a charge multiplier of $1/3$. In the IC, each capacitor has a value of 1.15 nF and each switch has a resistance of 4.75Ω . Thus, the predicted output impedances in both limits, calculated using (9) and (14), are

$$R_{SSL} = \frac{2(1/3)^2/1.15\text{nF}}{f_{sw}} = \frac{193 \text{ M}\Omega}{f_{sw}} \quad (37)$$

$$R_{FSL} = 2 \cdot 7(4.75(1/3)^2) = 7.39 \Omega. \quad (38)$$

Fig. 12 shows the output of the 3-2 converter for various switching frequencies and resistive loads. The plotted curves indicate the ideal output voltage calculated by considering a resistor divider between the load resistance and converter output impedance. The data indicate that the model accurately predicts the converter performance for low frequencies. At high

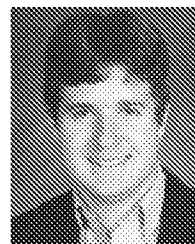
switching frequencies, parasitic losses hurt the converter's performance. The most influential parasitics, in this case the capacitor bottom-plate capacitance and drain-source capacitance, need to be carefully considered in any realization of switched-capacitor converters.

VIII. CONCLUSION

An analysis method has been presented to determine the performance of any switched-capacitor power converter using easily-determined charge multiplier vectors. The capacitors and semiconductor switches of the converter were optimized to minimize output impedance for several conditions and constraints. Five separate converter topologies were considered for their effectiveness in utilizing capacitors and switches. This comparison allows the use of an optimal topology suited to its application and implementation technology. Significantly, the performance (based on conduction loss) of a ladder-type converter was found to be superior to that of a conventional boost converter for medium to high conversion ratios.

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Table 1: Comparison with CCMPA and general CMPA

	General CMPA	Linear CCMPA	Circular CCMPA
Return losses	-33.8 dB	-29 dB	-15.15 dB
-10 dB return loss bandwidths	55.5 MHz (3.5%)	71 MHz (4.5%)	140 MHz (8.9%)
Gains at 0°	7.1 dBd	6.2 dBd	2.8 dBd
-3 dB beamwidths	<i>E</i> -plane	72°	61°
	<i>H</i> -plane	76°	72°
Axial ratio			1.3 dB (1.575 GHz)
Patch diameter	99 mm	91 mm (8.08%)	92 mm (7.07%)

Conclusion: A corrugated antenna showing a reduction of 8.08% in patch diameter and 21.12% in patch area by maintenance of the fundamental mode for general CMPA as well as shortening of current length has been reported. Each linearly-polarised CCMPA and circularly-polarised CCMPA has the improved characteristic of 71 MHz (4.5%) and 144 MHz (8.9%) in comparison with the general CMPA in the aspect of -10 dB return loss bandwidth at 1.575 GHz. Therefore, the CCMPA proposed in this Letter has been confirmed to be appropriate for the miniaturisation of antennas.

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Generalised analysis of switched-capacitor step-down quasi-resonant converter

Y.P.B. Yeung, K.W.E. Cheng, S.L. Ho and D. Sutanto

Conventional switched-capacitor converter circuits have a high current stress because of the short circuit when charging and discharging the capacitors. A novel switched-capacitor quasi-resonant step-down converter family with a generalised analysis that can improve the current stress problem is presented. Measured efficiency is high and all the circuits can operate under zero-current switching.

Introduction: Switched-capacitor DC-DC converters [1] use capacitors only for energy storage, therefore the main advantage is that it is not necessary to include an inductor in the design and circuit size can be reduced. The fabrication of the converter in an integrated circuit is also possible. However, the main drawback of the circuit is the momentary short circuit when charging and discharging the capacitor; this generates switching loss and electromagnetic interference. A zero-current switching method for the switched-capacitor quasi-resonant (SCQR) converter was proposed in [2]. In this Letter we describe extending the principle of the switched-capacitor resonant converter into a generalised method. A switching-capacitor cell is proposed to the switched-capacitor converter and hence a series of circuits of various conversion ratios and zero-current switching of all switches can be obtained.

Concepts: The proposed SCQR converters are shown in Fig. 1 and 2 where the circuits give the output voltage equal to $1/3$ and $1/n$ of input voltage, respectively. The output voltages of the proposed family of converters are dependent on the number of switched-capacitor cells, which is formed by C_2 , D_a , D_b , D_c as shown in Fig. 1. For other fractional output, this cell can be extended to the $1/n$ version as shown in Fig. 2. Each capacitor C_i ($i > 1$) in the switched-capacitor cells is for sharing the input voltage together in series and releasing the energy in parallel with C_1 to the load where the diodes decide the direction of the current flow to charge the capacitors. For a converter with $1/n$ conversion ratio ($1/n$ mode converter), there should be $(n-2)$ switched-capacitor cells. L_r is to ensure zero-current switching, and the current of all the transistors must pass through L_r .

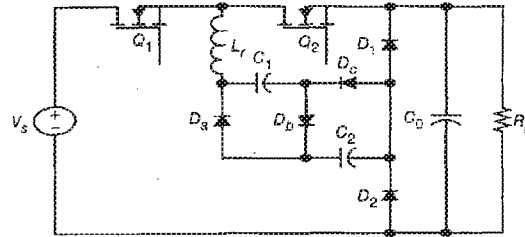


Fig. 1 $1/3$ mode SCQR converter

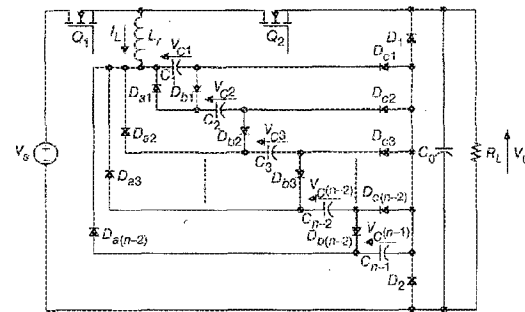


Fig. 2 $1/n$ mode zero-current switching SCQR converters

Principles of operations: The $1/3$ mode SCQR converter is analysed. There are a total of four states of operation in each switching period. Fig. 3 shows the computer simulation waveforms of the converter. Assuming the output capacitor C_o is very large to keep the output voltage constant, the capacitance of C_1 is equal to that of C_2 (called C), and there is no power loss in the circuit, the analysis of each state is as following.

(a) *States of operations:* (i) *State I* [$t_0 - t_1$]: At t_0 , Q_1 is switched on while Q_2 is switched off. C_1 and C_2 are charged in series with D_b , D_1 , L_r and the load. At this moment, C_1 and C_2 resonate with L_r in series. The charging current of the capacitors resonates from zero at t_0 to the peak and is back to the zero at t_1 in sinusoidal waveform so that Q_1 is switched on under zero-current switching condition. The equations of this state are:

$$V_s = L_r \frac{di_L}{dt} + (n-1)v_C + V_0 \quad (1)$$

$$i_L = C \frac{dv_C}{dt} \quad (2)$$

where V_s is the input voltage, V_0 the output voltage, v_C the resonant capacitors (C_1 or C_2) voltage, i_L the resonant inductor (L_r) current, and n represents the $1/n$ version of the circuits.

(ii) *State II* [$t_1 - t_2$]: Q_1 keeps to be on and Q_2 is still off. At t_1 , the inductor current reaches zero and the diodes D_b and D_1 stop the resonance. Both the inductor and input currents are equal to zero. C_0 is discharged to the load. v_C stays constant and $i_L = 0$.

(iii) *State III* [$t_2 - t_3$]: Q_1 is switched off and Q_2 is switched on at t_2 . C_1 and C_2 are discharged in parallel to L_r and the load. D_a , D_c and D_2 are conducting. These two capacitors resonate with L_r in sinusoidal wave which is similar to State I. Since the current of Q_1 is zero at t_2 , Q_1 is switched off in zero-current switching condition while Q_2 is zero-current switched on. At t_3 , the inductor current is resonated to be zero.

The equations of this state are:

$$v_C + L_r \frac{di_L}{dt} = V_0 \quad (3)$$

$$i_L = (n-1)C \frac{dv_C}{dt} \quad (4)$$

(iv) *State IV* [$t_3 - t_4$]: Similar to State II, the diodes stop the resonance at t_3 . Inductor current is equal to zero and only C_0 is discharged to the load. v_C stays constant and $i_L = 0$.

(b) *General equations of 1/n mode converter*: The above equations can be solved by using input and output power balancing, and the continuity of the four states and the inductor current.

For State I,

$$V_S \int_0^{T_S} i_L dt = V_0 I_0 T_S \quad (5)$$

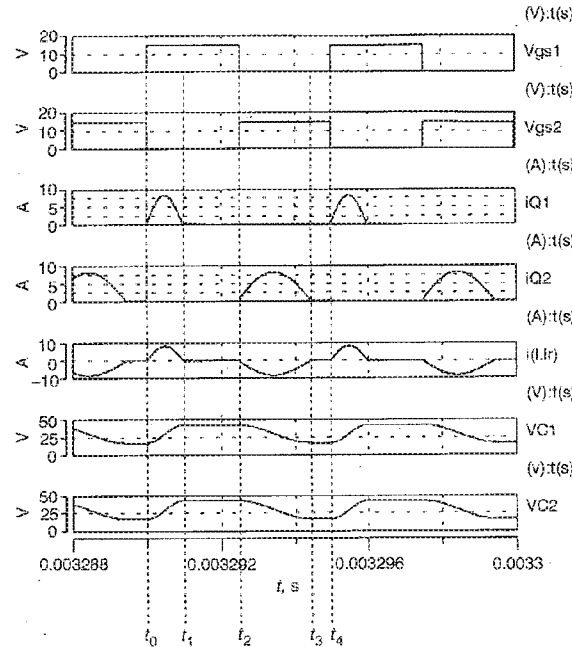


Fig. 3 Simulation waveforms of 1/3 mode SCQR converter

By solving (5), and for a 1/n mode converter,

$$v_C = V_0 - \frac{\pi I_0 T_S Z_0}{n(n-1)T_0} \cos \omega_0(t - t_0) \quad (6)$$

$$i_L = \frac{\pi I_0 T_S}{nT_0} \sin \omega_0(t - t_0) \quad (7)$$

where

$$\omega_0 = \sqrt{\frac{n-1}{L_r C}} \quad \text{and} \quad Z_0 = \sqrt{\frac{(n-1)L_r}{C}} \quad (8)$$

T_S is the switching frequency, T_0 is the resonant period in State I, and I_0 is the output current.

In State II,

$$v_C = V_0 + \frac{\pi I_0 T_S Z_0}{n(n-1)T_0} \quad (9)$$

In State III,

$$v_C = V_0 + \frac{(n-1)\pi I_0 T_S Z_1}{nT_1} \cos \omega_1(t - t_2) \quad (10)$$

$$i_L = -\frac{(n-1)\pi I_0 T_S}{nT_1} \sin \omega_1(t - t_2) \quad (11)$$

where

$$\omega_1 = \sqrt{\frac{1}{(n-1)L_r C}} \quad \text{and} \quad Z_1 = \sqrt{\frac{L_r}{(n-1)C}} \quad (12)$$

where T_1 is the resonant period of State III.

In State IV,

$$v_C = V_0 - \frac{(n-1)\pi I_0 T_S Z_1}{nT_1} \quad (13)$$

Experimental results: The 1/3 mode SCQR converter has been tested. The specification of the prototype is input voltage = 90 V, output voltage = 30 V, switching frequency = 200 kHz, $C_1 = C_2 = 0.22 \mu\text{F}$, $C_0 = 50 \mu\text{F}$ and $L_r = 1 \mu\text{H}$. The measured waveforms, the gate-source voltage, the drain current and drain-source voltage of Q_1 and Q_2 , of the converter in 80 W output power are shown in Fig. 4. From Figs. 4a and 4b it is clear that two transistors are operated under zero-current switching. The measured output voltage of the converter in 80 W operation is 25.8 V and is slightly lower than the theoretical output voltage of 30 V because of the loss. The measured efficiency varied between 80 and 93% when the load power varies between 5 and 80 W, respectively.

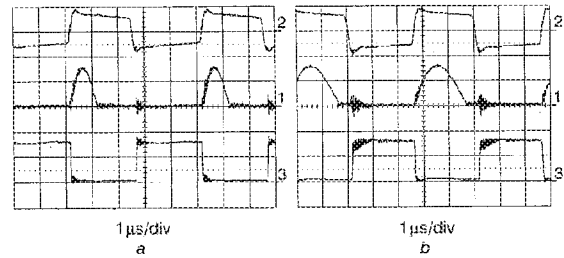


Fig. 4 Experimental waveforms of converters

a Upper trace: gate signal Q_1 , 20 V/div; middle trace: Q_1 current, 5 A/div; lower trace: $V_{ds} Q_1$, 40 V/div
b Upper trace: gate signal Q_1 , 20 V/div; middle trace: Q_1 current, 5 A/div; lower trace: $V_{ds} Q_1$, 40 V/div

Conclusion: A generalised analysis of switched-capacitor circuits is presented. They are operated under zero-current switching; the inherited disadvantage of high pulsational current has been eliminated, and the number of switches needed is low. The main contribution is that a number of fractional output voltage conversion zero-current switching converters and a generalised analysis has been produced. Experimental results confirmed that the operation agrees with the theoretical prediction. All the transistors can achieve zero-current switching, and efficiency is high.

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Zero-current switching switched-capacitor quasiresonant step-down converter

Y.P.B. Yeung, K.W.E. Cheng, D. Sutanto and S.L. Ho

Abstract: Current stress is usually high for conventional switched-capacitor converter circuits, because of short-circuiting when charging up the capacitors. In this paper, two zero-current switching switched-capacitor resonant step-down converter families are presented which can improve the current stress problem. They are able to provide $1/2$ to $1/n$, and -1 to $-1/n$ conversion ratios respectively by using only two switches. Principles of operation, computer simulation and experimental results of the proposed converters are presented. The measured efficiency is high and all parts of the circuits are under zero-current switching conditions.

1 Introduction

The switched-capacitor DC-DC converter [1–6] uses capacitors only for energy storage. Its novel feature is that only capacitors and switches are required in the power stage. The main advantage of this kind of converter is that no magnetic component is required for storing energy, making it possible to fabricate a smaller size and lighter weight converter in an integrated circuit.

The drawback of the switched-capacitor DC-DC converter is that the voltage conversion ratio is fixed. If the voltage conversion ratio of the circuit changes widely, more switches, capacitors and diodes are required. This means that more gate drive circuits will be required for the extra switches. Also, very high current spikes can be observed when the switched capacitors are being charged from the voltage source causing high EMI.

Another shortcoming of the switched-capacitor converter is voltage regulation. Because it is based on shorting switched-capacitors to a certain voltage, the voltage output depends on the number of switched-capacitor stages hence it is theoretically difficult to regulate. One way to overcome this is to use the equivalent series resistance (ESR) of the capacitor and the resistance of the transistor when it is on as an RC charging and discharging control [3]. The drawback of such a method is an increase in switching losses since the transistor does not operate in the saturation region [6]. These converters are therefore usually used only for low power and operating at lower efficiency (80% or less). The efficiency also deteriorates seriously as the switching frequency increases.

A family of switched-capacitor DC-DC converters, which can solve the high current spike problem has been proposed [2]. All the converters in this family use zero-current switching [7]. There is only a very small inductor in each circuit which can resonate with the switched capacitor, so the charging current is limited. The turn-off switching loss

can also be eliminated. However, this family of converters only includes the circuits which provide 0.5, -1 and 2 voltage conversion ratios.

In this paper, two switched-capacitor step-down DC-DC resonant converters are presented: noninverting mode and inverting mode. These two circuits are based on the inverting mode and the half mode converters [2], which are shown in Figs. 1 and 2. A simple switched-capacitor cell is proposed. This switched-capacitor cell consists of only one capacitor and three diodes. By adding this switched-capacitor cell to the converters, the voltage conversion ratios of the converters can decrease fractionally. No other

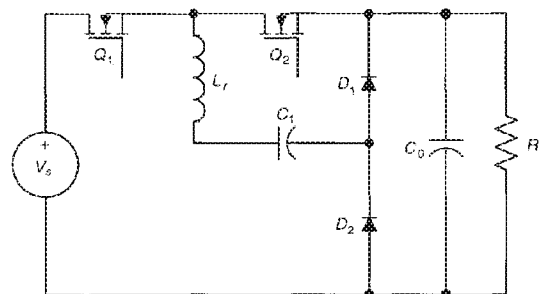


Fig. 1 Conventional half-mode zero-current switching switched-capacitor resonant converter

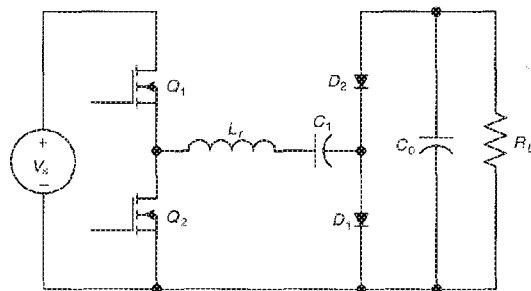


Fig. 2 Conventional inverting mode zero-current switching switched-capacitor resonant converter

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switch is required to change the voltage conversion ratio. Also, all the circuits of these two families are operated under zero-current switching conditions. They have no current spike problems and can operate at high frequency with high efficiency. Therefore, these converters have the potential to be used where DC isolation is unnecessary and fixed conversion is needed.

2 Concepts of proposed step-down switched-capacitor resonant converters

The proposed noninverting mode step-down switched-capacitor resonant converters are shown in Figs. 3–5 and the inverting mode step-down switched-capacitor resonant converters are shown in Figs. 6–8, where the circuits in Figs. 3–8 give an output voltage equal to $1/3$, $1/4$, $1/n$, $-1/2$, $-1/3$ and $-1/n$ input voltage, respectively.

The output voltages of both noninverting mode and inverting mode converters are dependent on the number of switched-capacitor cells, shown in Fig. 9, applied on the converter circuit. Referring to Figs. 3–5, in each cell for the proposed converters, ports a and b are connected across the capacitor of the former cell. Port c is connected to the node between D_1 and D_2 . Port d is connected to the anode of the D_b of the next cell. Port e is connected to the cathode of D_b of the former cell. Port f is connected to L_r .

Each capacitor in the switched-capacitor cells shares the input voltage in series and releases the energy in parallel with C_1 to the load where the diodes are used to decide the direction of the current flow which charges the capacitors.

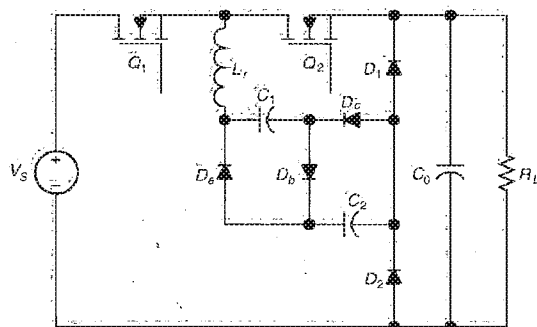


Fig. 3 Proposed noninverting zero-current switching switched-capacitor resonant step-down converter
1/3 mode converter

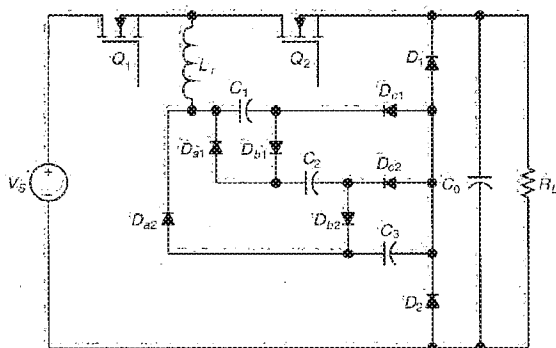


Fig. 4 Proposed noninverting zero-current switching switched-capacitor resonant step-down converter
1/4 mode converter

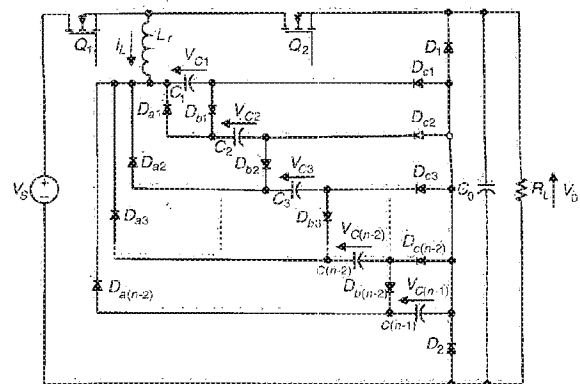


Fig. 5 Proposed noninverting zero-current switching switched-capacitor resonant step-down converter
1/n mode converter

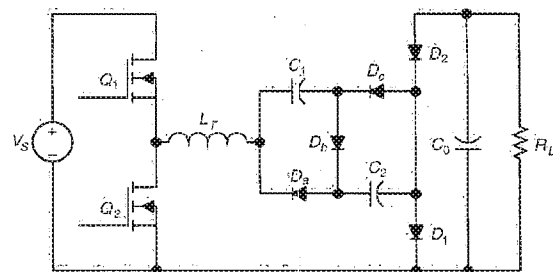


Fig. 6 Proposed inverting zero-current switching switched-capacitor resonant step-down converter
1/2 mode converter

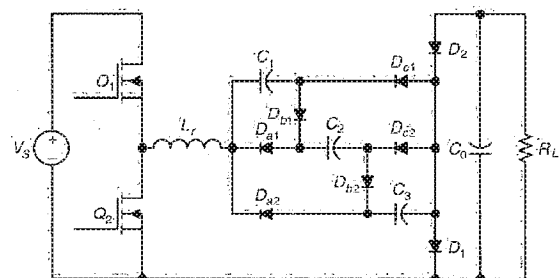


Fig. 7 Proposed inverting zero-current switching switched-capacitor resonant step-down converters
-1/3 mode converter

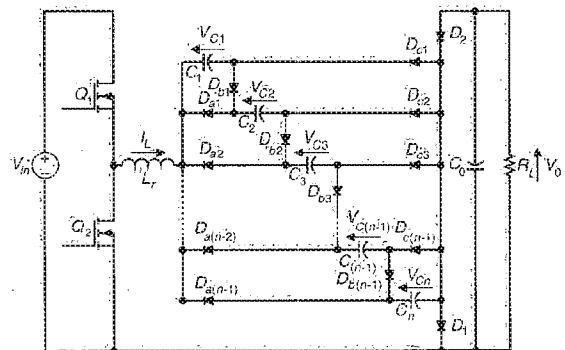


Fig. 8 Proposed inverting zero-current switching switched-capacitor resonant step-down converters
-1/n mode converter

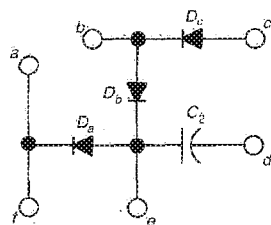


Fig. 9 Switched-capacitor cell

Using this method, the converter can step down the input voltage fractionally. For a converter with $1/n$ voltage conversion ratio ($1/n$ mode converter), there should be $(n-2)$ switched-capacitor cells.

3 Principles of operations

3.1 Noninverting mode step-down resonant converters

The noninverting $1/3$ mode switched-capacitor resonant step-down converter is analysed. There are four states of operation in each switching period. Figs. 10–13 shows the equivalent circuits of each state of operations of the noninverting $1/3$ mode converter and Fig. 14 shows the computer simulation waveforms of the converter. Assuming the output capacitor C_0 is very large to keep the output voltage constant, the capacitance of C_1 is equal to that of C_2 (called C), and there is no power loss in the circuit, the analysis of each state is given below.

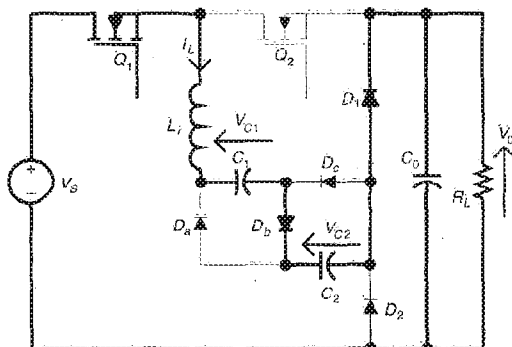


Fig. 10 State I of noninverting $1/3$ mode switched-capacitor resonant step-down converter

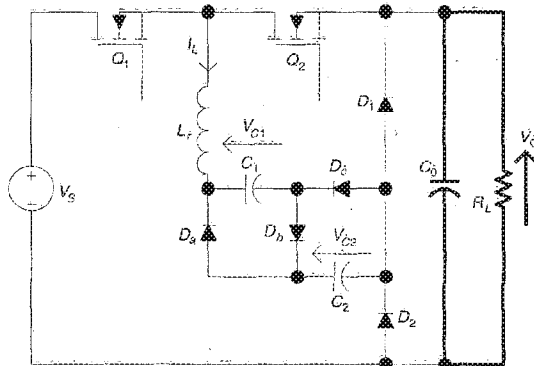


Fig. 11 State II of noninverting $1/3$ mode switched-capacitor resonant step-down converter

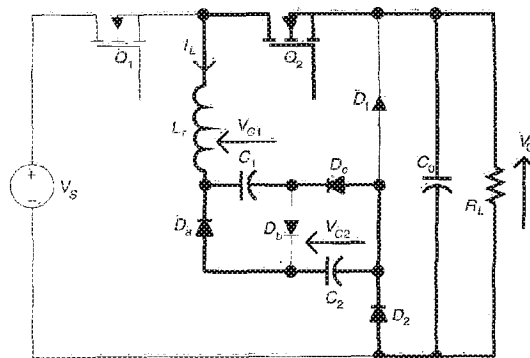


Fig. 12 State III of noninverting $1/3$ mode switched-capacitor resonant step-down converter

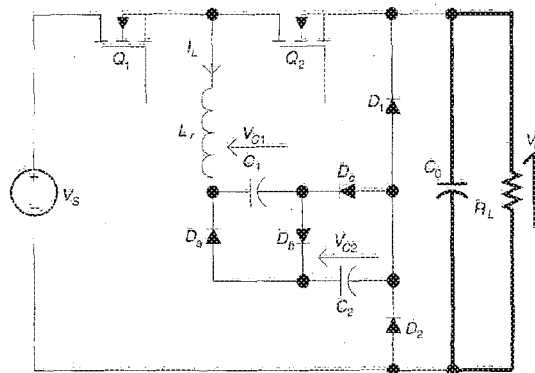


Fig. 13 State IV of noninverting $1/3$ mode switched-capacitor resonant step-down converter

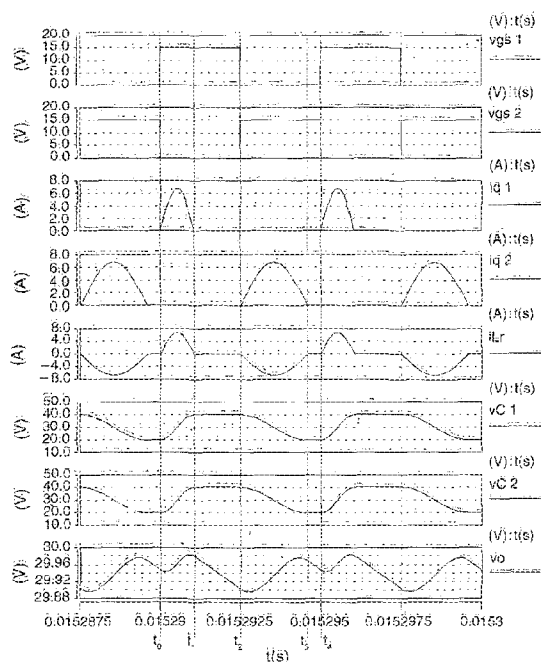


Fig. 14 Idealised simulation waveforms of the noninverting $1/3$ mode switched-capacitor resonant step-down converter

3.1.1 Operating states

(i) State I [t_0-t_1]

At t_0 , Q_1 is switched on while Q_2 is switched off. C_1 and C_2 are charged in series with D_b , D_1 , L_r and the load. At this moment, C_1 and C_2 resonate with L_r in series. The charging current of the capacitors resonates from zero at t_0 to the peak and the back to the zero at t_1 in sinusoidal waveform so that Q_1 is switched on under the zero-current switching condition. The equations of this state are

$$V_S = L_r \frac{di_L}{dt} + 2v_C + V_0 \quad (1)$$

$$i_L = C \frac{dv_C}{dt} \quad (2)$$

where V_S is the input voltage, V_0 is the output voltage, V_C is the resonant capacitors (C_1 or C_2) voltage, and i_L is the resonant inductor (L_r) current.

Let V_{C0} be the value of V_C at t_0 , the solutions of (1) and (2) are

$$v_C = \frac{V_S - V_0}{2} - \left(\frac{V_S - V_0 - 2V_{C0}}{2} \right) \cos \omega_0(t - t_0) \quad (3)$$

$$i_L = \left(\frac{V_S - V_0 - 2V_{C0}}{Z_0} \right) \sin \omega_0(t - t_0) \quad (4)$$

where the resonant angular frequency of this state is

$$\omega_0 = \sqrt{\frac{2}{L_r C}} \quad (5)$$

The resonant impedance of this state is

$$Z_0 = \sqrt{\frac{2L_r}{C}} \quad (6)$$

(ii) State II [t_1-t_2]

Q_1 remains on and Q_2 remains off. At t_1 , the inductor current reaches zero and the diodes D_b and D_1 stop the resonance. Both the inductor current and input current are equal to zero. C_0 is discharged to the load. The equations of this state are

$$v_C = V_{C1} \quad (7)$$

$$i_L = 0 \quad (8)$$

where V_{C1} is the voltage of C at t_1 .

(iii) State III [t_2-t_3]

Q_1 is switched off and Q_2 is switched on at t_2 . C_1 and C_2 are discharged in parallel to L_r and the load. D_a and D_2 and D_2 are conducting. These two capacitors resonate with L_r in sinusoidal manner, which is similar to state I. Since the current of Q_1 is zero at t_2 , Q_1 is switched off in the zero-current switching condition while Q_2 is zero-current switched on. At t_3 , the inductor current is resonated to zero. The equations of this state are

$$v_C + L_r \frac{di_L}{dt} = V_0 \quad (9)$$

$$i_L = 2C \frac{dv_C}{dt} \quad (10)$$

The solutions of (9) and (10) are

$$v_C = V_0 - (V_0 - V_{C2}) \cos \omega_1(t - t_2) \quad (11)$$

$$i_L = \frac{V_0 - V_{C2}}{Z_1} \sin \omega_1(t - t_2) \quad (12)$$

where V_{C2} is the voltage of C at t_2 .

The resonant angular frequency of this state is

$$\omega_1 = \sqrt{\frac{1}{2L_r C}} \quad (13)$$

The resonant impedance of this state is

$$Z_1 = \sqrt{\frac{L_r}{2C}} \quad (14)$$

(iv) State IV [t_3-t_4]

Similar to state II, the diodes stop the resonance at t_3 . The inductor current is equal to zero and only C_0 is discharged to the load. The equations of this state are

$$v_C = V_{C3} \quad (15)$$

$$i_L = 0 \quad (16)$$

where V_{C3} is the voltage of C at t_3 .

3.1.2 Generalised equations of 1/3 mode converter

By using input and output power balancing

$$V_S I_m = V_0 I_0 \quad (17)$$

For state I

$$V_S \int_{t_0}^{t_1} i_L dt = V_0 I_0 T_S \quad (18)$$

By solving (18), and assuming the voltage conversion ratio is 1/3, then

$$v_C = V_0 - \frac{\pi I_0 T_S Z_0}{6 T_0} \cos \omega_0(t - t_0) \quad (19)$$

$$i_L = \frac{\pi I_0 T_S}{3 T_0} \sin \omega_0(t - t_0) \quad (20)$$

where T_S is the switching frequency, T_0 is the resonant period in state I, and I_0 is the output current.

For state II

$$v_C = V_0 + \frac{\pi I_0 T_S Z_0}{6 T_0} \quad (21)$$

$$i_L = 0 \quad (22)$$

For state III

$$V_S \int_{t_2}^{t_3} i_L dt = V_0 (I_0 - I_m) T_S \quad (23)$$

By solving (23)

$$v_C = V_0 + \frac{2\pi I_0 T_S Z_1}{3 T_1} \cos \omega_1(t - t_2) \quad (24)$$

$$i_L = -\frac{2\pi I_0 T_S}{3 T_1} \sin \omega_1(t - t_2) \quad (25)$$

where T_1 is the resonant period of state III.

For state IV,

$$v_C = V_0 - \frac{2\pi I_0 T_S Z_1}{3 T_1} \quad (26)$$

$$i_L = 0 \quad (27)$$

3.1.3 Generalised equations of 1/n mode converter

For a 1/n mode converter, in state I

$$v_C = V_0 - \frac{\pi I_0 T_S Z_0}{n(n-1)T_0} \cos \omega_0(t - t_0) \quad (28)$$

$$i_L = \frac{\pi I_0 T_S}{n T_0} \sin \omega_0 (t - t_0) \quad (29)$$

where

$$\omega_0 = \sqrt{\frac{n-1}{L_r C}} \quad (30)$$

$$Z_0 = \sqrt{\frac{(n-1)L_r}{C}} \quad (31)$$

In state II

$$v_C = V_0 + \frac{\pi I_0 T_S Z_0}{n(n-1)T_0} \quad (32)$$

$$i_L = 0 \quad (33)$$

In state III

$$v_C = V_0 + \frac{(n-1)\pi I_0 T_S Z_1}{n T_1} \cos \omega_1 (t - t_2) \quad (34)$$

$$i_L = -\frac{(n-1)\pi I_0 T_S}{n T_1} \sin \omega_1 (t - t_2) \quad (35)$$

where

$$\omega_1 = \sqrt{\frac{1}{(n-1)L_r C}} \quad (36)$$

$$Z_1 = \sqrt{\frac{L_r}{(n-1)C}} \quad (37)$$

In state IV

$$v_C = V_0 - \frac{(n-1)\pi I_0 T_S Z_1}{n T_1} \quad (38)$$

$$i_L = 0 \quad (39)$$

3.2 Inverting mode step-down resonant converters

Each state of the operation of the inverting $-1/2$ mode step-down converter is analysed. The equivalent circuits of each state of the converter are shown in Figs. 15–18. Fig. 19 shows the simulation waveforms of this converter. Assuming the converter is lossless and the output filtering capacitor, C_o , is very large to maintain a constant output voltage, the calculations of the converter are given below:

3.2.1 Operating states

(i) State I [$t_0 - t_1$]

Q_1 is switched on and Q_2 is switched off. The current passing through L_r , D_b and D_1 from the source is charging

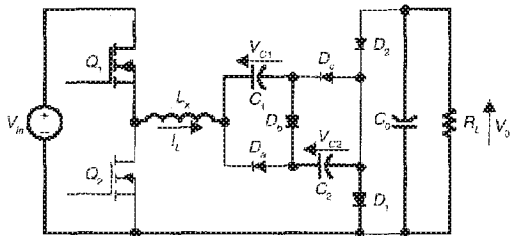


Fig. 15 State I of inverting $-1/2$ mode switched-capacitor resonant step-down converter

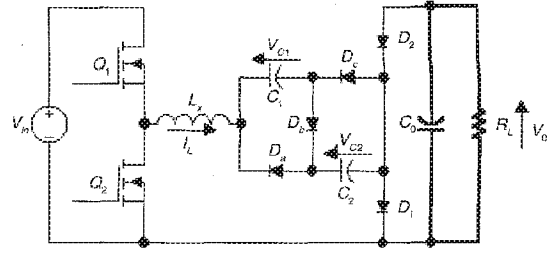


Fig. 16 State II of inverting $-1/2$ mode switched-capacitor resonant step-down converter

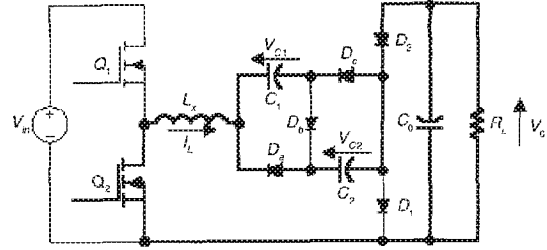


Fig. 17 State III of inverting $-1/2$ mode switched capacitor resonant step-down converter

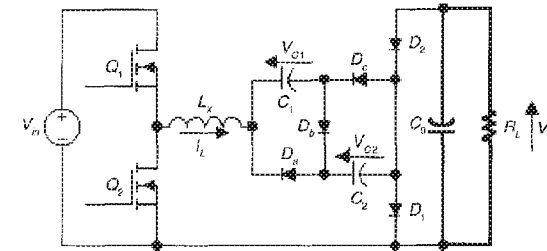


Fig. 18 State IV of inverting $-1/2$ mode switched capacitor resonant step-down converter

the capacitors C_1 and C_2 in series. The output capacitor C_o is discharged to the load. The inductor current i_L starts resonating with C_1 and C_2 in sinusoidal manner so that zero-current switching-on can be achieved. The resonant inductor current increases from zero at t_0 and then decreases from the peak. It reaches zero again at t_1 . Since Q_2 is switched off at t_0 while the current is zero, Q_2 is switched off under zero-current switching conditions. The equations of this state are

$$V_S = L_r \frac{di_L}{dt} + 2v_C \quad (40)$$

$$i_L = C \frac{dv_C}{dt} \quad (41)$$

By solving (40) and (41)

$$v_C = \frac{V_S}{2} - \frac{V_S - 2V_{C0}}{2} \cos \omega_0 (t - t_0) \quad (42)$$

$$i_L = \frac{V_S - 2V_{C0}}{Z_0} \sin \omega_0 (t - t_0) \quad (43)$$

where V_{C0} is the voltage of C at t_0 .

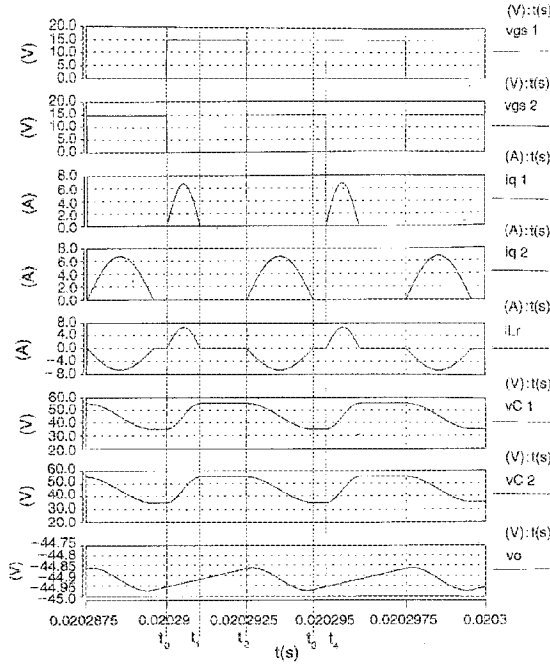


Fig. 19 Idealised simulation waveforms of the inverting $-1/2$ mode switched-capacitor resonant step-down converter

The resonant angular frequency of this state is

$$\omega_0 = \sqrt{\frac{2}{L_r C}} \quad (44)$$

The resonant impedance of this state is

$$Z_0 = \sqrt{\frac{2L_r}{C}} \quad (45)$$

(ii) State II [$t_1 - t_2$]

Q_1 is on and Q_2 is off. At t_1 , the inductor current reaches zero. The resonance is stopped by the reverse bias of the diodes. The current is kept to be zero. C_0 is still being discharged to the load in this state. The equations of this state are

$$v_C = V_{C1} \quad (46)$$

$$i_L = 0 \quad (47)$$

where V_{C1} is the voltage of C at t_1 .

(iii) State III [$t_2 - t_3$]

Q_1 is switched off and Q_2 is switched on. Since the inductor current is zero at t_2 when Q_1 is switched off, Q_1 is switched off under zero-current conditions. At t_2 , C_1 and C_2 are discharged to the load through L_r , D_a , D_b and D_2 . The current starts resonating at t_2 in sinusoidal manner which is similar to state I and hence, Q_2 is switched on under zero-current switching conditions. The inductor current i_L starts decreasing from zero at t_2 to the peak and then increases to zero at t_3 . The equations of this state are

$$v_C + L_r \frac{di_L}{dt} + V_0 = 0 \quad (48)$$

$$i_L = 2C \frac{dv_C}{dt} \quad (49)$$

The solution of (48) and (49) are

$$v_C = -V_0 + (V_0 + V_{C2}) \cos \omega_1(t - t_2) \quad (50)$$

$$i_L = -\frac{V_0 + V_{C2}}{Z_1} \sin \omega_1(t - t_2) \quad (51)$$

where V_{C2} is the voltage of C at t_2 .

The resonant angular frequency of this state is

$$\omega_1 = \sqrt{\frac{1}{2L_r C}} \quad (52)$$

and the resonant impedance is

$$Z_1 = \sqrt{\frac{L_r}{2C}} \quad (53)$$

(iv) State IV [$t_3 - t_4$]

Q_1 is off and Q_2 is on. Similarly to state II, the resonance of the inductor current is stopped by the reverse bias of the diodes. The inductor current is zero in this state. The output capacitor C_0 is discharged to the load. The equations of this state are

$$v_C = V_{C3} \quad (54)$$

$$i_L = 0 \quad (55)$$

where V_{C3} is the voltage of C at t_3 .

3.2.2 General equations of $-1/2$ mode converter

By using input and output power balancing, for state I

$$V_S \int_{t_0}^{t_1} i_L dt = V_0 I_0 T_S \quad (56)$$

By solving (56), and assuming the voltage conversion ratio is $-1/2$, then

$$v_C = -V_0 + \frac{\pi I_0 T_S Z_0}{4T_0} \cos \omega_0(t - t_0) \quad (57)$$

$$i_L = -\frac{\pi I_0 T_S}{2T_0} \sin \omega_0(t - t_0) \quad (58)$$

where T_S is the switching frequency, T_0 is the resonant period in state I, and I_0 is the output current.

For state II

$$v_C = -V_0 - \frac{\pi I_0 T_S Z_0}{4T_0} \quad (59)$$

$$i_L = 0 \quad (60)$$

For state III

$$\int_{t_2}^{t_3} i_L dt = I_0 T_S \quad (61)$$

By solving (61),

$$v_C = -V_0 - \frac{\pi I_0 T_S Z_1}{T_1} \cos \omega_1(t - t_2) \quad (62)$$

$$i_L = \frac{\pi I_0 T_S}{T_1} \sin \omega_1(t - t_2) \quad (63)$$

where T_1 is the resonant period of state III.

For state IV

$$v_C = -V_0 - \frac{\pi I_0 T_S Z_1}{T_1} \quad (64)$$

$$i_L = 0 \quad (65)$$

3.2.3 General equations of $-1/n$ mode converter

For a $1/n$ mode converter, in state I

$$v_C = -V_0 + \frac{\pi I_0 T_S Z_0}{n^2 T_0} \cos \omega_0 (t - t_0) \quad (66)$$

$$i_L = -\frac{\pi I_0 T_S}{n T_0} \sin \omega_0 (t - t_0) \quad (67)$$

where

$$\omega_0 = \sqrt{\frac{n}{L_r C}} \quad (68)$$

$$Z_0 = \sqrt{\frac{n L_r}{C}} \quad (69)$$

In state II

$$v_C = -V_0 - \frac{\pi I_0 T_S Z_0}{n^2 T_0} \quad (70)$$

$$i_L = 0 \quad (71)$$

In state III

$$v_C = -V_0 - \frac{\pi I_0 T_S Z_1}{T_1} \cos \omega_1 (t - t_2) \quad (72)$$

$$i_L = -\frac{\pi I_0 T_S}{T_1} \sin \omega_1 (t - t_2) \quad (73)$$

where

$$\omega_1 = \sqrt{\frac{1}{n L_r C}} \quad (74)$$

$$Z_1 = \sqrt{\frac{L_r}{n C}} \quad (75)$$

In state IV

$$v_C = -V_0 - \frac{\pi I_0 T_S Z_1}{T_1} \quad (76)$$

$$i_L = 0 \quad (77)$$

4 Simulation results

4.1 Comparison of generalized equations and idealized computer simulation results

4.1.1 Noninverting $1/3$ mode switched capacitor converter

Fig. 14 shows the idealised computer simulation waveforms of a noninverting $1/3$ mode switched capacitor converter. The simulation was done by using the Saber Designer program. All switches and diodes in the simulation were ideal components. The specification of the simulated circuit is:

Input voltage	90 V
Output power	80 W
Switching frequency	200 kHz

C_1, C_2	0.22 μ F
C_0	50 μ F
L_r	1 μ H

From the simulation waveforms, the output voltage is about 30 V. It also shows that the inductor current and the voltage of C_1 and C_2 resonate in sinusoidal manner on the regions t_0-t_1 and t_2-t_3 . The voltages of C_1 and C_2 are the same all the time. For 80 W output power, the output current I_0 is equal to 2.67 A. According to (19) and (20), v_c at t_0 is 19.9 V and at t_1 is 40.1 V. The highest peak value of i_L is 6.7 A in the period t_0-t_1 . According to (24) and (25), v_c at t_2 is 40.1 V and at t_3 is 19.9 V. The lowest peak value of i_L is -6.7 A. All the calculated results are equal to the simulation results. It is evident that the generalised equations of the noninverting converter can describe the waveforms of the converter correctly.

4.1.2 Inverting $-1/2$ mode switched capacitor converter

Fig. 19 shows the idealised computer simulation waveforms of inverting $-1/2$ mode switched capacitor converter. The simulation was also done by using a Saber Designer program. All switches and diodes in the simulation were ideal components. The specification of the simulated circuit is:

Input voltage	90 V
Output power	80 W
Switching frequency	200 kHz
C_1, C_2	0.22 μ F
C_0	50 μ F
L_r	1 μ H

From the simulation waveforms, the output voltage is about -45 V. It also shows that the inductor current and the voltages of C_1 and C_2 resonate in sinusoidally between t_0-t_1 and t_2-t_3 . The voltages of C_1 and C_2 are equal all the time. For 80 W output power, the output current I_0 is equal to -1.78 A. According to (57) and (58), v_c at t_0 is 34.9 V and at t_1 is 55.1 V. The highest peak value of i_L is 6.7 A in the period t_0-t_1 . According to (62) and (63), v_c at t_2 is 55.1 V and at t_3 is 34.9 V. The lowest peak value of i_L is -6.7 A. All the calculated results are equal to the simulation results. It is evident that the generalised equations of the noninverting converter can describe the waveforms of the converter correctly.

4.2 Comparison of idealised and practical computer simulation results

The specifications of the circuit for practical computer simulations are the same as those for idealised computer simulations in both converters except the former ones use MOSFETs IRF 630 and Schottky diodes MBR10100 for the switches and diodes, respectively. The practical simulation results of the noninverting $1/3$ mode converter and the inverting $-1/2$ mode converter are shown in Figs. 20 and 21, respectively.

Comparing Fig. 14 with Fig. 20 and Fig. 19 with Fig. 21, they show that when using practical switches (Figs. 20 and 21) instead of ideal switches (Figs. 14 and 19), the output voltages of the converters in the practical circuits are lower than those of the ideal circuits. This is because of the drain-to-source resistance (R_{ds}) of the MOSFET. For IRF 630, R_{ds} is about 0.4 Ω . When the MOSFETs are turned on and conduct, there will be a voltage

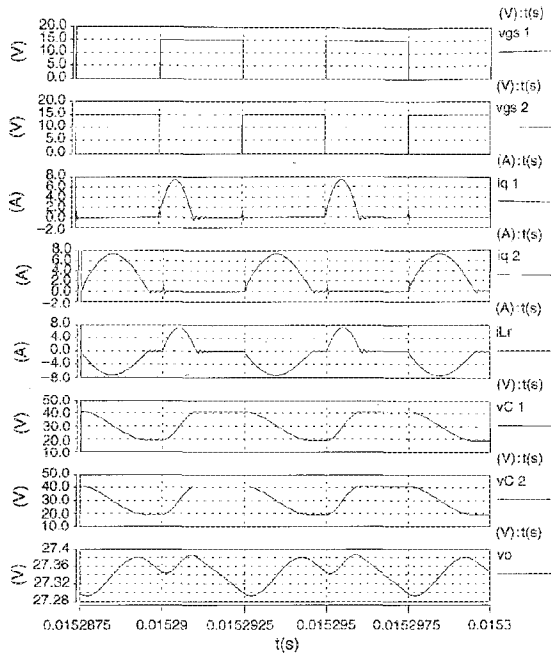


Fig. 20 Practical simulation waveforms of the noninverting 1/3 mode switched-capacitor resonant step-down converter

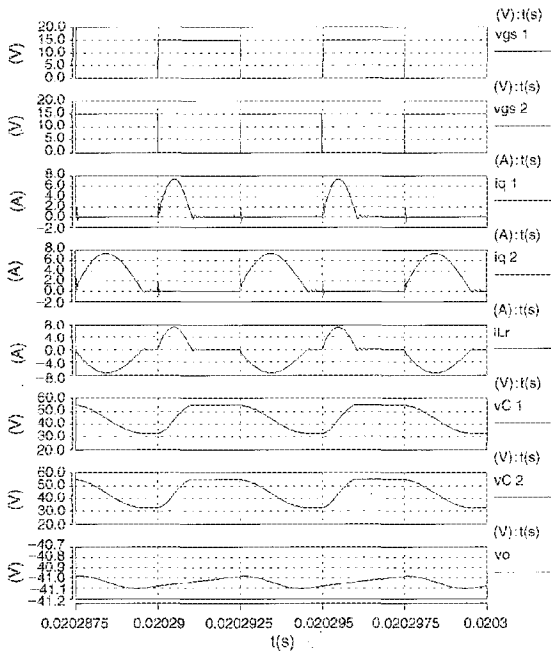


Fig. 21 Practical simulation waveforms of the inverting -1/2 mode switched-capacitor resonant step-down converter

across them. The practical diodes also have forward voltages (V_F). For MBR10100, V_F is about 0.8 V. These two components ensure that the output voltage will be lower than that in the ideal case. In fact, it is the main power loss in the circuit. It also makes the voltages of the switched capacitors V_{C1} and V_{C2} lower than those in the ideal case.

Figs. 9 and 10 show that when either transistor Q_1 or transistor Q_2 is switched off, the transistor currents I_{Q1} and I_{Q2} give short pulses at that moment. All practical transistors have junction capacitors across their legs. Since Q_1 and Q_2 are connected together, when one transistor is switched off, its junction capacitor will be charged through another transistor and hence, there will be a short pulse charging current. The junction capacitor will discharge internally when that transistor is switched on so that it gives power loss. This power loss is directly proportional to the switching frequency. Also, since both the transistors and the Schottky diodes have junction capacitors and the junction capacitors resonate with the resonant inductor in the practical case, when the main resonance stops, the currents still ring at a very small amplitude.

To maintain the output power at 80 W, the input power should be higher to compensate the power loss practically. Since the input voltage of each converter is constant, in practice the input current is higher than that in the ideal case. As a result, the amplitudes of all the transistor currents and inductor currents are higher than those in the ideal case.

5 Design

The design of a 1/3 mode step-down converter is shown in the following procedure as an example:

(i) Define the specification:

$V_s = 90$ V, $P_o = 10$ –80 W, switching frequency = 200 kHz

(ii) The resonant frequency is chosen to be higher than the switching frequency so that a zero-current switching can be naturally achieved. Because states I and III have two frequencies, the worst case, i.e. ω_1 , is used for the calculation. A 20% higher frequency is used to ensure zero current switching

$$\omega_1 = 1.2(2\pi f_s) \quad (78)$$

This gives a resonant frequency $f_o = 240$ kHz.

(iii) The ripple voltage on the resonant capacitor C_{1a} and C_{1b} must be small compared to their DC level. The peak-to-peak ripple voltage is derived from (21) and (26), or (32) and (38)

$$\Delta v_{C1} = 0.943 \frac{\pi I_o T_s}{T_1} \sqrt{\frac{L_r}{C}} \quad (79)$$

A maximum voltage ripple of 60% of the DC component is allowed on the resonant capacitor. Hence, it gives $Z = \sqrt{L_r/C} = 1.96 \Omega$

(iv) The resonant components can therefore be calculated by angular frequency and impedance from (13) or (36)

$$C = \frac{1}{\sqrt{2}\omega_1 Z} = 0.24 \mu\text{F} \quad (80)$$

$$L_r = \frac{Z}{\sqrt{2}\omega_1} = 0.92 \mu\text{H} \quad (81)$$

C and L_r are chosen to be $0.22 \mu\text{F}$ and $1 \mu\text{H}$ in the experiment.

(v) C_o is a larger capacitance and is to maintain constant voltage for the output stage. Its value can be estimated by the basic capacitor ripple voltage calculation

$$\Delta V_{Co} = \frac{I_o}{C_o} \left(\frac{\cos \theta}{f_s} - \frac{1}{f_o} \left(1 - 2 \frac{\theta}{\pi} \right) \right) \quad (82)$$

where $\theta = \sin^{-1} f_s / \pi f_0$ and $I_0 = P_0 / V_0$. For a 0.1 V ripple voltage, and 25% overload reservation, this gives $C_0 = 50 \mu\text{F}$.

(vi) The transistor rating voltage is at least equal to V_s . The diode rating voltage is also equal to V_s/n , but with a tolerance of ripple voltage on C_1 . Therefore a rating of at least 90 V is chosen.

6 Experimental results

The noninverting 1/3 mode step-down resonant converter as shown in Fig. 22 has been tested in the laboratory. The circuit is very similar to that shown in Fig. 7 but only a filtering capacitor C_m was added to ensure a good input voltage for experimental purposes. The specification for the prototype is

input voltage	90 V
output power	10 W 80 W
output voltage	30 V
switching frequency	200 kHz
Q_1 and Q_2	IRF630
D_1, D_2, D_m, D_b and D_c	MBR10100
C_1, C_2	0.22 μF
C_m	100 μF
C_0	50 μF
L_r	1 μH

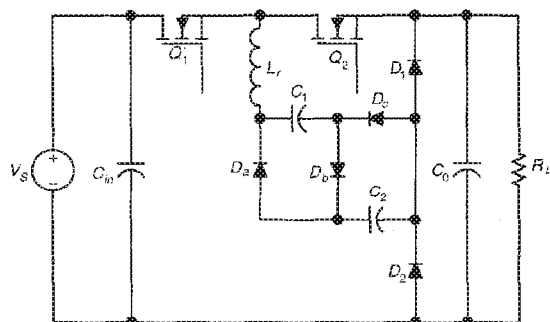


Fig. 22 Circuit of prototype for noninverting 1/3 mode switched-capacitor resonant step-down converter

The measured waveforms of the converter operating with 80 W output power are shown in Figs. 23–27. Fig. 23 shows the resonant inductor current while Figs. 25 and 26 show the currents and drain-to-source voltages of the transistors. When compared with Figs. 25 and 26, it can be observed that the resonant inductor current consists of the Q_1 and Q_2 currents.

The output voltage of the converter is shown in Fig. 24. The measured output voltage of the converter when operating at 80 W is 25.84 V. The measured output voltage is lower than the expected output voltage (30 V) and the practical simulated output voltage because of the drain-to-source resistance of the transistors and the voltage drop of the diodes in forward bias. The copper loss of the wires and the inductors, and also the equivalent series resistance (ESR) of all capacitors give output voltage drop and power loss in the circuit. To minimise the influence of the ESR of the capacitors, polypropylene capacitors, which have low ESR, are used for the switched capacitors.

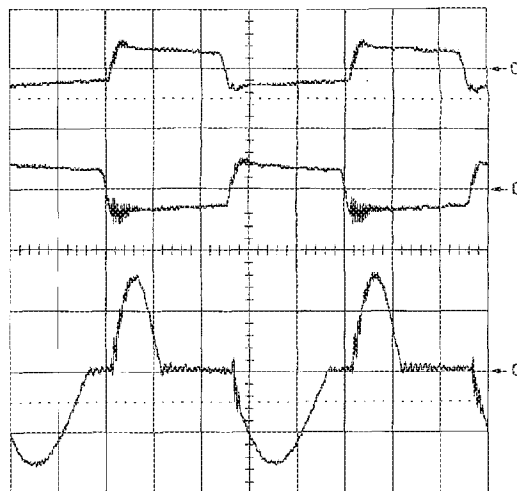


Fig. 23 Measured waveforms of converter output power 80 W
upper trace: gate signal of Q_1 , 40 V/div
middle trace: gate signal of Q_2 , 40 V/div
lower trace: i_L , 5 A/div

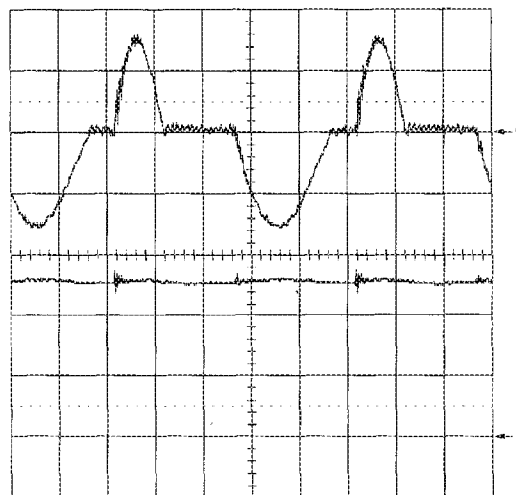


Fig. 24 Measured waveforms of converter
output power 80 W
upper trace: i_L , 5 A/div
lower trace: output voltage: 10 V/div

Figs. 25 and 26 show that the transistor currents resonate sinusoidally. The resonance stops when the current reaches zero while the drain-to-source voltage of the corresponding transistors is zero. The transistors are switched off when the currents are equal to zero so that a zero-current switching condition is obtained. This is very similar to the simulated waveforms. Because of the power loss of the circuit and to obtain 80 W output power, the input current and hence the transistors and inductor currents are higher than in the simulated waveforms.

Fig. 27 shows that the resonant inductor current and the voltages of the switched capacitors resonate together. The voltage waveform of C_1 is almost equal to that of C_2 . The amplitude is lower than those in simulated waveforms.

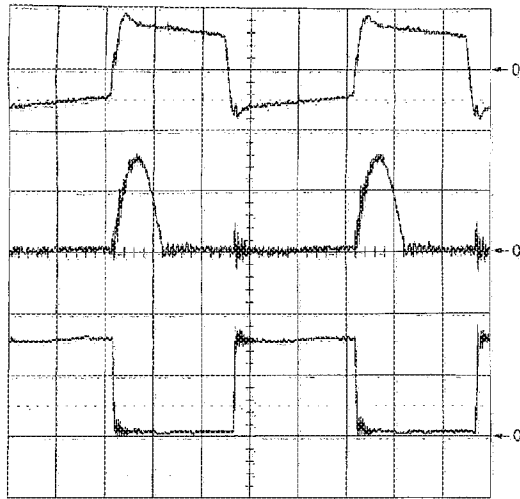


Fig. 25 Measured waveforms of converter
output power 80 W
upper trace: gate signal of Q_1 , 20 V/div
middle trace: Q_1 current, 5 A/div
lower trace: V_{ds} of Q_1 , 40 V/div

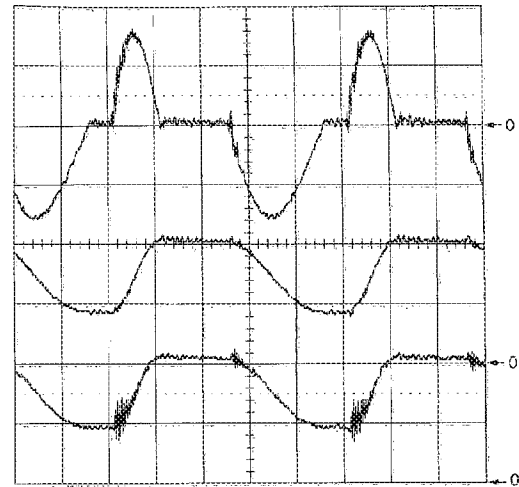


Fig. 27 Measured waveforms of converter
output power 80 W
upper trace: I_L , 5 A/div
middle trace: V_{C1} , 20 V/div
lower trace: V_{C2} , 20 V/div

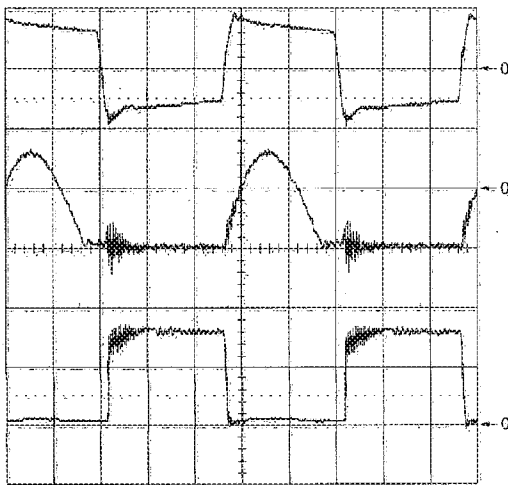


Fig. 26 Measured waveforms of converter
output power 80 W
upper trace: gate signal of Q_1 , 20 V/div
middle trace: Q_1 current, 5 A/div
lower trace: V_{ds} of Q_1 , 40 V/div

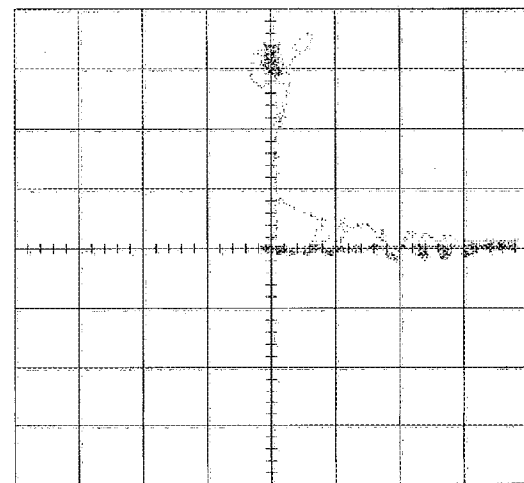


Fig. 28 Noninverting H3 mode switched-capacitor resonant step-down converter
 $v-i$ trajectory of Q_1
x-axis: Q_1 current, 2 A/div
y-axis: V_{ds} of Q_1 , 20 V/div

because of the voltage drop of the transistors, diodes and the wires.

The diagrams of the $v-i$ trajectory of each transistor are shown in Figs. 28 and 29. Figs. 23 and 24 again confirm that both transistors are in zero-current switching condition.

Fig. 30 plots the efficiency against the output power of the converter. By testing the circuit with the output power between 10–80 W, the efficiency varied between 85.5% at both 10 W and 80 W output power to 93.2% at 22 W output power. Fig. 31 plots the voltage conversion ratio against the output power of the converter. The voltage

conversion ratio varies between 0.325 at 10 W output power to 0.287 at 80 W output power. Figs. 23–27 show that the efficiency was highest when the output power was 22 W. At light load, power loss in the circuit is mainly due to the diodes, which have voltage drop in forward bias. At high load, the power loss is mainly due to the drain-to-source resistance of the transistors. On the other hand, Figs. 28 and 29 show that the voltage conversion ratio decreases when the output power increases. Other than the voltage drop on the diodes, when the current increases, the voltage drop on both transistors increases and hence, the voltage conversion ratio decreases.

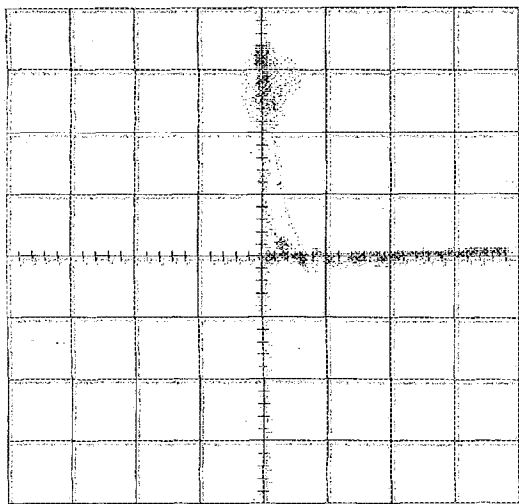


Fig. 29 Noninverting $1/3$ mode switched-capacitor resonant step-down converter
 v - i trajectory of Q_2
 x-axis: Q_2 current, 2 A/div.
 y-axis: V_{ds} of Q_2 , 20 V/div

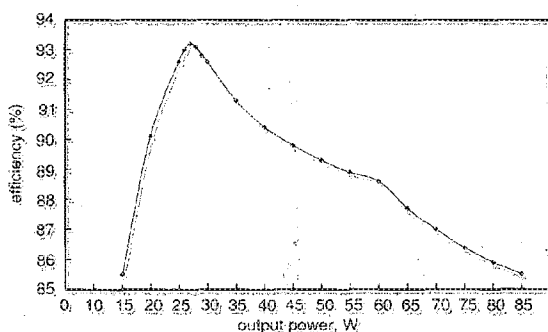


Fig. 30 Measured efficiency and output power of noninverting $1/3$ mode switched-capacitor resonant step-down converter

7 Conclusions

This paper has proposed two families of switched-capacitor circuits. One is the noninverting power converter and the other is the inverting converter. Both families operate under zero-current switching. The inherent disadvantage of a high

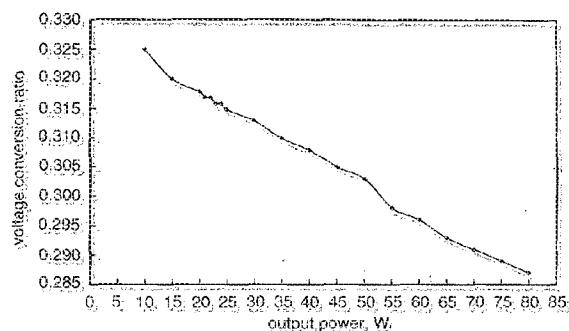


Fig. 31 Measured voltage conversion ratio and output power of noninverting $1/3$ mode switched-capacitor resonant step-down converter

pulsational current has been eliminated. The number of switches needed is low. The complete analysis of the converters has also been presented. Each family has a number of fractional output voltage conversion converters. Experimental results have confirmed that the operation agrees with the theoretical prediction in that all transistors can achieve zero-current switching. The efficiency is also very high.

8 Acknowledgment

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Load Adaptive, High Efficiency, Switched Capacitor Intermediate Bus Converter

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Abstract – A new non-isolated Intermediate Bus Converter is proposed using a divide-by-4 switched capacitor network. A load adaptive scheme is designed that maximizes the efficiency for all load currents. Detailed analysis using steady-state charge transfer is provided with a static charge constraint concept for degenerate switched capacitor networks. Simple average models for output impedance, capacitor and switch RMS currents, and efficiency are defined. A bus stability criterion is established, and a novel current splitting technique is introduced that reduces the capacitors' RMS currents. A 300W 48V to 12V prototype is designed using a provided efficiency-specified design procedure that achieves 96%-97% over the full range of load current, 1A-25A.

I. INTRODUCTION

This paper introduces a new approach to 48V to 12V Intermediate Bus Architectures. Commonly, unregulated, isolated half bridge topologies are used as bus converters. The proposed innovative approach targets applications where isolation is not required between a 48V backplane and a 12V bus. Removing the isolation transformer saves considerable board area and reduces converter cost. Reliability analyses can be conducted for a particular application to assess the risk of current faults on the 48V side propagating to the 12V side of the system. In many applications, this risk should be acceptable; therefore a non-isolated Intermediate Bus Architecture has merit in many practical systems.

Some unregulated 300W isolated bus converters that are commercially available claim peak efficiencies up to 95-96% [1-3]. This paper suggests using a high efficiency, switched capacitor voltage divider that adapts to the load current, as shown in Fig. 1. The proposed non-isolated topology achieves up to 96%-97% efficiency over the entire load range. A particular improvement in efficiency is achieved at light load conditions, which is advantageous for reducing energy consumption in systems where usage is light. Conventional, magnetic-based, switching power converters have poor light load efficiency, which is a significant disadvantage since switching losses dominate when systems are lightly used. The time integral of light load power losses can amount to a tremendous energy waste.

Characteristic high efficiency and high power density are key advantages of switched cap networks over magnetic-based power converters. A 98% peak efficiency, 60W 12V to 6V switched capacitor divider was demonstrated in [4] with

very high power density. An automotive application in [5] showed a 98% peak efficiency with a 600W 42V to 14V switched capacitor converter. The converter proposed in Fig. 1 divides the input by 4; where as these other application only divide by two or three. It requires more switches; hence the losses are greater. 98% efficiency is possible with this topology if the parasitic resistance and capacitances of the MOSFETs are small enough and if larger capacitance is used in the network. A tradeoff between converter cost and efficiency must be made. A 97% peak efficiency is a reasonable balance with cost and can compete with commercially available isolated converters [1].

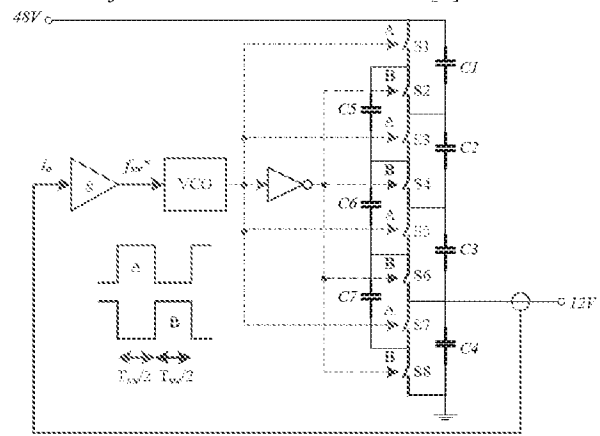


Figure 1. Proposed load adaptive non-isolated bus converter.

The proposed converter in Fig. 1 is a two phase switched capacitor network. The phases are switched with a fixed 50% duty cycle. The proposed load adaptation scheme modulates the switching frequency such that the efficiency is maximized as the load current varies. Efficiency is determined by losses in the network. Switched-capacitor energy losses dominate when the load current is high, and MOSFET capacitor charging and discharging losses dominate at light load. The loss incurred from switching the capacitors is proportional to the load current but inversely related to the switching frequency. Specifically, the converter has a DC output impedance that is inversely proportional to the switching frequency. The adaptation scheme reduces the output resistance by increasing the switching frequency at heavy loads. Conversely, MOSFET capacitor losses are proportional to switching frequency. Thus a low switching frequency reduces MOSFET capacitor losses. Hence a balance must be found between switched cap and MOSFET cap losses. A

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simple formula for the optimal switching frequency is provided in this paper that minimizes the sum of these losses, hence maximizing efficiency. The optimal switching frequency is found to be proportional to the load current.

A simple charge transfer analysis is described and used to formulate useful models for output resistance, switch currents, and capacitor currents. These models are used as design equations. The output resistance determines the conversion efficiency and the deviation in the output voltage from the ideal voltage. The models for the currents are used to select appropriate devices. The capacitor RMS currents are critical design parameters, which primarily determine the capacitor selections.

II. CHARGE TRANSFER ANALYSIS

This section develops an analysis framework that simplifies modeling steady state behavior. The framework is based on a simple KCL analysis. The technique is preferred over other modeling techniques, which are possibly more complicated and do not scale well for higher order networks. The switched capacitor network in Fig. 1 is 7th order. Periodic steady state analysis of such a high order system via solving state equations is very cumbersome numerically and impractical, if at all possible, symbolically. State space averaging techniques have been established. [6] proposes a state space averaging approach, but illustrates the method with only a 2nd order system. Unfortunately the approach does not scale well to a 7th order network. [5] provides a power loss analysis based on a detailed capacitor voltage study, which would be difficult to generalize. [7] presents a simple time-constant analysis, which is intuitive for 2nd or 3rd order systems.

A generalized analysis framework is needed that can address the scalability of switched capacitor networks. An analysis technique that is based on steady state charge balance is most appropriate. The charge transfer analysis technique described in [8] and elaborated in [9] provides a complete steady-state framework for modeling switched capacitor networks. From this simple approach, models for steady state output resistance, switch currents, and capacitor currents can be obtained. These models are useful for calculating conversion efficiency and selecting suitable devices. Moreover, the relationships are quite simple as compared with the complexity and ambiguity of state space models. The derived models are intuitive and are very practical design equations.

A. KCL System with a Static Charge Constraint

The charge transfer analysis begins by identifying the circuits of the converter in each phase. The phases are named 'A' for the circuit (Fig. 2a) when the odd numbered switches are on, and 'B' for the circuit (Fig. 2b) during the on-state of the even numbered switches. The charge balance mechanism requires that the net charge entering each capacitor is zero. In other words, all cycle-by-cycle average capacitor currents must equal zero, i.e. amp-second balance. The charge entering each branch (e.g. q_{inA} , q_1 , q_2 , etc.) is labeled in Fig. 2. Note

that for charge balance, the phase charges of each capacitor, C_i , must be equal and opposite, i.e., $q_{C1A} = -q_{C1B} = q_1$.

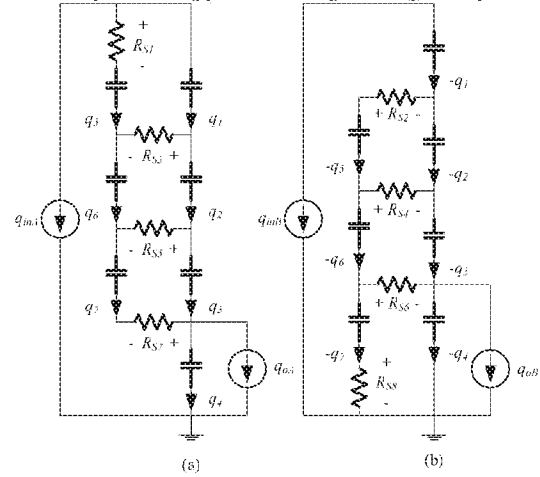


Figure 2 Circuits during each phase. The polarity of each conducting switch is indicated.

The KCL equations for the network are

$$\begin{aligned} 0 &= q_1 + q_5 + q_{inA} \\ 0 &= q_5 - q_6 + q_1 - q_2 \\ 0 &= q_6 - q_7 + q_2 - q_3 \\ 0 &= q_3 + q_7 - q_4 - q_{outA} \\ 0 &= -q_1 + q_{inB} \\ 0 &= -q_1 + q_5 + q_2 \\ 0 &= -q_5 + q_6 - q_2 + q_3 \\ 0 &= -q_6 + q_7 + q_4 - q_3 - q_{outB} \end{aligned} \quad (1)$$

As mentioned in [8,9], the proposed switched capacitor divider is a degenerate case. That is to say that (1) cannot be solved as it is indeterminate. However, there is a static charge constraint equation from a particular KVL loop. Note that voltage sum across the stationary capacitors, ($C1$, $C2$, $C3$, $C4$), must be equal to input voltage, V_{in} . Assume that the input voltage is an ideal DC source, then the derivatives of the capacitor voltages must sum to zero. Consequently, the change in capacitors charges must also sum to zero. An interpretation of this behavior is that there is no net charge absorbed by the stack of stationary capacitors during a switching cycle. Rather, the switching capacitors transfer the charge from the input source to the load. Further, this explains why the switching capacitors carry much higher RMS currents than the stationary capacitors, which is described in a later section. If $C1=C2=C3=C4$, then the static charge constraint is stated

$$0 = q_1 + q_2 + q_3 + q_4 \quad (2)$$

If the capacitances are not equal, (2) is still a reasonable approximation. The steady state solution does not vary significantly with variation in the stationary capacitances.

B. Charge Transfer Vector

A normalized charge transfer vector, $a = q / q_{in}$, is formed where q is the solution of the constrained KCL system, (1) and (2). The vector describes the relative amount of charge

transferred to the load, absorbed by each capacitor, and drawn from the input source. These vectors are arranged as follows

$$\begin{aligned} a_A &= \begin{bmatrix} a_{aA} & (a_c)^T & a_{inA} \end{bmatrix}^T \\ a_B &= \begin{bmatrix} a_{aB} & (-a_c)^T & a_{inB} \end{bmatrix}^T \end{aligned} \quad (3)$$

The terms a_{aA} , a_{aB} , a_{inA} and a_{inB} are scalar quantities. The term $a_c = [a_{c1} \ a_{c2} \ a_{c3} \ a_{c4} \ a_{c5} \ a_{c6} \ a_{c7}]^T$ is a vector with components for each capacitor. The capacitor vector component values are characteristic of a particular network topology. The capacitor vector is also the unique factor in determining the output resistance, the switch currents, and the capacitor currents. Moreover, these relationships are generalized in terms of this unique charge transfer vector. The relationships are the same for all switch capacitor networks, only the length and values of the charge transfer vector is different. Hence, charge transfer analysis is widely applicable to other switched capacitor networks. Though, as in the case of the switched capacitor divider presented in this paper, the extension of the methodology to degenerate networks requires identifying necessary static charge constraints. The vector components for the proposed topology in Fig. 1 are

$$\begin{aligned} a_{aA} &= a_{aB} = \frac{1}{2}, & a_{inA} &= -\frac{1}{2}, & a_{inB} &= \frac{1}{4} \\ a_c &= \begin{bmatrix} \frac{1}{4} & 0 & -\frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{2} & \frac{3}{4} \end{bmatrix}^T \end{aligned} \quad (4)$$

These components uniquely characterize the charge transfer between the capacitors, the input source, and the load. They are useful for deriving the steady state output resistance of the switched capacitor network.

C. Output Resistance Model

A model for the network is an ideal transformer with a turns ratio equal to the number of stationary capacitors, N [8,9]. All switched capacitor energy losses, hard-switching losses, and parasitic resistive losses are modeled as a lumped output resistance, R_o , as depicted in Fig. 3.

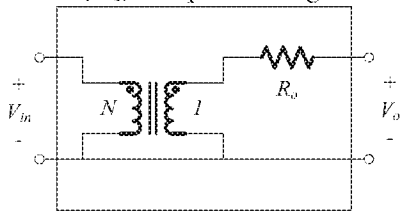


Figure 3 Ideal transformer model of switched capacitor network with losses represented by the output resistance R_o [8,9].

The output resistance varies with switching frequency and has two distinct limits. A slow switching limit, R_{SSL} , quantifies the resistance at low switching frequencies. The output resistance is inversely proportional to slow switching frequencies, where the proportionality constant is a function of the network capacitances. Conversely for high switching frequency, the output resistance approaches a minimum, R_{FSL} . This minimum is determined primarily by the on-state resistances, R_{dson} , of the switches and secondarily by the equivalent series resistances, R_c , of the capacitors. Typically the R_{dson} is much greater than the R_c , since each capacitor is

composed of many parallel MLCCs in order to carry large RMS currents. The output resistance and the two limits are illustrated in Fig. 4.

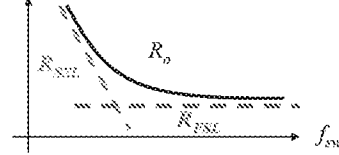


Figure 4 Variation of output resistance with the switching frequency, f_{sw} .

The output resistance approaches the fast switching limit (R_{FSL}) at high frequency and the slow switching limit (R_{SSL}) at low frequency.

The effective output resistance, R_o , is equal to the Euclidian distance between the fast switching limit and the slow switching limit. Specifically,

$$R_o = \sqrt{R_{SSL}^2 + R_{FSL}^2} \quad (5)$$

D. Slow Switching Limit

The slow switching limit describes the circuit behavior when the switching frequency is low enough such that each pair of capacitors in parallel will have equal voltages at the end of a half cycle. The output resistance for the slow switching limit is determined by considering the capacitor voltage deviation as it varies with the amount of charge transferred. By Tellegen's Theorem, a vector of branch voltages that satisfies KVL is orthogonal to a vector of branch currents that satisfies KCL [9]. This theorem is applicable to the switch capacitor network, since the average currents in each phase are the charge transferred per half cycle. Hence, the charge transfer vectors are orthogonal to the respective voltages.

$$\begin{aligned} a_A \cdot v_A &= 0 \\ a_B \cdot v_B &= 0 \end{aligned} \quad (6)$$

where v_A and v_B are the respective voltage vectors with components for the output voltage and each capacitor voltage. For analyzing the output impedance, zero input voltage can be assumed, $v_{in}=0$. The inner products can be rewritten as sums

$$(a_{aA} + a_{aB})v_o + \sum_i (a_{cA}v_{cA} + a_{cB}v_{cB}) = 0 \quad (7)$$

Or equally

$$v_o + \sum_i a_{ci} \Delta v_{ci} = 0 \quad (8)$$

$$\Delta v_{ci} = (v_{cA} - v_{cB}) = \frac{q_{ci}}{C_i} = \frac{q_o a_{ci}}{C_i}$$

The total charge delivered to the load is constant, hence $(a_{aA} + a_{aB}) = 1$. Dividing (8) by the total output charge, q_o , it follows that

$$\frac{v_o}{q_o} + \sum_i \frac{(a_{ci})^2}{C_i} = 0 \quad (9)$$

Further (9) describes the output resistance for the slow switching limit, noting that the output current is $i_o = q_o f_{sw}$. Making this substitution and rearranging (9), the slow switching limit is found to be

$$R_{SSL} = -\frac{v_o}{i_o} = \sum_i \frac{(a_{ci})^2}{C_i f_{sw}} = \frac{1}{f_{sw} C_{eq}} \quad (10)$$

where C_{eq} describes the equivalent capacitance of the network

$$C_{eq} = \left(\sum_i \frac{(a_{ci})^2}{C_i} \right)^{-1} \quad (11)$$

The equivalent capacitance is a useful concept. Particularly it creates a compact representation for the output resistance and also simplifies the notation when considering optimal switching frequency, which is described in a later section. The equivalent capacitance also can be a design parameter. The output resistance of the network represents the switched-cap losses in the circuit. Lower output resistance results in lower losses, and hence produces a higher efficiency. Therefore it is desirable that the equivalent capacitance be as large as possible. Of course the amount of capacitance will be constrained in a practical design based on cost and board area. The equivalent capacitance for the network in Fig. 1 is

$$C_{eq} = \frac{16(C_1 C_3 C_5 C_6 C_7)}{\left(C_3 C_5 C_6 C_7 + C_1 C_5 C_6 C_7 + C_1 C_3 C_6 C_7 \right.} \quad (12)$$

$$\left. + 4 C_1 C_3 C_5 C_7 + 9 C_1 C_3 C_5 C_6 \right)$$

E. Fast Switching Limit

The fast switching limit models the circuit behavior, where the parallel capacitors do not have equal voltages at the end of each half cycle. The circuit switches between phases much faster than RC time constants of the paralleled capacitors. It is expected from the slow switching limit that, R_{SSL} , goes to zero as the switching frequency goes to infinity. In other words the effect of the capacitances on the output resistance diminishes as the switching frequency increases. Accordingly, the parasitic resistances will tend to dominate the output resistance at high frequency. It is reasonable to assume that the on-state resistances of the switches are much larger than the equivalent series resistances (ESRs) of the capacitors. The capacitors will be constructed from many parallel devices in order to carry large RMS currents. The ESRs can be included in the analysis, but considering ideal capacitors is sufficient and simplifies the analysis greatly.

Define a charge vector for the switches, $a_s = [a_{s1} \ a_{s2} \ a_{s3} \ a_{s4} \ a_{s5} \ a_{s6} \ a_{s7} \ a_{s8}]^T$, which describes the charge transferred through the switches in each phase. This vector has components that are functions of the capacitor charge vector components. By consistently defining the polarity of the switches in each phase, the switch charge vector for the switched capacitor divider is found to be

$$a_s = \begin{bmatrix} a_{s1-s4} & a_{s5-s8} \end{bmatrix}^T$$

$$a_{s1-s4} = [a_{c5} \ a_{c5} \ (a_{c1} - a_{c2}) \ (a_{c6} - a_{c5})]^T \quad (13)$$

$$a_{s5-s8} = [(a_{c2} - a_{c3}) \ (a_{c7} - a_{c6}) \ -a_{c7} \ -a_{c7}]^T$$

$$a_s = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & -\frac{3}{4} & -\frac{3}{4} \end{bmatrix}^T$$

The average value of the switch currents during the on-interval of the respective switches are derived from the switch charge vector. Consider that each switch is operated at a 50% duty cycle, D_i . The switch currents are equal to the charge transferred per phase interval divided by the duration of the

interval. The phase interval is the product of the duty cycle and the switching period. Explicitly,

$$I_{si} = \frac{q_{si} f_{sw}}{D_i} = 2q_{si} f_{sw} = 2a_{si} I_o \quad (14)$$

where $I_o = q_o f_{sw}$ and $a_{si} = q_{si} / q_o$. The average switch currents are

$$I_s = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{3}{2} & -\frac{3}{2} \end{bmatrix}^T I_o \quad (15)$$

The RMS switch currents are calculated from the average currents. Note that the average current is carried in the duration of the switching period when each switch is turned on. At high switching frequencies, the switch currents can be approximated to square waveforms switching between zero and I_{si} . For a 50% duty cycle the switch RMS currents are

$$I_{SRMS} = (\sqrt{D_i}) |I_{si}| = \frac{|I_{si}|}{\sqrt{2}} = \sqrt{2} a_{si} I_o \quad (16)$$

This approximation is valid only for fast switching frequencies. The actual RMS currents will be larger for slow switching frequencies, since the peak currents will be larger. A typical design will select an adequately large switching frequency such that efficiency is preserved at heavy loads. Therefore the maximum switch RMS current can be calculated from the above approximations.

The fast switching limit for the output resistance is derived from the switch RMS currents. Consider the power losses due to the R_{don} of each switch. The sum of these losses is the total switching loss.

$$P_{FSL} = \sum_i (R_{don} I_{SRMS}^2) \quad (17)$$

$$= 2R_{don} \left(\sum_i (a_{si}^2) \right) I_o^2$$

The fast switching limit resistance is equal to the total loss divided by the square of the load current.

$$R_{FSL} = 2R_{don} \left(\sum_i (a_{si}^2) \right) = 3R_{don} \quad (18)$$

This is a very simple relationship and quite useful for selecting switches. Certainly switches with low on-state resistance should be selected, otherwise efficiency will be low at heavy loads.

The capacitor RMS currents at high switching frequencies can be calculated following an approach similar to the switch currents. In short, the capacitor current waveforms are square waves with peaks $\pm 2a_{ci} I_o$. Thus RMS currents are

$$I_{CRMS} = |2a_{ci} I_o| = \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 1 & \frac{3}{2} \end{bmatrix}^T I_o \quad (19)$$

As mentioned earlier, the switching capacitors ($C5$, $C6$, $C7$) carry more RMS current than the stationary capacitors ($C1$, $C2$, $C3$, $C4$). This signifies that the charge is moved from the input source to the load via the switching capacitors. The stationary capacitor stack does not transfer any net charge per cycle to the load. The components with values of zero in the RMS current vector are not exactly zero in an actual switching circuit. A zero value indicates that the RMS current of a

device does not charge with the load current. The importance of the particular values in the RMS current vector is that they can be used to select suitable capacitors. The RMS current ratings of the capacitors are the critical factors in the design, more so than the capacitance values. Supporting large RMS currents requires using many devices in parallel. The RMS current vector indicates the relative number of parallel capacitors that should be used.

Charge transfer analysis is a simple methodology that completely characterizes a switch capacitor network. The formulation yields intuitive relationships, which are well suited as design equations. The following section, regarding load adaptation, exemplifies such a design process, which would be difficult with more complicated models. The models developed with charge transfer analysis are used to determine the optimal switching frequency for a particular load current. A load adaptation scheme is proposed to track the optimal efficiency as the load varies.

III. LOAD ADAPTATION

A load adaptation scheme is devised that maximizes the efficiency for all load currents. It will be shown that the optimal switching frequency is proportional to the load current. In a hardware implementation, optimality can be maintained by using a voltage control oscillator (VCO) for generating the MOSFET switching signal. A sensor measurement of the load current is scaled and applied as the input to the VCO. Hence, the switching frequency adjusts dynamically to changes in the load current. The adaptation gain, g , of the sensed current is set equal to the optimal proportionality constant, which is derived in this section. The load adaptive control action tracks the maximum efficiency point of the converter.

The efficiency of the switched capacitor divider decreases as the load current increases. The switched cap losses, which are lumped in the output resistance (R_o) derived in the previous section, increase with the square of the load current. Therefore it is favorable to increase the switching frequency at heavy loads. Conversely, there are additional losses due to charging and discharging of the input and output capacitances of the MOSFET switches. These losses dominate the efficiency at light loads and are proportional to the switching frequency. Thus it is best to decrease the switching frequency at light loads. The general trends of switched capacitor losses, P_{SC} , and MOSFET capacitor losses, P_{MC} , are shown in Fig. 5.

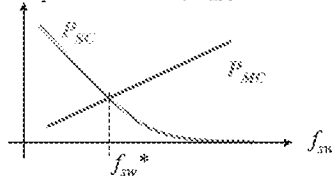


Figure 5 The switched capacitor losses, P_{SC} , decrease with frequency; whereas, the MOSFET capacitor losses, P_{MC} , increase with frequency.

The MOSFET capacitor losses are determined by the charge stored in the gate and output capacitances, Q_g and Q_{oss} . These values are typically provided in MOSFET datasheets or can be calculated from the parasitic capacitances and the

applied voltages. The losses are proportional to the switching frequency and the applied voltages. The total losses are summations of the losses from all switches. In total, there are $2N$ switches where N is the number of stationary caps in the divider stack. In this case, $N=4$, so there are 8 switches. Explicitly the MOSFET cap losses are

$$P_{MC} = 2(Q_g + Q_{oss})V_{in}f_{sw} = 2Q_M V_{in}f_{sw} \quad (20)$$

which assumes that the MOSFETs gates are driven by a (V_{in}/N) voltage. For compactness, the two charges are summed to form the quantity, $Q_M = Q_g + Q_{oss}$. Note in (20) that the MOSFET cap losses are independent of the number, N , of caps in the stack, if the gate drive voltage is equal to V_{in}/N . This is significant since the MOSFET losses would not increase if the topology was scaled to divide by $N > 4$. Hence efficiency does not decrease by adding more switches.

The switched capacitor losses in the network are modeled using the output resistance and are proportional to the square of the load current.

$$P_{SC} = R_o I_o^2 \quad (21)$$

Ideally, the power transferred to the load, P , is determined by the input voltage, the load current, and the number of stationary caps in the divider stack.

$$P = \frac{V_{in}}{N} I_o \quad (22)$$

The actual power delivered to the load is P less the switched cap losses. The input power is P more the MOSFET cap losses. The efficiency, η , is calculated using (20) (21), and (22):

$$\eta = \frac{P_o}{P_{in}} = \frac{P - P_{SC}}{P + P_{MC}} \quad (23)$$

F. Optimal Policy

The efficiency is maximized by selecting a particular switching frequency. This optimal switching frequency, f_{sw}^* , varies with the load current. Clearly, the efficiency as a function of the switching frequency has extrema where its first derivative with respect to frequency is zero. The maximum occurs at the extrema where the second derivative is negative. The optimal policy is

$$f_{sw}^* = \arg \max \eta = \left\{ f_{sw} : \frac{\partial \eta}{\partial f_{sw}} = 0, \frac{\partial^2 \eta}{\partial f_{sw}^2} < 0 \right\} \quad (24)$$

Solving this optimization problem symbolically is rather involved and actually not necessary. One can argue that the efficiency is maximized when the switched cap losses are equal to the MOSFET cap losses, as illustrated in Fig. 5. Intuitively this is reasonable, because P_{SC} varies inversely with frequency and P_{MC} varies directly. In other words, minimizing the sum of the losses maximizes the efficiency. The minimum of the sum occurs when the two losses are equal. Specifically, the problem simplifies to finding the switching frequency such that

$$f_{sw}^* = \arg \max \eta = \{ f_{sw} : P_{SC}(f_{sw}) = P_{MC}(f_{sw}) \} \quad (25)$$

Equating the losses (20) and (21), the following relationship must hold

$$I_o^4 + (3R_{don} C_{eq})^2 f_{sw}^*{}^2 = (2C_{eq} Q_M V_{in})^2 f_{sw}^*{}^4 \quad (26)$$

The roots of this quartic equation are rather complicated. The positive real root is

$$f_{sw}^* = \sqrt{\frac{2C_{eq} \sqrt{9R_{don}^2 C_{eq}^2 + (81R_{don}^4 C_{eq}^2 + 16Q_M^2 V_{in}^2 I_o^4)}}{4C_{eq} Q_M V_{in}}} \quad (27)$$

This solution is not very practical for implementing an adaptive control. Instead an approximation can be made. One can either argue that $R_{don} \ll 1$ and so the resistive terms in (27) can be omitted, or that the quartic terms dominate (26) such that

$$I_o^4 \approx (2C_{eq} Q_M V_{in})^2 f_{sw}^*{}^4 \quad (28)$$

Solving this equation yields a more practical result. Though the solution is strictly suboptimal, the approximation is reasonably accurate for use in the load adaptive scheme. The suboptimal solution is

$$f_{sw}^* \approx \frac{I_o}{\sqrt{2C_{eq} Q_M V_{in}}} = g I_o \quad (29)$$

where g is the adaptation gain that maps the load current to the optimal switching frequency. The load adaptation scheme maintains this optimal relationship between the load current and the switching frequency. Hence by applying this simple load adaptive control, the efficiency is maximized.

IV. CURRENT SPLITTING TECHNIQUE

The RMS current ratings of the switching capacitors pose a significant design challenge. The capacitor currents for the proposed network, (19), have ratings up to 150% of the load current. A novel circuit technique can be introduced that reduces the RMS rating of certain capacitors. Using this technique the max RMS current can be reduced to 50% of the load current. The technique involves adding additional capacitors across several capacitors in the network in Fig. 1. The method presented in this paper discusses adding caps across the switching caps ($C5, C6, C7$), since these carry the largest RMS currents. But the technique could be equally applied to the stationary caps ($C1, C2, C3, C4$). The proposed embodiment is shown in Fig. 6.

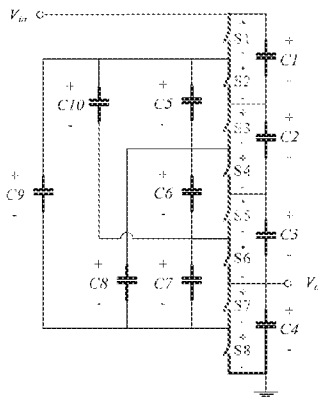


Figure 6 Proposed topology using current splitting technique to reduce the RMS current ratings of the switching capacitors. The additional capacitors, $C8$, $C9$, and $C10$, reduce the original $C6$ and $C7$ currents. The currents through the

switches are split amongst the additional capacitor branches. This potentially reduces the current through a single branch. Using the proposed configuration, the maximum RMS current of all capacitors in the network is reduced to 50% of the load current. Other variations could be used, such as including only $C8$ or $C9$, but using the topology in Fig. 6 is most effective. The charge transfer vectors and the RMS currents for using different splitting configurations are summarized in Table 1.

Table 1 Charge transfer vectors and capacitor RMS currents for different configurations of splitting capacitors.

Splitting Caps Used	a_c	I_{CRMS}/I_o
none	$\begin{bmatrix} 1 & 0 & -1/4 & 0 & 1/4 & 1/2 & 3/4 \\ 1/4 & 0 & -1/4 & 0 & 1/4 & 1/2 & 3/4 \end{bmatrix}^T$	$\begin{bmatrix} 1 & 0 & 1/2 & 0 & 1/2 & 1 & 3/2 \\ 1/2 & 0 & 1/2 & 0 & 1/2 & 1 & 3/2 \end{bmatrix}^T$
$C8, C9$	$\begin{bmatrix} 1 & 0 & -1/4 & 0 & 0 & 1/4 & 1/4 \\ 1/4 & 0 & -1/4 & 0 & 0 & 1/4 & 1/4 \end{bmatrix}^T$	$\begin{bmatrix} 1 & 0 & 1/2 & 0 & 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 & 0 & 0 & 1/2 & 1/2 \end{bmatrix}^T$
$C8, C9, C10$	$\begin{bmatrix} 1 & 0 & -1/4 & 0 & 0 & 1/4 & 1/4 & 0 \\ 1/4 & 0 & -1/4 & 0 & 0 & 1/4 & 1/4 & 0 \end{bmatrix}^T$	$\begin{bmatrix} 1 & 0 & 1/2 & 0 & 0 & 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 & 1/2 & 1/2 & 0 \end{bmatrix}^T$

The data in Table 1 indicate that including $C8$ and $C9$ greatly reduces the current ratings of all capacitors. Moreover it is not necessary to add $C10$, including only ($C8, C9$) yields the same currents as ($C8, C9, C10$). The current splitting technique reduces the maximum RMS current of all caps to half the load current. This facilitates an even distribution of capacitors. The basic network in Fig. 1 would have an uneven distribution, since two capacitors have a really large RMS current rating while the rest have much smaller ratings. One might suspect that adding more capacitors for this splitting technique would increase the converter cost and complicate the board layout. Actually since the current distribution is even among all capacitors, the layout can be more compact and hence possibly less complicated. If the equivalent capacitance is examined, the converter cost will not increase. Table 2 provides the equivalent capacitance for each configuration.

Table 2 Equivalent capacitance for different configurations of splitting capacitors.

Splitting Caps Used	C_{eq}
none	$\frac{16(C_1 C_2 C_3 C_4)}{(C_1 C_2 C_3 C_4 + C_1 C_2 C_3 C_5 + C_1 C_2 C_4 C_5 + C_1 C_3 C_4 C_5 + 4 C_2 C_3 C_4 C_5 + 9 C_1 C_2 C_3 C_4)}$
$C8, C9$	$\frac{16(C_1 C_2 C_3 C_4)}{(C_1 C_2 C_3 C_4 + C_1 C_2 C_3 C_5 + C_1 C_2 C_4 C_5 + C_1 C_3 C_4 C_5 + C_2 C_3 C_4 C_5 + C_1 C_2 C_3 C_4)}$
$C8, C9, C10$	$\frac{16(C_1 C_2 C_3 C_4)}{(C_1 C_2 C_3 C_4 + C_1 C_2 C_3 C_5 + C_1 C_2 C_4 C_5 + C_1 C_3 C_4 C_5 + C_2 C_3 C_4 C_5 + C_1 C_2 C_3 C_4)}$

Suppose all capacitors have equal capacitance, C (i.e. $C1 = C2 = C3 = \dots = C$). Substitute C for all C_i in Table 2. The equivalent capacitance using $C8$ and $C9$ is $3.2C$; whereas the equivalent capacitance with no splitting caps equals only C . Therefore the total capacitance in the network could be reduced 70% using splitting caps and still maintain the same equivalent capacitance as network with out the splitting caps. Thus the converter cost could actually decrease if splitting capacitors are used.

V. BUS STABILITY

Output impedance of a bus converter is important to the stability of a cascaded system, Fig. 7. The stability depends on the interaction between the output impedance of the bus

converter and the input impedance of the down stream point of load (POL) converters.

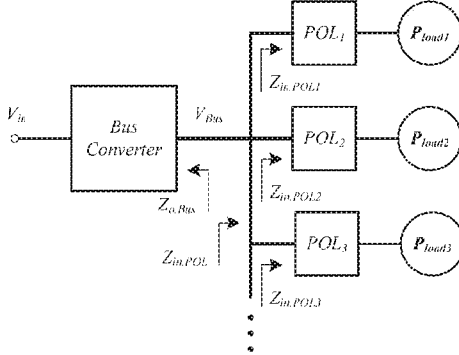


Figure 7 Cascaded system of an Intermediate Bus Architecture.

The output impedance of the bus converter and the input impedance of the POLs form voltage dividers. The transfer function from the input voltage to the bus voltage is

$$\frac{V_{bus}(s)}{V_{in}(s)} = \frac{1}{1 + \frac{Z_{o,Bus}(s)}{Z_{in,POL}(s)}} \quad (30)$$

where $Z_{in,POL}$ is the parallel combination of all POL input impedances.

$$Z_{in,POL} = Z_{in,POL1} // Z_{in,POL2} // Z_{in,POL3} // \dots \quad (31)$$

The POL converters are regulated and thus exhibit negative impedance at low frequency. This negative impedance can lead to instability at the bus. A necessary condition for stability is the output impedance magnitude of the bus converter must be less than the input impedance of all POLs.

$$|Z_{o,Bus}(j\omega)| < |Z_{in,POL}(j\omega)| \quad \forall \omega \quad (32)$$

A sufficient condition for stability is the maximum output impedance magnitude of the bus converter is less than minimum input impedance of all POLs. It can be shown that the input impedance of a POL has its minimum at low frequency. An approximation for the minimum is the DC impedance, which is determined by the power delivered to the load.

$$\min(Z_{in,POL1}) = \frac{V_{bus}^2}{\max(P_{load1})} \quad (33)$$

Further, the effective input impedance of all parallel POLs is determined by the sum of all loads.

$$\min(Z_{in,POL}) = \frac{V_{bus}^2}{\sum_i \max(P_{loadi})} = \frac{V_{bus}^2}{\max(P_{bus})} = \frac{V_{bus}}{\max(I_{bus})} \quad (34)$$

The output impedance of the bus converter can be approximated by the parallel combination of the output resistance, R_o , and the equivalent capacitance, C_{eq} , of the network. (11) is a valid approximation of the equivalent capacitance for impedance frequencies less than the switching frequency. At frequencies exceeding f_{sw} , the equivalent capacitance of the output impedance is determined by the stationary capacitors, $C_{eq} \rightarrow (C1//C2//C3)+C4$. The complete average model for the switched capacitor bus converter is illustrated in Fig. 8, including power losses and the approximate output impedance model.

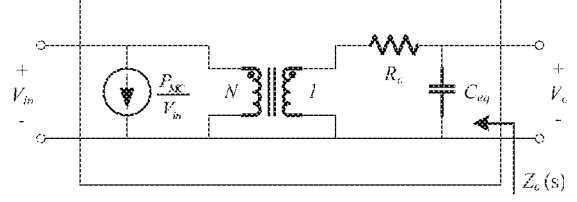


Figure 8 Complete average model for the switched capacitor bus converter.

The complete average model describes steady state and transient behavior and captures the efficiency of the converter. Notably, the current source at the primary side of the ideal transformer models the power losses in the MOSFET capacitors. The output resistance models the switched-capacitor losses. Both of these depend on the switching frequency.

The first order model for the bus converter output impedance is

$$Z_{o,Bus}(s) = \frac{R_o}{sR_oC_{eq} + 1} \quad (35)$$

This has a maximum at DC,

$$\max(Z_{o,Bus}) = R_o \quad (36)$$

Using (34) and (36), bus stability, (32), is ensured if

$$R_o < \frac{V_{bus}}{\max(I_{bus})} \quad (37)$$

Since R_o is limited by the $R_{ds(on)}$ of the switches, (37) is a useful design equation for selecting suitable MOSFETs.

VI. DESIGN PROCEDURE

An appropriate design approach is to specify a desired efficiency at a particular load. This procedure describes how the capacitors in the network must be sized in order to achieve a specified efficiency. Assume C8 and C9 are used.

1. Use a_C (Table 1) to determine the relative ratios of the capacitances. Define a vector of the network capacitances, $C_{Network}$. The capacitances are defined in proportion to a nominal value, C , which is the capacitance of each MLCC used.

$$C_{Network} = [\max(k a_C N_C, 1)] C \quad (38)$$

The maximum operator is used so that $C_{Network}$ does not have zero elements, since a_C may contain zero elements. As such one MLCC will be used for each capacitor with a zero element in a_C . k is a normalizing factor that scales all capacitors relative to the largest capacitance.

$$k = \frac{1}{\max(a_C)} \quad (39)$$

2. Use the I_{CRMS} vector to determine the number, N_C , of parallel MLCCs for the largest capacitor to meet the maximum current rating. Assume that each MLCC can carry 2A to 3A RMS, $I_{MLCC} = 2A-3A$.

$$N_C = \frac{\max(I_{CRMS})}{I_{MLCC}} \quad (40)$$

3. Solve for the optimal equivalent capacitance, C_{eq}^* . Assume optimal switching frequency, (30).

$$C_{eq}^* = \frac{2V_m N^2 Q_m (\eta - 1) \left(V_m^2 (\eta^3 - \eta^2 + \eta - 1) + V_{R_s}^2 (\eta + 1) - 2\eta V_m \sqrt{(\eta - 1) (\eta (V_{R_s}^2 + V_m^2) + V_{R_s}^2 - V_m^2)} \right)}{(V_m (1 - \eta) + V_{R_s})^2 (V_m (\eta - 1) + V_{R_s})^2} \quad (41)$$

where

$$V_{R_s} = I_o (3R_{ds(on)})N \quad (42)$$

4. Determine the equivalent capacitance as a function of the network capacitance, $C_{eq}(C_{Network})$. Solve for the nominal capacitance C .

$$C_{eq} = \frac{16}{5} N_c C \quad (43)$$

5. Select an appropriate MLCC by equating (41) = (43).

VII. PROTOTYPE

A 300W prototype was designed and constructed using the proposed design procedure. The design achieves 96% efficiency at full load, $I_{o,max}=25A$. The MOSFETs selected are IRLU7843, which have $R_{ds(on)}=3.3m$ and $Q_m=110nC$ for $V_{gs}=4.5V$ and $V_{ds}=12V$. Four floating ISL6605 drivers drive the gates of adjacent MOSFET pairs (e.g. S1 and S2). The voltage across each stationary capacitor provides a 12V supply voltage for the drivers. The 12V is stepped down to 4.5V using ADP3331 LDOs. The efficiency of the converter is greatly improved using this 4.5V gate drive voltage rather than using the full 12V. The gate charge is more significant than driving the gate at a higher voltage to reduce the $R_{ds(on)}$. The PWM signal is AC coupled to the floating drivers. A MM74HC4046 PLL is used as a VCO to generate the PWM signal. A 0.002 current sense resistor is in series with the output current. The sensed current is applied as the input to the VCO to implement the load adaptive scheme.

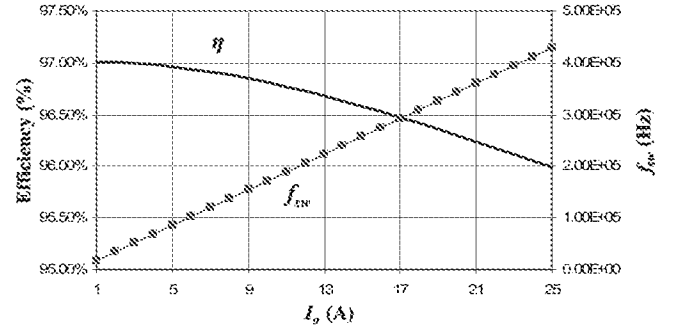
Using C8 and C9 splitting caps, the design values are compiled in Table 3. Design 1 is the exact calculation using the proposed procedure from section VI. The second is an equivalent solution that uses standard 10uF MLCC capacitors.

Table 3 Design values for a 48V to 12V 300W load adaptive switched capacitor bus converter using C8 and C9 splitting capacitors.

	Design 1	Design 2
$C_{Network} / (N_c C)$	$[1\ 0\ 1\ 0\ 0\ 0\ 1\ 1\ 1]^T$	$[1\ 0\ 1\ 0\ 0\ 0\ 1\ 1\ 1]^T$
k	4	4
I_{MLCC}	2A RMS	2A RMS
N_c	7	10
C_{eq}^*	320uF	320uF
C	14.3uF	10uF
Total MLCCs	51	72
g	17.2 kHz/A	17.2 kHz/A

Using the load adaptive scheme, the efficiency is 96% at full load and peaks to 97% at light load. The efficiency and optimal switching frequencies are plotted in Fig. 9.

Table 4 Efficiency and optimal switching frequency for a Load Adaptive Switched Capacitor Bus Converter using C8 and C9 splitting caps.



VIII. CONCLUSION

This paper presented a new approach to Intermediate Bus Architectures. A 300W 48V to 12V load adaptive switched capacitor bus converter was shown to achieve 96%-97% efficiency over the entire load range, using an efficiency-specified design procedure. A charge transfer analysis was described in detail and used to derive simple steady-state average models for output impedance, capacitor and switch RMS currents, and efficiency. A novel current splitting techniques was introduced that reduces the capacitors' RMS currents, and criterion for bus stability was examined.

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Attorney Docket Number:	00614-0180001
Receipt Date:	24-MAY-2018
Filing Date:	07-JUN-2017
Time Stamp:	15:32:41
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Information Disclosure Statement (IDS) Form (SB08)	00614-0180001_Transmittal.pdf	69961 b0c45da1c01fb2f360c64a031e69095900d83b3d	no	1

Warnings:

Information:					
This is not an USPTO supplied IDS fillable form					
2	Information Disclosure Statement (IDS) Form (SB08)	00614-0180001_1449.pdf	126914	no	1
			863b6905e3d1b07a796ba0e1d600e1aa7dfdc992		
Warnings:					
Information:					
This is not an USPTO supplied IDS fillable form					
3	Non Patent Literature	Cao_AFamilyoZero.pdf	652583	no	8
			e4a54125b047cdbb20fa11b9e6e9d38da30066d7		
Warnings:					
Information:					
4	Non Patent Literature	Cao_ZeroCurrentSwitching_46_2536-2544.pdf	2360035	no	9
			0155f2801717a3eff6beab19faaa4eeff898c8f41		
Warnings:					
Information:					
5	Non Patent Literature	Seeman_AnalysisandOptimization_23_841-851.pdf	366222	no	11
			b0501ec4ec59a73f80cabad314d81409217b0d0d		
Warnings:					
Information:					
6	Non Patent Literature	Yeung_GeneralisedAnalysis_38_263-264.pdf	388419	no	2
			8c0ca237b0ef2da193b9a8db52c6193b042b9840		
Warnings:					
Information:					
7	Non Patent Literature	Yeung_ZerocurrentSwitching_149_111-121.pdf	1768182	no	11
			cd32e1afeaa5844bd04e0c3b02bb90e8d4f5d7b		
Warnings:					
Information:					
8	Non Patent Literature	Oraw_LoadAdaptive1.pdf	5616993	no	8
			c3e2425f7a91e48ae5c46bc2c6094f7e16aaac7b		
Warnings:					

Information:	
Total Files Size (in bytes):	11349309
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.	



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/616,288	06/07/2017	Patrizio Vinciarelli	00614-0180001	7438

26214 7590 06/29/2018
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

COX, CASSANDRA F

ART UNIT	PAPER NUMBER
----------	--------------

2842

NOTIFICATION DATE	DELIVERY MODE
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06/29/2018

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

PATDOCTC@fr.com

Office Action Summary	Application No. 15/616,288	Applicant(s) VINCIARELLI ET AL.	
	Examiner CASSANDRA COX	Art Unit 2842	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 2 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 06/07/17.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.
- 2a) ☐ This action is **FINAL**. 2b) ☐ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

- 5) ☒ Claim(s) 1-194 is/are pending in the application.
 5a) Of the above claim(s) 1-131 is/are withdrawn from consideration.
- 6) ☐ Claim(s) ____ is/are allowed.
- 7) ☐ Claim(s) ____ is/are rejected.
- 8) ☐ Claim(s) ____ is/are objected to.
- 9) ☒ Claim(s) 132-194 are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☐ The drawing(s) filed on ____ is/are: a) ☐ accepted or b) ☐ objected to by the Examiner.
 Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
 Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
2. ☐ Certified copies of the priority documents have been received in Application No. ____.
3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|---|---|
| 1) ☐ Notice of References Cited (PTO-892) | 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. ____. |
| 2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date ____. | 4) ☐ Other: ____. |

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

Election/Restrictions

2. This application contains claims directed to the following patentably distinct species I. claims 132-164 and 193 and II. Claims 165-192 and 194. The species are independent or distinct because *the claims to the different species recite the mutually exclusive characteristics of such species--*, .I recites one or more driver circuits, one or more output circuits, a plurality of output rails, one or more rail selection circuits, and II recites first and third power conversion stages and a point-of-load circuit. In addition, these species are not obvious variants of each other based on the current record.

Applicant is required under 35 U.S.C. 121 to elect a single disclosed species, or a single grouping of patentably indistinct species, for prosecution on the merits to which the claims shall be restricted if no generic claim is finally held to be allowable. Currently, no claims are generic.

There is a search and/or examination burden for the patentably distinct species as set forth above because at least the following reason(s) apply:

- *the species or groupings of patentably indistinct species require a different field of search (e.g., searching different classes/subclasses or electronic resources, or employing different search strategies or search queries).*

Applicant is advised that the reply to this requirement to be complete must include (i) an election of a species to be examined even though the requirement may be traversed (37 CFR 1.143) **and (ii) identification of the claims encompassing the elected species or grouping of patentably indistinct species**, including any claims subsequently added. An argument that a claim is allowable or that all claims are generic is considered nonresponsive unless accompanied by an election.

The election may be made with or without traverse. To preserve a right to petition, the election must be made with traverse. If the reply does not distinctly and specifically point out supposed errors in the election of species requirement, the election shall be treated as an election without traverse. Traversal must be presented at the time of election in order to be considered timely. Failure to timely traverse the requirement will result in the loss of right to petition under 37 CFR 1.144. If claims are added after the election, applicant must indicate which of these claims are readable on the elected species or grouping of patentably indistinct species.

Should applicant traverse on the ground that the species, or groupings of patentably indistinct species from which election is required, are not patentably distinct, applicant should submit evidence or identify such evidence now of record showing them to be obvious variants or clearly admit on the record that this is the case. In either instance, if the examiner finds one of the species unpatentable over the prior art, the evidence or admission may be used in a rejection under 35 U.S.C. 103 or pre-AIA 35 U.S.C. 103(a) of the other species.

3. Upon the allowance of a generic claim, applicant will be entitled to consideration of claims to additional species which depend from or otherwise require all the limitations of an allowable generic claim as provided by 37 CFR 1.141. Applicant is reminded that upon the cancellation of claims to a non-elected invention, the inventorship must be corrected in compliance with 37 CFR 1.48(a) if one or more of the currently named inventors is no longer an inventor of at least one claim remaining in the application. A request to correct inventorship under 37 CFR 1.48(a) must be accompanied by an application data sheet in accordance with 37 CFR 1.76 that identifies each inventor by his or her legal name and by the processing fee required under 37 CFR 1.17(i).

Any inquiry concerning this communication or earlier communications from the examiner should be directed to CASSANDRA COX whose telephone number is (571)272-1741. The examiner can normally be reached on Monday-Thursday from 6:00 AM to 3:30 PM and alternate Fridays from 7:00 AM to 3:30 PM.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Donovan Lincoln can be reached on 571-272-1988. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Art Unit: 2842

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/CASSANDRA COX/
Primary Examiner, Art Unit 2842
4.

<i>Index of Claims</i> 	Application/Control No. 15616288	Applicant(s)/Patent Under Reexamination VINCIARELLI ET AL.
	Examiner CASSANDRA COX	Art Unit 2842

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant		☐ CPA		☐ T.D.		☐ R.1.47			
CLAIM		DATE							
Final	Original	06/25/2018							
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	3	-							
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<i>Index of Claims</i> 	Application/Control No. 15616288	Applicant(s)/Patent Under Reexamination VINCIARELLI ET AL.
	Examiner CASSANDRA COX	Art Unit 2842

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant		☐ CPA		☐ T.D.		☐ R.1.47			
CLAIM		DATE							
Final	Original	06/25/2018							
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<i>Index of Claims</i> 	Application/Control No. 15616288	Applicant(s)/Patent Under Reexamination VINCIARELLI ET AL.
	Examiner CASSANDRA COX	Art Unit 2842

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant		☐ CPA		☐ T.D.		☐ R.1.47			
CLAIM		DATE							
Final	Original	06/25/2018							
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<i>Index of Claims</i> 	Application/Control No. 15616288	Applicant(s)/Patent Under Reexamination VINCIARELLI ET AL.
	Examiner CASSANDRA COX	Art Unit 2842

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant		☐ CPA		☐ T.D.		☐ R.1.47			
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<i>Index of Claims</i> 	Application/Control No. 15616288	Applicant(s)/Patent Under Reexamination VINCIARELLI ET AL.
	Examiner CASSANDRA COX	Art Unit 2842

✓	Rejected
=	Allowed

-	Cancelled
÷	Restricted

N	Non-Elected
I	Interference

A	Appeal
O	Objected

☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	06/25/2018								
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<i>Index of Claims</i> 	Application/Control No. 15616288	Applicant(s)/Patent Under Reexamination VINCIARELLI ET AL.
	Examiner CASSANDRA COX	Art Unit 2842

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant		☐ CPA		☐ T.D.		☐ R.1.47			
CLAIM		DATE							
Final	Original	06/25/2018							
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	191	÷							
	192	÷							
	193	÷							
	194	÷							

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Jun 29, 2018 04:28:55 AM

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Application	Document	Mailroom Date	Attorney Docket No.
15616288	CTRS	06/29/2018	00614-0180001

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Thank you for prompt attention to this notice,

UNITED STATES PATENT AND TRADEMARK OFFICE
PATENT APPLICATION INFORMATION RETRIEVAL SYSTEM

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	2842
Serial No. :	15/616,288	Examiner :	Cassandra Cox
Filed :	June 7, 2017	Conf. No. :	7438
Title :	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS		

MAIL STOP AMENDMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Please consider the documents listed on the disclosure form submitted with this statement. Copies of the documents are being submitted with this statement to the extent required under 37 C.F.R. § 1.98(a)(2).

This statement is being filed before the receipt of a first action on the merits.

Apply any necessary charges or credits applicable under 37 C.F.R. § 1.17(p) to deposit account 06-1050, referencing the above attorney docket number.

Please contact the undersigned if there are any questions regarding this statement.

Respectfully submitted,

Date: September 20, 2018_____

/Rex I. Huang/_____

Rex I. Huang
Reg. No. 57,661

Customer Number 26214
Fish & Richardson P.C.
Telephone: (617) 542-5070
Facsimile: (877) 769-7945

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0180001	Application No. 15/616,288
Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Applicant VLT, Inc.	
	Filing Date June 7, 2017	Group Art Unit 2842

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1.	5,438,294	8/1/1995	Smith			
	2.	9,087,656	7/21/2015	Vinciarelli			
	3.	2013/0030534	1/31/2013	DeLurio et al.			

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	4.	Notice of Allowance in U.S. Application No. 15/091,346, dated May 21, 2018, 9 pages
	5.	Notice of Allowance in U.S. Application No. 15/091,346, dated August 13, 2018, 8 pages

Examiner Signature	Date Considered
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	



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NOTICE OF ALLOWANCE AND FEE(S) DUE

26214 7590 05/21/2018
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

COX, CASSANDRA F

ART UNIT

PAPER NUMBER

2842

DATE MAILED: 05/21/2018

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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15/091,346

04/05/2016

Patrizio Vinciarelli

00614-0178001

8817

TITLE OF INVENTION: METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0	\$0	\$1000	08/21/2018

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

26214 7590 05/21/2018
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/091,346	04/05/2016	Patrizio Vinciarelli	00614-0178001	8817

TITLE OF INVENTION: METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0	\$0	\$1000	08/21/2018

EXAMINER	ART UNIT	CLASS-SUBCLASS
COX, CASSANDRA F	2842	327-110000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363). ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached. ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.	2. For printing on the patent front page, list (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____ (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____ 3 _____
---	---

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
☐ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
☐ Payment by credit card. Form PTO-2038 is attached.
☐ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
☐ Applicant asserting small entity status. See 37 CFR 1.27
☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____ Date _____
 Typed or printed name _____ Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/091,346	04/05/2016	Patrizio Vinciarelli	00614-0178001	8817

26214	7590	05/21/2018
FISH & RICHARDSON P.C. (VLT, INC)		
P.O. BOX 1022		
MINNEAPOLIS, MN 55440-1022		

EXAMINER
COX, CASSANDRA F

ART UNIT	PAPER NUMBER
2842	

DATE MAILED: 05/21/2018

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

<i>Notice Requiring Inventor's Oath or Declaration</i>	Application No. 15/091,346	Applicant(s) Patrizio Vinciarelli	
	Examiner COX, CASSANDRA F	Art Unit 2842	

This notice is an attachment to the Notice of Allowability (PTOL-37), or the Notice of Allowability For A Design Application (PTOL-37D).

An inventor's oath or declaration in compliance with 37 CFR 1.63 or 1.64 executed by or with respect to each inventor has not yet been submitted.

An oath or declaration in compliance with 37 CFR 1.63, or a substitute statement in compliance with 37 CFR 1.64, executed by or with respect to each inventor (for any inventor for which a compliant oath, declaration, or substitute statement has not yet been submitted) **MUST** be filed no later than the date on which the issue fee is paid. See 35 U.S.C. 115(f). Failure to timely comply will result in ABANDONMENT of this application.

A properly executed inventor's oath to declaration has not been received for the following inventor(s):

If applicant previously filed one or more oaths, declarations, or substitute statements, applicant may have received an informational notice regarding deficiencies therein.

The following deficiencies are noted:

INFORMAL ACTION PROBLEMS

- A properly executed inventor's oath or declaration has not been received for the following inventor(s): **Patrizio Vinciarelli**.

Applicant may submit the inventor's oath or declaration at any time before the Notice of Allowance and Fee(s) Due, PTOL-85, is mailed.

Questions relating to this Notice should be directed to the Application Assistance Unit at 571-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 15/091,346	Applicant(s) VINCIARELLI ET AL.	
	Examiner CASSANDRA COX	Art Unit 2842	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 02/08/18.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1-56,121-127 and 131. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying Indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|--|--|
| 1. ☒ Notice of References Cited (PTO-892) | 5. ☐ Examiner's Amendment/Comment |
| 2. ☒ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date _____ | 6. ☒ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | 7. ☐ Other _____. |
| 4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date _____. | |

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1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

Allowable Subject Matter

2. Claims 1-56, 121-127- and 131 are allowed.
3. The following is an examiner's statement of reasons for allowance: Claims 1-29, 121-127- and 131 are allowed because the closest prior art of record fails to disclose a method of supplying power received from an input source, comprising :providing one or more output circuits for deployment in the semiconductor package, each output circuit including a printed circuit board, a power transformer including a first winding and a second winding, an input connected to the first winding for receiving AC power from the driver output, a rectification circuit connected to the second winding for rectifying power received from the transformer, and an output connected to the rectification circuit for supplying DC power to the one or more semiconductor chips; providing a power bus for carrying AC power from the driver output to the input of each of the one or more output circuits in the semiconductor package; operating the driver circuit to drive the transformer in a series of converter operating cycles, each converter operating cycle characterized by two power transfer phases of essentially equal duration during which one or more switches in the driver circuit are ON and power is transferred from the input to the output via the transformer; wherein the bus voltage is a multiple, X, times greater than a voltage required by the one or more semiconductor chips in combination with the

Art Unit: 2842

rest of the limitations of the base claims and any intervening claims. Claims 30-56 are allowed because the closest prior art of record fails to disclose an apparatus comprising: one or more output circuits housed in the semiconductor package, each output circuit including a printed circuit board, a power transformer including a first winding and a second winding, an input connected to the first winding for receiving AC power, a rectification circuit connected to the second winding for rectifying power received from the transformer, and an output connected to the rectification circuit for supplying DC power to the one or more semiconductor chips; and a power bus connected to selected interface connections for carrying AC power at a bus voltage from the system board to the input of each of the one or more output circuits in the semiconductor package; wherein the bus voltage is a multiple, X, times greater than a voltage required by the one or more semiconductor chips in combination with the rest of the limitations of the base claims and any intervening claims.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to CASSANDRA COX whose telephone number is (571)272-1741. The examiner can normally be reached on Monday-Thursday from 6:00 AM to 3:30 PM and alternate Fridays from 7:00 AM to 3:30 PM.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Donovan Lincoln can be reached on 571-272-1988. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/CASSANDRA COX/
Primary Examiner, Art Unit 2842



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

26214 7590 08/13/2018
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

COX, CASSANDRA F

ART UNIT

PAPER NUMBER

2842

DATE MAILED: 08/13/2018

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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15/091,346

04/05/2016

Patrizio Vinciarelli

00614-0178001

8817

TITLE OF INVENTION: METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0	\$0	\$1000	11/13/2018

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

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III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), to: **Mail** Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

26214 7590 08/13/2018
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/091,346	04/05/2016	Patrizio Vinciarelli	00614-0178001	8817

TITLE OF INVENTION: METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0	\$0	\$1000	11/13/2018

EXAMINER	ART UNIT	CLASS-SUBCLASS
COX, CASSANDRA F	2842	327-110000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
- ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) The names of up to 3 registered patent attorneys or agents OR, alternatively,
- (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 _____
 2 _____
 3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
- ☐ Publication Fee (No small entity discount permitted)
- ☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
- ☐ Payment by credit card. Form PTO-2038 is attached.
- ☐ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
- ☐ Applicant asserting small entity status. See 37 CFR 1.27
- ☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
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Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/091,346	04/05/2016	Patrizio Vinciarelli	00614-0178001	8817

26214 7590 08/13/2018
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

COX, CASSANDRA F

ART UNIT PAPER NUMBER

2842

DATE MAILED: 08/13/2018

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability

Application No.

15/091,346

Applicant(s)

VINCIARELLI ET AL.

Examiner

CASSANDRA COX

Art Unit

2842

**AtA (First Inventor to File)
Status**

Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 05/30/18.
☐ A declaration(s)/affidavit(s) under 37 CFR 1.130(b) was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1-56, 121-127 and 131. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892)
2. ☒ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date _____
3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material
4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date _____
5. ☐ Examiner's Amendment/Comment
6. ☒ Examiner's Statement of Reasons for Allowance
7. ☐ Other _____

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

Allowable Subject Matter

2. Claims 1-56, 121-127- and 131 are allowed.
3. The following is an examiner's statement of reasons for allowance: Claims 1-29, 121-127- and 131 are allowed because the closest prior art of record fails to disclose a method of supplying power received from an input source, comprising :providing one or more output circuits for deployment in the semiconductor package, each output circuit including a printed circuit board, a power transformer including a first winding and a second winding, an input connected to the first winding for receiving AC power from the driver output, a rectification circuit connected to the second winding for rectifying power received from the transformer, and an output connected to the rectification circuit for supplying DC power to the one or more semiconductor chips; providing a power bus for carrying AC power from the driver output to the input of each of the one or more output circuits in the semiconductor package; operating the driver circuit to drive the transformer in a series of converter operating cycles, each converter operating cycle characterized by two power transfer phases of essentially equal duration during which one or more switches in the driver circuit are ON and power is transferred from the input to the output via the transformer; wherein the bus voltage is a multiple, X, times greater than a voltage required by the one or more semiconductor chips in combination with the

Art Unit: 2842

rest of the limitations of the base claims and any intervening claims. Claims 30-56 are allowed because the closest prior art of record fails to disclose an apparatus comprising: one or more output circuits housed in the semiconductor package, each output circuit including a printed circuit board, a power transformer including a first winding and a second winding, an input connected to the first winding for receiving AC power, a rectification circuit connected to the second winding for rectifying power received from the transformer, and an output connected to the rectification circuit for supplying DC power to the one or more semiconductor chips; and a power bus connected to selected interface connections for carrying AC power at a bus voltage from the system board to the input of each of the one or more output circuits in the semiconductor package; wherein the bus voltage is a multiple, X, times greater than a voltage required by the one or more semiconductor chips in combination with the rest of the limitations of the base claims and any intervening claims.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to CASSANDRA COX whose telephone number is (571)272-1741. The examiner can normally be reached on Monday-Thursday from 6:00 AM to 3:30 PM and alternate Fridays from 7:00 AM to 3:30 PM.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Donovan Lincoln can be reached on 571-272-1988. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/CASSANDRA COX/
Primary Examiner, Art Unit 2842

Electronic Acknowledgement Receipt

EFS ID:	33783651
Application Number:	15616288
International Application Number:	
Confirmation Number:	7438
Title of Invention:	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Michelle Leidall
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0180001
Receipt Date:	20-SEP-2018
Filing Date:	07-JUN-2017
Time Stamp:	16:50:03
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Information Disclosure Statement (IDS) Form (SB08)	00614-0180001_Transmittal.pdf	70623 fb1d489e183e5eae8dea971985ded97806497c28	no	1

Warnings:

Information:					
This is not an USPTO supplied IDS fillable form					
2	Information Disclosure Statement (IDS) Form (SB08)	00614-0180001_1449.pdf	120954	no	1
			a9d1f6266ec3611a332b8dbc3f0042672f0db 7179		
Warnings:					
Information:					
This is not an USPTO supplied IDS fillable form					
3	Non Patent Literature	NOA_1780_15091346_521201 8.pdf	1010825	no	9
			d2e2d67af900c8b5408f76ecf781ce2835a4 9ce1		
Warnings:					
Information:					
4	Non Patent Literature	NOA_1780_15091346_813201 8.pdf	10169479	no	8
			90476849c09fbbb6b50b2ae2b3de6b2bc dfe78		
Warnings:					
Information:					
			Total Files Size (in bytes):	11371881	
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	2842
Application No. :	15/616,288	Examiner :	Cassandra Cox
Filed :	June 7, 2017	Conf. No. :	7438
Title :	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS		

Mail Stop Amendment

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

REPLY TO RESTRICTION REQUIREMENT

In response to the restriction requirement made in the action dated June 29, 2018, identified Group I (claims 132-164 and 193) is elected for examination. The election is made without traverse.

Please apply any necessary charges or credits to Deposit Account No. 06-1050, referencing the above attorney docket number.

Respectfully submitted,

Date: September 25, 2018_____

/Rex I. Huang/

Rex I. Huang
Reg. No. 57,661

Customer Number 26214
Fish & Richardson P.C.
Telephone: (617) 542-5070
Facsimile: (877) 769-7945

Electronic Patent Application Fee Transmittal

Application Number:	15616288			
Filing Date:	07-Jun-2017			
Title of Invention:	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS			
First Named Inventor/Applicant Name:	Patrizio Vinciarelli			
Filer:	Rex I. Huang/Rita Liston			
Attorney Docket Number:	00614-0180001			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension - 1 month with \$0 paid	1251	1	200	200
Miscellaneous:				
Total in USD (\$)				200

Electronic Acknowledgement Receipt

EFS ID:	33815581
Application Number:	15616288
International Application Number:	
Confirmation Number:	7438
Title of Invention:	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Rita Liston
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0180001
Receipt Date:	25-SEP-2018
Filing Date:	07-JUN-2017
Time Stamp:	10:29:29
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$200
RAM confirmation Number	092518INTEFSW00009170061050
Deposit Account	061050
Authorized User	Rita Liston

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:
37 CFR 1.21 (Miscellaneous fees and charges)

--

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Response to Election / Restriction Filed	00614_0180001_Resp_RR.pdf	72328	no	1
			8a897737062d754610ab9d7dea67a62e7b400091		

Warnings:

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	30808	no	2
			b71e82d863788364c59bb2c3720b30c5f501be9f		

Warnings:

Information:

Total Files Size (in bytes):			103136
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

26214 7590 12/11/2018
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

EXAMINER

COX, CASSANDRA F

ART UNIT

PAPER NUMBER

2842

DATE MAILED: 12/11/2018

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
-----------------	-------------	----------------------	---------------------	------------------

15/616,288

06/07/2017

Patrizio Vinciarelli

00614-0180001

7438

TITLE OF INVENTION: METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0	\$0	\$1000	03/11/2019

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web.

**By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450**

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

26214 7590 12/11/2018
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date indicated below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/616,288	06/07/2017	Patrizio Vinciarelli	00614-0180001	7438

TITLE OF INVENTION: METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0	\$0	\$1000	03/11/2019

EXAMINER	ART UNIT	CLASS-SUBCLASS
COX, CASSANDRA F	2842	327-109000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363). ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached. ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-09 or more recent) attached. Use of a Customer Number is required.	2. For printing on the patent front page, list (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____ (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____ 3 _____
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3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies _____

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)
☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
- ☐ Applicant asserting small entity status. See 37 CFR 1.27
- ☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____ Date _____
 Typed or printed name _____ Registration No. _____



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/616,288	06/07/2017	Patrizio Vinciarelli	00614-0180001	7438
26214	7590	12/11/2018	EXAMINER	
FISH & RICHARDSON P.C. (VLT, INC)			COX, CASSANDRA F	
P.O. BOX 1022			ART UNIT	
MINNEAPOLIS, MN 55440-1022			PAPER NUMBER	
			2842	
DATE MAILED: 12/11/2018				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 15/616,288	Applicant(s) VINCIARELLI ET AL.	
	Examiner CASSANDRA COX	Art Unit 2842	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 09/25/18.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 132-164 and 193. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
- Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).**
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|--|--|
| 1. ☒ Notice of References Cited (PTO-892) | 5. ☐ Examiner's Amendment/Comment |
| 2. ☒ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date _____ | 6. ☒ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | 7. ☐ Other _____. |
| 4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date _____. | |

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

Allowable Subject Matter

2. Claims 132-164 and 193 are allowed.
3. The following is an examiner's statement of reasons for allowance: Claims 132-150 are allowed because the closest prior art of record fails to disclose a method of supplying power received from an input source at an input voltage to one or more circuit loads on one or more semiconductor chips mounted in a semiconductor package at a DC output voltage, comprising: providing one or more rail selection circuits on the one or more semiconductor chips, each having inputs connected to the output rails and an output connected to deliver power to a respective load at a respective load voltage, each rail selection circuit having a plurality of rail switches connected to provide a selectable path for current flow from a selected one of the rails to the output; wherein each rail selection circuit is operative to enable a respective rail switch to conduct current from selected output rail to its respective output in combination with the rest of the limitations of the base claims and any intervening claims. Claims 151-164 are allowed because the closest prior art of record fails to disclose a method of supplying power received from an input source at an input voltage to one or more circuit loads on one or more semiconductor chips mounted in a semiconductor package at a DC output voltage, comprising providing one or more driver circuits; providing one or more output

Art Unit: 2842

circuits; providing a power bus; and providing a first capacitor; wherein the driver circuit is adapted to drive the transformer with an average voltage of essentially zero volts; wherein the one or more output circuits are adapted to receive power from the driver circuit at a peak-to-peak bus voltage and supply power to the one or more semiconductor chips at a load voltage, wherein the peak-to-peak bus voltage is a multiple of the load voltage and that multiple is at least six in combination with the rest of the limitations of the base claims and any intervening claims. Claim 164 is allowed because the closest prior art of record fails to disclose an apparatus comprising: one or more rail selection circuits disposed on the one or more semiconductor chips, each having inputs electrically coupled to the output rails and an output electrically coupled to deliver power to a respective load at a respective load voltage, each rail selection circuit having a plurality of rail switches electrically coupled to provide a selectable path for current to flow from a selected one of the output rails to the respective output in combination with the rest of the limitations of the base claim and any intervening claims.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

4. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to CASSANDRA COX whose telephone number is (571)272-1741. The examiner can normally be reached on Monday-Thursday from 6:00 AM to 3:30 PM and alternate Fridays from 7:00 AM to 3:30 PM.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Donovan Lincoln can be reached on 571-272-1988. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Art Unit: 2842

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/CASSANDRA COX/
Primary Examiner, Art Unit 2842

Notice of References Cited	Application/Control No. 15/616,288		Applicant(s)/Patent Under Reexamination VINCIARELLI ET AL.	
	Examiner CASSANDRA COX		Art Unit 2842	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-6,118,186 A	09-2000	Scott; Harold C.	H02P9/04	290/1A
*	B	US-2006/0291265 A1	12-2006	Schrom; Gerhard	G11C17/18	365/94
*	C	US-7,245,509 B1	07-2007	Chan; Chun-Kong	H02M7/538	363/17
*	D	US-2015/0188407 A1	07-2015	Golder; Barnaby	H02J1/00	307/31
*	E	US-9,765,750 B2	09-2017	Ferguson; Tod	F02P5/15	1/1
	F	US-				
	G	US-				
	H	US-				
	I	US-				
	J	US-				
	K	US-				
	L	US-				
	M	US-				

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	283	vinciarelli-patrizio.in.	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/06 11:22
L2	33690	switch\$3 with capacitor with transformer	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/06 11:22
L3	71	L1 and L2	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/06 11:22
S1	259144	semiconductor adj chip	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/06/20 13:01
S2	150649	driver adj circuit	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/06/20 13:01
S3	4686	output adj rail	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/06/20 13:02
S4	216	rail adj selection	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/06/20 13:02
S5	4114	rail adj switch\$3	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/06/20 13:02
S6	0	S1 and S2 and S3 and S4 and S5	US-PGPUB; USPAT;	OR	ON	2018/06/20 14:08

			USOCR; EPO; JPO; DERWENT; IBM_TDB			
S7	0	S1 and S2 and S3 and S5	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/06/20 14:08
S8	0	S2 and S3 and S4 and S5	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/06/20 14:08
S9	3	S2 and S3 and S5	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/06/20 14:09
S10	30	("8582333" "9660537" "6031743" "6035261" "6275958" "6300749" "6421262" "6466458" "6603292" "6700361" "6788033" "6911848" "6934166" "6969909" "6985341" "7145786" "7521909" "7561446" "7768807" "8232783" "RE40072" "0014731" "0014822" "0014874" "20130069608" "6930893" "6975098" "6984965" "9112422" "9166481").PN.	US-PGPUB; USPAT	OR	ON	2018/06/25 10:09
S11	28151	(H02M1/08 or H02M3/155 or H01L23/645 or H01L23/5385 or H01L23/5383 or H01L23/5386 or H01L25/18 or H01L23/5384).cpc.	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/06/25 10:12
S12	4691	output adj rail	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/06/25 10:15
S13	216	rail adj selection	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/06/25 10:15
S15	4124	rail adj switch\$3	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/06/25 10:16
S16	78	S12 and S15	US-PGPUB; USPAT; USOCR; EPO; JPO;	OR	ON	2018/06/25 10:16

			DERWENT; IBM_TDB			
S17	235	point adj load adj circuit	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/06/25 10:29
S18	4747	output adj rail	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/11/28 13:04
S19	220	rail adj selection	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/11/28 13:04
S20	11	S18 and S19	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/11/28 13:04
S21	64553	rail\$1 with switch\$3	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/11/28 13:04
S22	9311	rail adj voltage	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/11/28 13:05
S23	264359	semiconductor adj chip	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/11/28 13:06
S24	154228	driver adj circuit	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/11/28 13:06
S25	94	S23 and S24 and S21	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/11/28 13:06
S26	33	("20130030534" "5438294" "9087656" "8582333" "9660537" "6031743" "6035261" "6275958" "6300749" "6421262" "6466458"	US-PGPUB; USPAT	OR	ON	2018/12/03 11:46

		"6603292" "6700361" "6788033" "6911848" "6934166" "6969909" "6985341" "7145786" "7521909" "7561446" "7768807" "8232783" "RE40072" "0014731" "0014822" "0014874" "20130069608" "6930893" "6975098" "6984965" "9112422" "9166481").PN.				
S27	32900	power adj bus	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/03 11:54
S28	64620	rail\$1 with switch\$3	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/03 11:54
S29	264463	semiconductor adj chip	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/03 11:54
S30	154295	driver adj circuit	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/03 11:54
S31	94	S29 and S30 and S28	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/03 11:54
S32	0	S27 and S31	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/03 11:54
S33	31113	(H02M1/08 or H02M3/155 or H01L23/645 or H01L23/5385 or H01L23/5383 or H01L23/5386 or H01L25/18 or H01L23/5384).cpc.	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/03 11:55
S34	25	S33 and S31	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/03 11:55
S35	16992	voltage adj driver	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT;	OR	ON	2018/12/03 15:01

			IBM_TDB			
S36	9315	rail adj voltage	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/03 15:33
S37	448	(dc or direct adj current) adj rail adj voltage	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/03 15:33
S38	41	plural\$3 adj3 rail adj voltage	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/03 15:34
S39	107	(plural\$3 or multiple) adj3 rail adj voltage	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/03 15:34
S40	0	(plural\$3 or multiple) adj3 rail adj voltage6 and S39	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/03 15:35
S41	154364	driver adj circuit	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/04 08:24
S42	108	(plural\$3 or multiple) adj3 rail adj voltage	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/04 08:24
S43	11	(plural\$3 or multiple) adj3 rail adj voltage and S41 and S42	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/04 08:24
S44	25	generat\$3 near3 (plural\$3 or multiple) adj3 rail adj voltage	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/04 09:17
S45	5	S41 and S42 and S44	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT;	OR	ON	2018/12/04 09:17

			IBM_TDB			
S46	108	(plural\$3 or multiple) adj3 rail adj voltages	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/04 09:18
S47	52	driver and S46	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/04 09:31
S48	1592	(generat\$3 or output\$4) near3 rail adj voltage\$1	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/04 09:31
S49	26	S47 and S48	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/04 09:32
S50	1475	half adj bridge adj driver	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/04 13:59
S51	33673	switch\$3 with capacitor with transformer	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/04 13:59
S52	136	S50 and S51	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/04 13:59
S53	18	S50 same S51	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/04 13:59
S54	31140	(H02M1/08 or H02M3/155 or H01L23/645 or H01L23/5385 or H01L23/5383 or H01L23/5386 or H01L25/18 or H01L23/5384).cpc.	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/04 14:00
S55	1	S52 and S54	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT;	OR	ON	2018/12/04 14:00

			IBM_TDB			
S56	283	vinciarelli-patrizio.in.	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/06 06:59
S57	9	damico-andrew-t.in.	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/06 06:59
S58	33690	switch\$3 with capacitor with transformer	US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/12/06 07:00

12/ 6/ 2018 11:34:34 AM
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Sheet 1 of 1

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0180001	Application No. 15/616,288
	Applicant VLT, Inc.	
	Filing Date June 7, 2017	Group Art Unit 2842

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	1.	Cao et al., "A Family of Zero Current Switching Switched-Capacitor DC-DC converters," Applied Power Electronics Conference and Exposition (APEC), 2010 Twenty-Fifth Annual IEEE, Mar. 18, 2010
	2.	Cao et al., "Zero-Current-Switching Multilevel Modular Switched-Capacitor DC-DC Converter" IEEE Transactions on Industry Applications, 46(6):2536-2544, Sep. 2010
	3.	Oraw et al., "Load Adaptive, High Efficiency, Switched Capacitor Intermediate Bus Converter," Telecommunications Energy Conference, IEEE 2007, pp. 628-635 Sept. 30 – Oct. 4, 2007
	4.	Seeman et al., "Analysis and optimization of switched-capacitor dc-dc converters," Power Electronics, IEEE Transactions on, vol. 23, pp. 841–851, March 2008
	5.	Yeung et al., "Generalised analysis of switched-capacitor step-down quasi-resonant converter," Electronics Letters, vol. 38, pp. 263-264, 2002
	6.	Yeung et al., "Zerocurrent switching switched-capacitor quasiresonant step-down converter," Electric Power Applications, IEE Proceedings -, vol. 149, pp. 111-121, 2002

Examiner Signature /CASSANDRA F COX/	Date Considered 12/04/2018
EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

Substitute Disclosure Form

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Sheet 1 of 1

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0180001	Application No. 15/616,288
	Applicant VLT, Inc.	
	Filing Date June 7, 2017	Group Art Unit 2842

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1.	5,438,294	8/1/1995	Smith			
	2.	9,087,656	7/21/2015	Vinciarelli			
	3.	2013/0030534	1/31/2013	DeLurio et al.			

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	4.	Notice of Allowance in U.S. Application No. 15/091,346, dated May 21, 2018, 9 pages
	5.	Notice of Allowance in U.S. Application No. 15/091,346, dated August 13, 2018, 8 pages

Examiner Signature /CASSANDRA F COX/	Date Considered 12/04/2018
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Substitute Disclosure Form

Issue Classification 	Application/Control No. 15616288	Applicant(s)/Patent Under Reexamination VINCIARELLI ET AL.	
	Examiner CASSANDRA COX	Art Unit 2842	

CPC						
Symbol					Type	Version
H02M		1	/	08	F	2013-01-01
H01L		25	/	18	I	2013-01-01
H01L		23	/	5386	I	2013-01-01
H01L		23	/	5383	I	2013-01-01
H01L		23	/	5384	I	2013-01-01
H01L		23	/	5385	I	2013-01-01
H01L		23	/	645	I	2013-01-01
H02M		3	/	155	I	2013-01-01
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CPC Combination Sets				
Symbol	Type	Set	Ranking	Version

NONE		Total Claims Allowed:	
(Assistant Examiner)		34	
(Date)			
/CASSANDRA COX/	12/6/18	O.G. Print Claim(s)	O.G. Print Figure
Primary Examiner.Art Unit 2842		193	21B
(Primary Examiner)	(Date)		

Search Notes 	Application/Control No. 15616288	Applicant(s)/Patent Under Reexamination VINCIARELLI ET AL.
	Examiner CASSANDRA COX	Art Unit 2842

CPC- SEARCHED		
Symbol	Date	Examiner

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

SEARCH NOTES		
Search Notes	Date	Examiner
EAST Search (USPAT, USPGPUB, USOCR, EPO, JPO, DERWENT, IBM_TDB) see search history	12/6/18	cfc
H02M1/08 or H02M3/155 or H01L23/645 or H01L23/5385 or H01L23/5383 or H01L23/5386 or H01L25/18 or H01L23/5384 (text only - see search history)	12/6/18	cfc
inventor search	12/6/18	cfc

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner
H02M	1/08, 3/155	12/6/18	cfc
H01L	23/645, 23/5385, 23/5383, 23/5386, 25/18, 23/5384	12/6/18	cfc

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Bibliographic Data

Application No: 15/616,288

Foreign Priority claimed: ☐ Yes ☒ No

35 USC 119 (a-d) conditions met: ☐ Yes ☒ No ☐ Met After Allowance

Verified and Acknowledged: /CASSANDRA F COX/

Examiner's Signature

Initials

Title: METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
06/07/2017	327	2842	
RULE			

APPLICANTS

VLT, Inc., Andover, MA, UNITED STATES

INVENTORS

Patrizio Vinciarelli Boston, MA, UNITED STATES

Andrew T. D'Amico Beverly Hills, CA, UNITED STATES

CONTINUING DATA

This application is a CIP of 15091346 04/05/2016 PAT 10158357

FOREIGN APPLICATIONS

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Sheet 1 of 2

Substitute Disclosure Form Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0180001	Application No. 15/616,288
	Applicant VLT, Inc.		
	Filing Date June 7, 2017	Group Art Unit	

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1.	6,031,743	02/29/2000	Carpenter et al.			
	2.	6,035,261	03/07/2000	Carpenter et al.			
	3.	6,275,958	08/14/2001	Carpenter et al.			
	4.	6,300,749	10/01/2001	Castelli et al.			
	5.	6,421,262	07/16/2002	Saxelby et al.			
	6.	6,466,458	10/15/2002	Zhang et al.			
	7.	6,603,292	08/01/2003	Schouten et al.			
	8.	6,700,361	03/01/2004	Gregorius, Peter			
	9.	6,788,033	09/07/2004	Vinciarelli			
	10.	6,911,848	06/28/2005	Vinciarelli			
	11.	6,934,166	08/23/2005	Vinciarelli			
	12.	6,969,909	11/29/2005	Briere			
	13.	6,985,341	01/10/2006	Vinciarelli, et al.			
	14.	7,145,786	12/05/2006	Vinciarelli			
	15.	7,521,909	04/01/2009	Dow et al.			
	16.	7,561,446	07/14/2009	Vinciarelli			
	17.	7,768,807	08/01/2010	Chen, et al.			
	18.	8,232,783	07/01/2012	Yanagawa, Kenji			
	19.	RE40,072	02/19/2008	Prager, et al.			

Foreign Patent Documents or Published Foreign Patent Applications							
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation
							Yes No

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	20.	Amendment after Allowance Pursuant to 37 C.F.R. §1.312 in U.S. Application No. 13/044,063, dated March 3, 2015, 17 pages

Examiner Signature /CASSANDRA F COX/	Date Considered 12/04/2018
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EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /C.F.C/

Substitute Disclosure Form

Sheet 2 of 2

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office	Attorney Docket No. 00614-0180001	Application No. 15/616,288
Information Disclosure Statement by Applicant (Use several sheets if necessary)	Applicant VLT, Inc.	
	Filing Date June 7, 2017	Group Art Unit
(37 CFR §1.98(b))		

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	21.	AN2738, Application Note AN2738, "L6390 half-bridge gate driver," STMicroelectronics, August 2009
	22.	Burton et al., "FIVR - Fully Integrated Voltage Regulators on 4th Generation Intel ® Core™ SoCs," Applied Power Electronics Conference and Exposition (APEC), 2014 Twenty-Ninth Annual IEEE, March 16, 2014, 432-439
	23.	ISL6210, Datasheet ISL6210, "Dual Synchronous Rectified MOSFET Drivers," Intersil Americas Inc., December 2008
	24.	Notice of Allowance in U.S. Application No. 13/044,063, Examiner Nguyen, dated February 20, 2015, 15 pages
	25.	Notice of Allowance in U.S. Application No. 13/044,063, Examiner Nguyen, dated June 5, 2015, 11 pages
	26.	Notice of Allowance in U.S. Application No. 14/682,187, dated January 17, 2017, 22 pages
	27.	Office Action in U.S. Application No. 13/044,063, Examiner Nguyen, dated May 23, 2014, 24 pages
	28.	Reply to Action of May 23, 2014 in U.S. Application No. 13/044,063, dated October 24, 2014, 39 pages
	29.	Supplemental Notice of Allowability in U.S. Application No. 13/044,063, Examiner Nguyen, dated April 20, 2015, 5 pages

Examiner Signature /CASSANDRA F COX/	Date Considered 12/04/2018
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Sheet 1 of 1

Substitute Disclosure Form U.S. Department of Commerce Patent and Trademark Office Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	Attorney Docket No. 00614-0180001	Application No.
	Applicant VL T, Inc.	
	Filing Date	Group Art Unit

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1.	14/731,287		Vinciarelli et al.			06/04/2015
	2.	14/822,561		Vinciarelli			08/10/2015
	3.	14/874,054		Vinciarelli			10/02/2015
	4.	6,930,893	08/16/2005	Vinciarelli			
	5.	6,975,098	12/13/2005	Vinciarelli			
	6.	6,984,965	01/10/2006	Vinciarelli			
	7.	2013/0069608	03-2013	Gakhar et al.			
	8.	9,112,422	08/18/2015	Vinciarelli			
	9.	9,166,481	10/15/2015	Vinciarelli et al.			

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No
	10.	JPH02280666	11/16/1990	JP			English Abstract	
	11.	EP2863531	4/22/2015	EP				

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document
	12.	DiBene II et al., "A 400 Amp Fully Integrated Silicon Voltage Regulator with in-die Magnetically Coupled Embedded Inductors," retrieved from http://www.psma.com/sites/default/files/uploads/tech-forums-nanotechnology/resources/400a-fully-integrated-silicon-voltage-regulator.pdf , retrieved on August 3, 2016, 25 pages.
	13.	European Office Communication in European Application No. 16173089.0, dated April 10, 2017, 2 pages
	14.	Extended Search Report in European Application No. 16173089.0, dated February 14, 2017, 8 pages
	15.	Miller, "12V PowerStage in Embedded Die System-in-Package," APEC 2015, 33 pages
	16.	Okudaira and Matsuse, "A New Quasi-Resonant Inverter With Two-way Short-circuit Switch Across A Resonant Capacitor," Power Conversion Conference 2002, 1496-1501
	17.	Watson, "New Techniques in the Design of Distributed Power Systems," Dissertation, Virginia Polytechnic Institute and State University, 1998, 32 pages

Examiner Signature /CASSANDRA F COX/	Date Considered 12/04/2018
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EXAMINER: Initials citation considered. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

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EAST Search History

EAST Search History (I nterference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L4	18701	(H02M1/08 or H02M3/155 or H01L23/645 or H01L23/5385 or H01L23/5383 or H01L23/5386 or H01L25/18 or H01L23/5384).cpc.	US-PGPUB; USPAT	OR	ON	2018/12/06 11:26
L5	20717	switch\$3 with capacitor with transformer	US-PGPUB; USPAT	OR	ON	2018/12/06 11:26
L6	8571	rail adj voltage	US-PGPUB; USPAT	OR	ON	2018/12/06 11:27
L7	28360	rail\$1 with switch\$3	US-PGPUB; USPAT	OR	ON	2018/12/06 11:27
L8	27971	power adj bus	US-PGPUB; USPAT	OR	ON	2018/12/06 11:27
L9	734	l4 and l5	US-PGPUB; USPAT	OR	ON	2018/12/06 11:28
L10	0	l6 and l8 and l9	US-PGPUB; USPAT	OR	ON	2018/12/06 11:28
L11	251	l6 and l8	US-PGPUB; USPAT	OR	ON	2018/12/06 11:28
L12	29	l6 and l9	US-PGPUB; USPAT	OR	ON	2018/12/06 11:28
L13	0	l6 and l8 and l4	US-PGPUB; USPAT	OR	ON	2018/12/06 11:29
L14	60	l7 and l11	US-PGPUB; USPAT	OR	ON	2018/12/06 11:29

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Sheet 1 of 1

Substitute Disclosure Form Information Disclosure Statement by Applicant (Use several sheets if necessary) (37 CFR §1.98(b))	U.S. Department of Commerce Patent and Trademark Office		Attorney Docket No. 00614-0180001	Application No. 15/616,288
	Applicant VLT, Inc.			
	Filing Date June 7, 2017		Group Art Unit 2842	

U.S. Patent Documents							
Examiner Initial	Desig. ID	Document Number	Publication Date	Patentee	Class	Subclass	Filing Date If Appropriate
	1.	8,582,333	11/12/2013	Oraw et al.			
	2.	9,660,537	05/23/2017	Vinciarelli			

Foreign Patent Documents or Published Foreign Patent Applications								
Examiner Initial	Desig. ID	Document Number	Publication Date	Country or Patent Office	Class	Subclass	Translation	
							Yes	No

Other Documents (include Author, Title, Date, and Place of Publication)		
Examiner Initial	Desig. ID	Document

Examiner Signature /CASSANDRA F COX/	Date Considered 12/04/2018
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Cc: PAIR_eOfficeAction@uspto.gov
Subject: Private PAIR Correspondence Notification for Customer Number 26214

Dec 11, 2018 04:01:12 AM

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P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022
UNITED STATES

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Application	Document	Mailroom Date	Attorney Docket No.
15616288	NOA	12/11/2018	00614-0180001
	892	12/11/2018	00614-0180001
	1449	12/11/2018	00614-0180001
	1449	12/11/2018	00614-0180001
	1449	12/11/2018	00614-0180001
	1449	12/11/2018	00614-0180001
	1449	12/11/2018	00614-0180001

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PATENT APPLICATION INFORMATION RETRIEVAL SYSTEM

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26214 7590 12/11/2018

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(Typed or printed name)

(Signature)

(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/616,288	06/07/2017	Patrizio Vinciarelli	00614-0180001	7438

TITLE OF INVENTION: METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	LARGE	\$1000	\$0		\$1000	03/11/2019
EXAMINER		ART UNIT	CLASS-SUBCLASS			
COX, CASSANDRA F.		2842				

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
- ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-09 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, 1 Fish & Richardson P.C.
- (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____
- 3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE
 VLT, Inc.

(B) RESIDENCE: (CITY and STATE OR COUNTRY)
 Andover, MA

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. Fees Submitted: ☒ Issue Fee ☐ Publication Fee (if required) ☒ Advance Order - # of Copies 5

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5. Change of Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
- ☐ Applicant asserting small entity status. See 37 CFR 1.27.
- ☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid Certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken as a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken as a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature /Rex I. Huang/

Date March 11, 2019

Typed or printed name Rex I. Huang

Registration No. 57,661

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor :	Patrizio Vinciarelli	Art Unit :	2842
Application No. :	15/616,288	Examiner :	Cassandra Cox
Filed :	June 7, 2017	Confirmation No. :	7438
		Notice of Allowance Date:	December 11, 2018
Title :	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS		

MAIL STOP ISSUE FEE

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

REPLY TO NOTICE OF ALLOWANCE

In response to the Notice of Allowance dated December 11, 2018, enclosed is a completed Part B - Fee(s) Transmittal.

The issue fee is being paid with this reply on the Electronic Filing System. Apply the fee and any other necessary charges or credits to Deposit Account 06-1050, referencing the above attorney docket number.

COMMENTS ON EXAMINER'S REASONS FOR ALLOWANCE

It is recognized that in accordance with M.P.E.P. § 1302.14, the Examiner's reasons for allowance need not set forth all of the details as to why the claims are allowed. In the above-referenced application, it is not conceded that the Examiner's stated reasons for allowance are the only reasons for which the claims are allowable. The Examiner's reasons for allowance indicate that particular claim elements are not disclosed or suggested by the prior art of record, yet the claims may be patentable for other reasons as well, including the inventive combination of all of the recited claim elements. It is not conceded that the specific limitations identified by the Examiner are necessary to distinguish the art of record or to satisfy the requirements of 35 U.S.C. § 112. Moreover, the Examiner does not assert, and it would not be conceded, that the Examiner's reasons have any bearing on the patentability of claims in any other applications directed to the disclosed subject matter.

First Named Inventor : Patrizio Vinciarelli
Application No. : 15/616,288
Filed : June 7, 2017
Page : 2 of 2

Attorney Docket: 00614-0180001 / 0254-US-1

In addition, each dependent claim stands on its own and may be allowable on its own merits. In particular, each dependent claim may be allowable on the basis of a combination of some of the features recited in the dependent claim and its base claim(s), which combination of features may not include all of the limitations identified in the Examiner's reasons for allowance.

Respectfully submitted,

Date: March 11, 2019_____

/Rex I. Huang/_____

Rex I. Huang
Reg. No. 57,661

Customer Number 26214
Fish & Richardson P.C.
Telephone: (617) 542-5070
Facsimile: (877) 769-7945

23865243.doc

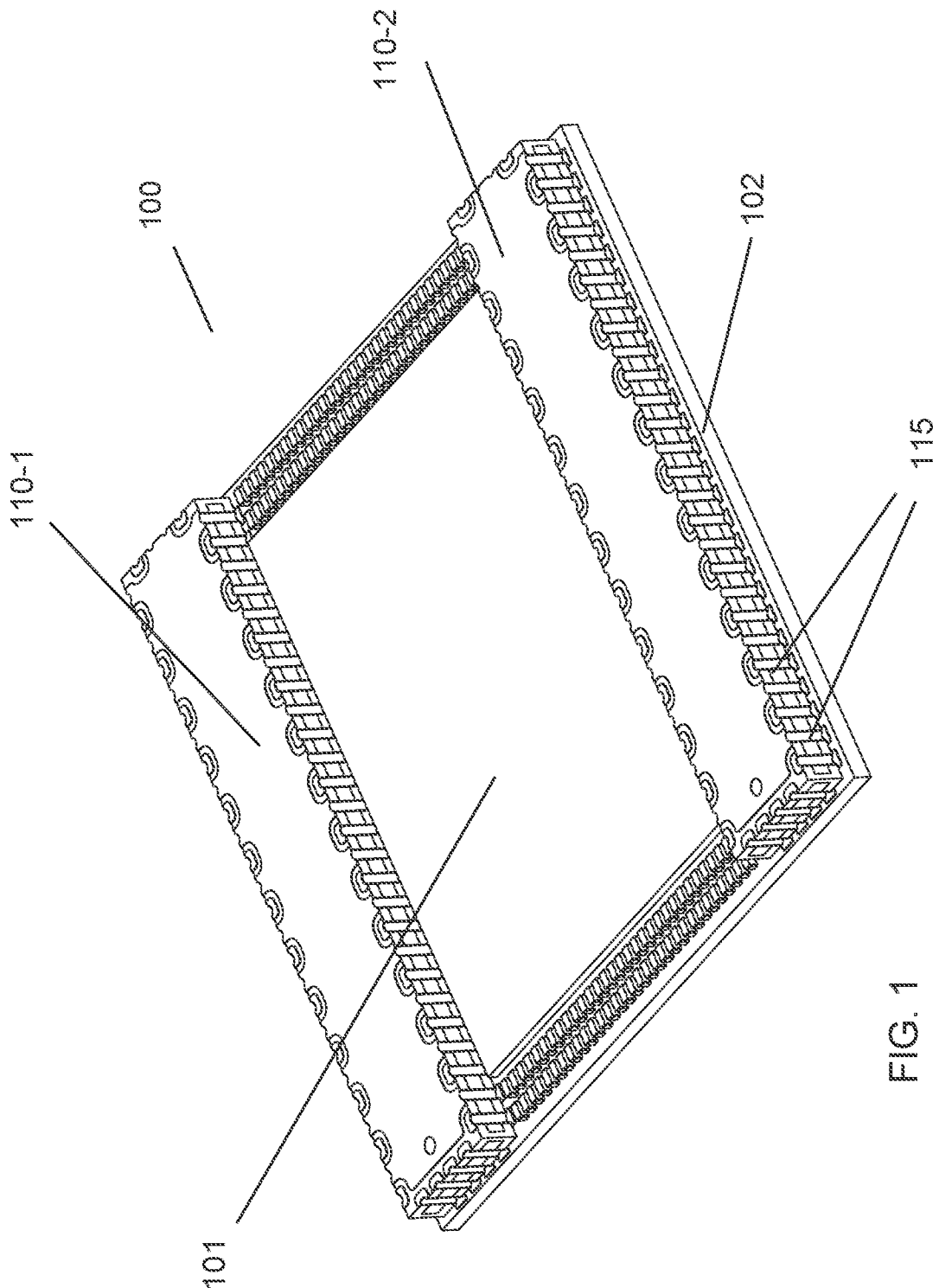
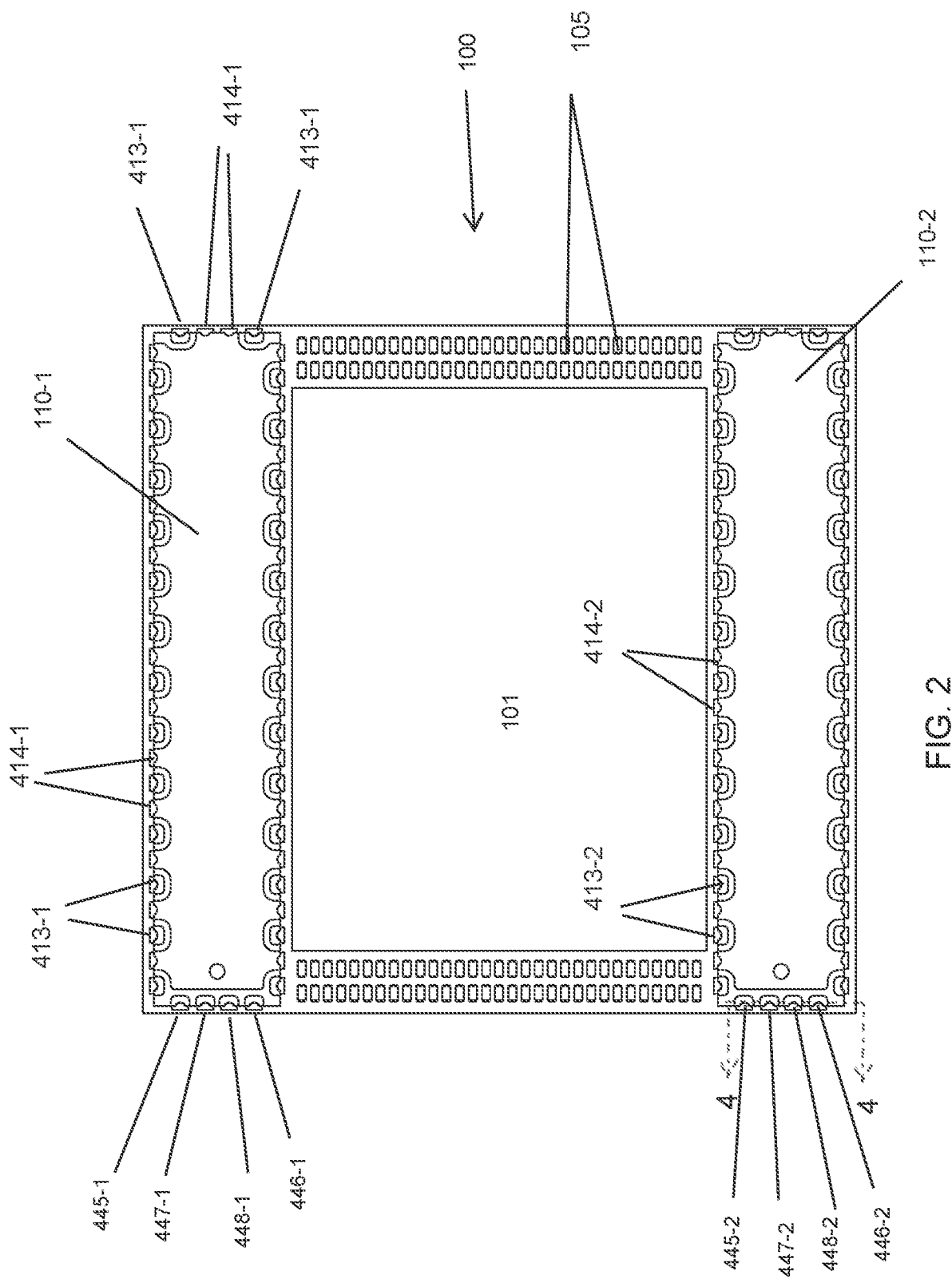


FIG. 1



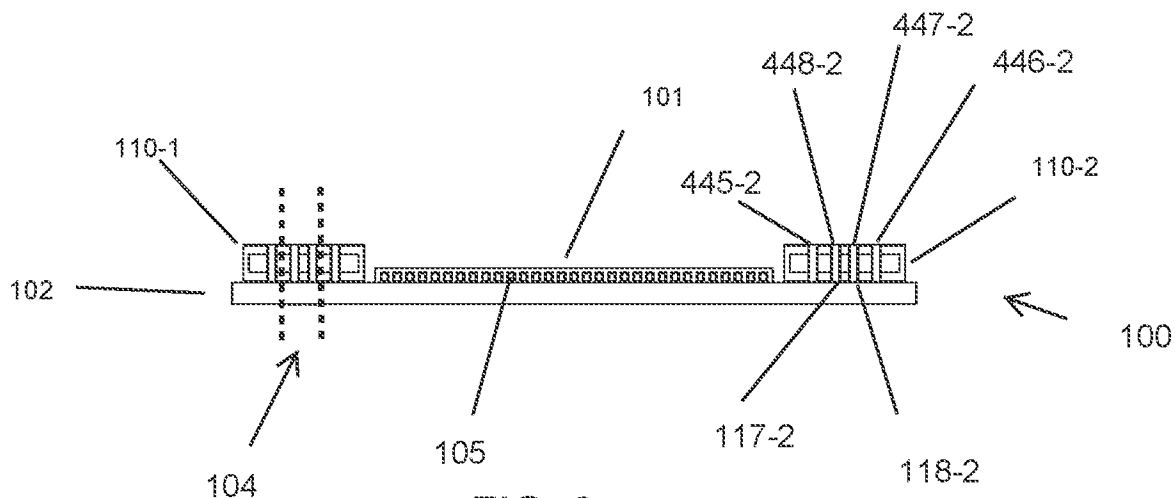


FIG. 3

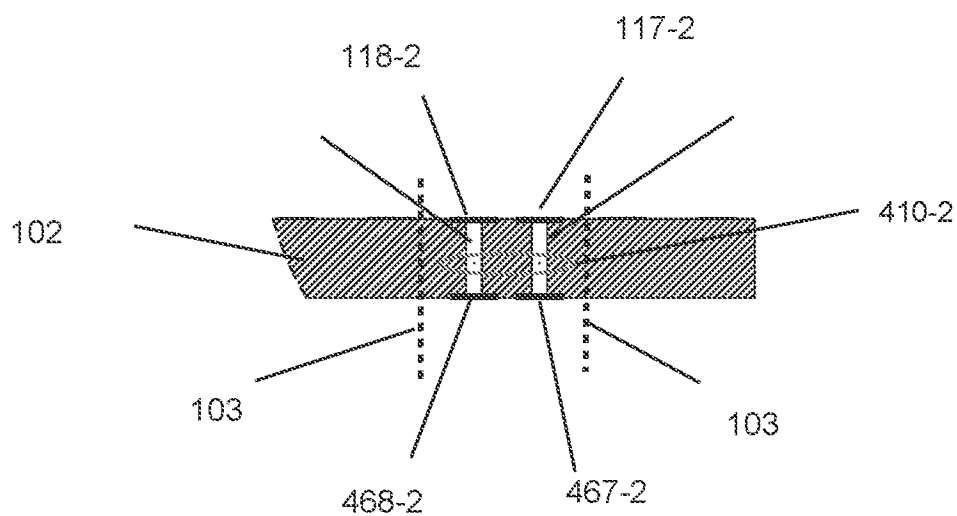


FIG. 4

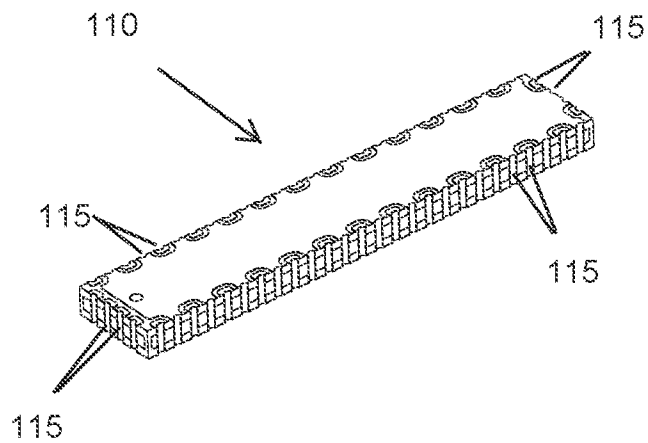


FIG. 8

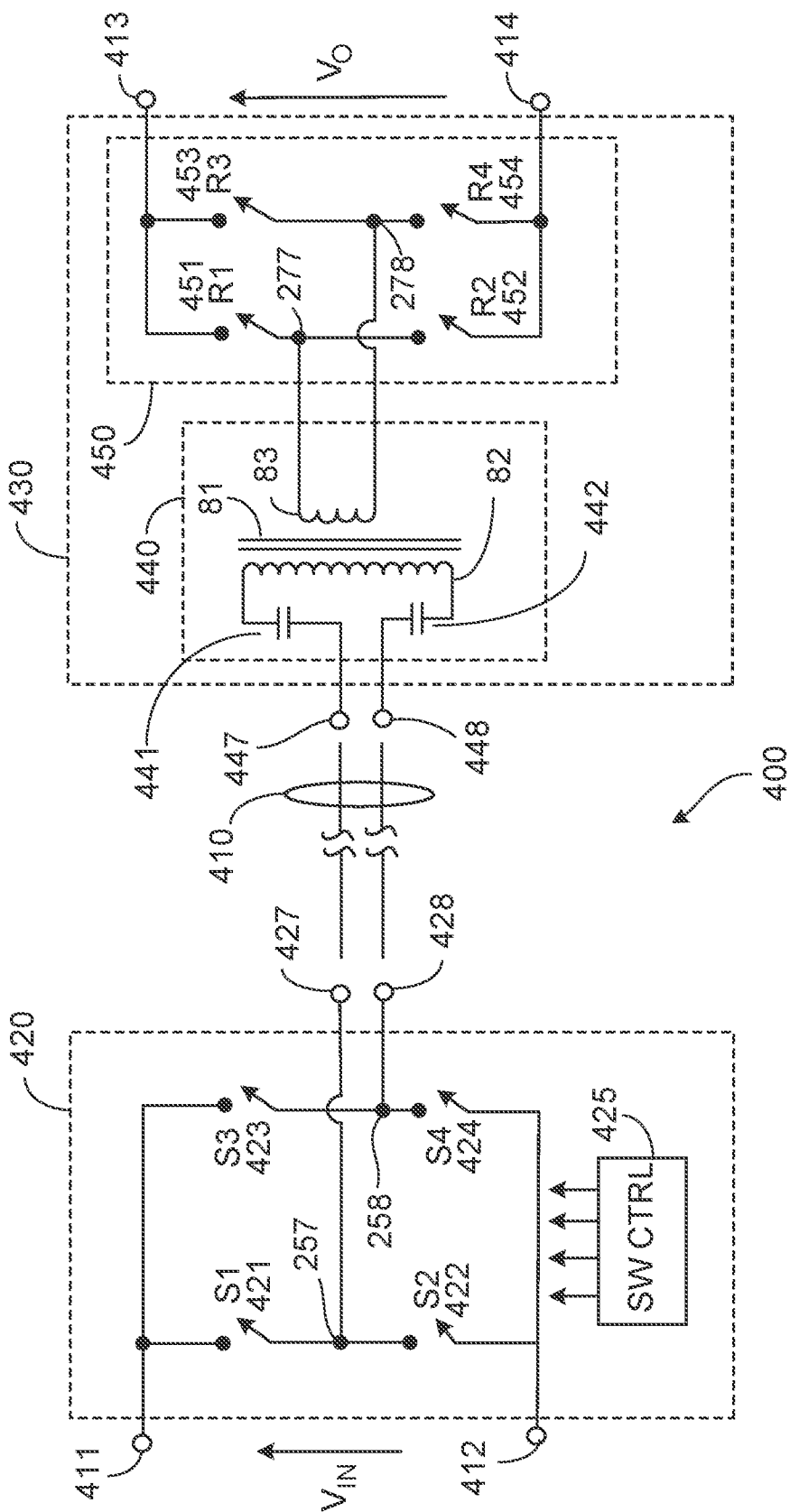


FIG. 5

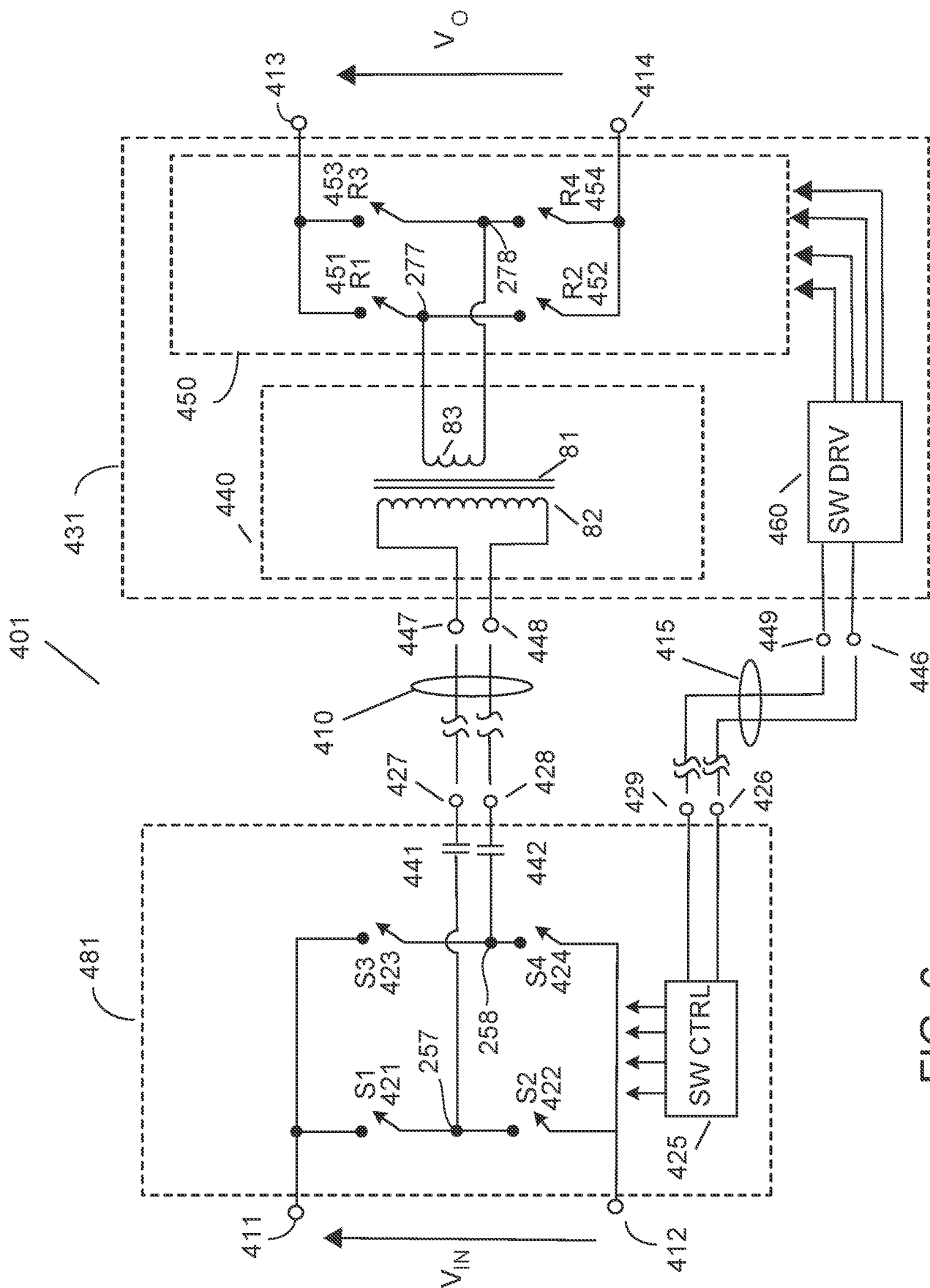


FIG. 6



7

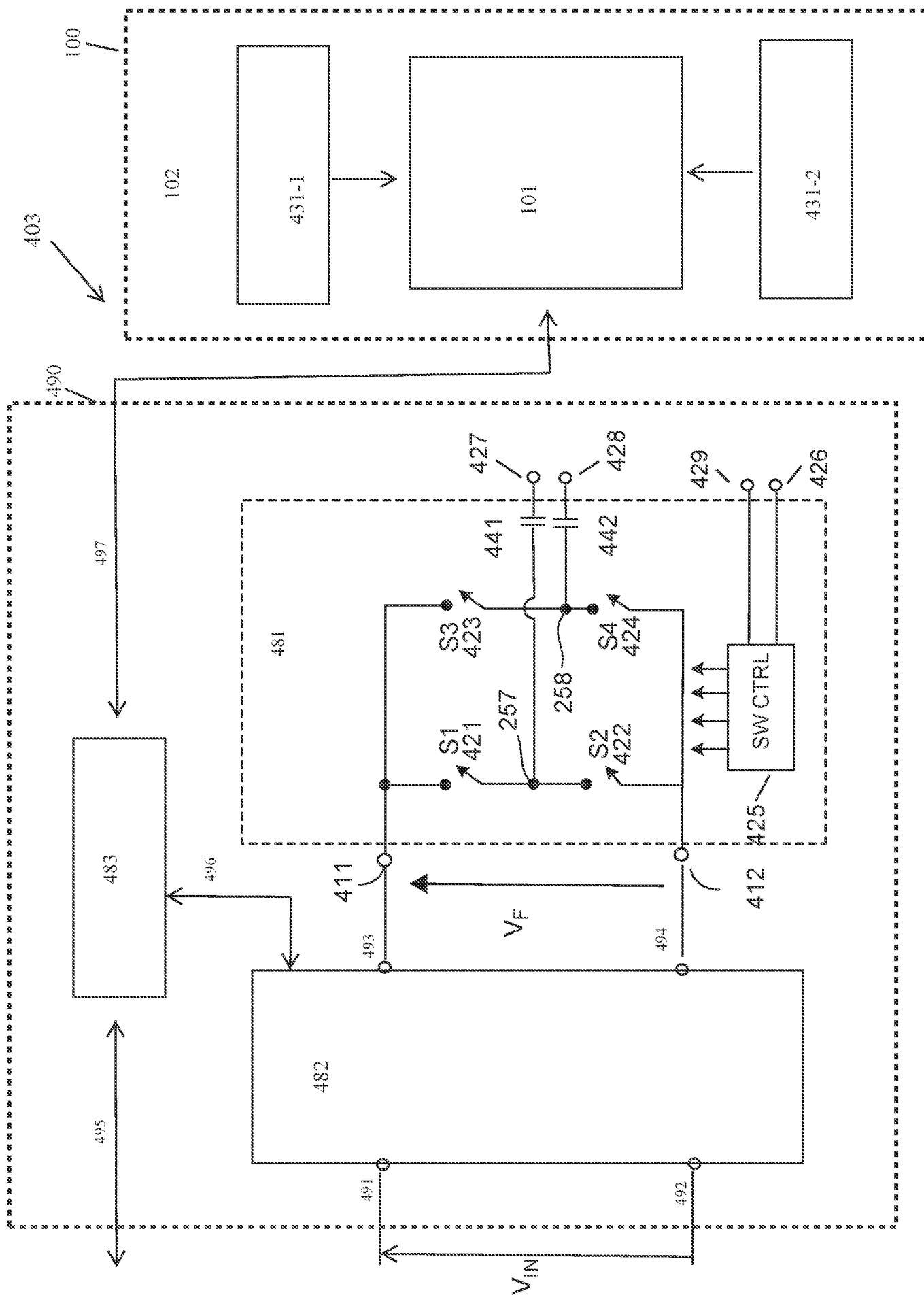


FIG. 9

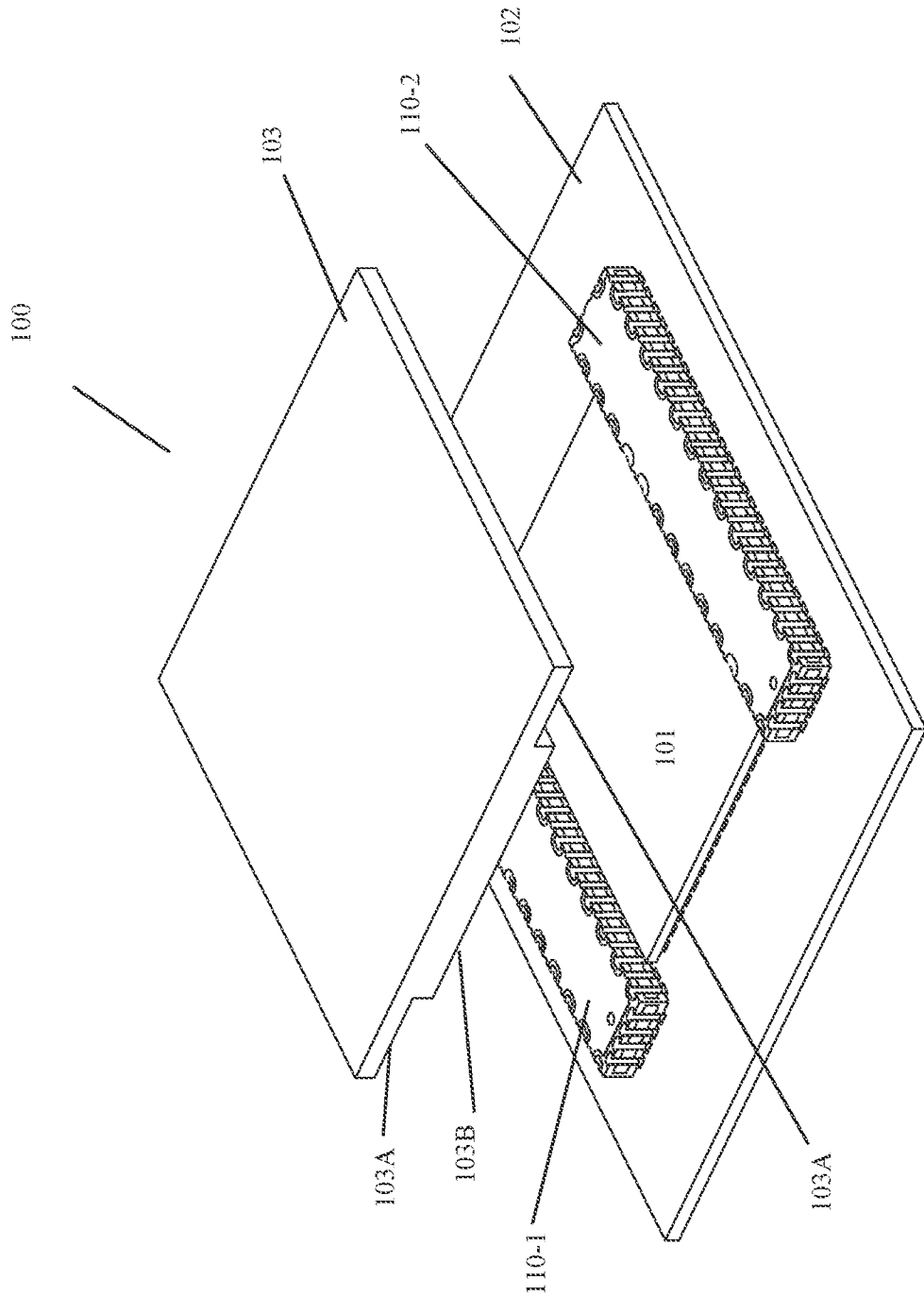
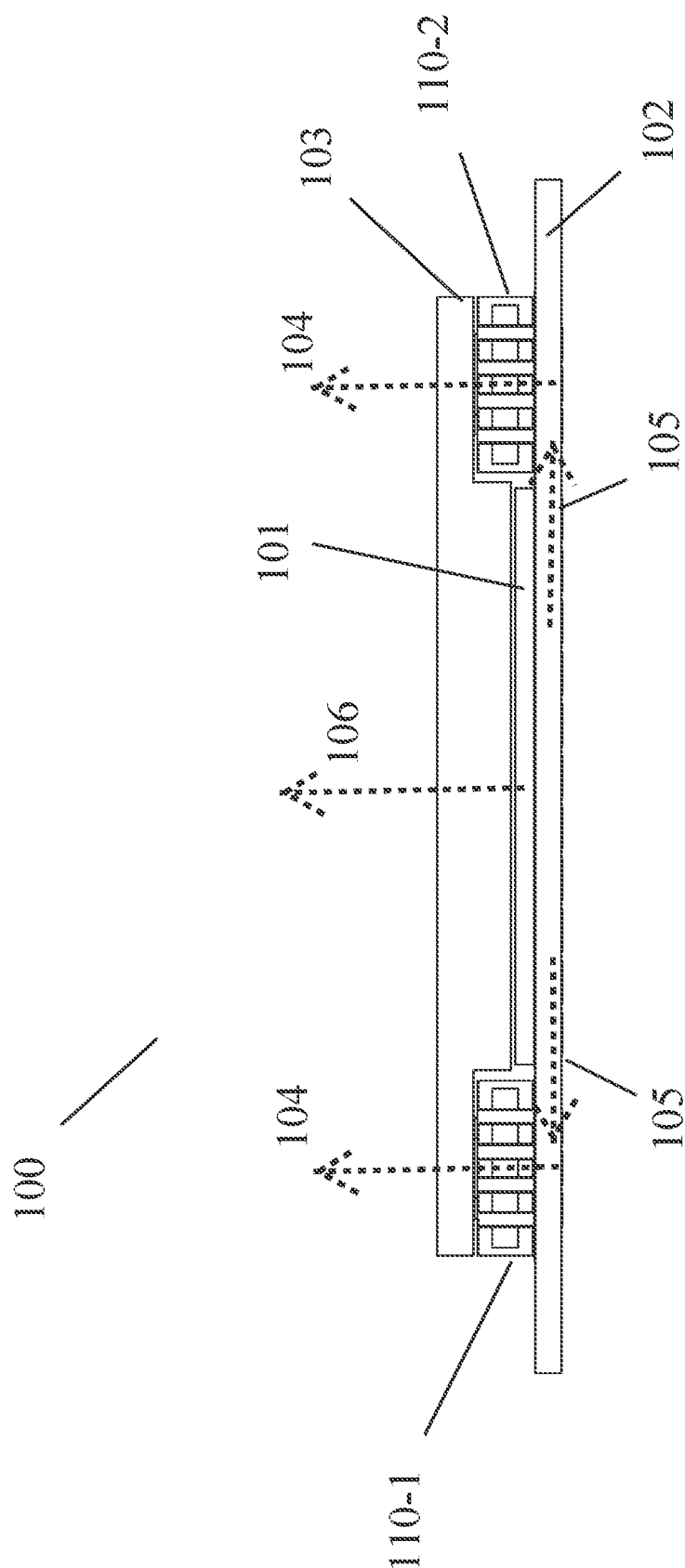


FIG. 10



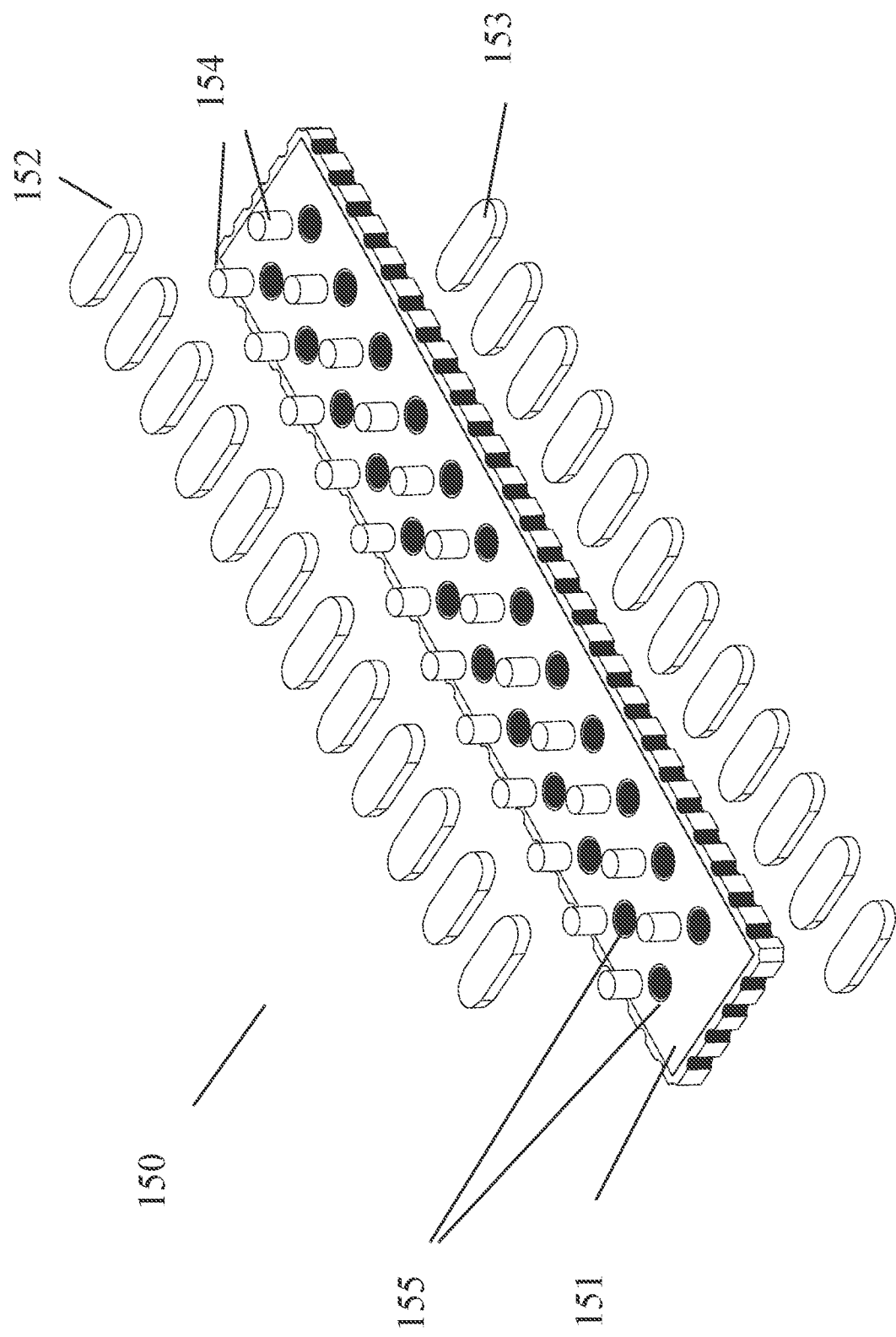


FIG. 12

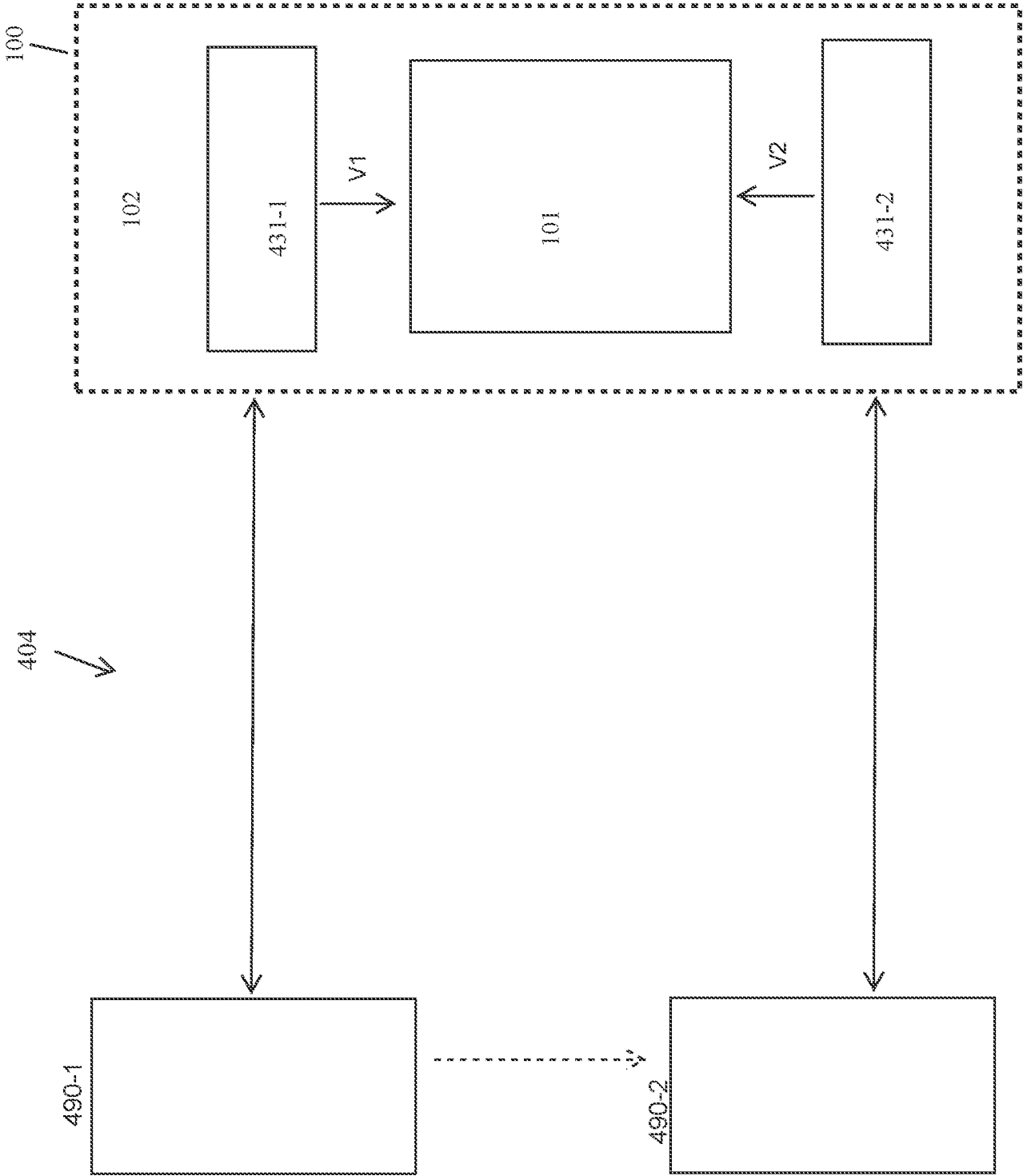


FIG. 13

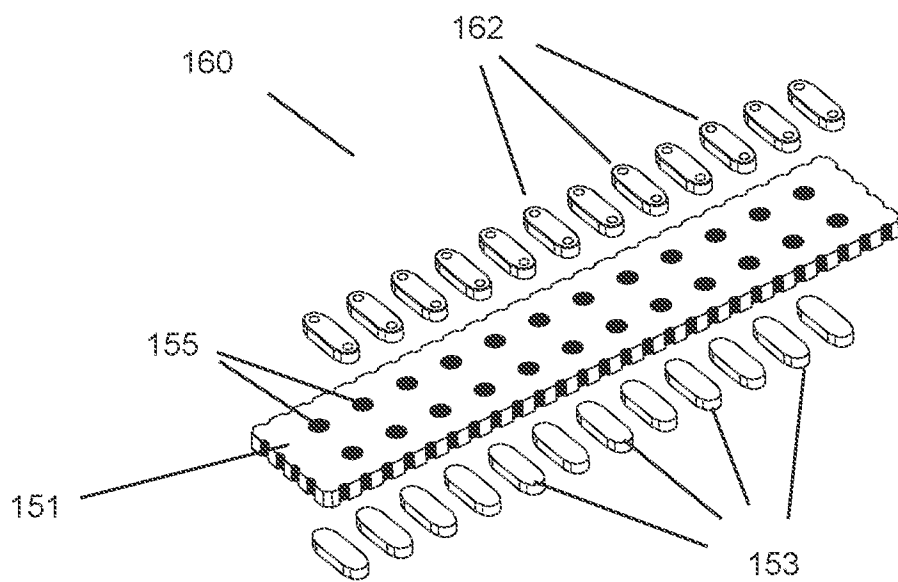


FIG. 14A

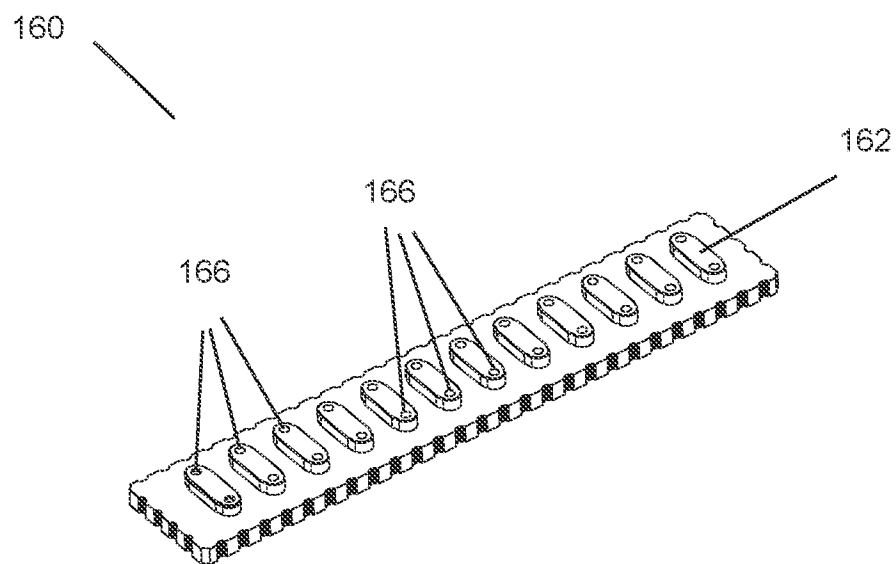


FIG. 14B

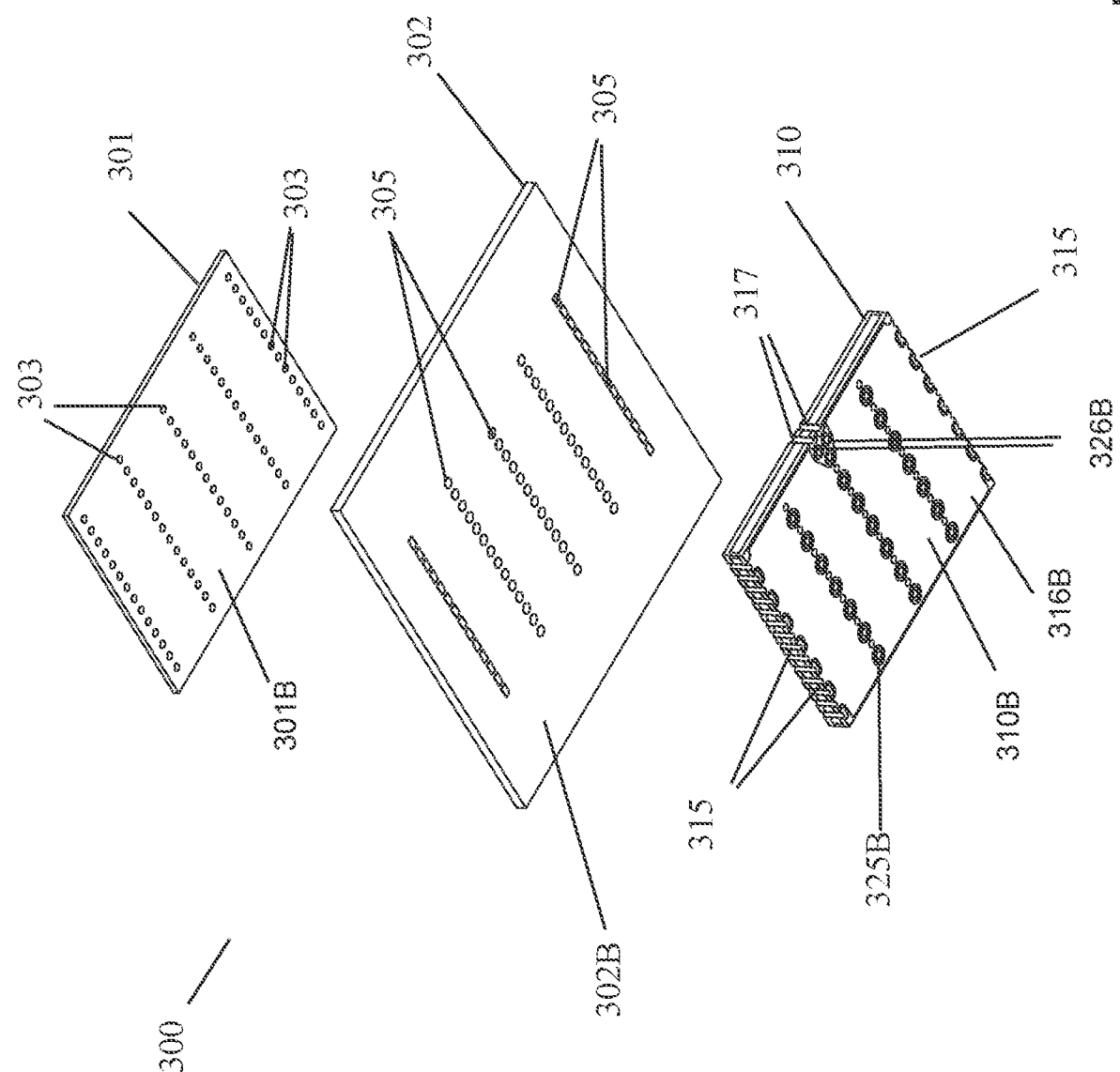


FIG. 15A

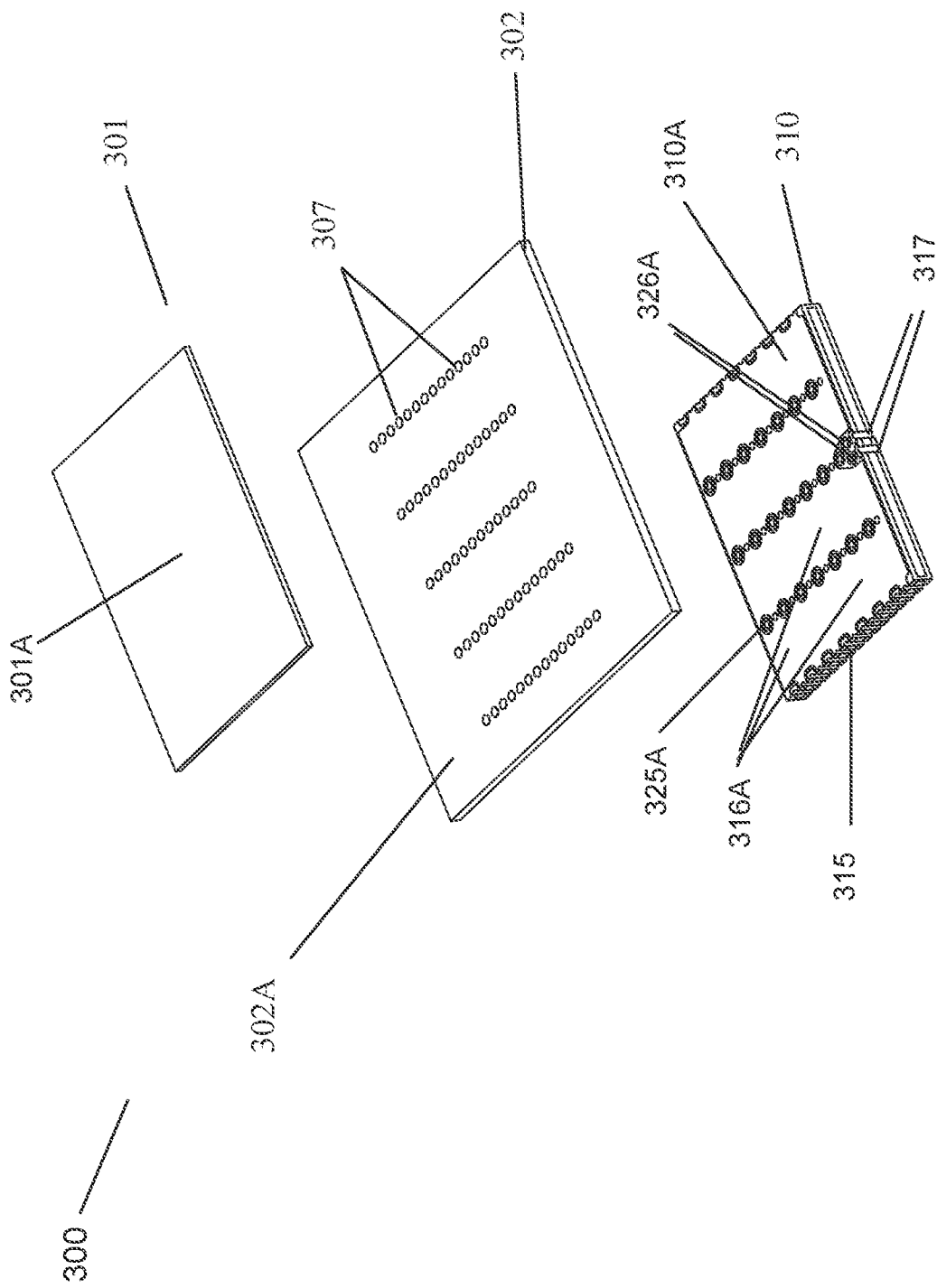


FIG. 15B

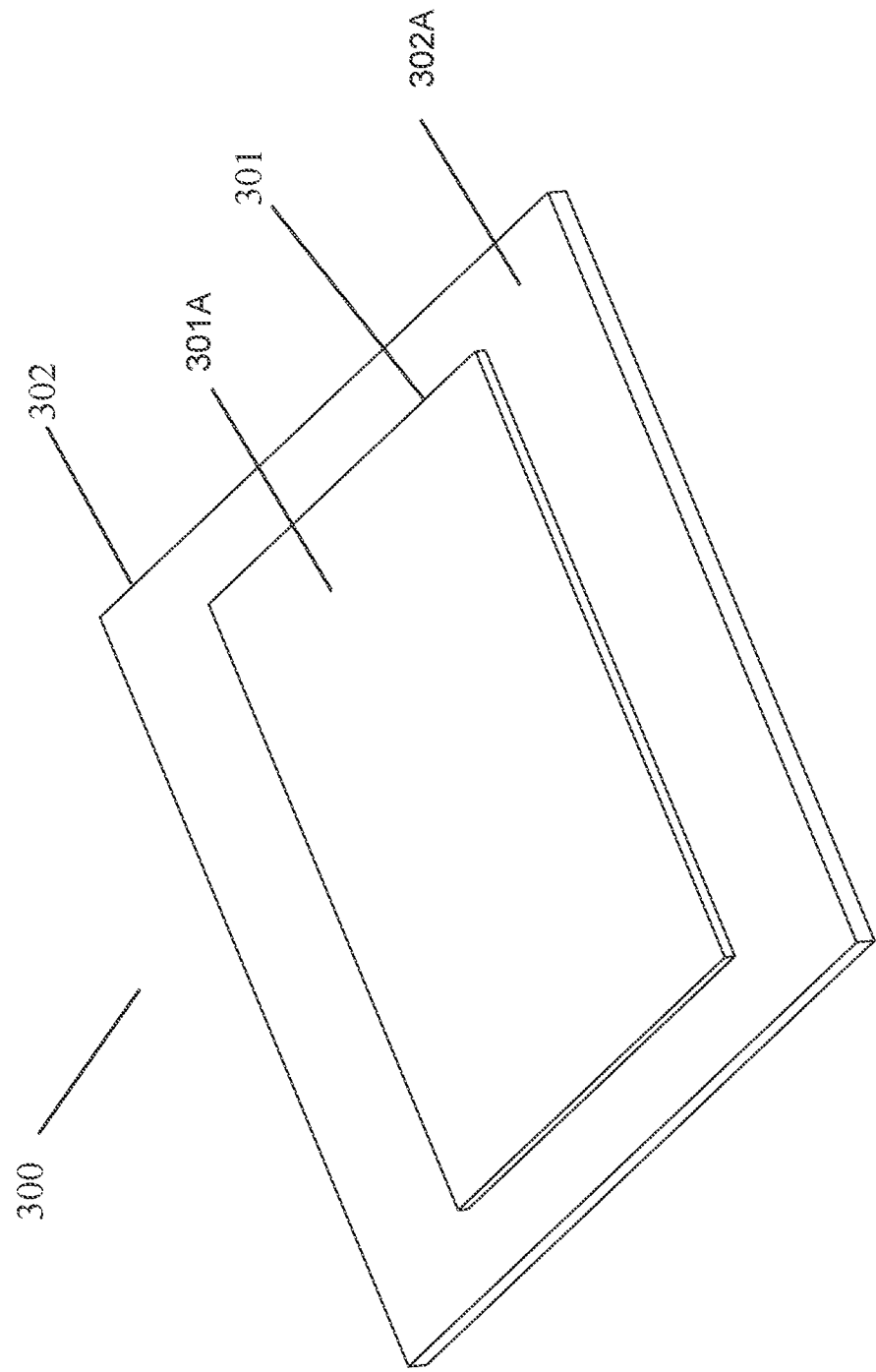


FIG. 16A

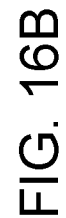
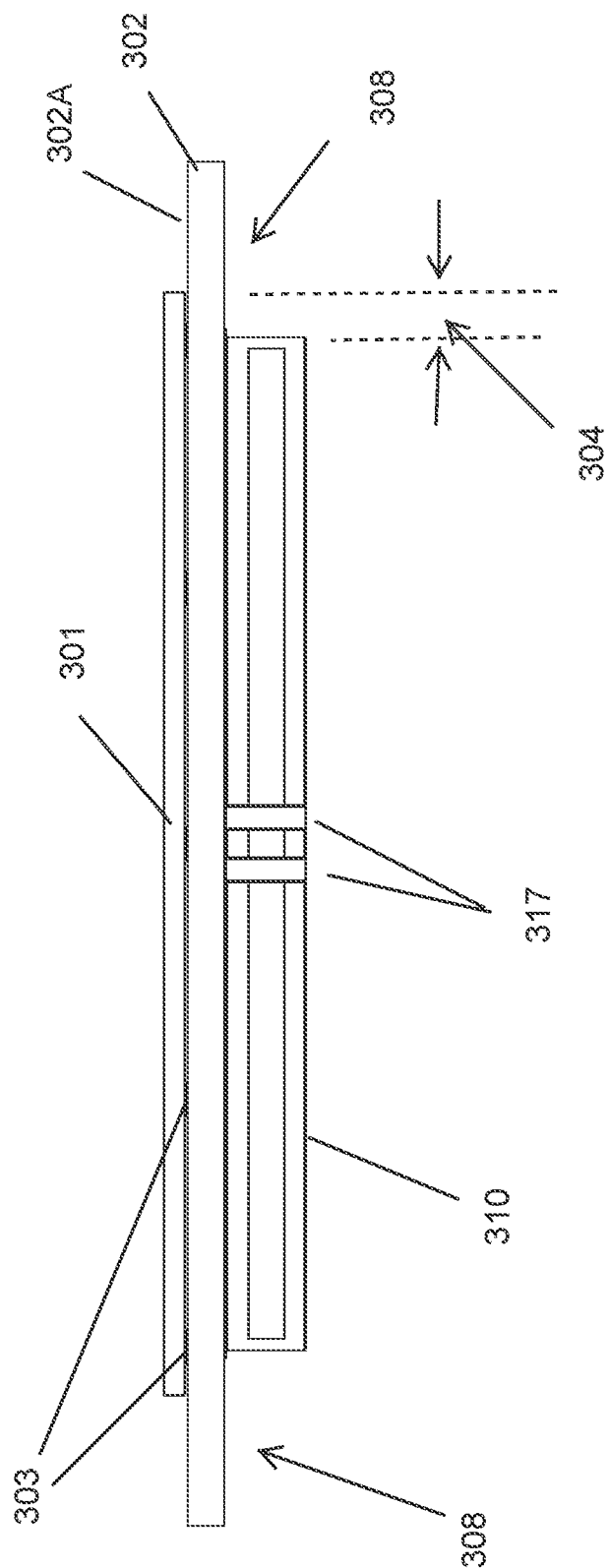


FIG. 16B



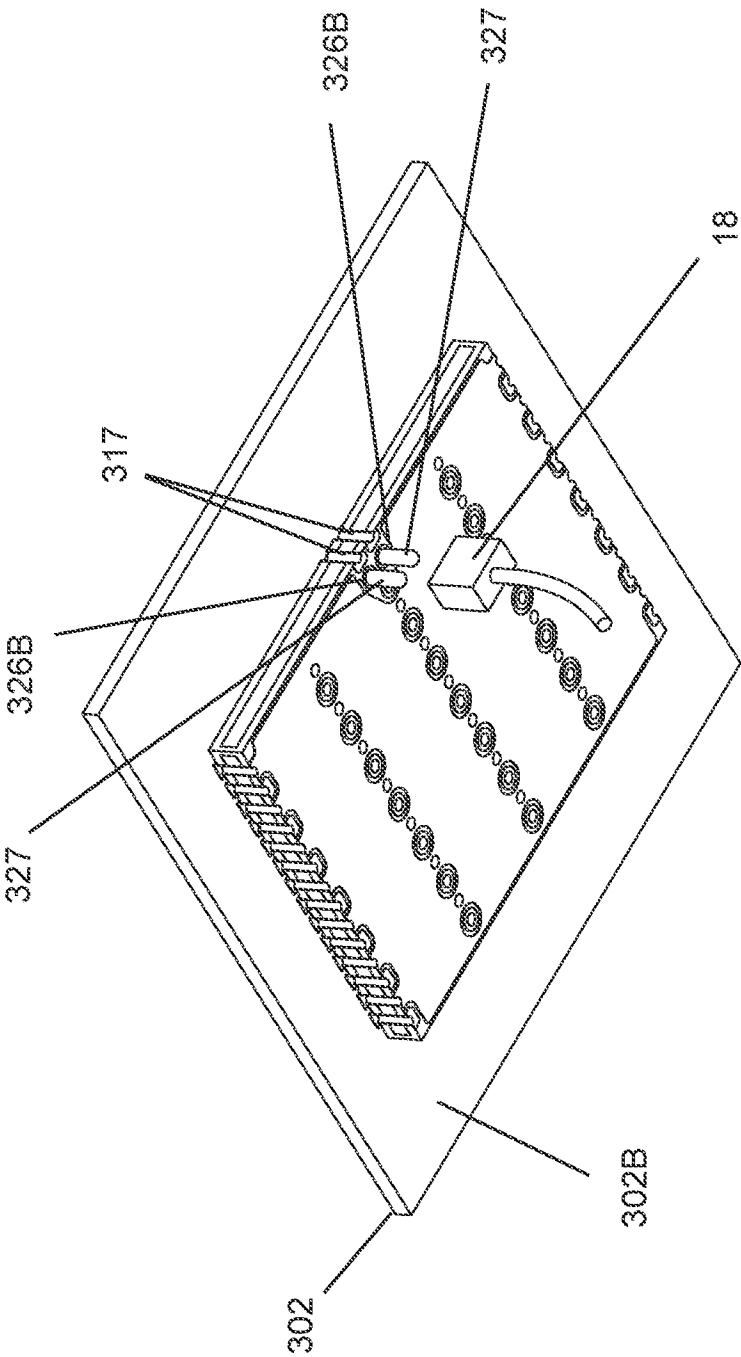


FIG. 18

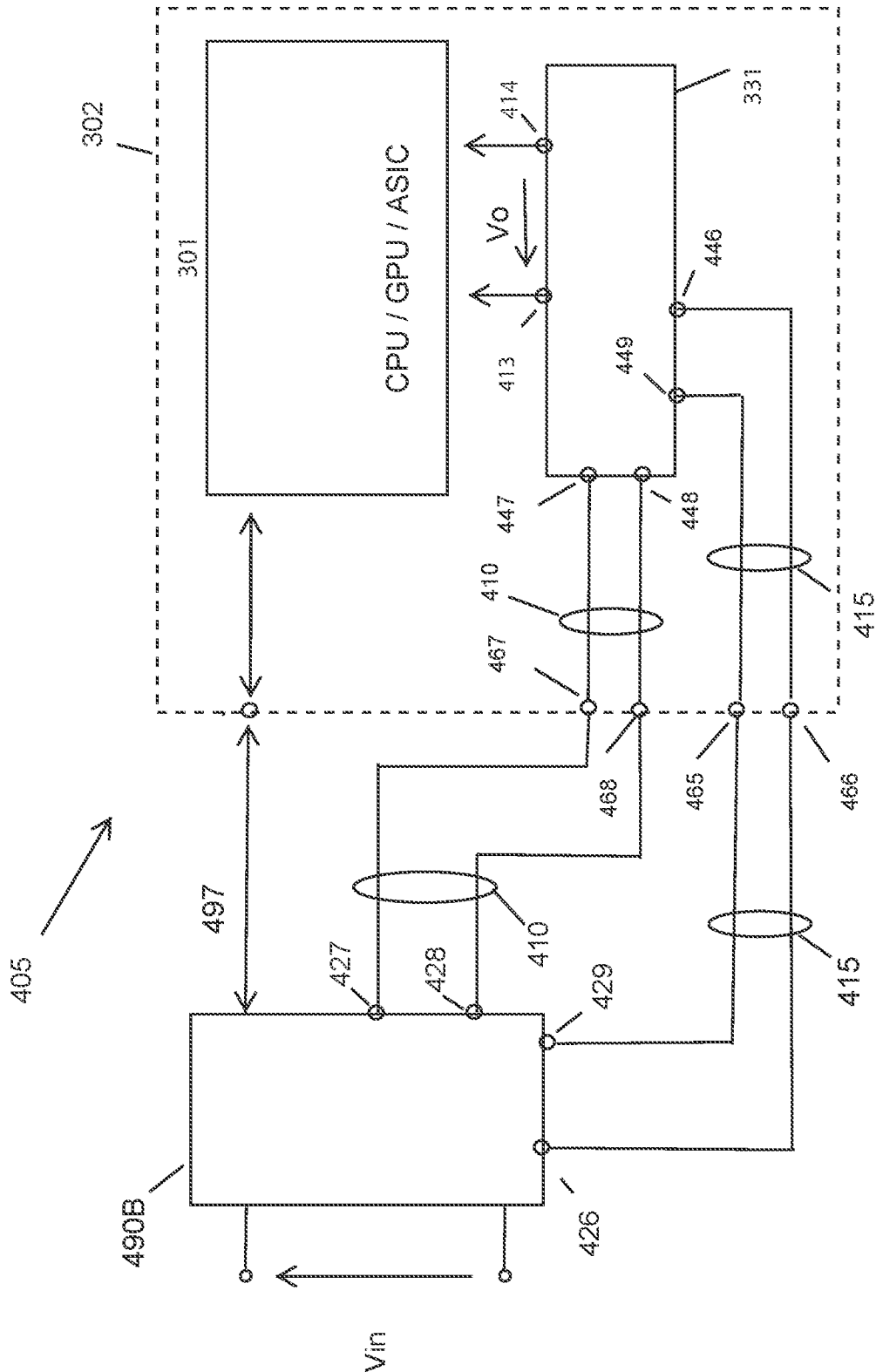


FIG. 19

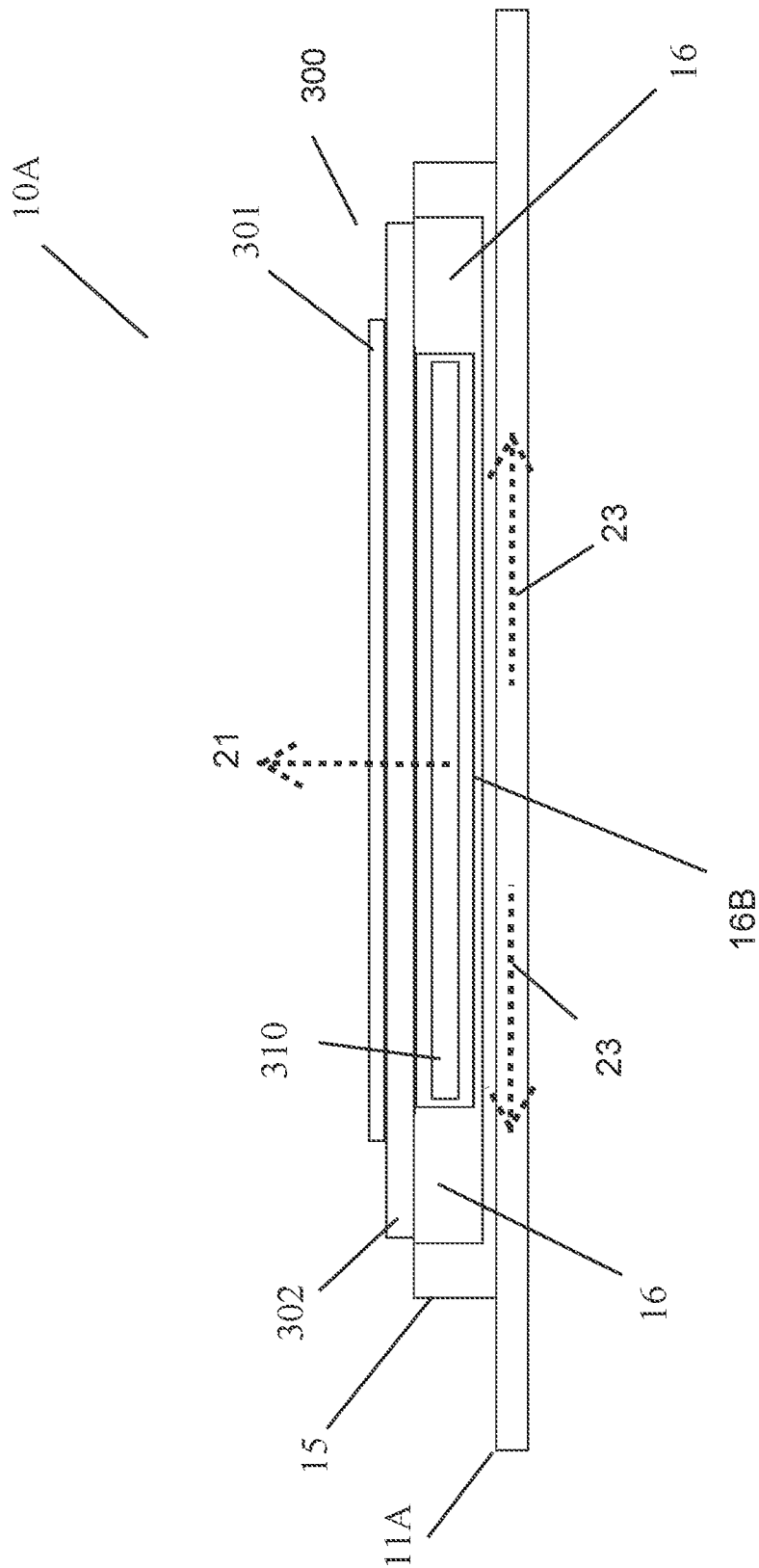


FIG. 20A

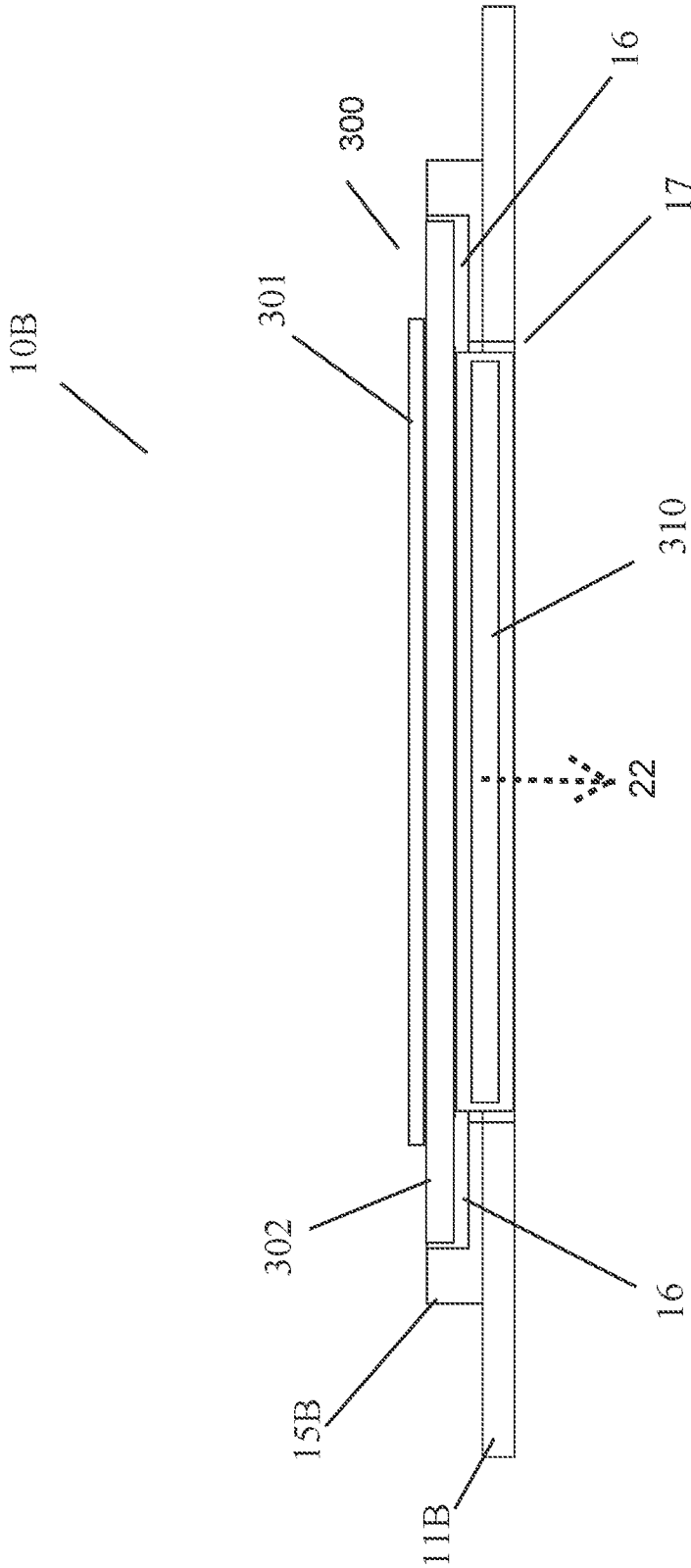


FIG. 20B

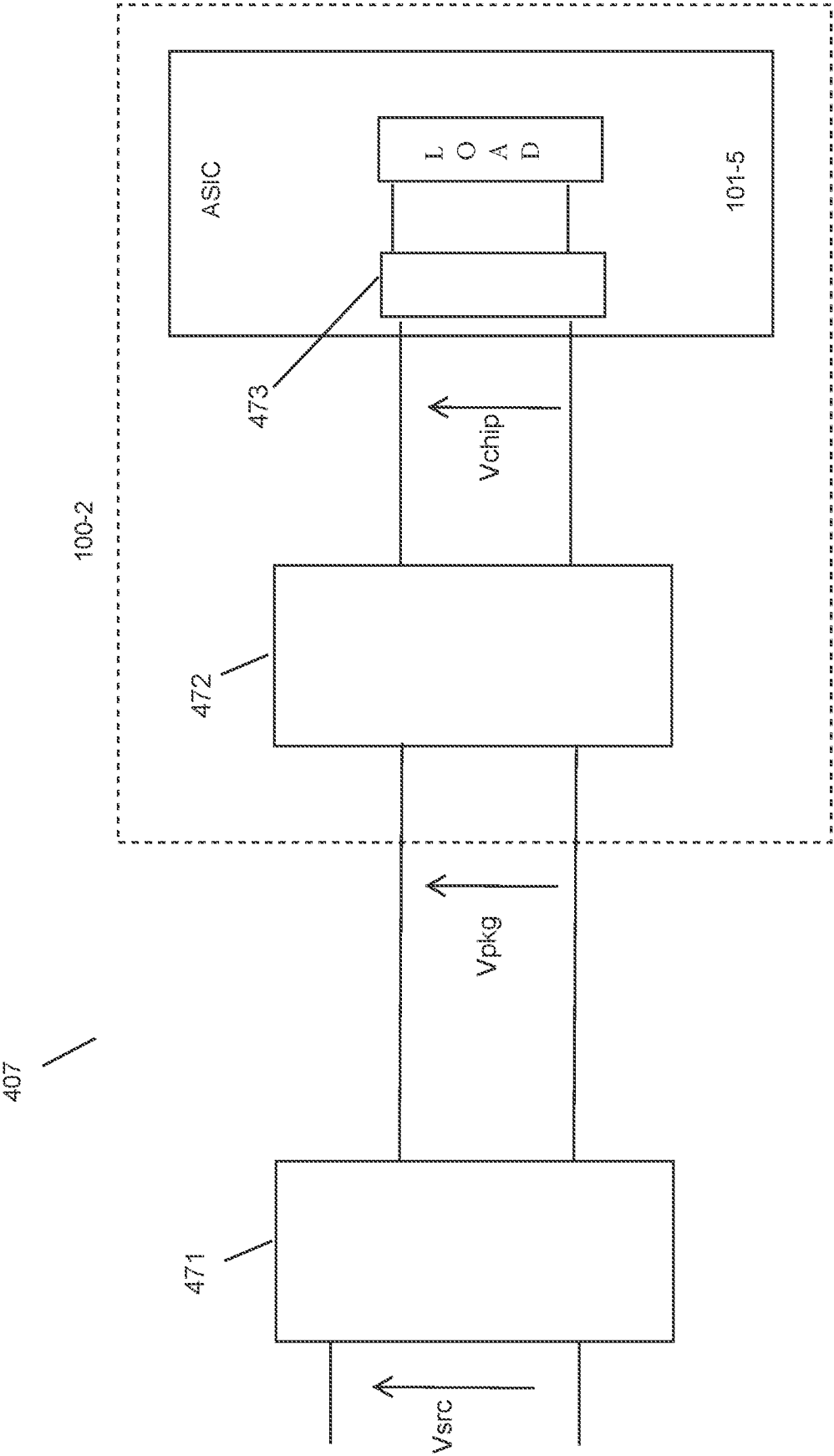


FIG. 21A

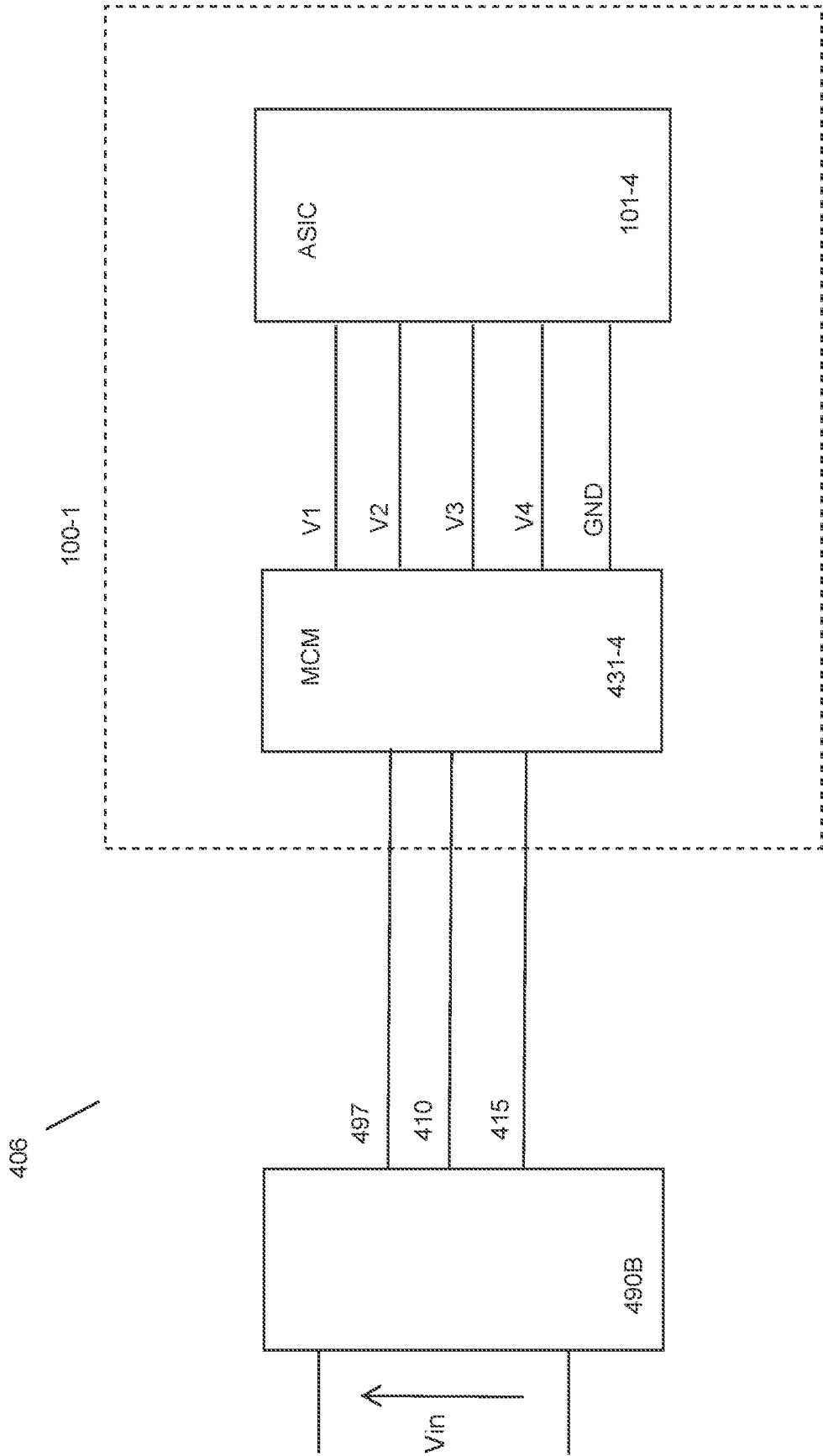


FIG. 21B

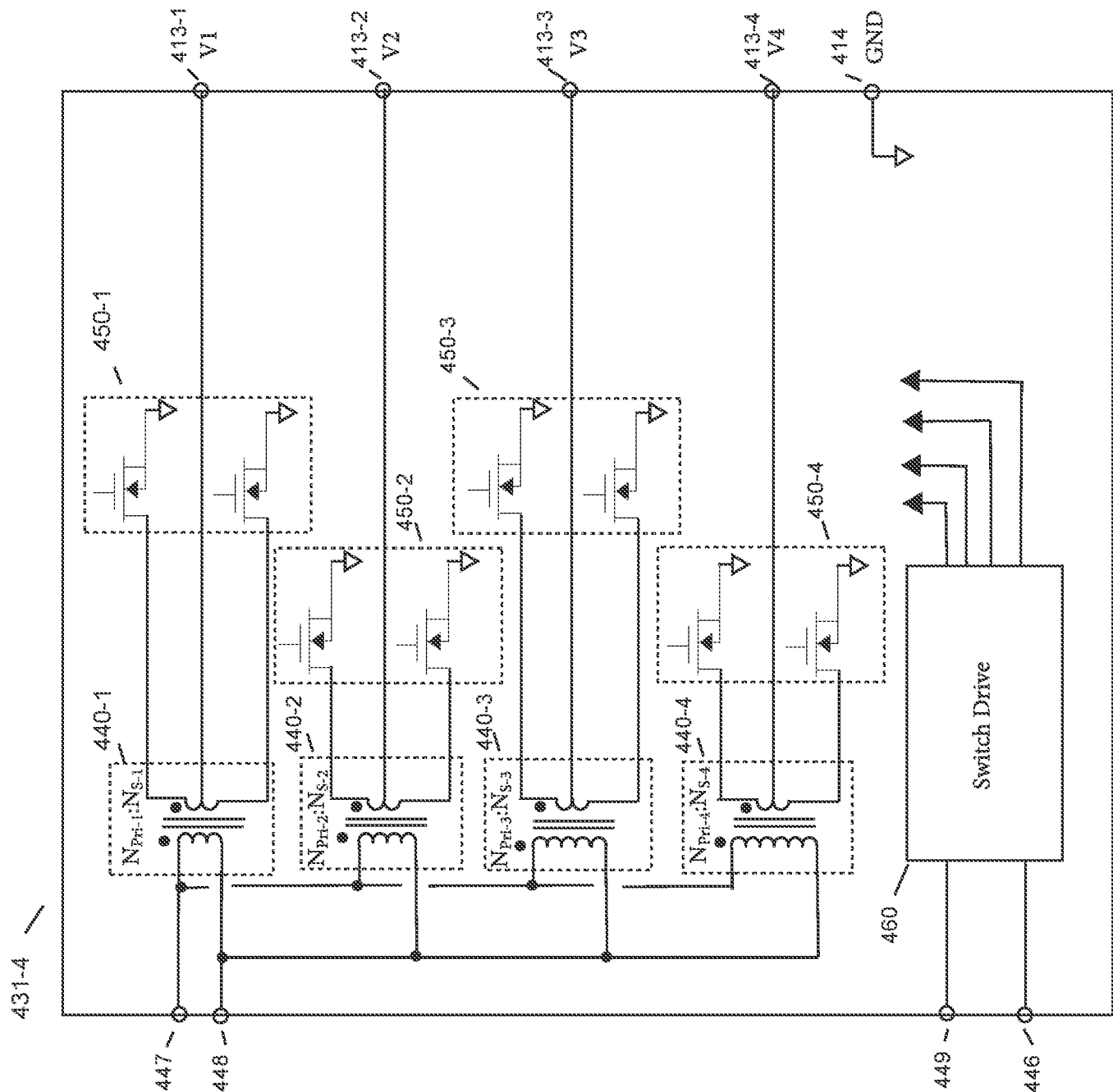


FIG. 22

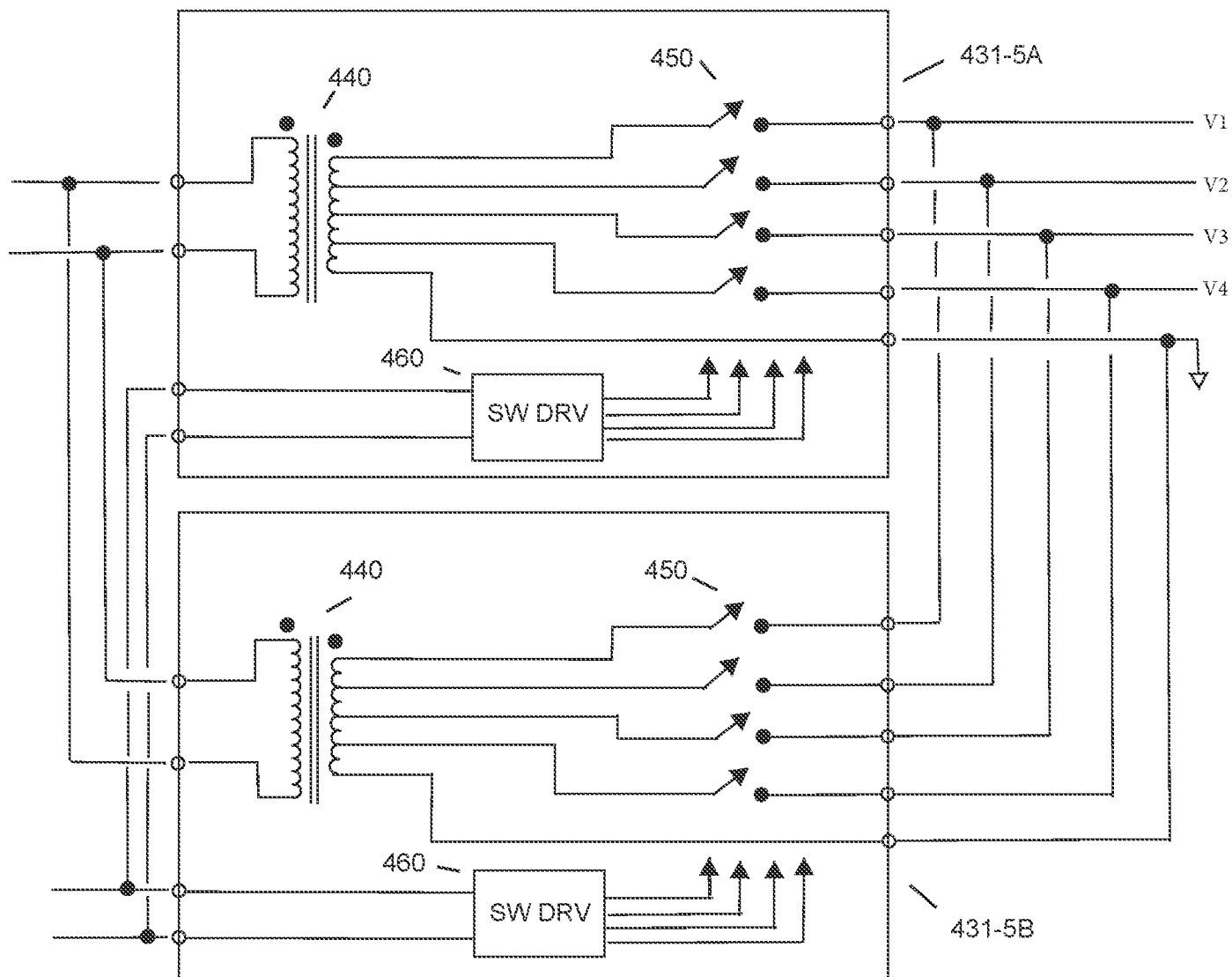


FIG. 23

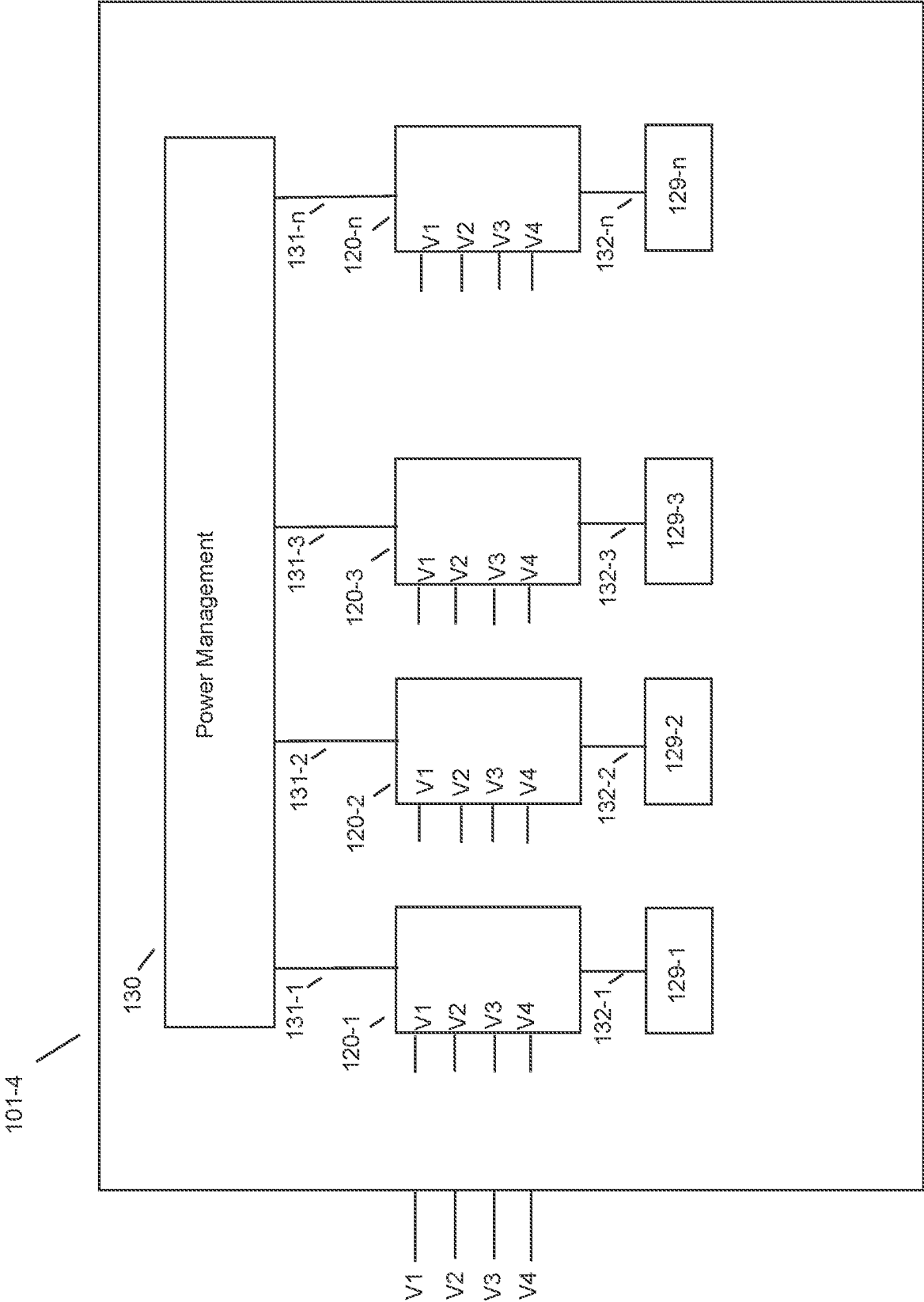


FIG. 24

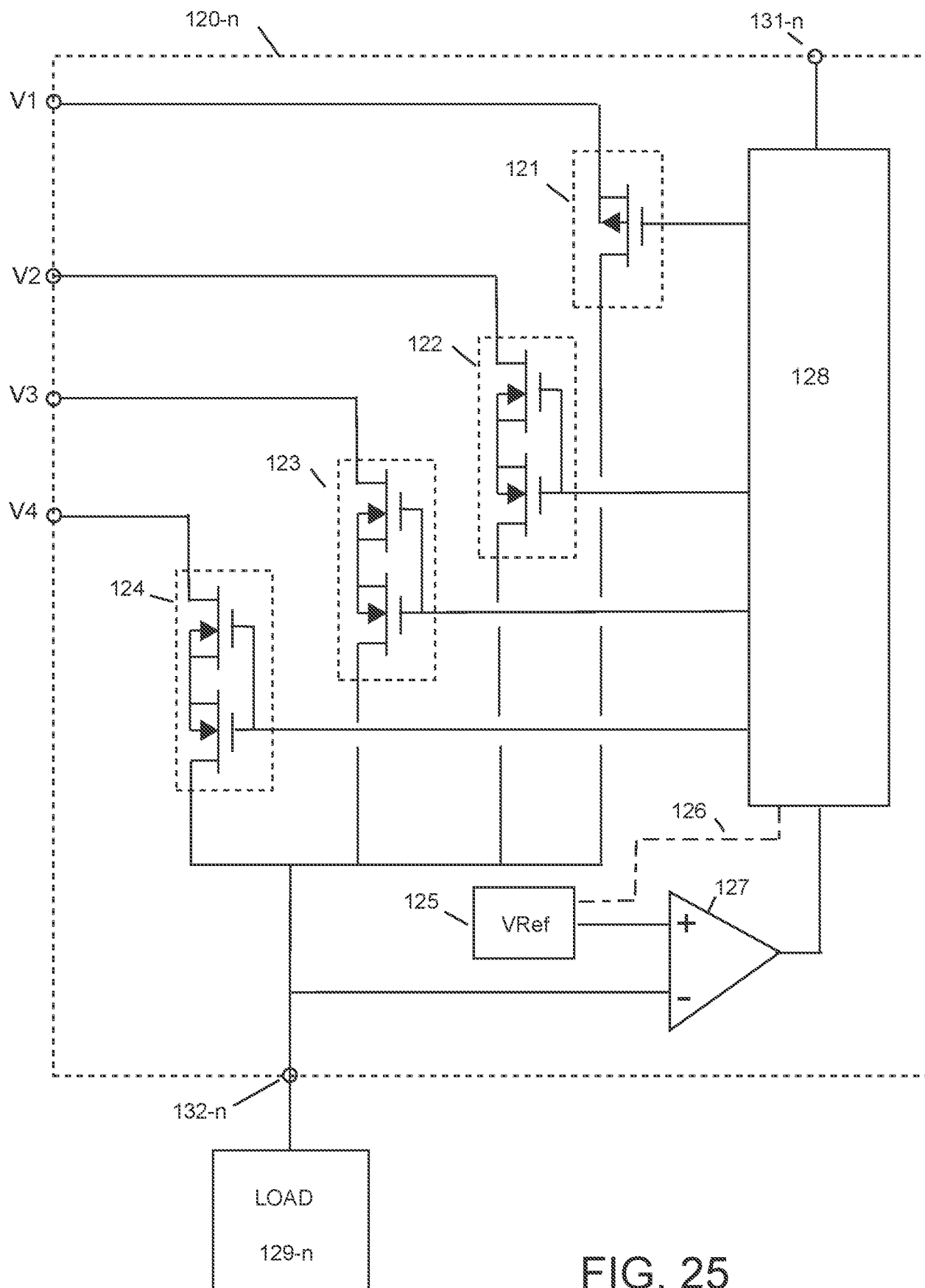


FIG. 25

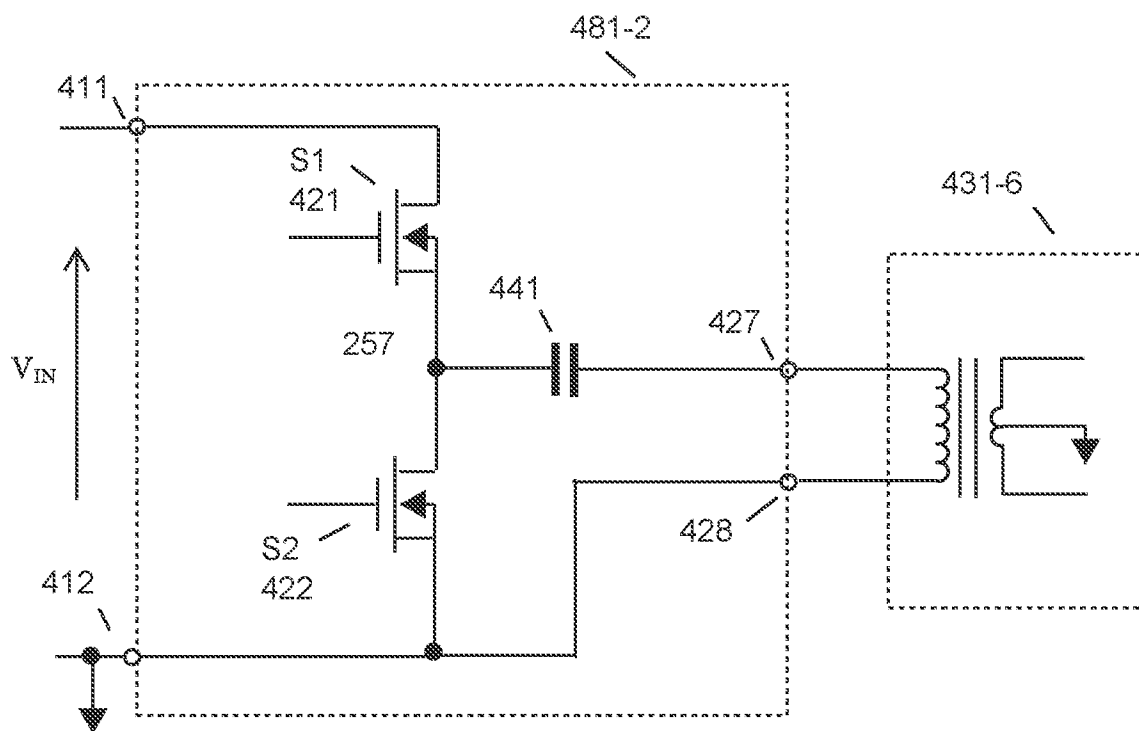


FIG. 26A

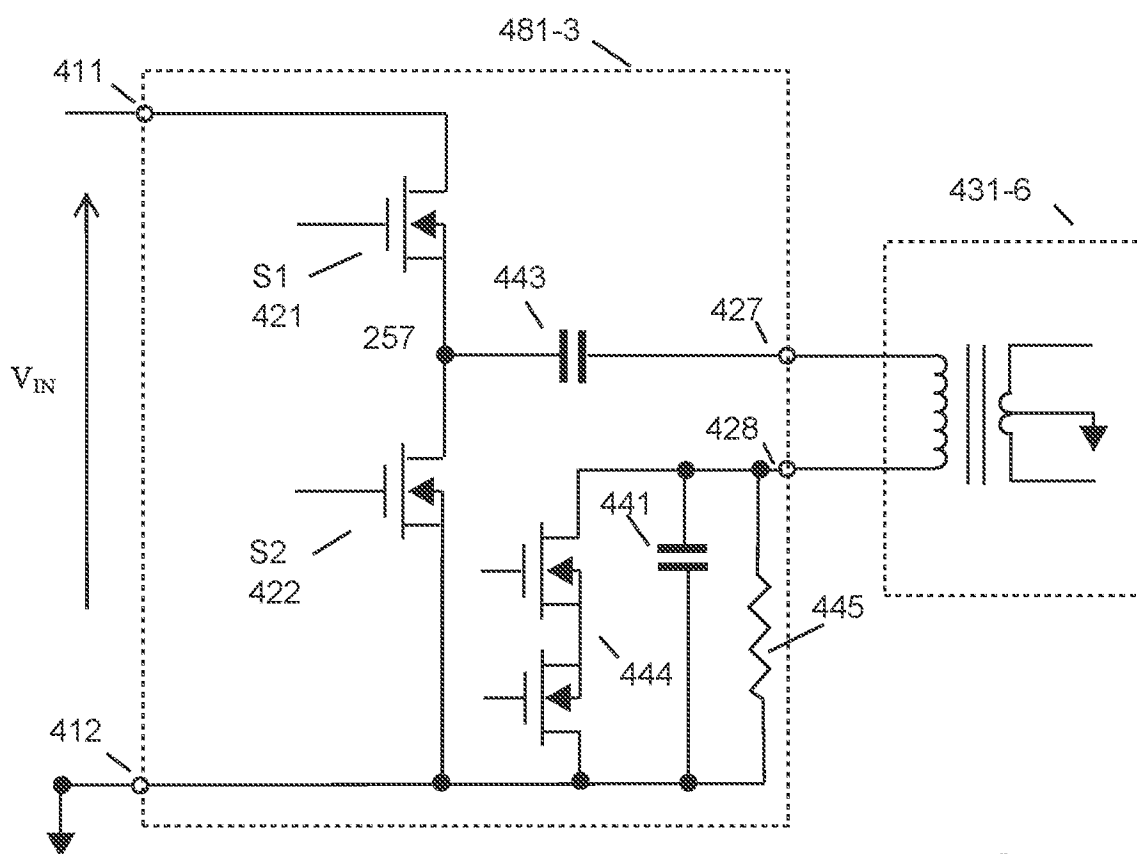


FIG. 26B

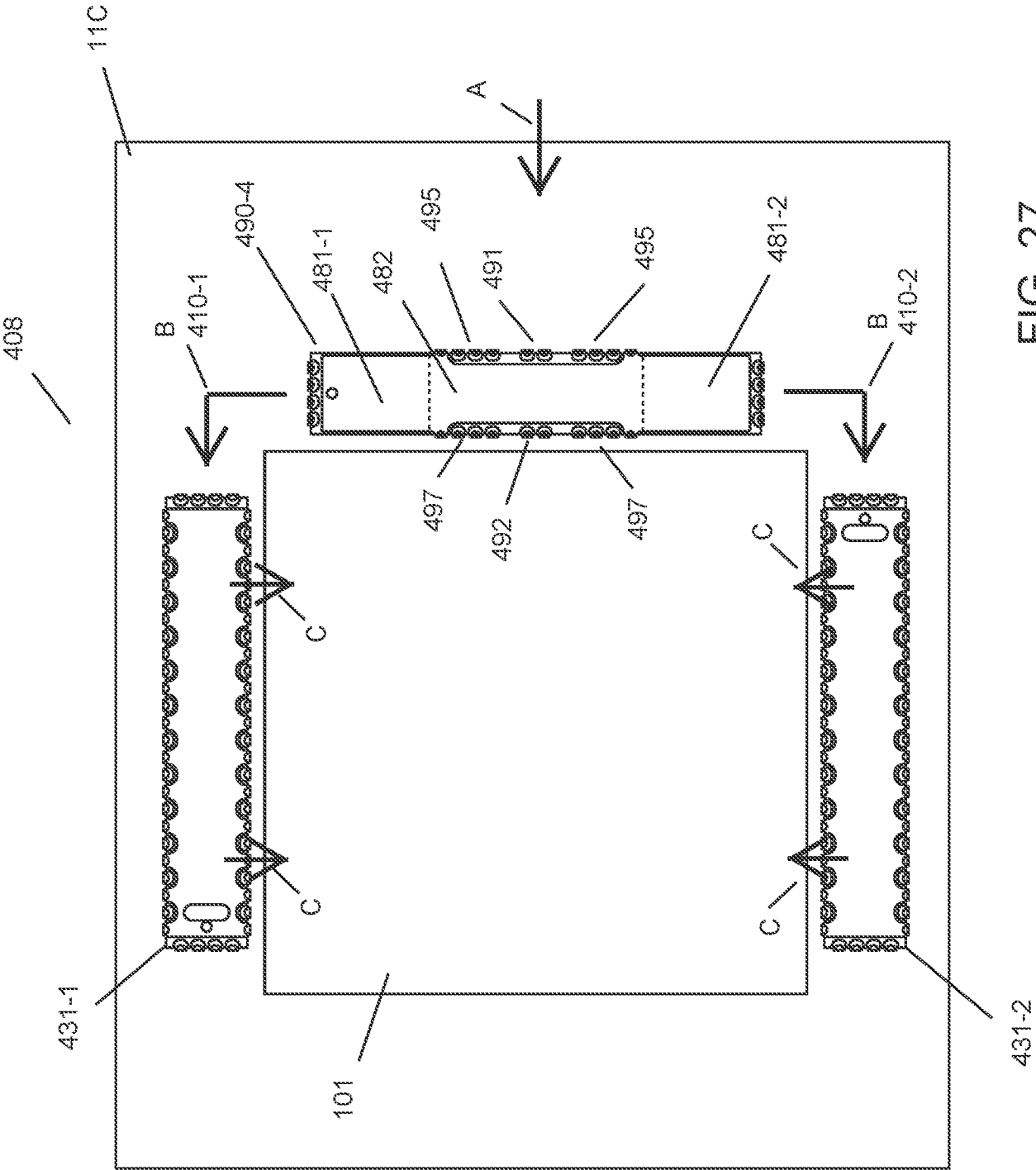


FIG. 27

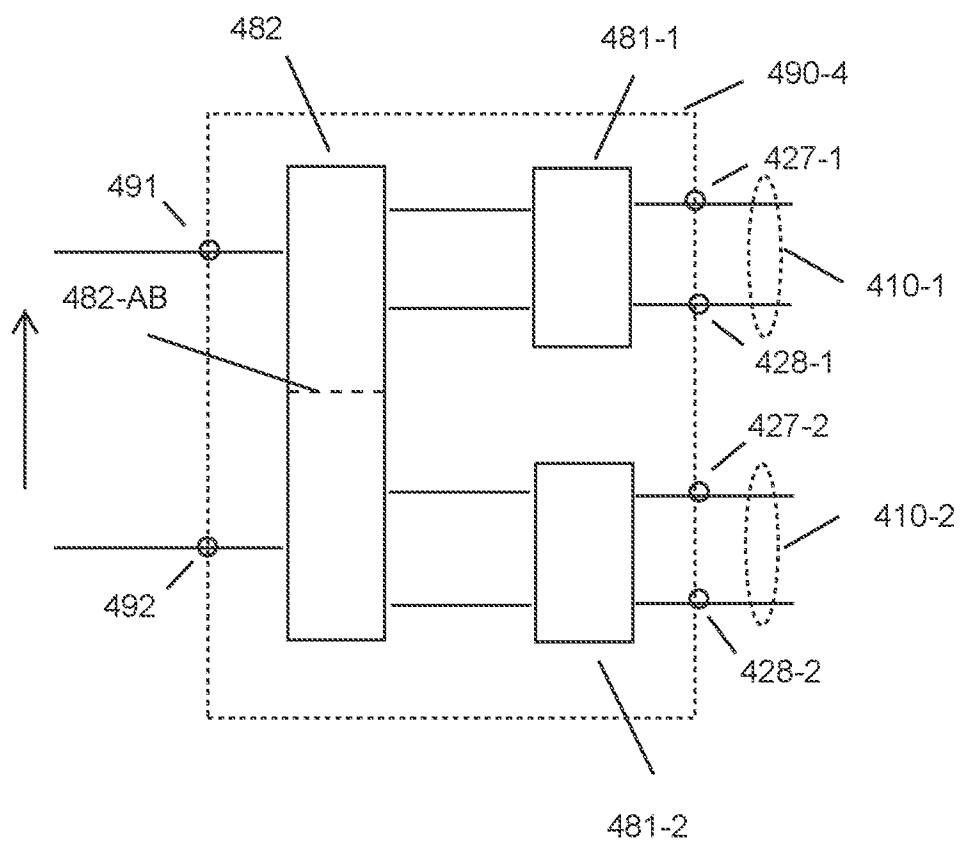


FIG. 28

Electronic Patent Application Fee Transmittal

Application Number:	15616288			
Filing Date:	07-Jun-2017			
Title of Invention:	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS			
First Named Inventor/Applicant Name:	Patrizio Vinciarelli			
Filer:	Rex I. Huang/Rita Liston			
Attorney Docket Number:	00614-0180001			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
UTILITY APPL ISSUE FEE	1501	1	1000	1000

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
PRINTED COPY OF PATENT - NO COLOR	8001	5	3	15
Total in USD (\$)				1015

Electronic Acknowledgement Receipt

EFS ID:	35384664
Application Number:	15616288
International Application Number:	
Confirmation Number:	7438
Title of Invention:	METHOD AND APPARATUS FOR DELIVERING POWER TO SEMICONDUCTORS
First Named Inventor/Applicant Name:	Patrizio Vinciarelli
Customer Number:	26214
Filer:	Rex I. Huang/Elizabeth Doherty
Filer Authorized By:	Rex I. Huang
Attorney Docket Number:	00614-0180001
Receipt Date:	11-MAR-2019
Filing Date:	07-JUN-2017
Time Stamp:	16:43:17
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$ 1015
RAM confirmation Number	031219INTEFSW00003899061050
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1		00614_0180001_Resp_NOA.pdf	236627	yes	3
			a424226ead5cae9a5e4eb6cac627c24ebbc8fc50		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Issue Fee Payment (PTO-85B)		3	3	
	Post Allowance Communication - Incoming		1	2	
Warnings:					
Information:					
2	Drawings-only black and white line drawings	00614_0180001_RevDwgs.pdf	900398	no	30
			2e1d3aa68a0e23e4295fc38a33c31d8b2527bca4		
Warnings:					
Information:					
3	Fee Worksheet (SB06)	fee-info.pdf	31971	no	2
			e44c638021dadcf4524a896f0791387edce62f3de		
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Total Files Size (in bytes):			1168996		

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If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

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If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/616,288	04/30/2019	10277105	00614-0180001	7438

26214 7590 04/10/2019
FISH & RICHARDSON P.C. (VLT, INC)
P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

VLT, Inc., Andover, MA
Patrizio Vinciarelli, Boston, MA;
Andrew T. D'Amico, Beverly Hills, CA;

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To: PATDOCTC@fr.com,,
From: PAIR_eOfficeAction@uspto.gov
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Subject: Private PAIR Correspondence Notification for Customer Number 26214

Apr 11, 2019 03:56:57 AM

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P.O. BOX 1022
MINNEAPOLIS, MN 55440-1022
UNITED STATES

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The official date of notification of the outgoing correspondence will be indicated on the form PTOL-90 accompanying the correspondence.

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Application	Document	Mailroom Date	Attorney Docket No.
15616288	ISSUE.NTF	04/10/2019	00614-0180001

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